Water Tank Controller ? STA Report

RTL-to-GDSII implementation using Verilog HDL

Technology: Sky130
Flow: OpenLane 1.0.2
Run: RUN_2026.10.10_02.11.36
Source: generated OpenLane reports

This document organizes the available implementation timing reports. SS, TT and FF are not claimed unless confirmed by source report data.

Report Scope

The report includes synthesis and physical-design timing artifacts where available, maximum and minimum path reports, timing summaries, checks, power and skew reports, plus signoff artifacts. Report names and contents are retained as the source of truth.

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/cts/13-cts_sta.checks.rpt ? part 1
==========

```
=======================================================================
report_checks -unconstrained
=======================================================================
====================== Typical Corner ===================================

Startpoint: rst_n (input port clocked by clk)
Endpoint: _20_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
Path Type: max

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                0.02      0.01    4.01 ^ rst_n (in)
     1    0.00                           rst_n (net)
                0.02      0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                0.17      0.17    4.18 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.01                           net4 (net)
                0.17      0.00    4.18 ^ _20_/SET_B (sky130_fd_sc_hd__dfstp_1)
                                  4.18   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock source latency
                0.06      0.03   20.03 ^ clk (in)
     1    0.01                           clk (net)
                0.06      0.00   20.03 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                0.04      0.12   20.16 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                0.04      0.00   20.16 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                0.03      0.10   20.26 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                0.03      0.00   20.26 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_1)
                         -0.25   20.01   clock uncertainty
                          0.00   20.01   clock reconvergence pessimism
                          0.12   20.13   library recovery time
                                 20.13   data required time
---------------------------------------------------------------------------
                                 20.13   data required time
                                 -4.18   data arrival time
---------------------------------------------------------------------------
                                 15.95   slack (MET)
```

```
Startpoint: level_full (input port clocked by clk)
Endpoint: pump_en (output port clocked by clk)
Path Group: clk
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/cts/13-cts_sta.checks.rpt ? part 2
==

Path Type: max

```
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 v input external delay
                   0.01   0.00    4.00 v level_full (in)
     1    0.00                           level_full (net)
                   0.01   0.00    4.00 v input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
                   0.08   0.17    4.18 v input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
     3    0.01                           net2 (net)
                   0.08   0.00    4.18 v _09_/A2 (sky130_fd_sc_hd__a21oi_2)
                   0.18   0.23    4.40 ^ _09_/Y (sky130_fd_sc_hd__a21oi_2)
     4    0.01                           _03_ (net)
                   0.18   0.00    4.40 ^ _18_/B (sky130_fd_sc_hd__and2_1)
                   0.05   0.15    4.56 ^ _18_/X (sky130_fd_sc_hd__and2_1)
     1    0.00                           _08_ (net)
                   0.05   0.00    4.56 ^ _19_/A (sky130_fd_sc_hd__clkbuf_1)
                   0.05   0.09    4.65 ^ _19_/X (sky130_fd_sc_hd__clkbuf_1)
     1    0.00                           net6 (net)
                   0.05   0.00    4.65 ^ output6/A (sky130_fd_sc_hd__clkbuf_4)
                   0.11   0.19    4.84 ^ output6/X (sky130_fd_sc_hd__clkbuf_4)
     1    0.03                           pump_en (net)
                   0.11   0.00    4.84 ^ pump_en (out)
                                  4.84   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock network delay (propagated)
                         -0.25   19.75   clock uncertainty
                          0.00   19.75   clock reconvergence pessimism
                         -4.00   15.75   output external delay
                                 15.75   data required time
---------------------------------------------------------------------------
                                 15.75   data required time
                                 -4.84   data arrival time
---------------------------------------------------------------------------
                                 10.91   slack (MET)
```

```
===========================================================================
report_checks --slack_max -0.01
===========================================================================
```

========== Typical Corner ==========

No paths found.

```
================================================================================


================================================================================
 report_check_types -max_slew -max_cap -max_fanout -violators
================================================================================
======================== Typical Corner ====================================



================================================================================
report_parasitic_annotation -report_unannotated
================================================================================
Found 28 unannotated drivers.
 clk
 dry_run_fault
 level_full
 level_low
 rst_n
 _09_/Y
 _10_/Y
 _11_/Y
 _12_/X
 _13_/Y
 _14_/Y
 _15_/Y
 _16_/Y
 _17_/X
 _18_/X
 _19_/X
 _20_/Q
 _21_/Q
 _22_/Q
 clkbuf_0_clk/X
 clkbuf_1_0__f_clk/X
 clkbuf_1_1__f_clk/X
 input1/X
 input2/X
 input3/X
 input4/X
 output5/X
 output6/X
Found 0 partially unannotated drivers.

================================================================================
max slew violation count 0
```

```
max fanout violation count 0
max cap violation count 0
================================================================================
```

==========

==========
check_setup -verbose -unconstrained_endpoints -multiple_clock -no_clock -no_input_delay
-loops -generated_clocks
==========

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/cts/13-cts_sta.max.rpt ? part 1
==========

```
============================================================================
report_checks -path_delay max (Setup)
============================================================================
======================== Typical Corner ===================================

Startpoint: level_full (input port clocked by clk)
Endpoint: pump_en (output port clocked by clk)
Path Group: clk
Path Type: max

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 v input external delay
                   0.01   0.00    4.00 v level_full (in)
     1    0.00                           level_full (net)
                   0.01   0.00    4.00 v input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
                   0.08   0.17    4.18 v input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
     3    0.01                           net2 (net)
                   0.08   0.00    4.18 v _09_/A2 (sky130_fd_sc_hd__a21oi_2)
                   0.18   0.23    4.40 ^ _09_/Y (sky130_fd_sc_hd__a21oi_2)
     4    0.01                           _03_ (net)
                   0.18   0.00    4.40 ^ _18_/B (sky130_fd_sc_hd__and2_1)
                   0.05   0.15    4.56 ^ _18_/X (sky130_fd_sc_hd__and2_1)
     1    0.00                           _08_ (net)
                   0.05   0.00    4.56 ^ _19_/A (sky130_fd_sc_hd__clkbuf_1)
                   0.05   0.09    4.65 ^ _19_/X (sky130_fd_sc_hd__clkbuf_1)
     1    0.00                           net6 (net)
                   0.05   0.00    4.65 ^ output6/A (sky130_fd_sc_hd__clkbuf_4)
                   0.11   0.19    4.84 ^ output6/X (sky130_fd_sc_hd__clkbuf_4)
     1    0.03                           pump_en (net)
                   0.11   0.00    4.84 ^ pump_en (out)
                          4.84    4.84   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock network delay (propagated)
                         -0.25   19.75   clock uncertainty
                          0.00   19.75   clock reconvergence pessimism
                         -4.00   15.75   output external delay
                                 15.75   data required time
---------------------------------------------------------------------------
                                 15.75   data required time
                                 -4.84   data arrival time
---------------------------------------------------------------------------
```

10.91 slack (MET)

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/cts/13-cts_sta.max.rpt ? part 2
==========

Startpoint: level_full (input port clocked by clk)
Endpoint: alarm (output port clocked by clk)
Path Group: clk
Path Type: max

Fanout	Cap	Slew	Delay	Time		Description
			0.00	0.00		clock clk (rise edge)
			0.00	0.00		clock network delay (propagated)
			4.00	4.00	v	input external delay
		0.01	0.00	4.00	v	level_full (in)
1	0.00					level_full (net)
		0.01	0.00	4.00	v	input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
		0.08	0.17	4.18	v	input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
3	0.01					net2 (net)
		0.08	0.00	4.18	v	_09_/A2 (sky130_fd_sc_hd__a21oi_2)
		0.18	0.23	4.40	^	_09_/Y (sky130_fd_sc_hd__a21oi_2)
4	0.01					_03_ (net)
		0.18	0.00	4.40	^	_10_/A (sky130_fd_sc_hd__inv_2)
		0.04	0.05	4.46	v	_10_/Y (sky130_fd_sc_hd__inv_2)
2	0.01					net5 (net)
		0.04	0.00	4.46	v	output5/A (sky130_fd_sc_hd__buf_2)
		0.09	0.19	4.65	v	output5/X (sky130_fd_sc_hd__buf_2)
1	0.03					alarm (net)
		0.09	0.00	4.65	v	alarm (out)
				4.65		data arrival time
			20.00	20.00		clock clk (rise edge)
			0.00	20.00		clock network delay (propagated)
			-0.25	19.75		clock uncertainty
			0.00	19.75		clock reconvergence pessimism
			-4.00	15.75		output external delay
				15.75		data required time
				15.75		data required time
				-4.65		data arrival time
				11.10		slack (MET)

Startpoint: level_full (input port clocked by clk)
Endpoint: _21_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

Fanout	Cap	Slew	Delay	Time	Description
			0.00	0.00	clock clk (rise edge)

```
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 v input external delay
                 0.01     0.00    4.00 v level_full (in)
     1    0.00                           level_full (net)
                 0.01     0.00    4.00 v input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
                 0.08     0.17    4.18 v input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
     3    0.01                           net2 (net)
                 0.08     0.00    4.18 v _09_/A2 (sky130_fd_sc_hd__a21oi_2)
                 0.18     0.23    4.40 ^ _09_/Y (sky130_fd_sc_hd__a21oi_2)
     4    0.01                           _03_ (net)
                 0.18     0.00    4.40 ^ _17_/B1 (sky130_fd_sc_hd__o211a_1)
                 0.04     0.18    4.58 ^ _17_/X (sky130_fd_sc_hd__o211a_1)
     1    0.00                           _01_ (net)
                 0.04     0.00    4.58 ^ _21_/D (sky130_fd_sc_hd__dfrtp_1)
                          4.58    4.58   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock source latency
                 0.06     0.03   20.03 ^ clk (in)
     1    0.01                           clk (net)
                 0.06     0.00   20.03 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                 0.04     0.12   20.16 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                 0.04     0.00   20.16 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                 0.03     0.10   20.26 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                 0.03     0.00   20.26 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                         -0.25   20.01   clock uncertainty
                          0.00   20.01   clock reconvergence pessimism
                         -0.06   19.95   library setup time
                                 19.95   data required time
-----------------------------------------------------------------------------
                                 19.95   data required time
                                 -4.58   data arrival time
-----------------------------------------------------------------------------
                                 15.37   slack (MET)


Startpoint: level_full (input port clocked by clk)
Endpoint: _22_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
```

```
Fanout     Cap     Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 v input external delay
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/cts/13-cts_sta.max.rpt ? part 4
==========

```
                  0.01    0.00    4.00 v level_full (in)
     1    0.00                           level_full (net)
                  0.01    0.00    4.00 v input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
                  0.08    0.17    4.18 v input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
     3    0.01                           net2 (net)
                  0.08    0.00    4.18 v _09_/A2 (sky130_fd_sc_hd__a21oi_2)
                  0.18    0.23    4.40 ^ _09_/Y (sky130_fd_sc_hd__a21oi_2)
     4    0.01                           _03_ (net)
                  0.18    0.00    4.40 ^ _10_/A (sky130_fd_sc_hd__inv_2)
                  0.04    0.05    4.46 v _10_/Y (sky130_fd_sc_hd__inv_2)
     2    0.01                           net5 (net)
                  0.04    0.00    4.46 v _14_/A (sky130_fd_sc_hd__nor2_1)
                  0.07    0.10    4.56 ^ _14_/Y (sky130_fd_sc_hd__nor2_1)
     1    0.00                           _02_ (net)
                  0.07    0.00    4.56 ^ _22_/D (sky130_fd_sc_hd__dfrtp_1)
                                  4.56   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock source latency
                  0.06    0.03   20.03 ^ clk (in)
     1    0.01                           clk (net)
                  0.06    0.00   20.03 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.04    0.12   20.16 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                  0.04    0.00   20.16 ^ clkbuf_1_1__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.03    0.10   20.26 ^ clkbuf_1_1__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     1    0.00                           clknet_1_1__leaf_clk (net)
                  0.03    0.00   20.26 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_1)
                         -0.25   20.01   clock uncertainty
                          0.00   20.01   clock reconvergence pessimism
                         -0.07   19.94   library setup time
                                 19.94   data required time
-----------------------------------------------------------------------------
                                 19.94   data required time
                                 -4.56   data arrival time
-----------------------------------------------------------------------------
                                 15.38   slack (MET)


Startpoint: level_full (input port clocked by clk)
Endpoint: _20_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
```

```
Fanout     Cap     Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
```

==========

```
                          4.00    4.00 v input external delay
                 0.01     0.00    4.00 v level_full (in)
     1    0.00                           level_full (net)
                 0.01     0.00    4.00 v input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
                 0.08     0.17    4.18 v input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
     3    0.01                           net2 (net)
                 0.08     0.00    4.18 v _09_/A2 (sky130_fd_sc_hd__a21oi_2)
                 0.18     0.23    4.40 ^ _09_/Y (sky130_fd_sc_hd__a21oi_2)
     4    0.01                           _03_ (net)
                 0.18     0.00    4.40 ^ _16_/B1 (sky130_fd_sc_hd__o21ai_1)
                 0.06     0.09    4.50 v _16_/Y (sky130_fd_sc_hd__o21ai_1)
     1    0.00                           _00_ (net)
                 0.06     0.00    4.50 v _20_/D (sky130_fd_sc_hd__dfstp_1)
                                  4.50   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock source latency
                 0.06     0.03   20.03 ^ clk (in)
     1    0.01                           clk (net)
                 0.06     0.00   20.03 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                 0.04     0.12   20.16 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                 0.04     0.00   20.16 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                 0.03     0.10   20.26 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                 0.03     0.00   20.26 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_1)
                         -0.25   20.01   clock uncertainty
                          0.00   20.01   clock reconvergence pessimism
                         -0.08   19.93   library setup time
                                 19.93   data required time
-----------------------------------------------------------------------------
                                 19.93   data required time
                                 -4.50   data arrival time
-----------------------------------------------------------------------------
                                 15.43   slack (MET)


Startpoint: rst_n (input port clocked by clk)
Endpoint: _20_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
Path Type: max

Fanout     Cap    Slew   Delay    Time   Description
```

```
-------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                 0.02     0.01    4.01 ^ rst_n (in)
```

==========

```
     1    0.00                           rst_n (net)
                0.02    0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                0.17    0.17    4.18 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.01                           net4 (net)
                0.17    0.00    4.18 ^ _20_/SET_B (sky130_fd_sc_hd__dfstp_1)
                                4.18   data arrival time

                       20.00   20.00   clock clk (rise edge)
                        0.00   20.00   clock source latency
                0.06    0.03   20.03 ^ clk (in)
     1    0.01                           clk (net)
                0.06    0.00   20.03 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                0.04    0.12   20.16 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                0.04    0.00   20.16 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                0.03    0.10   20.26 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                0.03    0.00   20.26 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_1)
                       -0.25   20.01   clock uncertainty
                        0.00   20.01   clock reconvergence pessimism
                        0.12   20.13   library recovery time
                               20.13   data required time
-----------------------------------------------------------------------------
                               20.13   data required time
                               -4.18   data arrival time
-----------------------------------------------------------------------------
                               15.95   slack (MET)


Startpoint: rst_n (input port clocked by clk)
Endpoint: _22_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
Path Type: max

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                        0.00    0.00   clock clk (rise edge)
                        0.00    0.00   clock network delay (propagated)
                        4.00    4.00 ^ input external delay
                0.02    0.01    4.01 ^ rst_n (in)
     1    0.00                           rst_n (net)
                0.02    0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                0.17    0.17    4.18 ^ input4/X (sky130_fd_sc_hd__buf_1)
```

```
3    0.01                           net4 (net)
             0.17    0.00    4.18 ^ _22_/RESET_B (sky130_fd_sc_hd__dfrtp_1)
                             4.18   data arrival time

                    20.00   20.00   clock clk (rise edge)
```

```
                          0.00   20.00   clock source latency
                 0.06     0.03   20.03 ^ clk (in)
     1    0.01                          clk (net)
                 0.06     0.00   20.03 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                 0.04     0.12   20.16 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                          clknet_0_clk (net)
                 0.04     0.00   20.16 ^ clkbuf_1_1__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                 0.03     0.10   20.26 ^ clkbuf_1_1__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     1    0.00                          clknet_1_1__leaf_clk (net)
                 0.03     0.00   20.26 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_1)
                         -0.25   20.01   clock uncertainty
                          0.00   20.01   clock reconvergence pessimism
                          0.19   20.20   library recovery time
                                 20.20   data required time
-------------------------------------------------------------------------
                                 20.20   data required time
                                 -4.18   data arrival time
-------------------------------------------------------------------------
                                 16.02   slack (MET)


Startpoint: rst_n (input port clocked by clk)
Endpoint: _21_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
Path Type: max

Fanout     Cap    Slew   Delay    Time   Description
-------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                 0.02     0.01    4.01 ^ rst_n (in)
     1    0.00                          rst_n (net)
                 0.02     0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                 0.17     0.17    4.18 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.01                          net4 (net)
                 0.17     0.00    4.18 ^ _21_/RESET_B (sky130_fd_sc_hd__dfrtp_1)
                                  4.18   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock source latency
                 0.06     0.03   20.03 ^ clk (in)
     1    0.01                          clk (net)
```

```
                0.06    0.00   20.03 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                0.04    0.12   20.16 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                         clknet_0_clk (net)
                0.04    0.00   20.16 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                0.03    0.10   20.26 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
```

```
     2    0.01                           clknet_1_0__leaf_clk (net)
                  0.03    0.00   20.26 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                         -0.25   20.01   clock uncertainty
                          0.00   20.01   clock reconvergence pessimism
                          0.19   20.20   library recovery time
                                 20.20   data required time
-----------------------------------------------------------------------------
                                 20.20   data required time
                                 -4.18   data arrival time
-----------------------------------------------------------------------------
                                 16.02   slack (MET)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/cts/13-cts_sta.min.rpt ? part 1
==========

```
=======================================================================
report_checks -path_delay min (Hold)
=======================================================================
======================== Typical Corner ===================================

Startpoint: _22_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _21_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                   0.06   0.03    0.03 ^ clk (in)
     1    0.01                           clk (net)
                   0.06   0.00    0.03 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                   0.04   0.12    0.16 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                   0.04   0.00    0.16 ^ clkbuf_1_1__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                   0.03   0.10    0.26 ^ clkbuf_1_1__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     1    0.00                           clknet_1_1__leaf_clk (net)
                   0.03   0.00    0.26 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_1)
                   0.09   0.33    0.59 ^ _22_/Q (sky130_fd_sc_hd__dfrtp_1)
     3    0.01                           state[2] (net)
                   0.09   0.00    0.59 ^ _17_/A2 (sky130_fd_sc_hd__o211a_1)
                   0.04   0.13    0.72 ^ _17_/X (sky130_fd_sc_hd__o211a_1)
     1    0.00                           _01_ (net)
                   0.04   0.00    0.72 ^ _21_/D (sky130_fd_sc_hd__dfrtp_1)
                                  0.72   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                   0.06   0.04    0.04 ^ clk (in)
     1    0.01                           clk (net)
                   0.06   0.00    0.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                   0.04   0.13    0.17 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                   0.04   0.00    0.17 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                   0.03   0.11    0.29 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
```

```
                  0.03    0.00    0.29 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                          0.25    0.54   clock uncertainty
                         -0.02    0.52   clock reconvergence pessimism
                         -0.04    0.49   library hold time
                                  0.49   data required time
-----------------------------------------------------------------------------
                                  0.49   data required time
```

```
================================================================================

                                -0.72   data arrival time
--------------------------------------------------------------------------------
                                 0.23   slack (MET)


Startpoint: _22_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _22_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

Fanout     Cap    Slew   Delay    Time   Description
--------------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                 0.06     0.03    0.03 ^ clk (in)
     1    0.01                           clk (net)
                 0.06     0.00    0.03 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                 0.04     0.12    0.16 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                 0.04     0.00    0.16 ^ clkbuf_1_1__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                 0.03     0.10    0.26 ^ clkbuf_1_1__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     1    0.00                           clknet_1_1__leaf_clk (net)
                 0.03     0.00    0.26 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_1)
                 0.09     0.33    0.59 ^ _22_/Q (sky130_fd_sc_hd__dfrtp_1)
     3    0.01                           state[2] (net)
                 0.09     0.00    0.59 ^ _13_/A2 (sky130_fd_sc_hd__a21oi_1)
                 0.03     0.07    0.65 v _13_/Y (sky130_fd_sc_hd__a21oi_1)
     1    0.00                           _06_ (net)
                 0.03     0.00    0.65 v _14_/B (sky130_fd_sc_hd__nor2_1)
                 0.07     0.07    0.72 ^ _14_/Y (sky130_fd_sc_hd__nor2_1)
     1    0.00                           _02_ (net)
                 0.07     0.00    0.72 ^ _22_/D (sky130_fd_sc_hd__dfrtp_1)
                                  0.72   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                 0.06     0.04    0.04 ^ clk (in)
     1    0.01                           clk (net)
                 0.06     0.00    0.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                 0.04     0.13    0.17 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                 0.04     0.00    0.17 ^ clkbuf_1_1__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
```

```
                0.03    0.11    0.28 ^ clkbuf_1_1__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     1    0.00                       clknet_1_1__leaf_clk (net)
                0.03    0.00    0.28 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_1)
                        0.25    0.53   clock uncertainty
                       -0.03    0.51   clock reconvergence pessimism
                       -0.04    0.46   library hold time
```

```
                                   0.46   data required time
-----------------------------------------------------------------------------
                                   0.46   data required time
                                  -0.72   data arrival time
-----------------------------------------------------------------------------
                                   0.26   slack (MET)


Startpoint: _20_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _20_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                  0.06    0.03    0.03 ^ clk (in)
     1    0.01                          clk (net)
                  0.06    0.00    0.03 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.04    0.12    0.16 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                          clknet_0_clk (net)
                  0.04    0.00    0.16 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.03    0.10    0.26 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                          clknet_1_0__leaf_clk (net)
                  0.03    0.00    0.26 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_1)
                  0.04    0.33    0.59 v _20_/Q (sky130_fd_sc_hd__dfstp_1)
     2    0.01                          state[0] (net)
                  0.04    0.00    0.59 v _15_/B1 (sky130_fd_sc_hd__a21oi_1)
                  0.09    0.11    0.70 ^ _15_/Y (sky130_fd_sc_hd__a21oi_1)
     1    0.00                          _07_ (net)
                  0.09    0.00    0.70 ^ _16_/A2 (sky130_fd_sc_hd__o21ai_1)
                  0.03    0.06    0.76 v _16_/Y (sky130_fd_sc_hd__o21ai_1)
     1    0.00                          _00_ (net)
                  0.03    0.00    0.76 v _20_/D (sky130_fd_sc_hd__dfstp_1)
                                  0.76   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                  0.06    0.04    0.04 ^ clk (in)
     1    0.01                          clk (net)
                  0.06    0.00    0.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.04    0.13    0.17 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
```

```
     2    0.02                           clknet_0_clk (net)
                  0.04    0.00    0.17 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.03    0.11    0.29 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                  0.03    0.00    0.29 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_1)
```

```
                           0.25    0.54   clock uncertainty
                          -0.03    0.51   clock reconvergence pessimism
                          -0.01    0.50   library hold time
                                   0.50   data required time
-----------------------------------------------------------------------------
                                   0.50   data required time
                                  -0.76   data arrival time
-----------------------------------------------------------------------------
                                   0.26   slack (MET)


Startpoint: rst_n (input port clocked by clk)
Endpoint: _21_ (removal check against rising-edge clock clk)
Path Group: asynchronous
Path Type: min

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                           0.00    0.00   clock clk (rise edge)
                           0.00    0.00   clock network delay (propagated)
                           4.00    4.00 ^ input external delay
                  0.02     0.01    4.01 ^ rst_n (in)
     1    0.00                            rst_n (net)
                  0.02     0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                  0.17     0.15    4.16 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.01                            net4 (net)
                  0.17     0.00    4.16 ^ _21_/RESET_B (sky130_fd_sc_hd__dfrtp_1)
                                   4.16   data arrival time

                           0.00    0.00   clock clk (rise edge)
                           0.00    0.00   clock source latency
                  0.06     0.04    0.04 ^ clk (in)
     1    0.01                            clk (net)
                  0.06     0.00    0.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.04     0.13    0.17 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                            clknet_0_clk (net)
                  0.04     0.00    0.17 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.03     0.11    0.29 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                            clknet_1_0__leaf_clk (net)
                  0.03     0.00    0.29 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                           0.25    0.54   clock uncertainty
                           0.00    0.54   clock reconvergence pessimism
                           0.35    0.89   library removal time
```

```
                                  0.89   data required time
-----------------------------------------------------------------------------
                                  0.89   data required time
                                 -4.16   data arrival time
-----------------------------------------------------------------------------
```

```
                                3.27   slack (MET)


Startpoint: rst_n (input port clocked by clk)
Endpoint: _22_ (removal check against rising-edge clock clk)
Path Group: asynchronous
Path Type: min

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                   0.02   0.01    4.01 ^ rst_n (in)
     1    0.00                           rst_n (net)
                   0.02   0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                   0.17   0.15    4.16 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.01                           net4 (net)
                   0.17   0.00    4.16 ^ _22_/RESET_B (sky130_fd_sc_hd__dfrtp_1)
                                  4.16   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                   0.06   0.04    0.04 ^ clk (in)
     1    0.01                           clk (net)
                   0.06   0.00    0.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                   0.04   0.13    0.17 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                   0.04   0.00    0.17 ^ clkbuf_1_1__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                   0.03   0.11    0.28 ^ clkbuf_1_1__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     1    0.00                           clknet_1_1__leaf_clk (net)
                   0.03   0.00    0.28 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_1)
                          0.25    0.53   clock uncertainty
                          0.00    0.53   clock reconvergence pessimism
                          0.35    0.88   library removal time
                                  0.88   data required time
-----------------------------------------------------------------------------
                                  0.88   data required time
                                 -4.16   data arrival time
-----------------------------------------------------------------------------
                                  3.28   slack (MET)
```

```
Startpoint: rst_n (input port clocked by clk)
Endpoint: _20_ (removal check against rising-edge clock clk)
Path Group: asynchronous
Path Type: min
```

```
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                  0.02    0.01    4.01 ^ rst_n (in)
     1    0.00                           rst_n (net)
                  0.02    0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                  0.17    0.15    4.16 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.01                           net4 (net)
                  0.17    0.00    4.16 ^ _20_/SET_B (sky130_fd_sc_hd__dfstp_1)
                                  4.16   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                  0.06    0.04    0.04 ^ clk (in)
     1    0.01                           clk (net)
                  0.06    0.00    0.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.04    0.13    0.17 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                  0.04    0.00    0.17 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.03    0.11    0.29 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                  0.03    0.00    0.29 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_1)
                          0.25    0.54   clock uncertainty
                          0.00    0.54   clock reconvergence pessimism
                          0.14    0.68   library removal time
                                  0.68   data required time
---------------------------------------------------------------------------
                                  0.68   data required time
                                 -4.16   data arrival time
---------------------------------------------------------------------------
                                  3.49   slack (MET)


Startpoint: _22_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: pump_en (output port clocked by clk)
Path Group: clk
Path Type: min

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
```

```
                   0.00    0.00   clock source latency
             0.06    0.03    0.03 ^ clk (in)
1    0.01                          clk (net)
             0.06    0.00    0.03 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
             0.04    0.12    0.16 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
```

```
     2    0.02                           clknet_0_clk (net)
                  0.04    0.00    0.16 ^ clkbuf_1_1__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.03    0.10    0.26 ^ clkbuf_1_1__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     1    0.00                           clknet_1_1__leaf_clk (net)
                  0.03    0.00    0.26 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_1)
                  0.09    0.33    0.59 ^ _22_/Q (sky130_fd_sc_hd__dfrtp_1)
     3    0.01                           state[2] (net)
                  0.09    0.00    0.59 ^ _18_/A (sky130_fd_sc_hd__and2_1)
                  0.05    0.12    0.70 ^ _18_/X (sky130_fd_sc_hd__and2_1)
     1    0.00                           _08_ (net)
                  0.05    0.00    0.70 ^ _19_/A (sky130_fd_sc_hd__clkbuf_1)
                  0.05    0.08    0.78 ^ _19_/X (sky130_fd_sc_hd__clkbuf_1)
     1    0.00                           net6 (net)
                  0.05    0.00    0.78 ^ output6/A (sky130_fd_sc_hd__clkbuf_4)
                  0.11    0.18    0.96 ^ output6/X (sky130_fd_sc_hd__clkbuf_4)
     1    0.03                           pump_en (net)
                  0.11    0.00    0.96 ^ pump_en (out)
                                  0.96   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          0.25    0.25   clock uncertainty
                          0.00    0.25   clock reconvergence pessimism
                         -4.00   -3.75   output external delay
                                 -3.75   data required time
-----------------------------------------------------------------------------
                                 -3.75   data required time
                                 -0.96   data arrival time
-----------------------------------------------------------------------------
                                  4.71   slack (MET)


Startpoint: dry_run_fault (input port clocked by clk)
Endpoint: alarm (output port clocked by clk)
Path Group: clk
Path Type: min

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                  0.02    0.01    4.01 ^ dry_run_fault (in)
```

```
1    0.00                           dry_run_fault (net)
             0.02    0.00    4.01 ^ input1/A (sky130_fd_sc_hd__clkbuf_1)
             0.07    0.08    4.09 ^ input1/X (sky130_fd_sc_hd__clkbuf_1)
1    0.00                           net1 (net)
             0.07    0.00    4.09 ^ _09_/B1 (sky130_fd_sc_hd__a21oi_2)
```

```
                0.04    0.05    4.15 v _09_/Y (sky130_fd_sc_hd__a21oi_2)
     4    0.01                           _03_ (net)
                0.04    0.00    4.15 v _10_/A (sky130_fd_sc_hd__inv_2)
                0.03    0.05    4.19 ^ _10_/Y (sky130_fd_sc_hd__inv_2)
     2    0.01                           net5 (net)
                0.03    0.00    4.19 ^ output5/A (sky130_fd_sc_hd__buf_2)
                0.17    0.18    4.37 ^ output5/X (sky130_fd_sc_hd__buf_2)
     1    0.03                           alarm (net)
                0.17    0.00    4.37 ^ alarm (out)
                                4.37   data arrival time

                        0.00    0.00   clock clk (rise edge)
                        0.00    0.00   clock network delay (propagated)
                        0.25    0.25   clock uncertainty
                        0.00    0.25   clock reconvergence pessimism
                       -4.00   -3.75   output external delay
                               -3.75   data required time
-----------------------------------------------------------------------------
                               -3.75   data required time
                               -4.37   data arrival time
-----------------------------------------------------------------------------
                                8.12   slack (MET)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/cts/13-cts_sta.power.rpt
==========

==========
 report_power
==========
======================= Typical Corner ===================================

Group	Internal Power	Switching Power	Leakage Power	Total Power (Watts)	
Sequential	6.59e-06	2.61e-07	3.33e-11	6.86e-06	17.5%
Combinational	2.64e-05	5.83e-06	1.86e-10	3.23e-05	82.5%
Macro	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Pad	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Total	3.30e-05	6.09e-06	2.19e-10	3.91e-05	100.0%
	84.4%	15.6%	0.0%		

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/cts/13-cts_sta.skew.rpt
==========

==========
report_clock_skew
==========
Clock clk
Latency   CRPR   Skew
_20_/CLK ^
   0.29
_22_/CLK ^
   0.26   -0.02   0.01

```
openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/cts/13-cts_sta.summary.rpt
================================================================================================

===========================================================================
report_tns
===========================================================================
tns 0.00

===========================================================================
report_wns
===========================================================================
wns 0.00

===========================================================================
report_worst_slack -max (Setup)
===========================================================================
worst slack 10.91

===========================================================================
report_worst_slack -min (Hold)
===========================================================================
worst slack 0.23
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/placement/11-dpl_sta.checks.rpt ? part 1
==

```
=======================================================================
report_checks -unconstrained
=======================================================================
======================== Typical Corner ===================================

Startpoint: rst_n (input port clocked by clk)
Endpoint: _20_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
Path Type: max

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          4.00    4.00 ^ input external delay
                  0.02    0.01    4.01 ^ rst_n (in)
     1    0.00                           rst_n (net)
                  0.02    0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                  0.17    0.17    4.18 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.01                           net4 (net)
                  0.17    0.00    4.18 ^ _20_/SET_B (sky130_fd_sc_hd__dfstp_1)
                                  4.18   data arrival time

                  0.15   20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock network delay (ideal)
                         -0.25   19.75   clock uncertainty
                          0.00   19.75   clock reconvergence pessimism
                                 19.75 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_1)
                          0.15   19.90   library recovery time
                                 19.90   data required time
---------------------------------------------------------------------------
                                 19.90   data required time
                                 -4.18   data arrival time
---------------------------------------------------------------------------
                                 15.72   slack (MET)


Startpoint: level_full (input port clocked by clk)
Endpoint: pump_en (output port clocked by clk)
Path Group: clk
Path Type: max

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.15    0.00    0.00   clock clk (rise edge)
```

```
          0.00    0.00   clock network delay (ideal)
          4.00    4.00 v input external delay
  0.01    0.00    4.00 v level_full (in)
```

```
=============================================================================

     1    0.00                           level_full (net)
                  0.01    0.00    4.00 v input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
                  0.08    0.17    4.18 v input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
     3    0.01                           net2 (net)
                  0.08    0.00    4.18 v _09_/A2 (sky130_fd_sc_hd__a21oi_2)
                  0.18    0.23    4.40 ^ _09_/Y (sky130_fd_sc_hd__a21oi_2)
     4    0.01                           _03_ (net)
                  0.18    0.00    4.40 ^ _18_/B (sky130_fd_sc_hd__and2_1)
                  0.05    0.15    4.56 ^ _18_/X (sky130_fd_sc_hd__and2_1)
     1    0.00                           _08_ (net)
                  0.05    0.00    4.56 ^ _19_/A (sky130_fd_sc_hd__clkbuf_1)
                  0.05    0.09    4.64 ^ _19_/X (sky130_fd_sc_hd__clkbuf_1)
     1    0.00                           net6 (net)
                  0.05    0.00    4.64 ^ output6/A (sky130_fd_sc_hd__clkbuf_4)
                  0.11    0.19    4.84 ^ output6/X (sky130_fd_sc_hd__clkbuf_4)
     1    0.03                           pump_en (net)
                  0.11    0.00    4.84 ^ pump_en (out)
                                  4.84   data arrival time

                  0.15   20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock network delay (ideal)
                         -0.25   19.75   clock uncertainty
                          0.00   19.75   clock reconvergence pessimism
                         -4.00   15.75   output external delay
                                 15.75   data required time
-----------------------------------------------------------------------------
                                 15.75   data required time
                                 -4.84   data arrival time
-----------------------------------------------------------------------------
                                 10.91   slack (MET)




=============================================================================
report_checks --slack_max -0.01
=============================================================================
====================== Typical Corner ===================================

No paths found.


=============================================================================
 report_check_types -max_slew -max_cap -max_fanout -violators
=============================================================================
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/placement/11-dpl_sta.checks.rpt ? part 3
==

==
report_parasitic_annotation -report_unannotated
==
Found 25 unannotated drivers.
 clk
 dry_run_fault
 level_full
 level_low
 rst_n
 _09_/Y
 _10_/Y
 _11_/Y
 _12_/X
 _13_/Y
 _14_/Y
 _15_/Y
 _16_/Y
 _17_/X
 _18_/X
 _19_/X
 _20_/Q
 _21_/Q
 _22_/Q
 input1/X
 input2/X
 input3/X
 input4/X
 output5/X
 output6/X
Found 0 partially unannotated drivers.

==
max slew violation count 0
max fanout violation count 0
max cap violation count 0
==

==
check_setup -verbose -unconstrained_endpoints -multiple_clock -no_clock -no_input_delay
-loops -generated_clocks
==

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/placement/11-dpl_sta.max.rpt ? part 1
==========

```
===========================================================================
report_checks -path_delay max (Setup)
===========================================================================
======================== Typical Corner ===================================

Startpoint: level_full (input port clocked by clk)
Endpoint: pump_en (output port clocked by clk)
Path Group: clk
Path Type: max

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          4.00    4.00 v input external delay
                  0.01    0.00    4.00 v level_full (in)
     1    0.00                           level_full (net)
                  0.01    0.00    4.00 v input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
                  0.08    0.17    4.18 v input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
     3    0.01                           net2 (net)
                  0.08    0.00    4.18 v _09_/A2 (sky130_fd_sc_hd__a21oi_2)
                  0.18    0.23    4.40 ^ _09_/Y (sky130_fd_sc_hd__a21oi_2)
     4    0.01                           _03_ (net)
                  0.18    0.00    4.40 ^ _18_/B (sky130_fd_sc_hd__and2_1)
                  0.05    0.15    4.56 ^ _18_/X (sky130_fd_sc_hd__and2_1)
     1    0.00                           _08_ (net)
                  0.05    0.00    4.56 ^ _19_/A (sky130_fd_sc_hd__clkbuf_1)
                  0.05    0.09    4.64 ^ _19_/X (sky130_fd_sc_hd__clkbuf_1)
     1    0.00                           net6 (net)
                  0.05    0.00    4.64 ^ output6/A (sky130_fd_sc_hd__clkbuf_4)
                  0.11    0.19    4.84 ^ output6/X (sky130_fd_sc_hd__clkbuf_4)
     1    0.03                           pump_en (net)
                  0.11    0.00    4.84 ^ pump_en (out)
                                  4.84   data arrival time

                  0.15   20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock network delay (ideal)
                         -0.25   19.75   clock uncertainty
                          0.00   19.75   clock reconvergence pessimism
                         -4.00   15.75   output external delay
                                 15.75   data required time
---------------------------------------------------------------------------
                                 15.75   data required time
                                 -4.84   data arrival time
---------------------------------------------------------------------------
```

10.91 slack (MET)

```
================================================================================

Startpoint: level_full (input port clocked by clk)
Endpoint: alarm (output port clocked by clk)
Path Group: clk
Path Type: max

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          4.00    4.00 v input external delay
                  0.01    0.00    4.00 v level_full (in)
     1    0.00                           level_full (net)
                  0.01    0.00    4.00 v input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
                  0.08    0.17    4.18 v input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
     3    0.01                           net2 (net)
                  0.08    0.00    4.18 v _09_/A2 (sky130_fd_sc_hd__a21oi_2)
                  0.18    0.23    4.40 ^ _09_/Y (sky130_fd_sc_hd__a21oi_2)
     4    0.01                           _03_ (net)
                  0.18    0.00    4.40 ^ _10_/A (sky130_fd_sc_hd__inv_2)
                  0.04    0.05    4.46 v _10_/Y (sky130_fd_sc_hd__inv_2)
     2    0.01                           net5 (net)
                  0.04    0.00    4.46 v output5/A (sky130_fd_sc_hd__buf_2)
                  0.09    0.19    4.65 v output5/X (sky130_fd_sc_hd__buf_2)
     1    0.03                           alarm (net)
                  0.09    0.00    4.65 v alarm (out)
                                  4.65   data arrival time

                  0.15   20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock network delay (ideal)
                         -0.25   19.75   clock uncertainty
                          0.00   19.75   clock reconvergence pessimism
                         -4.00   15.75   output external delay
                                 15.75   data required time
---------------------------------------------------------------------------
                                 15.75   data required time
                                 -4.65   data arrival time
---------------------------------------------------------------------------
                                 11.10   slack (MET)


Startpoint: level_full (input port clocked by clk)
Endpoint: _21_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
```

Fanout	Cap	Slew	Delay	Time	Description
		0.15	0.00	0.00	clock clk (rise edge)

```
                          0.00    0.00   clock network delay (ideal)
                          4.00    4.00 v input external delay
                 0.01     0.00    4.00 v level_full (in)
     1    0.00                           level_full (net)
                 0.01     0.00    4.00 v input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
                 0.08     0.17    4.18 v input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
     3    0.01                           net2 (net)
                 0.08     0.00    4.18 v _09_/A2 (sky130_fd_sc_hd__a21oi_2)
                 0.18     0.23    4.40 ^ _09_/Y (sky130_fd_sc_hd__a21oi_2)
     4    0.01                           _03_ (net)
                 0.18     0.00    4.40 ^ _17_/B1 (sky130_fd_sc_hd__o211a_1)
                 0.04     0.18    4.59 ^ _17_/X (sky130_fd_sc_hd__o211a_1)
     1    0.00                           _01_ (net)
                 0.04     0.00    4.59 ^ _21_/D (sky130_fd_sc_hd__dfrtp_1)
                          4.59   data arrival time

                 0.15    20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock network delay (ideal)
                         -0.25   19.75   clock uncertainty
                          0.00   19.75   clock reconvergence pessimism
                                 19.75 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                         -0.04   19.71   library setup time
                                 19.71   data required time
-----------------------------------------------------------------------------
                                 19.71   data required time
                                 -4.59   data arrival time
-----------------------------------------------------------------------------
                                 15.12   slack (MET)


Startpoint: level_full (input port clocked by clk)
Endpoint: _22_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                 0.15     0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          4.00    4.00 v input external delay
                 0.01     0.00    4.00 v level_full (in)
     1    0.00                           level_full (net)
                 0.01     0.00    4.00 v input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
                 0.08     0.17    4.18 v input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
     3    0.01                           net2 (net)
```

```
              0.08    0.00    4.18 v _09_/A2 (sky130_fd_sc_hd__a21oi_2)
              0.18    0.23    4.40 ^ _09_/Y (sky130_fd_sc_hd__a21oi_2)
4    0.01                           _03_ (net)
```

```
                  0.18    0.00    4.40 ^ _10_/A (sky130_fd_sc_hd__inv_2)
                  0.04    0.05    4.46 v _10_/Y (sky130_fd_sc_hd__inv_2)
     2    0.01                           net5 (net)
                  0.04    0.00    4.46 v _14_/A (sky130_fd_sc_hd__nor2_1)
                  0.07    0.10    4.56 ^ _14_/Y (sky130_fd_sc_hd__nor2_1)
     1    0.00                           _02_ (net)
                  0.07    0.00    4.56 ^ _22_/D (sky130_fd_sc_hd__dfrtp_1)
                                  4.56   data arrival time

                  0.15   20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock network delay (ideal)
                         -0.25   19.75   clock uncertainty
                          0.00   19.75   clock reconvergence pessimism
                                 19.75 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_1)
                         -0.05   19.70   library setup time
                                 19.70   data required time
-----------------------------------------------------------------------------
                                 19.70   data required time
                                 -4.56   data arrival time
-----------------------------------------------------------------------------
                                 15.14   slack (MET)


Startpoint: level_full (input port clocked by clk)
Endpoint: _20_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          4.00    4.00 v input external delay
                  0.01    0.00    4.00 v level_full (in)
     1    0.00                           level_full (net)
                  0.01    0.00    4.00 v input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
                  0.08    0.17    4.18 v input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
     3    0.01                           net2 (net)
                  0.08    0.00    4.18 v _09_/A2 (sky130_fd_sc_hd__a21oi_2)
                  0.18    0.23    4.40 ^ _09_/Y (sky130_fd_sc_hd__a21oi_2)
     4    0.01                           _03_ (net)
                  0.18    0.00    4.40 ^ _16_/B1 (sky130_fd_sc_hd__o21ai_1)
                  0.06    0.10    4.50 v _16_/Y (sky130_fd_sc_hd__o21ai_1)
     1    0.00                           _00_ (net)
                  0.06    0.00    4.50 v _20_/D (sky130_fd_sc_hd__dfstp_1)
```

```
                  4.50   data arrival time

0.15   20.00   20.00   clock clk (rise edge)
```

```
                           0.00   20.00   clock network delay (ideal)
                          -0.25   19.75   clock uncertainty
                           0.00   19.75   clock reconvergence pessimism
                                  19.75 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_1)
                          -0.05   19.70   library setup time
                                  19.70   data required time
-----------------------------------------------------------------------------
                                  19.70   data required time
                                  -4.50   data arrival time
-----------------------------------------------------------------------------
                                  15.19   slack (MET)


Startpoint: rst_n (input port clocked by clk)
Endpoint: _20_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
Path Type: max

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          4.00    4.00 ^ input external delay
                  0.02    0.01    4.01 ^ rst_n (in)
     1    0.00                           rst_n (net)
                  0.02    0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                  0.17    0.17    4.18 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.01                           net4 (net)
                  0.17    0.00    4.18 ^ _20_/SET_B (sky130_fd_sc_hd__dfstp_1)
                                  4.18   data arrival time

                  0.15   20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock network delay (ideal)
                         -0.25   19.75   clock uncertainty
                          0.00   19.75   clock reconvergence pessimism
                                 19.75 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_1)
                          0.15   19.90   library recovery time
                                 19.90   data required time
-----------------------------------------------------------------------------
                                 19.90   data required time
                                 -4.18   data arrival time
-----------------------------------------------------------------------------
                                 15.72   slack (MET)
```

```
Startpoint: rst_n (input port clocked by clk)
Endpoint: _21_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
```

```
================================================================================

Path Type: max

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          4.00    4.00 ^ input external delay
                  0.02    0.01    4.01 ^ rst_n (in)
     1    0.00                           rst_n (net)
                  0.02    0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                  0.17    0.17    4.18 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.01                           net4 (net)
                  0.17    0.00    4.18 ^ _21_/RESET_B (sky130_fd_sc_hd__dfrtp_1)
                                  4.18   data arrival time

                  0.15   20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock network delay (ideal)
                         -0.25   19.75   clock uncertainty
                          0.00   19.75   clock reconvergence pessimism
                                 19.75 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                          0.23   19.98   library recovery time
                                 19.98   data required time
---------------------------------------------------------------------------
                                 19.98   data required time
                                 -4.18   data arrival time
---------------------------------------------------------------------------
                                 15.80   slack (MET)


Startpoint: rst_n (input port clocked by clk)
Endpoint: _22_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
Path Type: max

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          4.00    4.00 ^ input external delay
                  0.02    0.01    4.01 ^ rst_n (in)
     1    0.00                           rst_n (net)
                  0.02    0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                  0.17    0.17    4.18 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.01                           net4 (net)
                  0.17    0.00    4.18 ^ _22_/RESET_B (sky130_fd_sc_hd__dfrtp_1)
```

```
                  4.18   data arrival time

 0.15   20.00   20.00   clock clk (rise edge)
```

```
                           0.00   20.00   clock network delay (ideal)
                          -0.25   19.75   clock uncertainty
                           0.00   19.75   clock reconvergence pessimism
                                  19.75 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_1)
                           0.23   19.98   library recovery time
                                  19.98   data required time
-----------------------------------------------------------------------------
                                  19.98   data required time
                                  -4.18   data arrival time
-----------------------------------------------------------------------------
                                  15.80   slack (MET)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/placement/11-dpl_sta.min.rpt ? part 1
==========

```
=========================================================================
report_checks -path_delay min (Hold)
=========================================================================
======================== Typical Corner ===================================

Startpoint: _22_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _21_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.15    0.00    0.00 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_1)
                  0.09    0.37    0.37 ^ _22_/Q (sky130_fd_sc_hd__dfrtp_1)
     3    0.01                           state[2] (net)
                  0.09    0.00    0.37 ^ _17_/A2 (sky130_fd_sc_hd__o211a_1)
                  0.04    0.13    0.50 ^ _17_/X (sky130_fd_sc_hd__o211a_1)
     1    0.00                           _01_ (net)
                  0.04    0.00    0.50 ^ _21_/D (sky130_fd_sc_hd__dfrtp_1)
                                  0.50   data arrival time

                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          0.25    0.25   clock uncertainty
                          0.00    0.25   clock reconvergence pessimism
                                  0.25 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                         -0.02    0.23   library hold time
                                  0.23   data required time
---------------------------------------------------------------------------
                                  0.23   data required time
                                 -0.50   data arrival time
---------------------------------------------------------------------------
                                  0.27   slack (MET)


Startpoint: _20_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _20_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.15    0.00    0.00   clock clk (rise edge)
```

```
        0.00    0.00   clock network delay (ideal)
0.15    0.00    0.00 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_1)
0.04    0.37    0.37 v _20_/Q (sky130_fd_sc_hd__dfstp_1)
```

```
     2    0.00                           state[0] (net)
                  0.04    0.00    0.37 v _15_/B1 (sky130_fd_sc_hd__a21oi_1)
                  0.09    0.11    0.48 ^ _15_/Y (sky130_fd_sc_hd__a21oi_1)
     1    0.00                           _07_ (net)
                  0.09    0.00    0.48 ^ _16_/A2 (sky130_fd_sc_hd__o21ai_1)
                  0.04    0.06    0.54 v _16_/Y (sky130_fd_sc_hd__o21ai_1)
     1    0.00                           _00_ (net)
                  0.04    0.00    0.54 v _20_/D (sky130_fd_sc_hd__dfstp_1)
                                  0.54   data arrival time

                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          0.25    0.25   clock uncertainty
                          0.00    0.25   clock reconvergence pessimism
                                  0.25 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_1)
                          0.01    0.26   library hold time
                                  0.26   data required time
-----------------------------------------------------------------------------
                                  0.26   data required time
                                 -0.54   data arrival time
-----------------------------------------------------------------------------
                                  0.28   slack (MET)


Startpoint: _22_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _22_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.15    0.00    0.00 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_1)
                  0.09    0.37    0.37 ^ _22_/Q (sky130_fd_sc_hd__dfrtp_1)
     3    0.01                           state[2] (net)
                  0.09    0.00    0.37 ^ _13_/A2 (sky130_fd_sc_hd__a21oi_1)
                  0.03    0.07    0.44 v _13_/Y (sky130_fd_sc_hd__a21oi_1)
     1    0.00                           _06_ (net)
                  0.03    0.00    0.44 v _14_/B (sky130_fd_sc_hd__nor2_1)
                  0.07    0.07    0.51 ^ _14_/Y (sky130_fd_sc_hd__nor2_1)
     1    0.00                           _02_ (net)
                  0.07    0.00    0.51 ^ _22_/D (sky130_fd_sc_hd__dfrtp_1)
                                  0.51   data arrival time
```

```
0.15    0.00    0.00   clock clk (rise edge)
        0.00    0.00   clock network delay (ideal)
        0.25    0.25   clock uncertainty
```

```
                          0.00    0.25   clock reconvergence pessimism
                                  0.25 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_1)
                         -0.03    0.22   library hold time
                                  0.22   data required time
-----------------------------------------------------------------------------
                                  0.22   data required time
                                 -0.51   data arrival time
-----------------------------------------------------------------------------
                                  0.28   slack (MET)


Startpoint: rst_n (input port clocked by clk)
Endpoint: _22_ (removal check against rising-edge clock clk)
Path Group: asynchronous
Path Type: min

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          4.00    4.00 ^ input external delay
                  0.02    0.01    4.01 ^ rst_n (in)
     1    0.00                           rst_n (net)
                  0.02    0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                  0.17    0.15    4.16 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.01                           net4 (net)
                  0.17    0.00    4.16 ^ _22_/RESET_B (sky130_fd_sc_hd__dfrtp_1)
                                  4.16   data arrival time

                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          0.25    0.25   clock uncertainty
                          0.00    0.25   clock reconvergence pessimism
                                  0.25 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_1)
                          0.38    0.63   library removal time
                                  0.63   data required time
-----------------------------------------------------------------------------
                                  0.63   data required time
                                 -4.16   data arrival time
-----------------------------------------------------------------------------
                                  3.53   slack (MET)


Startpoint: rst_n (input port clocked by clk)
Endpoint: _21_ (removal check against rising-edge clock clk)
```

Path Group: asynchronous
Path Type: min

```
==========================================================================================

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          4.00    4.00 ^ input external delay
                  0.02    0.01    4.01 ^ rst_n (in)
     1    0.00                          rst_n (net)
                  0.02    0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                  0.17    0.15    4.16 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.01                          net4 (net)
                  0.17    0.00    4.16 ^ _21_/RESET_B (sky130_fd_sc_hd__dfrtp_1)
                                  4.16   data arrival time

                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          0.25    0.25   clock uncertainty
                          0.00    0.25   clock reconvergence pessimism
                                  0.25 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                          0.38    0.63   library removal time
                                  0.63   data required time
---------------------------------------------------------------------------
                                  0.63   data required time
                                 -4.16   data arrival time
---------------------------------------------------------------------------
                                  3.53   slack (MET)


Startpoint: rst_n (input port clocked by clk)
Endpoint: _20_ (removal check against rising-edge clock clk)
Path Group: asynchronous
Path Type: min

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          4.00    4.00 ^ input external delay
                  0.02    0.01    4.01 ^ rst_n (in)
     1    0.00                          rst_n (net)
                  0.02    0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                  0.17    0.15    4.16 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.01                          net4 (net)
                  0.17    0.00    4.16 ^ _20_/SET_B (sky130_fd_sc_hd__dfstp_1)
                                  4.16   data arrival time
```

```
0.15    0.00    0.00   clock clk (rise edge)
        0.00    0.00   clock network delay (ideal)
        0.25    0.25   clock uncertainty
```

```
                          0.00    0.25   clock reconvergence pessimism
                                  0.25 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_1)
                          0.17    0.42   library removal time
                                  0.42   data required time
-----------------------------------------------------------------------------
                                  0.42   data required time
                                 -4.16   data arrival time
-----------------------------------------------------------------------------
                                  3.74   slack (MET)


Startpoint: _22_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: pump_en (output port clocked by clk)
Path Group: clk
Path Type: min

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.15    0.00    0.00 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_1)
                  0.09    0.37    0.37 ^ _22_/Q (sky130_fd_sc_hd__dfrtp_1)
     3    0.01                           state[2] (net)
                  0.09    0.00    0.37 ^ _18_/A (sky130_fd_sc_hd__and2_1)
                  0.04    0.12    0.49 ^ _18_/X (sky130_fd_sc_hd__and2_1)
     1    0.00                           _08_ (net)
                  0.04    0.00    0.49 ^ _19_/A (sky130_fd_sc_hd__clkbuf_1)
                  0.05    0.08    0.57 ^ _19_/X (sky130_fd_sc_hd__clkbuf_1)
     1    0.00                           net6 (net)
                  0.05    0.00    0.57 ^ output6/A (sky130_fd_sc_hd__clkbuf_4)
                  0.11    0.18    0.74 ^ output6/X (sky130_fd_sc_hd__clkbuf_4)
     1    0.03                           pump_en (net)
                  0.11    0.00    0.74 ^ pump_en (out)
                                  0.74   data arrival time

                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          0.25    0.25   clock uncertainty
                          0.00    0.25   clock reconvergence pessimism
                         -4.00   -3.75   output external delay
                                 -3.75   data required time
-----------------------------------------------------------------------------
                                 -3.75   data required time
                                 -0.74   data arrival time
-----------------------------------------------------------------------------
```

4.49 slack (MET)

```
==========================================================================================

Startpoint: dry_run_fault (input port clocked by clk)
Endpoint: alarm (output port clocked by clk)
Path Group: clk
Path Type: min

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          4.00    4.00 ^ input external delay
                  0.02    0.01    4.01 ^ dry_run_fault (in)
     1    0.00                           dry_run_fault (net)
                  0.02    0.00    4.01 ^ input1/A (sky130_fd_sc_hd__clkbuf_1)
                  0.07    0.08    4.09 ^ input1/X (sky130_fd_sc_hd__clkbuf_1)
     1    0.00                           net1 (net)
                  0.07    0.00    4.09 ^ _09_/B1 (sky130_fd_sc_hd__a21oi_2)
                  0.04    0.05    4.15 v _09_/Y (sky130_fd_sc_hd__a21oi_2)
     4    0.01                           _03_ (net)
                  0.04    0.00    4.15 v _10_/A (sky130_fd_sc_hd__inv_2)
                  0.03    0.05    4.19 ^ _10_/Y (sky130_fd_sc_hd__inv_2)
     2    0.01                           net5 (net)
                  0.03    0.00    4.19 ^ output5/A (sky130_fd_sc_hd__buf_2)
                  0.17    0.18    4.37 ^ output5/X (sky130_fd_sc_hd__buf_2)
     1    0.03                           alarm (net)
                  0.17    0.00    4.37 ^ alarm (out)
                                  4.37   data arrival time

                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          0.25    0.25   clock uncertainty
                          0.00    0.25   clock reconvergence pessimism
                         -4.00   -3.75   output external delay
                                 -3.75   data required time
---------------------------------------------------------------------------
                                 -3.75   data required time
                                 -4.37   data arrival time
---------------------------------------------------------------------------
                                  8.12   slack (MET)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/placement/11-dpl_sta.power.rpt
==

```
===========================================================================
 report_power
===========================================================================
======================== Typical Corner ===================================

Group                  Internal  Switching    Leakage      Total
                          Power      Power      Power      Power (Watts)
----------------------------------------------------------------
Sequential             6.55e-06   2.47e-07   3.33e-11   6.79e-06  70.9%
Combinational          1.14e-06   1.65e-06   1.49e-10   2.79e-06  29.1%
Macro                  0.00e+00   0.00e+00   0.00e+00   0.00e+00   0.0%
Pad                    0.00e+00   0.00e+00   0.00e+00   0.00e+00   0.0%
----------------------------------------------------------------
Total                  7.68e-06   1.90e-06   1.82e-10   9.58e-06 100.0%
                          80.2%      19.8%       0.0%
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/placement/11-dpl_sta.skew.rpt
==

==========================
report_clock_skew
==========================
Clock clk
Latency   CRPR   Skew
_20_/CLK ^
   0.02
_20_/CLK ^
   0.02   0.00   0.00

```
openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/placement/11-dpl_sta.summary.rpt
==========================================================================================

===========================================================================
report_tns
===========================================================================
tns 0.00

===========================================================================
report_wns
===========================================================================
wns 0.00

===========================================================================
report_worst_slack -max (Setup)
===========================================================================
worst slack 10.91

===========================================================================
report_worst_slack -min (Hold)
===========================================================================
worst slack 0.27
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/placement/8-gpl_sta.checks.rpt ? part 1
==

```
=========================================================================
report_checks -unconstrained
=========================================================================
======================== Typical Corner ===================================

Startpoint: rst_n (input port clocked by clk)
Endpoint: _20_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
Path Type: max

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          4.00    4.00 ^ input external delay
                  0.07    0.05    4.05 ^ rst_n (in)
     3    0.01                          rst_n (net)
                  0.07    0.00    4.05 ^ _20_/SET_B (sky130_fd_sc_hd__dfstp_2)
                                  4.05   data arrival time

                  0.15   20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock network delay (ideal)
                         -0.25   19.75   clock uncertainty
                          0.00   19.75   clock reconvergence pessimism
                                 19.75 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_2)
                          0.16   19.91   library recovery time
                                 19.91   data required time
---------------------------------------------------------------------------
                                 19.91   data required time
                                 -4.05   data arrival time
---------------------------------------------------------------------------
                                 15.86   slack (MET)


Startpoint: level_full (input port clocked by clk)
Endpoint: pump_en (output port clocked by clk)
Path Group: clk
Path Type: max

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          4.00    4.00 v input external delay
                  0.04    0.03    4.03 v level_full (in)
```

```
3    0.02                           level_full (net)
             0.04    0.00    4.03 v _09_/A2 (sky130_fd_sc_hd__a21oi_2)
             0.21    0.23    4.25 ^ _09_/Y (sky130_fd_sc_hd__a21oi_2)
```

```
================================================================================

     4    0.02                           _03_ (net)
                  0.21    0.00    4.25 ^ _18_/B (sky130_fd_sc_hd__and2_2)
                  0.04    0.18    4.43 ^ _18_/X (sky130_fd_sc_hd__and2_2)
     1    0.00                           _08_ (net)
                  0.04    0.00    4.43 ^ _19_/A (sky130_fd_sc_hd__buf_1)
                  0.41    0.35    4.78 ^ _19_/X (sky130_fd_sc_hd__buf_1)
     1    0.03                           pump_en (net)
                  0.41    0.00    4.78 ^ pump_en (out)
                                  4.78   data arrival time

                  0.15   20.00   20.00   clock clk (rise edge)
                         0.00   20.00   clock network delay (ideal)
                        -0.25   19.75   clock uncertainty
                         0.00   19.75   clock reconvergence pessimism
                        -4.00   15.75   output external delay
                                15.75   data required time
---------------------------------------------------------------------------
                                15.75   data required time
                                -4.78   data arrival time
---------------------------------------------------------------------------
                                10.97   slack (MET)




============================================================================
report_checks --slack_max -0.01
============================================================================
====================== Typical Corner ===================================

No paths found.


============================================================================
 report_check_types -max_slew -max_cap -max_fanout -violators
============================================================================
====================== Typical Corner ===================================



============================================================================
report_parasitic_annotation -report_unannotated
============================================================================
Found 19 unannotated drivers.
 clk
```

```
dry_run_fault
level_full
level_low
```

==========

 rst_n
 _09_/Y
 _10_/Y
 _11_/Y
 _12_/X
 _13_/Y
 _14_/Y
 _15_/Y
 _16_/Y
 _17_/X
 _18_/X
 _19_/X
 _20_/Q
 _21_/Q
 _22_/Q
Found 0 partially unannotated drivers.

==========
max slew violation count 0
max fanout violation count 0
max cap violation count 0
==========

==========
check_setup -verbose -unconstrained_endpoints -multiple_clock -no_clock -no_input_delay
-loops -generated_clocks
==========

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/placement/8-gpl_sta.max.rpt ? part 1
==========

```
===========================================================================
report_checks -path_delay max (Setup)
===========================================================================
======================== Typical Corner ===================================

Startpoint: level_full (input port clocked by clk)
Endpoint: pump_en (output port clocked by clk)
Path Group: clk
Path Type: max

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          4.00    4.00 v input external delay
                  0.04    0.03    4.03 v level_full (in)
     3    0.02                           level_full (net)
                  0.04    0.00    4.03 v _09_/A2 (sky130_fd_sc_hd__a21oi_2)
                  0.21    0.23    4.25 ^ _09_/Y (sky130_fd_sc_hd__a21oi_2)
     4    0.02                           _03_ (net)
                  0.21    0.00    4.25 ^ _18_/B (sky130_fd_sc_hd__and2_2)
                  0.04    0.18    4.43 ^ _18_/X (sky130_fd_sc_hd__and2_2)
     1    0.00                           _08_ (net)
                  0.04    0.00    4.43 ^ _19_/A (sky130_fd_sc_hd__buf_1)
                  0.41    0.35    4.78 ^ _19_/X (sky130_fd_sc_hd__buf_1)
     1    0.03                           pump_en (net)
                  0.41    0.00    4.78 ^ pump_en (out)
                                  4.78   data arrival time

                  0.15   20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock network delay (ideal)
                         -0.25   19.75   clock uncertainty
                          0.00   19.75   clock reconvergence pessimism
                         -4.00   15.75   output external delay
                                 15.75   data required time
---------------------------------------------------------------------------
                                 15.75   data required time
                                 -4.78   data arrival time
---------------------------------------------------------------------------
                                 10.97   slack (MET)


Startpoint: level_full (input port clocked by clk)
Endpoint: alarm (output port clocked by clk)
Path Group: clk
```

Path Type: max

Fanout	Cap	Slew	Delay	Time	Description

```
--------------------------------------------------------------------------------
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          4.00    4.00 v input external delay
                  0.04    0.03    4.03 v level_full (in)
     3    0.02                           level_full (net)
                  0.04    0.00    4.03 v _09_/A2 (sky130_fd_sc_hd__a21oi_2)
                  0.21    0.23    4.25 ^ _09_/Y (sky130_fd_sc_hd__a21oi_2)
     4    0.02                           _03_ (net)
                  0.21    0.00    4.25 ^ _10_/A (sky130_fd_sc_hd__inv_2)
                  0.11    0.15    4.41 v _10_/Y (sky130_fd_sc_hd__inv_2)
     2    0.04                           alarm (net)
                  0.11    0.00    4.41 v alarm (out)
                                  4.41   data arrival time

                  0.15   20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock network delay (ideal)
                         -0.25   19.75   clock uncertainty
                          0.00   19.75   clock reconvergence pessimism
                         -4.00   15.75   output external delay
                                 15.75   data required time
--------------------------------------------------------------------------------
                                 15.75   data required time
                                 -4.41   data arrival time
--------------------------------------------------------------------------------
                                 11.34   slack (MET)


Startpoint: level_full (input port clocked by clk)
Endpoint: _22_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

Fanout     Cap    Slew   Delay    Time   Description
--------------------------------------------------------------------------------
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          4.00    4.00 v input external delay
                  0.04    0.03    4.03 v level_full (in)
     3    0.02                           level_full (net)
                  0.04    0.00    4.03 v _09_/A2 (sky130_fd_sc_hd__a21oi_2)
                  0.21    0.23    4.25 ^ _09_/Y (sky130_fd_sc_hd__a21oi_2)
     4    0.02                           _03_ (net)
                  0.21    0.00    4.25 ^ _10_/A (sky130_fd_sc_hd__inv_2)
                  0.11    0.15    4.41 v _10_/Y (sky130_fd_sc_hd__inv_2)
```

```
2    0.04                           alarm (net)
             0.11    0.00    4.41 v _14_/A (sky130_fd_sc_hd__nor2_2)
             0.06    0.12    4.53 ^ _14_/Y (sky130_fd_sc_hd__nor2_2)
```

==========

```
     1    0.00                           _02_ (net)
                  0.06    0.00    4.53 ^ _22_/D (sky130_fd_sc_hd__dfrtp_2)
                                  4.53   data arrival time

                  0.15   20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock network delay (ideal)
                         -0.25   19.75   clock uncertainty
                          0.00   19.75   clock reconvergence pessimism
                                 19.75 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_2)
                         -0.05   19.70   library setup time
                                 19.70   data required time
-----------------------------------------------------------------------------
                                 19.70   data required time
                                 -4.53   data arrival time
-----------------------------------------------------------------------------
                                 15.17   slack (MET)


Startpoint: level_full (input port clocked by clk)
Endpoint: _21_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          4.00    4.00 v input external delay
                  0.04    0.03    4.03 v level_full (in)
     3    0.02                           level_full (net)
                  0.04    0.00    4.03 v _09_/A2 (sky130_fd_sc_hd__a21oi_2)
                  0.21    0.23    4.25 ^ _09_/Y (sky130_fd_sc_hd__a21oi_2)
     4    0.02                           _03_ (net)
                  0.21    0.00    4.25 ^ _17_/B1 (sky130_fd_sc_hd__o211a_2)
                  0.04    0.22    4.47 ^ _17_/X (sky130_fd_sc_hd__o211a_2)
     1    0.00                           _01_ (net)
                  0.04    0.00    4.47 ^ _21_/D (sky130_fd_sc_hd__dfrtp_2)
                                  4.47   data arrival time

                  0.15   20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock network delay (ideal)
                         -0.25   19.75   clock uncertainty
                          0.00   19.75   clock reconvergence pessimism
                                 19.75 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_2)
                         -0.05   19.70   library setup time
```

```
                                  19.70   data required time
-----------------------------------------------------------------------------
                                  19.70   data required time
```

```
================================================================================

                                -4.47   data arrival time
--------------------------------------------------------------------------------
                                15.23   slack (MET)


Startpoint: level_full (input port clocked by clk)
Endpoint: _20_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

Fanout     Cap    Slew   Delay    Time   Description
--------------------------------------------------------------------------------
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          4.00    4.00 v input external delay
                  0.04    0.03    4.03 v level_full (in)
     3    0.02                           level_full (net)
                  0.04    0.00    4.03 v _09_/A2 (sky130_fd_sc_hd__a21oi_2)
                  0.21    0.23    4.25 ^ _09_/Y (sky130_fd_sc_hd__a21oi_2)
     4    0.02                           _03_ (net)
                  0.21    0.00    4.25 ^ _16_/B1 (sky130_fd_sc_hd__o21ai_2)
                  0.05    0.08    4.33 v _16_/Y (sky130_fd_sc_hd__o21ai_2)
     1    0.00                           _00_ (net)
                  0.05    0.00    4.33 v _20_/D (sky130_fd_sc_hd__dfstp_2)
                                  4.33   data arrival time

                  0.15   20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock network delay (ideal)
                         -0.25   19.75   clock uncertainty
                          0.00   19.75   clock reconvergence pessimism
                                 19.75 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_2)
                         -0.05   19.70   library setup time
                                 19.70   data required time
--------------------------------------------------------------------------------
                                19.70   data required time
                                -4.33   data arrival time
--------------------------------------------------------------------------------
                                15.37   slack (MET)


Startpoint: rst_n (input port clocked by clk)
Endpoint: _20_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
Path Type: max
```

Fanout	Cap	Slew	Delay	Time	Description
		0.15	0.00	0.00	clock clk (rise edge)

```
                          0.00    0.00   clock network delay (ideal)
                          4.00    4.00 ^ input external delay
                 0.07     0.05    4.05 ^ rst_n (in)
     3    0.01                          rst_n (net)
                 0.07     0.00    4.05 ^ _20_/SET_B (sky130_fd_sc_hd__dfstp_2)
                                  4.05   data arrival time

                 0.15    20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock network delay (ideal)
                         -0.25   19.75   clock uncertainty
                          0.00   19.75   clock reconvergence pessimism
                                 19.75 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_2)
                          0.16   19.91   library recovery time
                                 19.91   data required time
-----------------------------------------------------------------------------
                                 19.91   data required time
                                 -4.05   data arrival time
-----------------------------------------------------------------------------
                                 15.86   slack (MET)


Startpoint: rst_n (input port clocked by clk)
Endpoint: _21_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
Path Type: max

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                 0.15     0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          4.00    4.00 ^ input external delay
                 0.07     0.05    4.05 ^ rst_n (in)
     3    0.01                          rst_n (net)
                 0.07     0.00    4.05 ^ _21_/RESET_B (sky130_fd_sc_hd__dfrtp_2)
                                  4.05   data arrival time

                 0.15    20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock network delay (ideal)
                         -0.25   19.75   clock uncertainty
                          0.00   19.75   clock reconvergence pessimism
                                 19.75 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_2)
                          0.24   19.99   library recovery time
                                 19.99   data required time
-----------------------------------------------------------------------------
                                 19.99   data required time
```

```
                                  -4.05   data arrival time
-----------------------------------------------------------------------------
                                  15.94   slack (MET)
```

```
================================================================================


Startpoint: rst_n (input port clocked by clk)
Endpoint: _22_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
Path Type: max

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          4.00    4.00 ^ input external delay
                  0.07    0.05    4.05 ^ rst_n (in)
     3    0.01                          rst_n (net)
                  0.07    0.00    4.05 ^ _22_/RESET_B (sky130_fd_sc_hd__dfrtp_2)
                                  4.05   data arrival time

                  0.15   20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock network delay (ideal)
                         -0.25   19.75   clock uncertainty
                          0.00   19.75   clock reconvergence pessimism
                                 19.75 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_2)
                          0.24   19.99   library recovery time
                                 19.99   data required time
---------------------------------------------------------------------------
                                 19.99   data required time
                                 -4.05   data arrival time
---------------------------------------------------------------------------
                                 15.94   slack (MET)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/placement/8-gpl_sta.min.rpt ? part 1
==========

```
===========================================================================
report_checks -path_delay min (Hold)
===========================================================================
======================== Typical Corner ===================================

Startpoint: _20_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _20_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.15    0.00    0.00 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_2)
                  0.03    0.38    0.38 v _20_/Q (sky130_fd_sc_hd__dfstp_2)
     2    0.01                           state[0] (net)
                  0.03    0.00    0.38 v _15_/B1 (sky130_fd_sc_hd__a21oi_2)
                  0.09    0.10    0.48 ^ _15_/Y (sky130_fd_sc_hd__a21oi_2)
     1    0.00                           _07_ (net)
                  0.09    0.00    0.48 ^ _16_/A2 (sky130_fd_sc_hd__o21ai_2)
                  0.03    0.05    0.53 v _16_/Y (sky130_fd_sc_hd__o21ai_2)
     1    0.00                           _00_ (net)
                  0.03    0.00    0.53 v _20_/D (sky130_fd_sc_hd__dfstp_2)
                                  0.53   data arrival time

                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          0.25    0.25   clock uncertainty
                          0.00    0.25   clock reconvergence pessimism
                                  0.25 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_2)
                          0.02    0.27   library hold time
                                  0.27   data required time
---------------------------------------------------------------------------
                                  0.27   data required time
                                 -0.53   data arrival time
---------------------------------------------------------------------------
                                  0.26   slack (MET)


Startpoint: _22_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _22_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
```

Fanout	Cap	Slew	Delay	Time	Description
		0.15	0.00	0.00	clock clk (rise edge)

```
                          0.00    0.00   clock network delay (ideal)
                 0.15     0.00    0.00 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_2)
                 0.07     0.39    0.39 ^ _22_/Q (sky130_fd_sc_hd__dfrtp_2)
     3    0.01                           state[2] (net)
                 0.07     0.00    0.39 ^ _13_/A2 (sky130_fd_sc_hd__a21oi_2)
                 0.02     0.06    0.44 v _13_/Y (sky130_fd_sc_hd__a21oi_2)
     1    0.00                           _06_ (net)
                 0.02     0.00    0.44 v _14_/B (sky130_fd_sc_hd__nor2_2)
                 0.06     0.05    0.50 ^ _14_/Y (sky130_fd_sc_hd__nor2_2)
     1    0.00                           _02_ (net)
                 0.06     0.00    0.50 ^ _22_/D (sky130_fd_sc_hd__dfrtp_2)
                          0.50   data arrival time

                 0.15     0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          0.25    0.25   clock uncertainty
                          0.00    0.25   clock reconvergence pessimism
                                  0.25 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_2)
                         -0.02    0.23   library hold time
                                  0.23   data required time
-----------------------------------------------------------------------------
                                  0.23   data required time
                                 -0.50   data arrival time
-----------------------------------------------------------------------------
                                  0.27   slack (MET)


Startpoint: _22_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _21_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                 0.15     0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                 0.15     0.00    0.00 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_2)
                 0.07     0.39    0.39 ^ _22_/Q (sky130_fd_sc_hd__dfrtp_2)
     3    0.01                           state[2] (net)
                 0.07     0.00    0.39 ^ _17_/A2 (sky130_fd_sc_hd__o211a_2)
                 0.04     0.15    0.54 ^ _17_/X (sky130_fd_sc_hd__o211a_2)
     1    0.00                           _01_ (net)
                 0.04     0.00    0.54 ^ _21_/D (sky130_fd_sc_hd__dfrtp_2)
                          0.54   data arrival time
```

```
0.15    0.00    0.00   clock clk (rise edge)
        0.00    0.00   clock network delay (ideal)
        0.25    0.25   clock uncertainty
```

```
                          0.00    0.25   clock reconvergence pessimism
                                  0.25 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_2)
                         -0.02    0.23   library hold time
                                  0.23   data required time
-----------------------------------------------------------------------------
                                  0.23   data required time
                                 -0.54   data arrival time
-----------------------------------------------------------------------------
                                  0.31   slack (MET)


Startpoint: rst_n (input port clocked by clk)
Endpoint: _22_ (removal check against rising-edge clock clk)
Path Group: asynchronous
Path Type: min

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          4.00    4.00 ^ input external delay
                  0.07    0.05    4.05 ^ rst_n (in)
     3    0.01                           rst_n (net)
                  0.07    0.00    4.05 ^ _22_/RESET_B (sky130_fd_sc_hd__dfrtp_2)
                                  4.05   data arrival time

                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          0.25    0.25   clock uncertainty
                          0.00    0.25   clock reconvergence pessimism
                                  0.25 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_2)
                          0.36    0.61   library removal time
                                  0.61   data required time
-----------------------------------------------------------------------------
                                  0.61   data required time
                                 -4.05   data arrival time
-----------------------------------------------------------------------------
                                  3.44   slack (MET)


Startpoint: rst_n (input port clocked by clk)
Endpoint: _21_ (removal check against rising-edge clock clk)
Path Group: asynchronous
Path Type: min
```

Fanout	Cap	Slew	Delay	Time	Description
		0.15	0.00	0.00	clock clk (rise edge)

```
                          0.00    0.00   clock network delay (ideal)
                          4.00    4.00 ^ input external delay
                  0.07    0.05    4.05 ^ rst_n (in)
     3    0.01                          rst_n (net)
                  0.07    0.00    4.05 ^ _21_/RESET_B (sky130_fd_sc_hd__dfrtp_2)
                                  4.05   data arrival time

                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          0.25    0.25   clock uncertainty
                          0.00    0.25   clock reconvergence pessimism
                                  0.25 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_2)
                          0.36    0.61   library removal time
                                  0.61   data required time
-----------------------------------------------------------------------------
                                  0.61   data required time
                                 -4.05   data arrival time
-----------------------------------------------------------------------------
                                  3.44   slack (MET)


Startpoint: rst_n (input port clocked by clk)
Endpoint: _20_ (removal check against rising-edge clock clk)
Path Group: asynchronous
Path Type: min

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          4.00    4.00 ^ input external delay
                  0.07    0.05    4.05 ^ rst_n (in)
     3    0.01                          rst_n (net)
                  0.07    0.00    4.05 ^ _20_/SET_B (sky130_fd_sc_hd__dfstp_2)
                                  4.05   data arrival time

                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          0.25    0.25   clock uncertainty
                          0.00    0.25   clock reconvergence pessimism
                                  0.25 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_2)
                          0.18    0.43   library removal time
                                  0.43   data required time
-----------------------------------------------------------------------------
                                  0.43   data required time
```

```
                                  -4.05   data arrival time
-----------------------------------------------------------------------------
                                   3.61   slack (MET)
```

```
Startpoint: _22_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: pump_en (output port clocked by clk)
Path Group: clk
Path Type: min

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.15    0.00    0.00 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_2)
                  0.06    0.42    0.42 v _22_/Q (sky130_fd_sc_hd__dfrtp_2)
     3    0.01                           state[2] (net)
                  0.06    0.00    0.42 v _18_/A (sky130_fd_sc_hd__and2_2)
                  0.03    0.16    0.58 v _18_/X (sky130_fd_sc_hd__and2_2)
     1    0.00                           _08_ (net)
                  0.03    0.00    0.58 v _19_/A (sky130_fd_sc_hd__buf_1)
                  0.20    0.21    0.79 v _19_/X (sky130_fd_sc_hd__buf_1)
     1    0.03                           pump_en (net)
                  0.20    0.00    0.79 v pump_en (out)
                                  0.79   data arrival time

                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          0.25    0.25   clock uncertainty
                          0.00    0.25   clock reconvergence pessimism
                         -4.00   -3.75   output external delay
                                 -3.75   data required time
---------------------------------------------------------------------------
                                 -3.75   data required time
                                 -0.79   data arrival time
---------------------------------------------------------------------------
                                  4.54   slack (MET)


Startpoint: dry_run_fault (input port clocked by clk)
Endpoint: alarm (output port clocked by clk)
Path Group: clk
Path Type: min

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
```

```
                    4.00    4.00 ^ input external delay
              0.03    0.02    4.02 ^ dry_run_fault (in)
1     0.01                         dry_run_fault (net)
```

```
                0.03    0.00    4.02 ^ _09_/B1 (sky130_fd_sc_hd__a21oi_2)
                0.04    0.04    4.06 v _09_/Y (sky130_fd_sc_hd__a21oi_2)
     4    0.01                        _03_ (net)
                0.04    0.00    4.06 v _10_/A (sky130_fd_sc_hd__inv_2)
                0.18    0.15    4.21 ^ _10_/Y (sky130_fd_sc_hd__inv_2)
     2    0.04                        alarm (net)
                0.18    0.00    4.21 ^ alarm (out)
                                4.21   data arrival time

                0.15    0.00    0.00   clock clk (rise edge)
                        0.00    0.00   clock network delay (ideal)
                        0.25    0.25   clock uncertainty
                        0.00    0.25   clock reconvergence pessimism
                       -4.00   -3.75   output external delay
                               -3.75   data required time
-----------------------------------------------------------------------------
                               -3.75   data required time
                               -4.21   data arrival time
-----------------------------------------------------------------------------
                                7.96   slack (MET)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/placement/8-gpl_sta.power.rpt
==

==
 report_power
==
======================== Typical Corner ===================================

Group	Internal Power	Switching Power	Leakage Power	Total Power (Watts)	
Sequential	6.65e-06	3.43e-07	3.18e-11	6.99e-06	74.9%
Combinational	9.21e-07	1.43e-06	1.19e-10	2.35e-06	25.1%
Macro	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Pad	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Total	7.57e-06	1.77e-06	1.51e-10	9.34e-06	100.0%
	81.0%	19.0%	0.0%		

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/placement/8-gpl_sta.skew.rpt
==

==========================
report_clock_skew
==========================
Clock clk
Latency   CRPR   Skew
_20_/CLK ^
   0.02
_20_/CLK ^
   0.02   0.00   0.00

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/placement/8-gpl_sta.summary.rpt
==

==
report_tns
==
tns 0.00

==
report_wns
==
wns 0.00

==
report_worst_slack -max (Setup)
==
worst slack 10.97

==
report_worst_slack -min (Hold)
==
worst slack 0.26

```
openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/routing/16-rsz_design_sta.checks.rpt ? par
========================================================================================

========================================================================
report_checks -unconstrained
========================================================================
======================= Typical Corner ===================================

Startpoint: rst_n (input port clocked by clk)
Endpoint: _20_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
Path Type: max

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                  0.02    0.01    4.01 ^ rst_n (in)
     1    0.00                           rst_n (net)
                  0.02    0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                  0.14    0.14    4.15 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.01                           net4 (net)
                  0.14    0.00    4.15 ^ _20_/SET_B (sky130_fd_sc_hd__dfstp_1)
                                  4.15   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock source latency
                  0.06    0.04   20.04 ^ clk (in)
     1    0.01                           clk (net)
                  0.06    0.00   20.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.04    0.12   20.16 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                  0.04    0.00   20.16 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.03    0.10   20.27 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                  0.03    0.00   20.27 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_1)
                         -0.25   20.02   clock uncertainty
                          0.00   20.02   clock reconvergence pessimism
                          0.12   20.14   library recovery time
                                 20.14   data required time
---------------------------------------------------------------------------
                                 20.14   data required time
                                 -4.15   data arrival time
---------------------------------------------------------------------------
                                 15.99   slack (MET)
```

```
Startpoint: level_full (input port clocked by clk)
Endpoint: pump_en (output port clocked by clk)
Path Group: clk
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/routing/16-rsz_design_sta.checks.rpt ? par
==

Path Type: max

```
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 v input external delay
                   0.02   0.01    4.01 v level_full (in)
     1    0.01                           level_full (net)
                   0.02   0.00    4.01 v input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
                   0.09   0.18    4.19 v input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
     3    0.01                           net2 (net)
                   0.09   0.00    4.19 v _09_/A2 (sky130_fd_sc_hd__a21oi_2)
                   0.20   0.24    4.43 ^ _09_/Y (sky130_fd_sc_hd__a21oi_2)
     4    0.01                           _03_ (net)
                   0.20   0.00    4.43 ^ _18_/B (sky130_fd_sc_hd__and2_1)
                   0.05   0.16    4.59 ^ _18_/X (sky130_fd_sc_hd__and2_1)
     1    0.00                           _08_ (net)
                   0.05   0.00    4.59 ^ _19_/A (sky130_fd_sc_hd__clkbuf_1)
                   0.05   0.09    4.68 ^ _19_/X (sky130_fd_sc_hd__clkbuf_1)
     1    0.00                           net6 (net)
                   0.05   0.00    4.68 ^ output6/A (sky130_fd_sc_hd__clkbuf_4)
                   0.11   0.20    4.87 ^ output6/X (sky130_fd_sc_hd__clkbuf_4)
     1    0.03                           pump_en (net)
                   0.11   0.00    4.87 ^ pump_en (out)
                                  4.87   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock network delay (propagated)
                         -0.25   19.75   clock uncertainty
                          0.00   19.75   clock reconvergence pessimism
                         -4.00   15.75   output external delay
                                 15.75   data required time
---------------------------------------------------------------------------
                                 15.75   data required time
                                 -4.87   data arrival time
---------------------------------------------------------------------------
                                 10.88   slack (MET)
```

```
===========================================================================
report_checks --slack_max -0.01
===========================================================================
```

========== Typical Corner ==========

No paths found.

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/routing/16-rsz_design_sta.checks.rpt ? par
==

```
=========================================================================
 report_check_types -max_slew -max_cap -max_fanout -violators
=========================================================================
======================== Typical Corner ===================================


=========================================================================
report_parasitic_annotation -report_unannotated
=========================================================================
Found 32 unannotated drivers.
 clk
 dry_run_fault
 level_full
 level_low
 rst_n
 _09_/Y
 _10_/Y
 _11_/Y
 _12_/X
 _13_/Y
 _14_/Y
 _15_/Y
 _16_/Y
 _17_/X
 _18_/X
 _19_/X
 _20_/Q
 _21_/Q
 _22_/Q
 clkbuf_0_clk/X
 clkbuf_1_0__f_clk/X
 clkbuf_1_1__f_clk/X
 hold1/X
 hold2/X
 hold3/X
 hold4/X
 input1/X
 input2/X
 input3/X
 input4/X
 output5/X
 output6/X
```

```
Found 0 partially unannotated drivers.

================================================================================
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/routing/16-rsz_design_sta.checks.rpt ? par
==

max slew violation count 0
max fanout violation count 0
max cap violation count 0
==

==
check_setup -verbose -unconstrained_endpoints -multiple_clock -no_clock -no_input_delay
-loops -generated_clocks
==

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/routing/16-rsz_design_sta.max.rpt ? part 1
==

```
=========================================================================
report_checks -path_delay max (Setup)
=========================================================================
======================== Typical Corner ===================================

Startpoint: level_full (input port clocked by clk)
Endpoint: pump_en (output port clocked by clk)
Path Group: clk
Path Type: max

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 v input external delay
                   0.02   0.01    4.01 v level_full (in)
     1    0.01                           level_full (net)
                   0.02   0.00    4.01 v input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
                   0.09   0.18    4.19 v input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
     3    0.01                           net2 (net)
                   0.09   0.00    4.19 v _09_/A2 (sky130_fd_sc_hd__a21oi_2)
                   0.20   0.24    4.43 ^ _09_/Y (sky130_fd_sc_hd__a21oi_2)
     4    0.01                           _03_ (net)
                   0.20   0.00    4.43 ^ _18_/B (sky130_fd_sc_hd__and2_1)
                   0.05   0.16    4.59 ^ _18_/X (sky130_fd_sc_hd__and2_1)
     1    0.00                           _08_ (net)
                   0.05   0.00    4.59 ^ _19_/A (sky130_fd_sc_hd__clkbuf_1)
                   0.05   0.09    4.68 ^ _19_/X (sky130_fd_sc_hd__clkbuf_1)
     1    0.00                           net6 (net)
                   0.05   0.00    4.68 ^ output6/A (sky130_fd_sc_hd__clkbuf_4)
                   0.11   0.20    4.87 ^ output6/X (sky130_fd_sc_hd__clkbuf_4)
     1    0.03                           pump_en (net)
                   0.11   0.00    4.87 ^ pump_en (out)
                          4.87   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock network delay (propagated)
                         -0.25   19.75   clock uncertainty
                          0.00   19.75   clock reconvergence pessimism
                         -4.00   15.75   output external delay
                                 15.75   data required time
---------------------------------------------------------------------------
                                 15.75   data required time
                                 -4.87   data arrival time
---------------------------------------------------------------------------
```

10.88 slack (MET)

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/routing/16-rsz_design_sta.max.rpt ? part 2
==========

Startpoint: level_full (input port clocked by clk)
Endpoint: alarm (output port clocked by clk)
Path Group: clk
Path Type: max

Fanout	Cap	Slew	Delay	Time		Description
			0.00	0.00		clock clk (rise edge)
			0.00	0.00		clock network delay (propagated)
			4.00	4.00	v	input external delay
		0.02	0.01	4.01	v	level_full (in)
1	0.01					level_full (net)
		0.02	0.00	4.01	v	input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
		0.09	0.18	4.19	v	input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
3	0.01					net2 (net)
		0.09	0.00	4.19	v	_09_/A2 (sky130_fd_sc_hd__a21oi_2)
		0.20	0.24	4.43	^	_09_/Y (sky130_fd_sc_hd__a21oi_2)
4	0.01					_03_ (net)
		0.20	0.00	4.43	^	_10_/A (sky130_fd_sc_hd__inv_2)
		0.05	0.06	4.49	v	_10_/Y (sky130_fd_sc_hd__inv_2)
2	0.01					net5 (net)
		0.05	0.00	4.49	v	output5/A (sky130_fd_sc_hd__buf_2)
		0.09	0.19	4.67	v	output5/X (sky130_fd_sc_hd__buf_2)
1	0.03					alarm (net)
		0.09	0.00	4.68	v	alarm (out)
				4.68		data arrival time
			20.00	20.00		clock clk (rise edge)
			0.00	20.00		clock network delay (propagated)
			-0.25	19.75		clock uncertainty
			0.00	19.75		clock reconvergence pessimism
			-4.00	15.75		output external delay
				15.75		data required time
				15.75		data required time
				-4.68		data arrival time
				11.07		slack (MET)

Startpoint: level_full (input port clocked by clk)
Endpoint: _21_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

Fanout	Cap	Slew	Delay	Time	Description
			0.00	0.00	clock clk (rise edge)

```
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 v input external delay
                 0.02     0.01    4.01 v level_full (in)
     1    0.01                           level_full (net)
                 0.02     0.00    4.01 v input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
                 0.09     0.18    4.19 v input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
     3    0.01                           net2 (net)
                 0.09     0.00    4.19 v _09_/A2 (sky130_fd_sc_hd__a21oi_2)
                 0.20     0.24    4.43 ^ _09_/Y (sky130_fd_sc_hd__a21oi_2)
     4    0.01                           _03_ (net)
                 0.20     0.00    4.43 ^ _17_/B1 (sky130_fd_sc_hd__o211a_1)
                 0.04     0.18    4.61 ^ _17_/X (sky130_fd_sc_hd__o211a_1)
     1    0.00                           _01_ (net)
                 0.04     0.00    4.61 ^ hold2/A (sky130_fd_sc_hd__dlygate4sd3_1)
                 0.06     0.56    5.17 ^ hold2/X (sky130_fd_sc_hd__dlygate4sd3_1)
     1    0.00                           net8 (net)
                 0.06     0.00    5.17 ^ _21_/D (sky130_fd_sc_hd__dfrtp_1)
                                  5.17   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock source latency
                 0.06     0.04   20.04 ^ clk (in)
     1    0.01                           clk (net)
                 0.06     0.00   20.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                 0.04     0.12   20.16 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                 0.04     0.00   20.16 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                 0.03     0.10   20.27 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                 0.03     0.00   20.27 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                         -0.25   20.02   clock uncertainty
                          0.00   20.02   clock reconvergence pessimism
                         -0.06   19.95   library setup time
                                 19.95   data required time
-----------------------------------------------------------------------------
                                 19.95   data required time
                                 -5.17   data arrival time
-----------------------------------------------------------------------------
                                 14.78   slack (MET)


Startpoint: level_full (input port clocked by clk)
Endpoint: _20_ (rising edge-triggered flip-flop clocked by clk)
```

Path Group: clk
Path Type: max

Fanout	Cap	Slew	Delay	Time	Description

```
==========================================================================================

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                 0.03     0.02    4.02 ^ level_full (in)
     1    0.01                           level_full (net)
                 0.03     0.00    4.02 ^ input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
                 0.17     0.18    4.20 ^ input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
     3    0.02                           net2 (net)
                 0.17     0.00    4.21 ^ _11_/A (sky130_fd_sc_hd__inv_2)
                 0.04     0.06    4.26 v _11_/Y (sky130_fd_sc_hd__inv_2)
     2    0.01                           _04_ (net)
                 0.04     0.00    4.26 v _15_/A1 (sky130_fd_sc_hd__a21oi_1)
                 0.12     0.13    4.40 ^ _15_/Y (sky130_fd_sc_hd__a21oi_1)
     1    0.00                           _07_ (net)
                 0.12     0.00    4.40 ^ hold4/A (sky130_fd_sc_hd__dlygate4sd3_1)
                 0.05     0.59    4.98 ^ hold4/X (sky130_fd_sc_hd__dlygate4sd3_1)
     1    0.00                           net10 (net)
                 0.05     0.00    4.98 ^ _16_/A2 (sky130_fd_sc_hd__o21ai_1)
                 0.07     0.06    5.04 v _16_/Y (sky130_fd_sc_hd__o21ai_1)
     1    0.00                           _00_ (net)
                 0.07     0.00    5.04 v _20_/D (sky130_fd_sc_hd__dfstp_1)
                           5.04    5.04   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock source latency
                 0.06     0.04   20.04 ^ clk (in)
     1    0.01                           clk (net)
                 0.06     0.00   20.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                 0.04     0.12   20.16 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                 0.04     0.00   20.16 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                 0.03     0.10   20.27 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                 0.03     0.00   20.27 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_1)
                         -0.25   20.02   clock uncertainty
                          0.00   20.02   clock reconvergence pessimism
                         -0.08   19.94   library setup time
                                 19.94   data required time
------------------------------------------------------------------------------
                                 19.94   data required time
                                 -5.04   data arrival time
------------------------------------------------------------------------------
                                 14.89   slack (MET)
```

Startpoint: level_full (input port clocked by clk)
Endpoint: _22_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk

```
================================================================================

Path Type: max

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 v input external delay
                   0.02   0.01    4.01 v level_full (in)
     1    0.01                           level_full (net)
                   0.02   0.00    4.01 v input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
                   0.09   0.18    4.19 v input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
     3    0.01                           net2 (net)
                   0.09   0.00    4.19 v _09_/A2 (sky130_fd_sc_hd__a21oi_2)
                   0.20   0.24    4.43 ^ _09_/Y (sky130_fd_sc_hd__a21oi_2)
     4    0.01                           _03_ (net)
                   0.20   0.00    4.43 ^ _10_/A (sky130_fd_sc_hd__inv_2)
                   0.05   0.06    4.49 v _10_/Y (sky130_fd_sc_hd__inv_2)
     2    0.01                           net5 (net)
                   0.05   0.00    4.49 v _14_/A (sky130_fd_sc_hd__nor2_1)
                   0.09   0.11    4.60 ^ _14_/Y (sky130_fd_sc_hd__nor2_1)
     1    0.00                           _02_ (net)
                   0.09   0.00    4.60 ^ _22_/D (sky130_fd_sc_hd__dfrtp_1)
                                  4.60   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock source latency
                   0.06   0.04   20.04 ^ clk (in)
     1    0.01                           clk (net)
                   0.06   0.00   20.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                   0.04   0.12   20.16 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                   0.04   0.00   20.16 ^ clkbuf_1_1__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                   0.03   0.10   20.26 ^ clkbuf_1_1__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     1    0.00                           clknet_1_1__leaf_clk (net)
                   0.03   0.00   20.26 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_1)
                         -0.25   20.01   clock uncertainty
                          0.00   20.01   clock reconvergence pessimism
                         -0.07   19.94   library setup time
                                 19.94   data required time
---------------------------------------------------------------------------
                                 19.94   data required time
                                 -4.60   data arrival time
---------------------------------------------------------------------------
```

15.35   slack (MET)

Startpoint: rst_n (input port clocked by clk)
Endpoint: _20_ (recovery check against rising-edge clock clk)

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/routing/16-rsz_design_sta.max.rpt ? part 6
==

Path Group: asynchronous
Path Type: max

```
Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                 0.02     0.01    4.01 ^ rst_n (in)
     1    0.00                           rst_n (net)
                 0.02     0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                 0.14     0.14    4.15 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.01                           net4 (net)
                 0.14     0.00    4.15 ^ _20_/SET_B (sky130_fd_sc_hd__dfstp_1)
                                  4.15   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock source latency
                 0.06     0.04   20.04 ^ clk (in)
     1    0.01                           clk (net)
                 0.06     0.00   20.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                 0.04     0.12   20.16 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                 0.04     0.00   20.16 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                 0.03     0.10   20.27 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                 0.03     0.00   20.27 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_1)
                         -0.25   20.02   clock uncertainty
                          0.00   20.02   clock reconvergence pessimism
                          0.12   20.14   library recovery time
                                 20.14   data required time
-----------------------------------------------------------------------------
                                 20.14   data required time
                                 -4.15   data arrival time
-----------------------------------------------------------------------------
                                 15.99   slack (MET)


Startpoint: rst_n (input port clocked by clk)
Endpoint: _22_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
Path Type: max
```

```
Fanout     Cap     Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
```

```
                  0.02    0.01    4.01 ^ rst_n (in)
     1    0.00                           rst_n (net)
                  0.02    0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                  0.14    0.14    4.15 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.01                           net4 (net)
                  0.14    0.00    4.15 ^ _22_/RESET_B (sky130_fd_sc_hd__dfrtp_1)
                                  4.15   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock source latency
                  0.06    0.04   20.04 ^ clk (in)
     1    0.01                           clk (net)
                  0.06    0.00   20.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.04    0.12   20.16 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                  0.04    0.00   20.16 ^ clkbuf_1_1__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.03    0.10   20.26 ^ clkbuf_1_1__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     1    0.00                           clknet_1_1__leaf_clk (net)
                  0.03    0.00   20.26 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_1)
                         -0.25   20.01   clock uncertainty
                          0.00   20.01   clock reconvergence pessimism
                          0.20   20.21   library recovery time
                                 20.21   data required time
-------------------------------------------------------------------------------
                                 20.21   data required time
                                 -4.15   data arrival time
-------------------------------------------------------------------------------
                                 16.06   slack (MET)


Startpoint: rst_n (input port clocked by clk)
Endpoint: _21_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
Path Type: max

Fanout     Cap    Slew   Delay    Time   Description
-------------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                  0.02    0.01    4.01 ^ rst_n (in)
     1    0.00                           rst_n (net)
                  0.02    0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
```

```
              0.14    0.14    4.15 ^ input4/X (sky130_fd_sc_hd__buf_1)
3    0.01                             net4 (net)
              0.14    0.00    4.15 ^ _21_/RESET_B (sky130_fd_sc_hd__dfrtp_1)
                              4.15   data arrival time
```

```
                          20.00   20.00   clock clk (rise edge)
                           0.00   20.00   clock source latency
                  0.06     0.04   20.04 ^ clk (in)
     1    0.01                            clk (net)
                  0.06     0.00   20.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.04     0.12   20.16 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                            clknet_0_clk (net)
                  0.04     0.00   20.16 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.03     0.10   20.27 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                            clknet_1_0__leaf_clk (net)
                  0.03     0.00   20.27 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                          -0.25   20.02   clock uncertainty
                           0.00   20.02   clock reconvergence pessimism
                           0.20   20.22   library recovery time
                                  20.22   data required time
-------------------------------------------------------------------------------
                                  20.22   data required time
                                  -4.15   data arrival time
-------------------------------------------------------------------------------
                                  16.07   slack (MET)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/routing/16-rsz_design_sta.min.rpt ? part 1
==

```
=========================================================================
report_checks -path_delay min (Hold)
=========================================================================
======================== Typical Corner ===================================

Startpoint: _21_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _22_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                  0.06    0.04    0.04 ^ clk (in)
     1    0.01                          clk (net)
                  0.06    0.00    0.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.04    0.12    0.16 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                          clknet_0_clk (net)
                  0.04    0.00    0.16 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.03    0.10    0.27 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                          clknet_1_0__leaf_clk (net)
                  0.03    0.00    0.27 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                  0.10    0.34    0.61 ^ _21_/Q (sky130_fd_sc_hd__dfrtp_1)
     3    0.01                          state[1] (net)
                  0.10    0.00    0.61 ^ _12_/A1 (sky130_fd_sc_hd__o21a_1)
                  0.05    0.13    0.74 ^ _12_/X (sky130_fd_sc_hd__o21a_1)
     1    0.00                          _05_ (net)
                  0.05    0.00    0.74 ^ _13_/B1 (sky130_fd_sc_hd__a21oi_1)
                  0.03    0.04    0.77 v _13_/Y (sky130_fd_sc_hd__a21oi_1)
     1    0.00                          _06_ (net)
                  0.03    0.00    0.77 v _14_/B (sky130_fd_sc_hd__nor2_1)
                  0.09    0.08    0.85 ^ _14_/Y (sky130_fd_sc_hd__nor2_1)
     1    0.00                          _02_ (net)
                  0.09    0.00    0.85 ^ _22_/D (sky130_fd_sc_hd__dfrtp_1)
                                  0.85   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                  0.06    0.04    0.04 ^ clk (in)
     1    0.01                          clk (net)
                  0.06    0.00    0.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.04    0.14    0.18 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
```

```
     2    0.02                           clknet_0_clk (net)
                  0.04    0.00    0.18 ^ clkbuf_1_1__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.03    0.11    0.29 ^ clkbuf_1_1__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     1    0.00                           clknet_1_1__leaf_clk (net)
                  0.03    0.00    0.29 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_1)
```

```
                           0.25    0.54   clock uncertainty
                          -0.02    0.52   clock reconvergence pessimism
                          -0.05    0.48   library hold time
                                   0.48   data required time
-----------------------------------------------------------------------------
                                   0.48   data required time
                                  -0.85   data arrival time
-----------------------------------------------------------------------------
                                   0.37   slack (MET)


Startpoint: _21_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _21_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                           0.00    0.00   clock clk (rise edge)
                           0.00    0.00   clock source latency
                  0.06     0.04    0.04 ^ clk (in)
     1    0.01                            clk (net)
                  0.06     0.00    0.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.04     0.12    0.16 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                            clknet_0_clk (net)
                  0.04     0.00    0.16 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.03     0.10    0.27 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                            clknet_1_0__leaf_clk (net)
                  0.03     0.00    0.27 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                  0.10     0.34    0.61 ^ _21_/Q (sky130_fd_sc_hd__dfrtp_1)
     3    0.01                            state[1] (net)
                  0.10     0.00    0.61 ^ _17_/A1 (sky130_fd_sc_hd__o211a_1)
                  0.04     0.15    0.76 ^ _17_/X (sky130_fd_sc_hd__o211a_1)
     1    0.00                            _01_ (net)
                  0.04     0.00    0.76 ^ hold2/A (sky130_fd_sc_hd__dlygate4sd3_1)
                  0.06     0.51    1.26 ^ hold2/X (sky130_fd_sc_hd__dlygate4sd3_1)
     1    0.00                            net8 (net)
                  0.06     0.00    1.26 ^ _21_/D (sky130_fd_sc_hd__dfrtp_1)
                                   1.26   data arrival time

                           0.00    0.00   clock clk (rise edge)
                           0.00    0.00   clock source latency
                  0.06     0.04    0.04 ^ clk (in)
```

```
     1    0.01                           clk (net)
                  0.06    0.00    0.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.04    0.14    0.18 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                  0.04    0.00    0.18 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
```

```
                  0.03    0.12    0.29 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                  0.03    0.00    0.29 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                          0.25    0.54   clock uncertainty
                         -0.03    0.52   clock reconvergence pessimism
                         -0.04    0.48   library hold time
                                  0.48   data required time
-----------------------------------------------------------------------------
                                  0.48   data required time
                                 -1.26   data arrival time
-----------------------------------------------------------------------------
                                  0.79   slack (MET)


Startpoint: _21_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _20_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                  0.06    0.04    0.04 ^ clk (in)
     1    0.01                           clk (net)
                  0.06    0.00    0.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.04    0.12    0.16 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                  0.04    0.00    0.16 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.03    0.10    0.27 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                  0.03    0.00    0.27 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                  0.10    0.34    0.61 ^ _21_/Q (sky130_fd_sc_hd__dfrtp_1)
     3    0.01                           state[1] (net)
                  0.10    0.00    0.61 ^ _15_/A2 (sky130_fd_sc_hd__a21oi_1)
                  0.03    0.07    0.68 v _15_/Y (sky130_fd_sc_hd__a21oi_1)
     1    0.00                           _07_ (net)
                  0.03    0.00    0.68 v hold4/A (sky130_fd_sc_hd__dlygate4sd3_1)
                  0.05    0.51    1.20 v hold4/X (sky130_fd_sc_hd__dlygate4sd3_1)
     1    0.00                           net10 (net)
                  0.05    0.00    1.20 v _16_/A2 (sky130_fd_sc_hd__o21ai_1)
                  0.08    0.11    1.31 ^ _16_/Y (sky130_fd_sc_hd__o21ai_1)
```

```
1    0.00                           _00_ (net)
             0.08    0.00    1.31 ^ _20_/D (sky130_fd_sc_hd__dfstp_1)
                             1.31   data arrival time

                     0.00    0.00   clock clk (rise edge)
                     0.00    0.00   clock source latency
```

```
                  0.06    0.04    0.04 ^ clk (in)
     1    0.01                           clk (net)
                  0.06    0.00    0.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.04    0.14    0.18 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                  0.04    0.00    0.18 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.03    0.12    0.29 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                  0.03    0.00    0.29 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_1)
                          0.25    0.54   clock uncertainty
                         -0.03    0.52   clock reconvergence pessimism
                         -0.04    0.47   library hold time
                                  0.47   data required time
-----------------------------------------------------------------------------
                                  0.47   data required time
                                 -1.31   data arrival time
-----------------------------------------------------------------------------
                                  0.84   slack (MET)


Startpoint: rst_n (input port clocked by clk)
Endpoint: _21_ (removal check against rising-edge clock clk)
Path Group: asynchronous
Path Type: min

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                  0.02    0.01    4.01 ^ rst_n (in)
     1    0.00                           rst_n (net)
                  0.02    0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                  0.14    0.13    4.14 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.01                           net4 (net)
                  0.14    0.00    4.14 ^ _21_/RESET_B (sky130_fd_sc_hd__dfrtp_1)
                                  4.14   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                  0.06    0.04    0.04 ^ clk (in)
     1    0.01                           clk (net)
                  0.06    0.00    0.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
```

```
                0.04    0.14    0.18 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                       clknet_0_clk (net)
                0.04    0.00    0.18 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                0.03    0.12    0.29 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                       clknet_1_0__leaf_clk (net)
```

```
                  0.03    0.00    0.29 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                          0.25    0.54   clock uncertainty
                          0.00    0.54   clock reconvergence pessimism
                          0.34    0.88   library removal time
                                  0.88   data required time
---------------------------------------------------------------------------
                                  0.88   data required time
                                 -4.14   data arrival time
---------------------------------------------------------------------------
                                  3.25   slack (MET)


Startpoint: rst_n (input port clocked by clk)
Endpoint: _22_ (removal check against rising-edge clock clk)
Path Group: asynchronous
Path Type: min

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                  0.02    0.01    4.01 ^ rst_n (in)
     1    0.00                           rst_n (net)
                  0.02    0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                  0.14    0.13    4.14 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.01                           net4 (net)
                  0.14    0.00    4.14 ^ _22_/RESET_B (sky130_fd_sc_hd__dfrtp_1)
                                  4.14   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                  0.06    0.04    0.04 ^ clk (in)
     1    0.01                           clk (net)
                  0.06    0.00    0.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.04    0.14    0.18 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                  0.04    0.00    0.18 ^ clkbuf_1_1__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.03    0.11    0.29 ^ clkbuf_1_1__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     1    0.00                           clknet_1_1__leaf_clk (net)
                  0.03    0.00    0.29 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_1)
                          0.25    0.54   clock uncertainty
                          0.00    0.54   clock reconvergence pessimism
```

```
                          0.34    0.88   library removal time
                                  0.88   data required time
-----------------------------------------------------------------------------
                                  0.88   data required time
                                 -4.14   data arrival time
```

```
================================================================================

--------------------------------------------------------------------------
                                3.26   slack (MET)


Startpoint: rst_n (input port clocked by clk)
Endpoint: _20_ (removal check against rising-edge clock clk)
Path Group: asynchronous
Path Type: min

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                         0.00    0.00   clock clk (rise edge)
                         0.00    0.00   clock network delay (propagated)
                         4.00    4.00 ^ input external delay
                  0.02   0.01    4.01 ^ rst_n (in)
     1    0.00                          rst_n (net)
                  0.02   0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                  0.14   0.13    4.14 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.01                          net4 (net)
                  0.14   0.00    4.14 ^ _20_/SET_B (sky130_fd_sc_hd__dfstp_1)
                                4.14   data arrival time

                         0.00    0.00   clock clk (rise edge)
                         0.00    0.00   clock source latency
                  0.06   0.04    0.04 ^ clk (in)
     1    0.01                          clk (net)
                  0.06   0.00    0.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.04   0.14    0.18 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                          clknet_0_clk (net)
                  0.04   0.00    0.18 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.03   0.12    0.29 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                          clknet_1_0__leaf_clk (net)
                  0.03   0.00    0.29 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_1)
                         0.25    0.54   clock uncertainty
                         0.00    0.54   clock reconvergence pessimism
                         0.14    0.69   library removal time
                                 0.69   data required time
---------------------------------------------------------------------------
                                 0.69   data required time
                                -4.14   data arrival time
---------------------------------------------------------------------------
                                 3.45   slack (MET)
```

```
Startpoint: _22_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: pump_en (output port clocked by clk)
Path Group: clk
Path Type: min
```

```
Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                  0.06    0.04    0.04 ^ clk (in)
     1    0.01                           clk (net)
                  0.06    0.00    0.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.04    0.12    0.16 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                  0.04    0.00    0.16 ^ clkbuf_1_1__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.03    0.10    0.26 ^ clkbuf_1_1__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     1    0.00                           clknet_1_1__leaf_clk (net)
                  0.03    0.00    0.26 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_1)
                  0.06    0.31    0.57 ^ _22_/Q (sky130_fd_sc_hd__dfrtp_1)
     2    0.01                           state[2] (net)
                  0.06    0.00    0.57 ^ _18_/A (sky130_fd_sc_hd__and2_1)
                  0.04    0.11    0.68 ^ _18_/X (sky130_fd_sc_hd__and2_1)
     1    0.00                           _08_ (net)
                  0.04    0.00    0.68 ^ _19_/A (sky130_fd_sc_hd__clkbuf_1)
                  0.05    0.08    0.76 ^ _19_/X (sky130_fd_sc_hd__clkbuf_1)
     1    0.00                           net6 (net)
                  0.05    0.00    0.76 ^ output6/A (sky130_fd_sc_hd__clkbuf_4)
                  0.11    0.18    0.94 ^ output6/X (sky130_fd_sc_hd__clkbuf_4)
     1    0.03                           pump_en (net)
                  0.11    0.00    0.94 ^ pump_en (out)
                                  0.94   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          0.25    0.25   clock uncertainty
                          0.00    0.25   clock reconvergence pessimism
                         -4.00   -3.75   output external delay
                                 -3.75   data required time
-----------------------------------------------------------------------------
                                 -3.75   data required time
                                 -0.94   data arrival time
-----------------------------------------------------------------------------
                                  4.69   slack (MET)


Startpoint: dry_run_fault (input port clocked by clk)
Endpoint: alarm (output port clocked by clk)
```

Path Group: clk
Path Type: min

Fanout	Cap	Slew	Delay	Time	Description

```
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                 0.02     0.01    4.01 ^ dry_run_fault (in)
     1    0.00                           dry_run_fault (net)
                 0.02     0.00    4.01 ^ input1/A (sky130_fd_sc_hd__clkbuf_1)
                 0.07     0.09    4.10 ^ input1/X (sky130_fd_sc_hd__clkbuf_1)
     1    0.01                           net1 (net)
                 0.07     0.00    4.10 ^ _09_/B1 (sky130_fd_sc_hd__a21oi_2)
                 0.04     0.05    4.16 v _09_/Y (sky130_fd_sc_hd__a21oi_2)
     4    0.01                           _03_ (net)
                 0.04     0.00    4.16 v _10_/A (sky130_fd_sc_hd__inv_2)
                 0.04     0.05    4.21 ^ _10_/Y (sky130_fd_sc_hd__inv_2)
     2    0.01                           net5 (net)
                 0.04     0.00    4.21 ^ output5/A (sky130_fd_sc_hd__buf_2)
                 0.17     0.18    4.39 ^ output5/X (sky130_fd_sc_hd__buf_2)
     1    0.03                           alarm (net)
                 0.17     0.00    4.39 ^ alarm (out)
                                  4.39   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          0.25    0.25   clock uncertainty
                          0.00    0.25   clock reconvergence pessimism
                         -4.00   -3.75   output external delay
                                 -3.75   data required time
-------------------------------------------------------------------------------
                                 -3.75   data required time
                                 -4.39   data arrival time
-------------------------------------------------------------------------------
                                  8.14   slack (MET)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/routing/16-rsz_design_sta.power.rpt
==

```
===========================================================================
 report_power
===========================================================================
======================== Typical Corner ===================================

Group                  Internal  Switching    Leakage      Total
                          Power      Power      Power      Power (Watts)
----------------------------------------------------------------
Sequential             6.60e-06   2.20e-07   3.33e-11   6.82e-06  16.8%
Combinational          2.73e-05   6.45e-06   2.03e-10   3.38e-05  83.2%
Macro                  0.00e+00   0.00e+00   0.00e+00   0.00e+00   0.0%
Pad                    0.00e+00   0.00e+00   0.00e+00   0.00e+00   0.0%
----------------------------------------------------------------
Total                  3.39e-05   6.67e-06   2.36e-10   4.06e-05 100.0%
                          83.6%      16.4%       0.0%
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/routing/16-rsz_design_sta.skew.rpt
==

```
============================================================================
report_clock_skew
============================================================================
Clock clk
Latency      CRPR       Skew
_20_/CLK ^
   0.29
_22_/CLK ^
   0.26     -0.02       0.01
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/routing/16-rsz_design_sta.summary.rpt
==

==
report_tns
==
tns 0.00

==
report_wns
==
wns 0.00

==
report_worst_slack -max (Setup)
==
worst slack 10.88

==
report_worst_slack -min (Hold)
==
worst slack 0.37

```
openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/routing/18-rsz_timing_sta.checks.rpt ? par
========================================================================================

===========================================================================
report_checks -unconstrained
===========================================================================
======================== Typical Corner ===================================

Startpoint: rst_n (input port clocked by clk)
Endpoint: _20_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
Path Type: max

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                  0.02    0.01    4.01 ^ rst_n (in)
     1    0.00                           rst_n (net)
                  0.02    0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                  0.14    0.14    4.15 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.01                           net4 (net)
                  0.14    0.00    4.15 ^ _20_/SET_B (sky130_fd_sc_hd__dfstp_1)
                                  4.15   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock source latency
                  0.06    0.04   20.04 ^ clk (in)
     1    0.01                           clk (net)
                  0.06    0.00   20.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.04    0.12   20.16 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                  0.04    0.00   20.16 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.03    0.10   20.27 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                  0.03    0.00   20.27 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_1)
                         -0.25   20.02   clock uncertainty
                          0.00   20.02   clock reconvergence pessimism
                          0.12   20.14   library recovery time
                                 20.14   data required time
---------------------------------------------------------------------------
                                 20.14   data required time
                                 -4.15   data arrival time
---------------------------------------------------------------------------
                                 15.99   slack (MET)
```

```
Startpoint: level_full (input port clocked by clk)
Endpoint: pump_en (output port clocked by clk)
Path Group: clk
```

```
openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/routing/18-rsz_timing_sta.checks.rpt ? par
================================================================================

Path Type: max

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 v input external delay
                   0.02   0.01    4.01 v level_full (in)
     1    0.01                           level_full (net)
                   0.02   0.00    4.01 v input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
                   0.09   0.18    4.19 v input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
     3    0.01                           net2 (net)
                   0.09   0.00    4.19 v _09_/A2 (sky130_fd_sc_hd__a21oi_2)
                   0.20   0.24    4.43 ^ _09_/Y (sky130_fd_sc_hd__a21oi_2)
     4    0.01                           _03_ (net)
                   0.20   0.00    4.43 ^ _18_/B (sky130_fd_sc_hd__and2_1)
                   0.05   0.16    4.59 ^ _18_/X (sky130_fd_sc_hd__and2_1)
     1    0.00                           _08_ (net)
                   0.05   0.00    4.59 ^ _19_/A (sky130_fd_sc_hd__clkbuf_1)
                   0.05   0.09    4.68 ^ _19_/X (sky130_fd_sc_hd__clkbuf_1)
     1    0.00                           net6 (net)
                   0.05   0.00    4.68 ^ output6/A (sky130_fd_sc_hd__clkbuf_4)
                   0.11   0.20    4.87 ^ output6/X (sky130_fd_sc_hd__clkbuf_4)
     1    0.03                           pump_en (net)
                   0.11   0.00    4.87 ^ pump_en (out)
                                  4.87   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock network delay (propagated)
                         -0.25   19.75   clock uncertainty
                          0.00   19.75   clock reconvergence pessimism
                         -4.00   15.75   output external delay
                                 15.75   data required time
---------------------------------------------------------------------------
                                 15.75   data required time
                                 -4.87   data arrival time
---------------------------------------------------------------------------
                                 10.88   slack (MET)




================================================================================
report_checks --slack_max -0.01
================================================================================
```

========== Typical Corner ==========

No paths found.

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/routing/18-rsz_timing_sta.checks.rpt ? par
==

===
 report_check_types -max_slew -max_cap -max_fanout -violators
===
======================== Typical Corner ===================================

===
report_parasitic_annotation -report_unannotated
===
Found 32 unannotated drivers.
 clk
 dry_run_fault
 level_full
 level_low
 rst_n
 _09_/Y
 _10_/Y
 _11_/Y
 _12_/X
 _13_/Y
 _14_/Y
 _15_/Y
 _16_/Y
 _17_/X
 _18_/X
 _19_/X
 _20_/Q
 _21_/Q
 _22_/Q
 clkbuf_0_clk/X
 clkbuf_1_0__f_clk/X
 clkbuf_1_1__f_clk/X
 hold1/X
 hold2/X
 hold3/X
 hold4/X
 input1/X
 input2/X
 input3/X
 input4/X
 output5/X
 output6/X

```
Found 0 partially unannotated drivers.

================================================================================
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/routing/18-rsz_timing_sta.checks.rpt ? par
==

max slew violation count 0
max fanout violation count 0
max cap violation count 0
==

==
check_setup -verbose -unconstrained_endpoints -multiple_clock -no_clock -no_input_delay
-loops -generated_clocks
==

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/routing/18-rsz_timing_sta.max.rpt ? part 1
==

```
=========================================================================
report_checks -path_delay max (Setup)
=========================================================================
======================== Typical Corner ===================================

Startpoint: level_full (input port clocked by clk)
Endpoint: pump_en (output port clocked by clk)
Path Group: clk
Path Type: max

Fanout     Cap    Slew   Delay    Time   Description
-------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 v input external delay
                0.02      0.01    4.01 v level_full (in)
     1    0.01                           level_full (net)
                0.02      0.00    4.01 v input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
                0.09      0.18    4.19 v input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
     3    0.01                           net2 (net)
                0.09      0.00    4.19 v _09_/A2 (sky130_fd_sc_hd__a21oi_2)
                0.20      0.24    4.43 ^ _09_/Y (sky130_fd_sc_hd__a21oi_2)
     4    0.01                           _03_ (net)
                0.20      0.00    4.43 ^ _18_/B (sky130_fd_sc_hd__and2_1)
                0.05      0.16    4.59 ^ _18_/X (sky130_fd_sc_hd__and2_1)
     1    0.00                           _08_ (net)
                0.05      0.00    4.59 ^ _19_/A (sky130_fd_sc_hd__clkbuf_1)
                0.05      0.09    4.68 ^ _19_/X (sky130_fd_sc_hd__clkbuf_1)
     1    0.00                           net6 (net)
                0.05      0.00    4.68 ^ output6/A (sky130_fd_sc_hd__clkbuf_4)
                0.11      0.20    4.87 ^ output6/X (sky130_fd_sc_hd__clkbuf_4)
     1    0.03                           pump_en (net)
                0.11      0.00    4.87 ^ pump_en (out)
                                  4.87   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock network delay (propagated)
                         -0.25   19.75   clock uncertainty
                          0.00   19.75   clock reconvergence pessimism
                         -4.00   15.75   output external delay
                                 15.75   data required time
-------------------------------------------------------------------------
                                 15.75   data required time
                                 -4.87   data arrival time
-------------------------------------------------------------------------
```

10.88 slack (MET)

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/routing/18-rsz_timing_sta.max.rpt ? part 2
==

Startpoint: level_full (input port clocked by clk)
Endpoint: alarm (output port clocked by clk)
Path Group: clk
Path Type: max

Fanout	Cap	Slew	Delay	Time		Description
			0.00	0.00		clock clk (rise edge)
			0.00	0.00		clock network delay (propagated)
			4.00	4.00	v	input external delay
		0.02	0.01	4.01	v	level_full (in)
1	0.01					level_full (net)
		0.02	0.00	4.01	v	input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
		0.09	0.18	4.19	v	input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
3	0.01					net2 (net)
		0.09	0.00	4.19	v	_09_/A2 (sky130_fd_sc_hd__a21oi_2)
		0.20	0.24	4.43	^	_09_/Y (sky130_fd_sc_hd__a21oi_2)
4	0.01					_03_ (net)
		0.20	0.00	4.43	^	_10_/A (sky130_fd_sc_hd__inv_2)
		0.05	0.06	4.49	v	_10_/Y (sky130_fd_sc_hd__inv_2)
2	0.01					net5 (net)
		0.05	0.00	4.49	v	output5/A (sky130_fd_sc_hd__buf_2)
		0.09	0.19	4.67	v	output5/X (sky130_fd_sc_hd__buf_2)
1	0.03					alarm (net)
		0.09	0.00	4.68	v	alarm (out)
				4.68		data arrival time
			20.00	20.00		clock clk (rise edge)
			0.00	20.00		clock network delay (propagated)
			-0.25	19.75		clock uncertainty
			0.00	19.75		clock reconvergence pessimism
			-4.00	15.75		output external delay
				15.75		data required time
				15.75		data required time
				-4.68		data arrival time
				11.07		slack (MET)

Startpoint: level_full (input port clocked by clk)
Endpoint: _21_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

Fanout	Cap	Slew	Delay	Time	Description
			0.00	0.00	clock clk (rise edge)

```
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 v input external delay
                 0.02     0.01    4.01 v level_full (in)
     1    0.01                           level_full (net)
                 0.02     0.00    4.01 v input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
                 0.09     0.18    4.19 v input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
     3    0.01                           net2 (net)
                 0.09     0.00    4.19 v _09_/A2 (sky130_fd_sc_hd__a21oi_2)
                 0.20     0.24    4.43 ^ _09_/Y (sky130_fd_sc_hd__a21oi_2)
     4    0.01                           _03_ (net)
                 0.20     0.00    4.43 ^ _17_/B1 (sky130_fd_sc_hd__o211a_1)
                 0.04     0.18    4.61 ^ _17_/X (sky130_fd_sc_hd__o211a_1)
     1    0.00                           _01_ (net)
                 0.04     0.00    4.61 ^ hold2/A (sky130_fd_sc_hd__dlygate4sd3_1)
                 0.06     0.56    5.17 ^ hold2/X (sky130_fd_sc_hd__dlygate4sd3_1)
     1    0.00                           net8 (net)
                 0.06     0.00    5.17 ^ _21_/D (sky130_fd_sc_hd__dfrtp_1)
                           0.00    5.17   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock source latency
                 0.06     0.04   20.04 ^ clk (in)
     1    0.01                           clk (net)
                 0.06     0.00   20.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                 0.04     0.12   20.16 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                 0.04     0.00   20.16 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                 0.03     0.10   20.27 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                 0.03     0.00   20.27 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                         -0.25   20.02   clock uncertainty
                          0.00   20.02   clock reconvergence pessimism
                         -0.06   19.95   library setup time
                                 19.95   data required time
-----------------------------------------------------------------------------
                                 19.95   data required time
                                 -5.17   data arrival time
-----------------------------------------------------------------------------
                                 14.78   slack (MET)


Startpoint: level_full (input port clocked by clk)
Endpoint: _20_ (rising edge-triggered flip-flop clocked by clk)
```

Path Group: clk
Path Type: max

Fanout	Cap	Slew	Delay	Time	Description

```
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                 0.03     0.02    4.02 ^ level_full (in)
     1    0.01                           level_full (net)
                 0.03     0.00    4.02 ^ input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
                 0.17     0.18    4.20 ^ input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
     3    0.02                           net2 (net)
                 0.17     0.00    4.21 ^ _11_/A (sky130_fd_sc_hd__inv_2)
                 0.04     0.06    4.26 v _11_/Y (sky130_fd_sc_hd__inv_2)
     2    0.01                           _04_ (net)
                 0.04     0.00    4.26 v _15_/A1 (sky130_fd_sc_hd__a21oi_1)
                 0.12     0.13    4.40 ^ _15_/Y (sky130_fd_sc_hd__a21oi_1)
     1    0.00                           _07_ (net)
                 0.12     0.00    4.40 ^ hold4/A (sky130_fd_sc_hd__dlygate4sd3_1)
                 0.05     0.59    4.98 ^ hold4/X (sky130_fd_sc_hd__dlygate4sd3_1)
     1    0.00                           net10 (net)
                 0.05     0.00    4.98 ^ _16_/A2 (sky130_fd_sc_hd__o21ai_1)
                 0.07     0.06    5.04 v _16_/Y (sky130_fd_sc_hd__o21ai_1)
     1    0.00                           _00_ (net)
                 0.07     0.00    5.04 v _20_/D (sky130_fd_sc_hd__dfstp_1)
                           5.04    5.04   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock source latency
                 0.06     0.04   20.04 ^ clk (in)
     1    0.01                           clk (net)
                 0.06     0.00   20.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                 0.04     0.12   20.16 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                 0.04     0.00   20.16 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                 0.03     0.10   20.27 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                 0.03     0.00   20.27 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_1)
                         -0.25   20.02   clock uncertainty
                          0.00   20.02   clock reconvergence pessimism
                         -0.08   19.94   library setup time
                                 19.94   data required time
-----------------------------------------------------------------------------
                                 19.94   data required time
                                 -5.04   data arrival time
-----------------------------------------------------------------------------
                                 14.89   slack (MET)
```

Startpoint: level_full (input port clocked by clk)
Endpoint: _22_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/routing/18-rsz_timing_sta.max.rpt ? part 5
==

Path Type: max

```
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 v input external delay
                   0.02   0.01    4.01 v level_full (in)
     1    0.01                           level_full (net)
                   0.02   0.00    4.01 v input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
                   0.09   0.18    4.19 v input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
     3    0.01                           net2 (net)
                   0.09   0.00    4.19 v _09_/A2 (sky130_fd_sc_hd__a21oi_2)
                   0.20   0.24    4.43 ^ _09_/Y (sky130_fd_sc_hd__a21oi_2)
     4    0.01                           _03_ (net)
                   0.20   0.00    4.43 ^ _10_/A (sky130_fd_sc_hd__inv_2)
                   0.05   0.06    4.49 v _10_/Y (sky130_fd_sc_hd__inv_2)
     2    0.01                           net5 (net)
                   0.05   0.00    4.49 v _14_/A (sky130_fd_sc_hd__nor2_1)
                   0.09   0.11    4.60 ^ _14_/Y (sky130_fd_sc_hd__nor2_1)
     1    0.00                           _02_ (net)
                   0.09   0.00    4.60 ^ _22_/D (sky130_fd_sc_hd__dfrtp_1)
                                  4.60   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock source latency
                   0.06   0.04   20.04 ^ clk (in)
     1    0.01                           clk (net)
                   0.06   0.00   20.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                   0.04   0.12   20.16 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                   0.04   0.00   20.16 ^ clkbuf_1_1__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                   0.03   0.10   20.26 ^ clkbuf_1_1__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     1    0.00                           clknet_1_1__leaf_clk (net)
                   0.03   0.00   20.26 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_1)
                         -0.25   20.01   clock uncertainty
                          0.00   20.01   clock reconvergence pessimism
                         -0.07   19.94   library setup time
                                 19.94   data required time
---------------------------------------------------------------------------
                                 19.94   data required time
                                 -4.60   data arrival time
---------------------------------------------------------------------------
```

15.35   slack (MET)

Startpoint: rst_n (input port clocked by clk)
Endpoint: _20_ (recovery check against rising-edge clock clk)

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/routing/18-rsz_timing_sta.max.rpt ? part 6
==

Path Group: asynchronous
Path Type: max

```
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                 0.02     0.01    4.01 ^ rst_n (in)
     1    0.00                           rst_n (net)
                 0.02     0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                 0.14     0.14    4.15 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.01                           net4 (net)
                 0.14     0.00    4.15 ^ _20_/SET_B (sky130_fd_sc_hd__dfstp_1)
                                  4.15   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock source latency
                 0.06     0.04   20.04 ^ clk (in)
     1    0.01                           clk (net)
                 0.06     0.00   20.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                 0.04     0.12   20.16 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                 0.04     0.00   20.16 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                 0.03     0.10   20.27 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                 0.03     0.00   20.27 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_1)
                         -0.25   20.02   clock uncertainty
                          0.00   20.02   clock reconvergence pessimism
                          0.12   20.14   library recovery time
                                 20.14   data required time
---------------------------------------------------------------------------
                                 20.14   data required time
                                 -4.15   data arrival time
---------------------------------------------------------------------------
                                 15.99   slack (MET)


Startpoint: rst_n (input port clocked by clk)
Endpoint: _22_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
Path Type: max
```

```
Fanout     Cap     Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
```

```
==========================================================================================

                  0.02    0.01    4.01 ^ rst_n (in)
     1    0.00                           rst_n (net)
                  0.02    0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                  0.14    0.14    4.15 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.01                           net4 (net)
                  0.14    0.00    4.15 ^ _22_/RESET_B (sky130_fd_sc_hd__dfrtp_1)
                                  4.15   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock source latency
                  0.06    0.04   20.04 ^ clk (in)
     1    0.01                           clk (net)
                  0.06    0.00   20.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.04    0.12   20.16 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                  0.04    0.00   20.16 ^ clkbuf_1_1__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.03    0.10   20.26 ^ clkbuf_1_1__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     1    0.00                           clknet_1_1__leaf_clk (net)
                  0.03    0.00   20.26 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_1)
                         -0.25   20.01   clock uncertainty
                          0.00   20.01   clock reconvergence pessimism
                          0.20   20.21   library recovery time
                                 20.21   data required time
------------------------------------------------------------------------------
                                 20.21   data required time
                                 -4.15   data arrival time
------------------------------------------------------------------------------
                                 16.06   slack (MET)


Startpoint: rst_n (input port clocked by clk)
Endpoint: _21_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
Path Type: max

Fanout     Cap    Slew   Delay    Time   Description
------------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                  0.02    0.01    4.01 ^ rst_n (in)
     1    0.00                           rst_n (net)
                  0.02    0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
```

```
              0.14    0.14    4.15 ^ input4/X (sky130_fd_sc_hd__buf_1)
3    0.01                             net4 (net)
              0.14    0.00    4.15 ^ _21_/RESET_B (sky130_fd_sc_hd__dfrtp_1)
                              4.15   data arrival time
```

```
                          20.00   20.00   clock clk (rise edge)
                           0.00   20.00   clock source latency
                  0.06     0.04   20.04 ^ clk (in)
     1    0.01                            clk (net)
                  0.06     0.00   20.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.04     0.12   20.16 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                            clknet_0_clk (net)
                  0.04     0.00   20.16 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.03     0.10   20.27 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                            clknet_1_0__leaf_clk (net)
                  0.03     0.00   20.27 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                          -0.25   20.02   clock uncertainty
                           0.00   20.02   clock reconvergence pessimism
                           0.20   20.22   library recovery time
                                  20.22   data required time
-------------------------------------------------------------------------------
                                  20.22   data required time
                                  -4.15   data arrival time
-------------------------------------------------------------------------------
                                  16.07   slack (MET)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/routing/18-rsz_timing_sta.min.rpt ? part 1
==

```
=========================================================================
report_checks -path_delay min (Hold)
=========================================================================
======================== Typical Corner ===================================

Startpoint: _21_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _22_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                   0.06    0.04    0.04 ^ clk (in)
     1    0.01                           clk (net)
                   0.06    0.00    0.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                   0.04    0.12    0.16 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                   0.04    0.00    0.16 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                   0.03    0.10    0.27 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                   0.03    0.00    0.27 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                   0.10    0.34    0.61 ^ _21_/Q (sky130_fd_sc_hd__dfrtp_1)
     3    0.01                           state[1] (net)
                   0.10    0.00    0.61 ^ _12_/A1 (sky130_fd_sc_hd__o21a_1)
                   0.05    0.13    0.74 ^ _12_/X (sky130_fd_sc_hd__o21a_1)
     1    0.00                           _05_ (net)
                   0.05    0.00    0.74 ^ _13_/B1 (sky130_fd_sc_hd__a21oi_1)
                   0.03    0.04    0.77 v _13_/Y (sky130_fd_sc_hd__a21oi_1)
     1    0.00                           _06_ (net)
                   0.03    0.00    0.77 v _14_/B (sky130_fd_sc_hd__nor2_1)
                   0.09    0.08    0.85 ^ _14_/Y (sky130_fd_sc_hd__nor2_1)
     1    0.00                           _02_ (net)
                   0.09    0.00    0.85 ^ _22_/D (sky130_fd_sc_hd__dfrtp_1)
                                   0.85   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                   0.06    0.04    0.04 ^ clk (in)
     1    0.01                           clk (net)
                   0.06    0.00    0.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                   0.04    0.14    0.18 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
```

```
     2    0.02                           clknet_0_clk (net)
                  0.04    0.00    0.18 ^ clkbuf_1_1__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.03    0.11    0.29 ^ clkbuf_1_1__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     1    0.00                           clknet_1_1__leaf_clk (net)
                  0.03    0.00    0.29 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_1)
```

```
                           0.25    0.54   clock uncertainty
                          -0.02    0.52   clock reconvergence pessimism
                          -0.05    0.48   library hold time
                                   0.48   data required time
-----------------------------------------------------------------------------
                                   0.48   data required time
                                  -0.85   data arrival time
-----------------------------------------------------------------------------
                                   0.37   slack (MET)


Startpoint: _21_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _21_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                           0.00    0.00   clock clk (rise edge)
                           0.00    0.00   clock source latency
                  0.06     0.04    0.04 ^ clk (in)
     1    0.01                            clk (net)
                  0.06     0.00    0.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.04     0.12    0.16 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                            clknet_0_clk (net)
                  0.04     0.00    0.16 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.03     0.10    0.27 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                            clknet_1_0__leaf_clk (net)
                  0.03     0.00    0.27 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                  0.10     0.34    0.61 ^ _21_/Q (sky130_fd_sc_hd__dfrtp_1)
     3    0.01                            state[1] (net)
                  0.10     0.00    0.61 ^ _17_/A1 (sky130_fd_sc_hd__o211a_1)
                  0.04     0.15    0.76 ^ _17_/X (sky130_fd_sc_hd__o211a_1)
     1    0.00                            _01_ (net)
                  0.04     0.00    0.76 ^ hold2/A (sky130_fd_sc_hd__dlygate4sd3_1)
                  0.06     0.51    1.26 ^ hold2/X (sky130_fd_sc_hd__dlygate4sd3_1)
     1    0.00                            net8 (net)
                  0.06     0.00    1.26 ^ _21_/D (sky130_fd_sc_hd__dfrtp_1)
                                   1.26   data arrival time

                           0.00    0.00   clock clk (rise edge)
                           0.00    0.00   clock source latency
                  0.06     0.04    0.04 ^ clk (in)
```

```
     1    0.01                           clk (net)
                  0.06    0.00    0.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.04    0.14    0.18 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                  0.04    0.00    0.18 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
```

```
                  0.03    0.12    0.29 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                  0.03    0.00    0.29 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                          0.25    0.54   clock uncertainty
                         -0.03    0.52   clock reconvergence pessimism
                         -0.04    0.48   library hold time
                                  0.48   data required time
-----------------------------------------------------------------------------
                                  0.48   data required time
                                 -1.26   data arrival time
-----------------------------------------------------------------------------
                                  0.79   slack (MET)


Startpoint: _21_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _20_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                  0.06    0.04    0.04 ^ clk (in)
     1    0.01                           clk (net)
                  0.06    0.00    0.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.04    0.12    0.16 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                  0.04    0.00    0.16 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.03    0.10    0.27 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                  0.03    0.00    0.27 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                  0.10    0.34    0.61 ^ _21_/Q (sky130_fd_sc_hd__dfrtp_1)
     3    0.01                           state[1] (net)
                  0.10    0.00    0.61 ^ _15_/A2 (sky130_fd_sc_hd__a21oi_1)
                  0.03    0.07    0.68 v _15_/Y (sky130_fd_sc_hd__a21oi_1)
     1    0.00                           _07_ (net)
                  0.03    0.00    0.68 v hold4/A (sky130_fd_sc_hd__dlygate4sd3_1)
                  0.05    0.51    1.20 v hold4/X (sky130_fd_sc_hd__dlygate4sd3_1)
     1    0.00                           net10 (net)
                  0.05    0.00    1.20 v _16_/A2 (sky130_fd_sc_hd__o21ai_1)
                  0.08    0.11    1.31 ^ _16_/Y (sky130_fd_sc_hd__o21ai_1)
```

```
1    0.00                           _00_ (net)
             0.08    0.00    1.31 ^ _20_/D (sky130_fd_sc_hd__dfstp_1)
                             1.31   data arrival time

                     0.00    0.00   clock clk (rise edge)
                     0.00    0.00   clock source latency
```

```
                  0.06    0.04    0.04 ^ clk (in)
     1    0.01                           clk (net)
                  0.06    0.00    0.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.04    0.14    0.18 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                  0.04    0.00    0.18 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.03    0.12    0.29 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                  0.03    0.00    0.29 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_1)
                          0.25    0.54   clock uncertainty
                         -0.03    0.52   clock reconvergence pessimism
                         -0.04    0.47   library hold time
                                  0.47   data required time
-----------------------------------------------------------------------------
                                  0.47   data required time
                                 -1.31   data arrival time
-----------------------------------------------------------------------------
                                  0.84   slack (MET)


Startpoint: rst_n (input port clocked by clk)
Endpoint: _21_ (removal check against rising-edge clock clk)
Path Group: asynchronous
Path Type: min

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                  0.02    0.01    4.01 ^ rst_n (in)
     1    0.00                           rst_n (net)
                  0.02    0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                  0.14    0.13    4.14 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.01                           net4 (net)
                  0.14    0.00    4.14 ^ _21_/RESET_B (sky130_fd_sc_hd__dfrtp_1)
                                  4.14   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                  0.06    0.04    0.04 ^ clk (in)
     1    0.01                           clk (net)
                  0.06    0.00    0.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
```

```
                0.04    0.14    0.18 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                       clknet_0_clk (net)
                0.04    0.00    0.18 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                0.03    0.12    0.29 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                       clknet_1_0__leaf_clk (net)
```

```
                  0.03    0.00    0.29 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                          0.25    0.54   clock uncertainty
                          0.00    0.54   clock reconvergence pessimism
                          0.34    0.88   library removal time
                                  0.88   data required time
-----------------------------------------------------------------------------
                                  0.88   data required time
                                 -4.14   data arrival time
-----------------------------------------------------------------------------
                                  3.25   slack (MET)


Startpoint: rst_n (input port clocked by clk)
Endpoint: _22_ (removal check against rising-edge clock clk)
Path Group: asynchronous
Path Type: min

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                  0.02    0.01    4.01 ^ rst_n (in)
     1    0.00                           rst_n (net)
                  0.02    0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                  0.14    0.13    4.14 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.01                           net4 (net)
                  0.14    0.00    4.14 ^ _22_/RESET_B (sky130_fd_sc_hd__dfrtp_1)
                                  4.14   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                  0.06    0.04    0.04 ^ clk (in)
     1    0.01                           clk (net)
                  0.06    0.00    0.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.04    0.14    0.18 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                  0.04    0.00    0.18 ^ clkbuf_1_1__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.03    0.11    0.29 ^ clkbuf_1_1__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     1    0.00                           clknet_1_1__leaf_clk (net)
                  0.03    0.00    0.29 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_1)
                          0.25    0.54   clock uncertainty
                          0.00    0.54   clock reconvergence pessimism
```

```
                          0.34    0.88   library removal time
                                  0.88   data required time
-----------------------------------------------------------------------------
                                  0.88   data required time
                                 -4.14   data arrival time
```

```
================================================================================

--------------------------------------------------------------------------------
                                3.26   slack (MET)


Startpoint: rst_n (input port clocked by clk)
Endpoint: _20_ (removal check against rising-edge clock clk)
Path Group: asynchronous
Path Type: min

Fanout     Cap    Slew   Delay    Time   Description
--------------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                  0.02    0.01    4.01 ^ rst_n (in)
     1    0.00                           rst_n (net)
                  0.02    0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                  0.14    0.13    4.14 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.01                           net4 (net)
                  0.14    0.00    4.14 ^ _20_/SET_B (sky130_fd_sc_hd__dfstp_1)
                                  4.14   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                  0.06    0.04    0.04 ^ clk (in)
     1    0.01                           clk (net)
                  0.06    0.00    0.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.04    0.14    0.18 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                  0.04    0.00    0.18 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.03    0.12    0.29 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                  0.03    0.00    0.29 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_1)
                          0.25    0.54   clock uncertainty
                          0.00    0.54   clock reconvergence pessimism
                          0.14    0.69   library removal time
                                  0.69   data required time
--------------------------------------------------------------------------------
                                  0.69   data required time
                                 -4.14   data arrival time
--------------------------------------------------------------------------------
                                  3.45   slack (MET)
```

```
Startpoint: _22_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: pump_en (output port clocked by clk)
Path Group: clk
Path Type: min
```

```
Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                  0.06    0.04    0.04 ^ clk (in)
     1    0.01                           clk (net)
                  0.06    0.00    0.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.04    0.12    0.16 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                  0.04    0.00    0.16 ^ clkbuf_1_1__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.03    0.10    0.26 ^ clkbuf_1_1__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     1    0.00                           clknet_1_1__leaf_clk (net)
                  0.03    0.00    0.26 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_1)
                  0.06    0.31    0.57 ^ _22_/Q (sky130_fd_sc_hd__dfrtp_1)
     2    0.01                           state[2] (net)
                  0.06    0.00    0.57 ^ _18_/A (sky130_fd_sc_hd__and2_1)
                  0.04    0.11    0.68 ^ _18_/X (sky130_fd_sc_hd__and2_1)
     1    0.00                           _08_ (net)
                  0.04    0.00    0.68 ^ _19_/A (sky130_fd_sc_hd__clkbuf_1)
                  0.05    0.08    0.76 ^ _19_/X (sky130_fd_sc_hd__clkbuf_1)
     1    0.00                           net6 (net)
                  0.05    0.00    0.76 ^ output6/A (sky130_fd_sc_hd__clkbuf_4)
                  0.11    0.18    0.94 ^ output6/X (sky130_fd_sc_hd__clkbuf_4)
     1    0.03                           pump_en (net)
                  0.11    0.00    0.94 ^ pump_en (out)
                                  0.94   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          0.25    0.25   clock uncertainty
                          0.00    0.25   clock reconvergence pessimism
                         -4.00   -3.75   output external delay
                                 -3.75   data required time
-----------------------------------------------------------------------------
                                 -3.75   data required time
                                 -0.94   data arrival time
-----------------------------------------------------------------------------
                                  4.69   slack (MET)


Startpoint: dry_run_fault (input port clocked by clk)
Endpoint: alarm (output port clocked by clk)
```

Path Group: clk
Path Type: min

Fanout	Cap	Slew	Delay	Time	Description

```
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                 0.02     0.01    4.01 ^ dry_run_fault (in)
     1    0.00                           dry_run_fault (net)
                 0.02     0.00    4.01 ^ input1/A (sky130_fd_sc_hd__clkbuf_1)
                 0.07     0.09    4.10 ^ input1/X (sky130_fd_sc_hd__clkbuf_1)
     1    0.01                           net1 (net)
                 0.07     0.00    4.10 ^ _09_/B1 (sky130_fd_sc_hd__a21oi_2)
                 0.04     0.05    4.16 v _09_/Y (sky130_fd_sc_hd__a21oi_2)
     4    0.01                           _03_ (net)
                 0.04     0.00    4.16 v _10_/A (sky130_fd_sc_hd__inv_2)
                 0.04     0.05    4.21 ^ _10_/Y (sky130_fd_sc_hd__inv_2)
     2    0.01                           net5 (net)
                 0.04     0.00    4.21 ^ output5/A (sky130_fd_sc_hd__buf_2)
                 0.17     0.18    4.39 ^ output5/X (sky130_fd_sc_hd__buf_2)
     1    0.03                           alarm (net)
                 0.17     0.00    4.39 ^ alarm (out)
                                  4.39   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          0.25    0.25   clock uncertainty
                          0.00    0.25   clock reconvergence pessimism
                         -4.00   -3.75   output external delay
                                 -3.75   data required time
-------------------------------------------------------------------------------
                                 -3.75   data required time
                                 -4.39   data arrival time
-------------------------------------------------------------------------------
                                  8.14   slack (MET)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/routing/18-rsz_timing_sta.power.rpt
==========

```
============================================================================
 report_power
============================================================================
======================== Typical Corner ===================================

Group                  Internal  Switching    Leakage      Total
                          Power      Power      Power      Power (Watts)
----------------------------------------------------------------
Sequential             6.60e-06   2.20e-07   3.33e-11   6.82e-06  16.8%
Combinational          2.73e-05   6.45e-06   2.03e-10   3.38e-05  83.2%
Macro                  0.00e+00   0.00e+00   0.00e+00   0.00e+00   0.0%
Pad                    0.00e+00   0.00e+00   0.00e+00   0.00e+00   0.0%
----------------------------------------------------------------
Total                  3.39e-05   6.67e-06   2.36e-10   4.06e-05 100.0%
                          83.6%      16.4%       0.0%
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/routing/18-rsz_timing_sta.skew.rpt
==

==
report_clock_skew
==
Clock clk
Latency   CRPR   Skew
_20_/CLK ^
   0.29
_22_/CLK ^
   0.26   -0.02   0.01

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/routing/18-rsz_timing_sta.summary.rpt
==

==
report_tns
==
tns 0.00

==
report_wns
==
wns 0.00

==
report_worst_slack -max (Setup)
==
worst slack 10.88

==
report_worst_slack -min (Hold)
==
worst slack 0.37

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/routing/21-grt_sta.checks.rpt ? part 1
==

```
=========================================================================
report_checks -unconstrained
=========================================================================
======================== Typical Corner ===================================

Startpoint: rst_n (input port clocked by clk)
Endpoint: _20_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
Path Type: max

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                  0.02    0.01    4.01 ^ rst_n (in)
     1    0.00                           rst_n (net)
                  0.02    0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                  0.14    0.14    4.15 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.01                           net4 (net)
                  0.14    0.00    4.15 ^ _20_/SET_B (sky130_fd_sc_hd__dfstp_1)
                                  4.15   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock source latency
                  0.04    0.03   20.03 ^ clk (in)
     1    0.01                           clk (net)
                  0.04    0.00   20.03 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.04    0.11   20.14 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                  0.04    0.00   20.14 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.03    0.10   20.24 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.00                           clknet_1_0__leaf_clk (net)
                  0.03    0.00   20.24 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_1)
                         -0.25   19.99   clock uncertainty
                          0.00   19.99   clock reconvergence pessimism
                          0.12   20.12   library recovery time
                                 20.12   data required time
---------------------------------------------------------------------------
                                 20.12   data required time
                                 -4.15   data arrival time
---------------------------------------------------------------------------
                                 15.97   slack (MET)
```

```
Startpoint: level_full (input port clocked by clk)
Endpoint: pump_en (output port clocked by clk)
Path Group: clk
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/routing/21-grt_sta.checks.rpt ? part 2
==

Path Type: max

```
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 v input external delay
                   0.01   0.00    4.00 v level_full (in)
     1    0.00                           level_full (net)
                   0.01   0.00    4.00 v input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
                   0.07   0.16    4.16 v input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
     3    0.01                           net2 (net)
                   0.07   0.00    4.16 v _09_/A2 (sky130_fd_sc_hd__a21oi_2)
                   0.16   0.20    4.37 ^ _09_/Y (sky130_fd_sc_hd__a21oi_2)
     4    0.01                           _03_ (net)
                   0.16   0.00    4.37 ^ _18_/B (sky130_fd_sc_hd__and2_1)
                   0.04   0.15    4.51 ^ _18_/X (sky130_fd_sc_hd__and2_1)
     1    0.00                           _08_ (net)
                   0.04   0.00    4.51 ^ _19_/A (sky130_fd_sc_hd__clkbuf_1)
                   0.04   0.08    4.59 ^ _19_/X (sky130_fd_sc_hd__clkbuf_1)
     1    0.00                           net6 (net)
                   0.04   0.00    4.59 ^ output6/A (sky130_fd_sc_hd__clkbuf_4)
                   0.11   0.19    4.78 ^ output6/X (sky130_fd_sc_hd__clkbuf_4)
     1    0.03                           pump_en (net)
                   0.11   0.00    4.78 ^ pump_en (out)
                          4.78    4.78   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock network delay (propagated)
                         -0.25   19.75   clock uncertainty
                          0.00   19.75   clock reconvergence pessimism
                         -4.00   15.75   output external delay
                                 15.75   data required time
---------------------------------------------------------------------------
                                 15.75   data required time
                                 -4.78   data arrival time
---------------------------------------------------------------------------
                                 10.97   slack (MET)
```

```
===========================================================================
report_checks --slack_max -0.01
===========================================================================
```

========== Typical Corner ==========

No paths found.

```
===========================================================================
 report_check_types -max_slew -max_cap -max_fanout -violators
===========================================================================
======================== Typical Corner ===================================


===========================================================================
report_parasitic_annotation -report_unannotated
===========================================================================
Found 32 unannotated drivers.
 clk
 dry_run_fault
 level_full
 level_low
 rst_n
 _09_/Y
 _10_/Y
 _11_/Y
 _12_/X
 _13_/Y
 _14_/Y
 _15_/Y
 _16_/Y
 _17_/X
 _18_/X
 _19_/X
 _20_/Q
 _21_/Q
 _22_/Q
 clkbuf_0_clk/X
 clkbuf_1_0__f_clk/X
 clkbuf_1_1__f_clk/X
 hold1/X
 hold2/X
 hold3/X
 hold4/X
 input1/X
 input2/X
 input3/X
 input4/X
 output5/X
 output6/X
```

```
Found 0 partially unannotated drivers.

================================================================================
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/routing/21-grt_sta.checks.rpt ? part 4
==

max slew violation count 0
max fanout violation count 0
max cap violation count 0
==

==
check_setup -verbose -unconstrained_endpoints -multiple_clock -no_clock -no_input_delay
-loops -generated_clocks
==

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/routing/21-grt_sta.max.rpt ? part 1
==========

```
============================================================================
report_checks -path_delay max (Setup)
============================================================================
======================== Typical Corner ===================================

Startpoint: level_full (input port clocked by clk)
Endpoint: pump_en (output port clocked by clk)
Path Group: clk
Path Type: max

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 v input external delay
                   0.01   0.00    4.00 v level_full (in)
     1    0.00                           level_full (net)
                   0.01   0.00    4.00 v input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
                   0.07   0.16    4.16 v input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
     3    0.01                           net2 (net)
                   0.07   0.00    4.16 v _09_/A2 (sky130_fd_sc_hd__a21oi_2)
                   0.16   0.20    4.37 ^ _09_/Y (sky130_fd_sc_hd__a21oi_2)
     4    0.01                           _03_ (net)
                   0.16   0.00    4.37 ^ _18_/B (sky130_fd_sc_hd__and2_1)
                   0.04   0.15    4.51 ^ _18_/X (sky130_fd_sc_hd__and2_1)
     1    0.00                           _08_ (net)
                   0.04   0.00    4.51 ^ _19_/A (sky130_fd_sc_hd__clkbuf_1)
                   0.04   0.08    4.59 ^ _19_/X (sky130_fd_sc_hd__clkbuf_1)
     1    0.00                           net6 (net)
                   0.04   0.00    4.59 ^ output6/A (sky130_fd_sc_hd__clkbuf_4)
                   0.11   0.19    4.78 ^ output6/X (sky130_fd_sc_hd__clkbuf_4)
     1    0.03                           pump_en (net)
                   0.11   0.00    4.78 ^ pump_en (out)
                          4.78   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock network delay (propagated)
                         -0.25   19.75   clock uncertainty
                          0.00   19.75   clock reconvergence pessimism
                         -4.00   15.75   output external delay
                                 15.75   data required time
---------------------------------------------------------------------------
                                 15.75   data required time
                                 -4.78   data arrival time
---------------------------------------------------------------------------
```

10.97   slack (MET)

```
==========================================================================================

Startpoint: level_full (input port clocked by clk)
Endpoint: alarm (output port clocked by clk)
Path Group: clk
Path Type: max

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 v input external delay
                   0.01   0.00    4.00 v level_full (in)
     1    0.00                           level_full (net)
                   0.01   0.00    4.00 v input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
                   0.07   0.16    4.16 v input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
     3    0.01                           net2 (net)
                   0.07   0.00    4.16 v _09_/A2 (sky130_fd_sc_hd__a21oi_2)
                   0.16   0.20    4.37 ^ _09_/Y (sky130_fd_sc_hd__a21oi_2)
     4    0.01                           _03_ (net)
                   0.16   0.00    4.37 ^ _10_/A (sky130_fd_sc_hd__inv_2)
                   0.04   0.05    4.41 v _10_/Y (sky130_fd_sc_hd__inv_2)
     2    0.00                           net5 (net)
                   0.04   0.00    4.41 v output5/A (sky130_fd_sc_hd__buf_2)
                   0.09   0.18    4.60 v output5/X (sky130_fd_sc_hd__buf_2)
     1    0.03                           alarm (net)
                   0.09   0.00    4.60 v alarm (out)
                                  4.60   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock network delay (propagated)
                         -0.25   19.75   clock uncertainty
                          0.00   19.75   clock reconvergence pessimism
                         -4.00   15.75   output external delay
                                 15.75   data required time
---------------------------------------------------------------------------
                                 15.75   data required time
                                 -4.60   data arrival time
---------------------------------------------------------------------------
                                 11.15   slack (MET)


Startpoint: level_full (input port clocked by clk)
Endpoint: _21_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
```

Fanout	Cap	Slew	Delay	Time	Description
			0.00	0.00	clock clk (rise edge)

```
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 v input external delay
                 0.01     0.00    4.00 v level_full (in)
     1    0.00                           level_full (net)
                 0.01     0.00    4.00 v input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
                 0.07     0.16    4.16 v input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
     3    0.01                           net2 (net)
                 0.07     0.00    4.16 v _09_/A2 (sky130_fd_sc_hd__a21oi_2)
                 0.16     0.20    4.37 ^ _09_/Y (sky130_fd_sc_hd__a21oi_2)
     4    0.01                           _03_ (net)
                 0.16     0.00    4.37 ^ _17_/B1 (sky130_fd_sc_hd__o211a_1)
                 0.04     0.17    4.53 ^ _17_/X (sky130_fd_sc_hd__o211a_1)
     1    0.00                           _01_ (net)
                 0.04     0.00    4.53 ^ hold2/A (sky130_fd_sc_hd__dlygate4sd3_1)
                 0.05     0.55    5.08 ^ hold2/X (sky130_fd_sc_hd__dlygate4sd3_1)
     1    0.00                           net8 (net)
                 0.05     0.00    5.08 ^ _21_/D (sky130_fd_sc_hd__dfrtp_1)
                           5.08   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock source latency
                 0.04     0.03   20.03 ^ clk (in)
     1    0.01                           clk (net)
                 0.04     0.00   20.03 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                 0.04     0.11   20.14 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                 0.04     0.00   20.14 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                 0.03     0.10   20.24 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.00                           clknet_1_0__leaf_clk (net)
                 0.03     0.00   20.24 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                         -0.25   19.99   clock uncertainty
                          0.00   19.99   clock reconvergence pessimism
                         -0.06   19.93   library setup time
                                 19.93   data required time
-----------------------------------------------------------------------------
                                 19.93   data required time
                                 -5.08   data arrival time
-----------------------------------------------------------------------------
                                 14.84   slack (MET)


Startpoint: level_full (input port clocked by clk)
Endpoint: _20_ (rising edge-triggered flip-flop clocked by clk)
```

Path Group: clk
Path Type: max

Fanout	Cap	Slew	Delay	Time	Description

```
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 v input external delay
                 0.01     0.00    4.00 v level_full (in)
     1    0.00                           level_full (net)
                 0.01     0.00    4.00 v input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
                 0.07     0.16    4.16 v input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
     3    0.01                           net2 (net)
                 0.07     0.00    4.16 v _11_/A (sky130_fd_sc_hd__inv_2)
                 0.04     0.06    4.23 ^ _11_/Y (sky130_fd_sc_hd__inv_2)
     2    0.00                           _04_ (net)
                 0.04     0.00    4.23 ^ _15_/A1 (sky130_fd_sc_hd__a21oi_1)
                 0.04     0.05    4.27 v _15_/Y (sky130_fd_sc_hd__a21oi_1)
     1    0.00                           _07_ (net)
                 0.04     0.00    4.27 v hold4/A (sky130_fd_sc_hd__dlygate4sd3_1)
                 0.05     0.57    4.84 v hold4/X (sky130_fd_sc_hd__dlygate4sd3_1)
     1    0.00                           net10 (net)
                 0.05     0.00    4.84 v _16_/A2 (sky130_fd_sc_hd__o21ai_1)
                 0.10     0.11    4.95 ^ _16_/Y (sky130_fd_sc_hd__o21ai_1)
     1    0.00                           _00_ (net)
                 0.10     0.00    4.95 ^ _20_/D (sky130_fd_sc_hd__dfstp_1)
                           4.95   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock source latency
                 0.04     0.03   20.03 ^ clk (in)
     1    0.01                           clk (net)
                 0.04     0.00   20.03 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                 0.04     0.11   20.14 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                 0.04     0.00   20.14 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                 0.03     0.10   20.24 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.00                           clknet_1_0__leaf_clk (net)
                 0.03     0.00   20.24 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_1)
                         -0.25   19.99   clock uncertainty
                          0.00   19.99   clock reconvergence pessimism
                         -0.07   19.92   library setup time
                                 19.92   data required time
-----------------------------------------------------------------------------
                                 19.92   data required time
                                 -4.95   data arrival time
-----------------------------------------------------------------------------
                                 14.97   slack (MET)
```

Startpoint: level_full (input port clocked by clk)
Endpoint: _22_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk

```
================================================================================

Path Type: max

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 v input external delay
                   0.01   0.00    4.00 v level_full (in)
     1    0.00                           level_full (net)
                   0.01   0.00    4.00 v input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
                   0.07   0.16    4.16 v input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
     3    0.01                           net2 (net)
                   0.07   0.00    4.16 v _09_/A2 (sky130_fd_sc_hd__a21oi_2)
                   0.16   0.20    4.37 ^ _09_/Y (sky130_fd_sc_hd__a21oi_2)
     4    0.01                           _03_ (net)
                   0.16   0.00    4.37 ^ _10_/A (sky130_fd_sc_hd__inv_2)
                   0.04   0.05    4.41 v _10_/Y (sky130_fd_sc_hd__inv_2)
     2    0.00                           net5 (net)
                   0.04   0.00    4.41 v _14_/A (sky130_fd_sc_hd__nor2_1)
                   0.07   0.09    4.50 ^ _14_/Y (sky130_fd_sc_hd__nor2_1)
     1    0.00                           _02_ (net)
                   0.07   0.00    4.50 ^ _22_/D (sky130_fd_sc_hd__dfrtp_1)
                          4.50    4.50   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock source latency
                   0.04   0.03   20.03 ^ clk (in)
     1    0.01                           clk (net)
                   0.04   0.00   20.03 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                   0.04   0.11   20.14 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                   0.04   0.00   20.14 ^ clkbuf_1_1__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                   0.03   0.10   20.24 ^ clkbuf_1_1__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     1    0.00                           clknet_1_1__leaf_clk (net)
                   0.03   0.00   20.24 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_1)
                         -0.25   19.99   clock uncertainty
                          0.00   19.99   clock reconvergence pessimism
                         -0.07   19.92   library setup time
                                 19.92   data required time
---------------------------------------------------------------------------
                                 19.92   data required time
                                 -4.50   data arrival time
---------------------------------------------------------------------------
```

15.42   slack (MET)

Startpoint: rst_n (input port clocked by clk)
Endpoint: _20_ (recovery check against rising-edge clock clk)

```
Path Group: asynchronous
Path Type: max

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                 0.02     0.01    4.01 ^ rst_n (in)
     1    0.00                           rst_n (net)
                 0.02     0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                 0.14     0.14    4.15 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.01                           net4 (net)
                 0.14     0.00    4.15 ^ _20_/SET_B (sky130_fd_sc_hd__dfstp_1)
                                  4.15   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock source latency
                 0.04     0.03   20.03 ^ clk (in)
     1    0.01                           clk (net)
                 0.04     0.00   20.03 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                 0.04     0.11   20.14 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                 0.04     0.00   20.14 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                 0.03     0.10   20.24 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.00                           clknet_1_0__leaf_clk (net)
                 0.03     0.00   20.24 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_1)
                         -0.25   19.99   clock uncertainty
                          0.00   19.99   clock reconvergence pessimism
                          0.12   20.12   library recovery time
                                 20.12   data required time
---------------------------------------------------------------------------
                                 20.12   data required time
                                 -4.15   data arrival time
---------------------------------------------------------------------------
                                 15.97   slack (MET)


Startpoint: rst_n (input port clocked by clk)
Endpoint: _22_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
Path Type: max
```

```
Fanout     Cap     Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
```

==========

```
                  0.02    0.01    4.01 ^ rst_n (in)
     1    0.00                           rst_n (net)
                  0.02    0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                  0.14    0.14    4.15 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.01                           net4 (net)
                  0.14    0.00    4.15 ^ _22_/RESET_B (sky130_fd_sc_hd__dfrtp_1)
                                  4.15   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock source latency
                  0.04    0.03   20.03 ^ clk (in)
     1    0.01                           clk (net)
                  0.04    0.00   20.03 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.04    0.11   20.14 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                  0.04    0.00   20.14 ^ clkbuf_1_1__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.03    0.10   20.24 ^ clkbuf_1_1__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     1    0.00                           clknet_1_1__leaf_clk (net)
                  0.03    0.00   20.24 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_1)
                         -0.25   19.99   clock uncertainty
                          0.00   19.99   clock reconvergence pessimism
                          0.20   20.19   library recovery time
                                 20.19   data required time
-----------------------------------------------------------------------------
                                 20.19   data required time
                                 -4.15   data arrival time
-----------------------------------------------------------------------------
                                 16.04   slack (MET)


Startpoint: rst_n (input port clocked by clk)
Endpoint: _21_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
Path Type: max

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                  0.02    0.01    4.01 ^ rst_n (in)
     1    0.00                           rst_n (net)
                  0.02    0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
```

```
              0.14    0.14    4.15 ^ input4/X (sky130_fd_sc_hd__buf_1)
3    0.01                             net4 (net)
              0.14    0.00    4.15 ^ _21_/RESET_B (sky130_fd_sc_hd__dfrtp_1)
                              4.15   data arrival time
```

```
                          20.00   20.00   clock clk (rise edge)
                           0.00   20.00   clock source latency
                  0.04     0.03   20.03 ^ clk (in)
     1    0.01                            clk (net)
                  0.04     0.00   20.03 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.04     0.11   20.14 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                            clknet_0_clk (net)
                  0.04     0.00   20.14 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.03     0.10   20.24 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.00                            clknet_1_0__leaf_clk (net)
                  0.03     0.00   20.24 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                          -0.25   19.99   clock uncertainty
                           0.00   19.99   clock reconvergence pessimism
                           0.20   20.19   library recovery time
                                  20.19   data required time
-----------------------------------------------------------------------------
                                  20.19   data required time
                                  -4.15   data arrival time
-----------------------------------------------------------------------------
                                  16.04   slack (MET)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/routing/21-grt_sta.min.rpt ? part 1
==========

```
============================================================================
report_checks -path_delay min (Hold)
============================================================================
======================== Typical Corner ===================================

Startpoint: _21_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _22_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                   0.04   0.03    0.03 ^ clk (in)
     1    0.01                          clk (net)
                   0.04   0.00    0.03 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                   0.04   0.11    0.14 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                          clknet_0_clk (net)
                   0.04   0.00    0.14 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                   0.03   0.10    0.24 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.00                          clknet_1_0__leaf_clk (net)
                   0.03   0.00    0.24 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                   0.08   0.32    0.56 ^ _21_/Q (sky130_fd_sc_hd__dfrtp_1)
     3    0.01                          state[1] (net)
                   0.08   0.00    0.56 ^ _12_/A1 (sky130_fd_sc_hd__o21a_1)
                   0.04   0.12    0.68 ^ _12_/X (sky130_fd_sc_hd__o21a_1)
     1    0.00                          _05_ (net)
                   0.04   0.00    0.68 ^ _13_/B1 (sky130_fd_sc_hd__a21oi_1)
                   0.02   0.03    0.71 v _13_/Y (sky130_fd_sc_hd__a21oi_1)
     1    0.00                          _06_ (net)
                   0.02   0.00    0.71 v _14_/B (sky130_fd_sc_hd__nor2_1)
                   0.06   0.06    0.77 ^ _14_/Y (sky130_fd_sc_hd__nor2_1)
     1    0.00                          _02_ (net)
                   0.06   0.00    0.77 ^ _22_/D (sky130_fd_sc_hd__dfrtp_1)
                          0.77   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                   0.04   0.03    0.03 ^ clk (in)
     1    0.01                          clk (net)
                   0.04   0.00    0.03 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                   0.04   0.13    0.16 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
```

```
     2    0.02                           clknet_0_clk (net)
                  0.04    0.00    0.16 ^ clkbuf_1_1__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.03    0.11    0.27 ^ clkbuf_1_1__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     1    0.00                           clknet_1_1__leaf_clk (net)
                  0.03    0.00    0.27 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_1)
```

```
                          0.25    0.52   clock uncertainty
                         -0.01    0.50   clock reconvergence pessimism
                         -0.04    0.46   library hold time
                                  0.46   data required time
-----------------------------------------------------------------------------
                                  0.46   data required time
                                 -0.77   data arrival time
-----------------------------------------------------------------------------
                                  0.31   slack (MET)


Startpoint: _21_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _21_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                 0.04     0.03    0.03 ^ clk (in)
     1    0.01                           clk (net)
                 0.04     0.00    0.03 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                 0.04     0.11    0.14 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                 0.04     0.00    0.14 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                 0.03     0.10    0.24 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.00                           clknet_1_0__leaf_clk (net)
                 0.03     0.00    0.24 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                 0.08     0.32    0.56 ^ _21_/Q (sky130_fd_sc_hd__dfrtp_1)
     3    0.01                           state[1] (net)
                 0.08     0.00    0.56 ^ _17_/A1 (sky130_fd_sc_hd__o211a_1)
                 0.03     0.14    0.70 ^ _17_/X (sky130_fd_sc_hd__o211a_1)
     1    0.00                           _01_ (net)
                 0.03     0.00    0.70 ^ hold2/A (sky130_fd_sc_hd__dlygate4sd3_1)
                 0.05     0.50    1.20 ^ hold2/X (sky130_fd_sc_hd__dlygate4sd3_1)
     1    0.00                           net8 (net)
                 0.05     0.00    1.20 ^ _21_/D (sky130_fd_sc_hd__dfrtp_1)
                                  1.20   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                 0.04     0.03    0.03 ^ clk (in)
```

```
     1    0.01                           clk (net)
                  0.04    0.00    0.03 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.04    0.13    0.16 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                  0.04    0.00    0.16 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
```

```
                  0.03    0.11    0.27 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.00                           clknet_1_0__leaf_clk (net)
                  0.03    0.00    0.27 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                          0.25    0.52   clock uncertainty
                         -0.03    0.49   clock reconvergence pessimism
                         -0.04    0.45   library hold time
                                  0.45   data required time
-----------------------------------------------------------------------------
                                  0.45   data required time
                                 -1.20   data arrival time
-----------------------------------------------------------------------------
                                  0.75   slack (MET)


Startpoint: _21_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _20_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                  0.04    0.03    0.03 ^ clk (in)
     1    0.01                           clk (net)
                  0.04    0.00    0.03 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.04    0.11    0.14 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                  0.04    0.00    0.14 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.03    0.10    0.24 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.00                           clknet_1_0__leaf_clk (net)
                  0.03    0.00    0.24 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                  0.08    0.32    0.56 ^ _21_/Q (sky130_fd_sc_hd__dfrtp_1)
     3    0.01                           state[1] (net)
                  0.08    0.00    0.56 ^ _15_/A2 (sky130_fd_sc_hd__a21oi_1)
                  0.02    0.06    0.62 v _15_/Y (sky130_fd_sc_hd__a21oi_1)
     1    0.00                           _07_ (net)
                  0.02    0.00    0.62 v hold4/A (sky130_fd_sc_hd__dlygate4sd3_1)
                  0.05    0.51    1.13 v hold4/X (sky130_fd_sc_hd__dlygate4sd3_1)
     1    0.00                           net10 (net)
                  0.05    0.00    1.13 v _16_/A2 (sky130_fd_sc_hd__o21ai_1)
                  0.07    0.10    1.24 ^ _16_/Y (sky130_fd_sc_hd__o21ai_1)
```

```
1    0.00                           _00_ (net)
             0.07    0.00    1.24 ^ _20_/D (sky130_fd_sc_hd__dfstp_1)
                             1.24   data arrival time

                     0.00    0.00   clock clk (rise edge)
                     0.00    0.00   clock source latency
```

```
================================================================================

                  0.04    0.03    0.03 ^ clk (in)
     1    0.01                           clk (net)
                  0.04    0.00    0.03 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.04    0.13    0.16 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                  0.04    0.00    0.16 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.03    0.11    0.27 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.00                           clknet_1_0__leaf_clk (net)
                  0.03    0.00    0.27 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_1)
                          0.25    0.52   clock uncertainty
                         -0.03    0.49   clock reconvergence pessimism
                         -0.04    0.45   library hold time
                                  0.45   data required time
-------------------------------------------------------------------------------
                                  0.45   data required time
                                 -1.24   data arrival time
-------------------------------------------------------------------------------
                                  0.79   slack (MET)


Startpoint: rst_n (input port clocked by clk)
Endpoint: _21_ (removal check against rising-edge clock clk)
Path Group: asynchronous
Path Type: min

Fanout     Cap    Slew   Delay    Time   Description
-------------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                  0.02    0.01    4.01 ^ rst_n (in)
     1    0.00                           rst_n (net)
                  0.02    0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                  0.14    0.13    4.13 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.01                           net4 (net)
                  0.14    0.00    4.13 ^ _21_/RESET_B (sky130_fd_sc_hd__dfrtp_1)
                                  4.13   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                  0.04    0.03    0.03 ^ clk (in)
     1    0.01                           clk (net)
                  0.04    0.00    0.03 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
```

```
                0.04    0.13    0.16 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                       clknet_0_clk (net)
                0.04    0.00    0.16 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                0.03    0.11    0.27 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.00                       clknet_1_0__leaf_clk (net)
```

```
                  0.03    0.00    0.27 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                          0.25    0.52   clock uncertainty
                          0.00    0.52   clock reconvergence pessimism
                          0.34    0.86   library removal time
                                  0.86   data required time
---------------------------------------------------------------------------
                                  0.86   data required time
                                 -4.13   data arrival time
---------------------------------------------------------------------------
                                  3.28   slack (MET)


Startpoint: rst_n (input port clocked by clk)
Endpoint: _22_ (removal check against rising-edge clock clk)
Path Group: asynchronous
Path Type: min

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                  0.02    0.01    4.01 ^ rst_n (in)
     1    0.00                           rst_n (net)
                  0.02    0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                  0.14    0.13    4.13 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.01                           net4 (net)
                  0.14    0.00    4.13 ^ _22_/RESET_B (sky130_fd_sc_hd__dfrtp_1)
                                  4.13   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                  0.04    0.03    0.03 ^ clk (in)
     1    0.01                           clk (net)
                  0.04    0.00    0.03 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.04    0.13    0.16 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                  0.04    0.00    0.16 ^ clkbuf_1_1__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.03    0.11    0.27 ^ clkbuf_1_1__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     1    0.00                           clknet_1_1__leaf_clk (net)
                  0.03    0.00    0.27 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_1)
                          0.25    0.52   clock uncertainty
                          0.00    0.52   clock reconvergence pessimism
```

```
                          0.34    0.86   library removal time
                                  0.86   data required time
-----------------------------------------------------------------------------
                                  0.86   data required time
                                 -4.13   data arrival time
```

```
================================================================================

--------------------------------------------------------------------------
                                3.28   slack (MET)


Startpoint: rst_n (input port clocked by clk)
Endpoint: _20_ (removal check against rising-edge clock clk)
Path Group: asynchronous
Path Type: min

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                         0.00    0.00   clock clk (rise edge)
                         0.00    0.00   clock network delay (propagated)
                         4.00    4.00 ^ input external delay
                  0.02   0.01    4.01 ^ rst_n (in)
     1    0.00                          rst_n (net)
                  0.02   0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                  0.14   0.13    4.13 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.01                          net4 (net)
                  0.14   0.00    4.13 ^ _20_/SET_B (sky130_fd_sc_hd__dfstp_1)
                                4.13   data arrival time

                         0.00    0.00   clock clk (rise edge)
                         0.00    0.00   clock source latency
                  0.04   0.03    0.03 ^ clk (in)
     1    0.01                          clk (net)
                  0.04   0.00    0.03 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.04   0.13    0.16 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                          clknet_0_clk (net)
                  0.04   0.00    0.16 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.03   0.11    0.27 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.00                          clknet_1_0__leaf_clk (net)
                  0.03   0.00    0.27 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_1)
                         0.25    0.52   clock uncertainty
                         0.00    0.52   clock reconvergence pessimism
                         0.14    0.66   library removal time
                                 0.66   data required time
---------------------------------------------------------------------------
                                 0.66   data required time
                                -4.13   data arrival time
---------------------------------------------------------------------------
                                 3.48   slack (MET)
```

```
Startpoint: _22_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: pump_en (output port clocked by clk)
Path Group: clk
Path Type: min
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/routing/21-grt_sta.min.rpt ? part 7
==========

```
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                  0.04    0.03    0.03 ^ clk (in)
     1    0.01                          clk (net)
                  0.04    0.00    0.03 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.04    0.11    0.14 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                          clknet_0_clk (net)
                  0.04    0.00    0.14 ^ clkbuf_1_1__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.03    0.10    0.24 ^ clkbuf_1_1__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     1    0.00                          clknet_1_1__leaf_clk (net)
                  0.03    0.00    0.24 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_1)
                  0.05    0.29    0.53 ^ _22_/Q (sky130_fd_sc_hd__dfrtp_1)
     2    0.00                          state[2] (net)
                  0.05    0.00    0.53 ^ _18_/A (sky130_fd_sc_hd__and2_1)
                  0.04    0.10    0.64 ^ _18_/X (sky130_fd_sc_hd__and2_1)
     1    0.00                          _08_ (net)
                  0.04    0.00    0.64 ^ _19_/A (sky130_fd_sc_hd__clkbuf_1)
                  0.04    0.07    0.71 ^ _19_/X (sky130_fd_sc_hd__clkbuf_1)
     1    0.00                          net6 (net)
                  0.04    0.00    0.71 ^ output6/A (sky130_fd_sc_hd__clkbuf_4)
                  0.11    0.17    0.88 ^ output6/X (sky130_fd_sc_hd__clkbuf_4)
     1    0.03                          pump_en (net)
                  0.11    0.00    0.88 ^ pump_en (out)
                                  0.88   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          0.25    0.25   clock uncertainty
                          0.00    0.25   clock reconvergence pessimism
                         -4.00   -3.75   output external delay
                                 -3.75   data required time
---------------------------------------------------------------------------
                                 -3.75   data required time
                                 -0.88   data arrival time
---------------------------------------------------------------------------
                                  4.63   slack (MET)


Startpoint: dry_run_fault (input port clocked by clk)
Endpoint: alarm (output port clocked by clk)
```

Path Group: clk
Path Type: min

Fanout	Cap	Slew	Delay	Time	Description

```
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                 0.02     0.01    4.01 ^ dry_run_fault (in)
     1    0.00                           dry_run_fault (net)
                 0.02     0.00    4.01 ^ input1/A (sky130_fd_sc_hd__clkbuf_1)
                 0.07     0.08    4.09 ^ input1/X (sky130_fd_sc_hd__clkbuf_1)
     1    0.00                           net1 (net)
                 0.07     0.00    4.09 ^ _09_/B1 (sky130_fd_sc_hd__a21oi_2)
                 0.04     0.05    4.14 v _09_/Y (sky130_fd_sc_hd__a21oi_2)
     4    0.01                           _03_ (net)
                 0.04     0.00    4.14 v _10_/A (sky130_fd_sc_hd__inv_2)
                 0.03     0.04    4.18 ^ _10_/Y (sky130_fd_sc_hd__inv_2)
     2    0.00                           net5 (net)
                 0.03     0.00    4.18 ^ output5/A (sky130_fd_sc_hd__buf_2)
                 0.17     0.18    4.36 ^ output5/X (sky130_fd_sc_hd__buf_2)
     1    0.03                           alarm (net)
                 0.17     0.00    4.36 ^ alarm (out)
                                  4.36   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          0.25    0.25   clock uncertainty
                          0.00    0.25   clock reconvergence pessimism
                         -4.00   -3.75   output external delay
                                 -3.75   data required time
---------------------------------------------------------------------------
                                 -3.75   data required time
                                 -4.36   data arrival time
---------------------------------------------------------------------------
                                  8.11   slack (MET)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/routing/21-grt_sta.power.rpt
==========

==========
 report_power
==========
======================= Typical Corner ===================================

Group	Internal Power	Switching Power	Leakage Power	Total Power (Watts)	
Sequential	6.60e-06	1.53e-07	3.33e-11	6.75e-06	17.3%
Combinational	2.73e-05	5.07e-06	2.03e-10	3.23e-05	82.7%
Macro	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Pad	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Total	3.38e-05	5.22e-06	2.36e-10	3.91e-05	100.0%
	86.6%	13.4%	0.0%		

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/routing/21-grt_sta.skew.rpt
==========

==========
report_clock_skew
==========
Clock clk
Latency   CRPR   Skew
_20_/CLK ^
   0.27
_22_/CLK ^
   0.24   -0.01   0.01

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/routing/21-grt_sta.summary.rpt
==

==
report_tns
==
tns 0.00

==
report_wns
==
wns 0.00

==
report_worst_slack -max (Setup)
==
worst slack 10.97

==
report_worst_slack -min (Hold)
==
worst slack 0.31

```
openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/26-mca/rcx_min_sta.checks.rpt ? pa
========================================================================================

========================================================================
report_checks -unconstrained
========================================================================
======================= Fastest Corner ===================================

Startpoint: rst_n (input port clocked by clk)
Endpoint: _20_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
Path Type: max
Corner: Fastest

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                0.01      0.01    4.01 ^ rst_n (in)
     1    0.00                           rst_n (net)
                0.01      0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                0.14      0.13    4.14 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.02                           net4 (net)
                0.14      0.00    4.14 ^ _20_/SET_B (sky130_fd_sc_hd__dfstp_1)
                                  4.14   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock source latency
                0.04      0.03   20.03 ^ clk (in)
     1    0.01                           clk (net)
                0.04      0.00   20.03 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                0.03      0.08   20.11 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                0.03      0.00   20.11 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                0.02      0.07   20.18 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                0.02      0.00   20.18 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_1)
                         -0.25   19.93   clock uncertainty
                          0.00   19.93   clock reconvergence pessimism
                          0.09   20.02   library recovery time
                                 20.02   data required time
---------------------------------------------------------------------------
                                 20.02   data required time
                                 -4.14   data arrival time
---------------------------------------------------------------------------
```

15.88   slack (MET)

Startpoint: level_full (input port clocked by clk)
Endpoint: pump_en (output port clocked by clk)

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/26-mca/rcx_min_sta.checks.rpt ? pa
==

Path Group: clk
Path Type: max
Corner: Fastest

Fanout	Cap	Slew	Delay	Time	Description
			0.00	0.00	clock clk (rise edge)
			0.00	0.00	clock network delay (propagated)
			4.00	4.00 v	input external delay
		0.01	0.00	4.00 v	level_full (in)
1	0.00				level_full (net)
		0.01	0.00	4.00 v	input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
		0.07	0.12	4.12 v	input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
3	0.02				net2 (net)
		0.07	0.00	4.12 v	_09_/A2 (sky130_fd_sc_hd__a21oi_2)
		0.16	0.18	4.30 ^	_09_/Y (sky130_fd_sc_hd__a21oi_2)
4	0.02				_03_ (net)
		0.16	0.00	4.30 ^	_18_/B (sky130_fd_sc_hd__and2_1)
		0.03	0.09	4.40 ^	_18_/X (sky130_fd_sc_hd__and2_1)
1	0.00				_08_ (net)
		0.03	0.00	4.40 ^	_19_/A (sky130_fd_sc_hd__clkbuf_1)
		0.03	0.06	4.45 ^	_19_/X (sky130_fd_sc_hd__clkbuf_1)
1	0.00				net6 (net)
		0.03	0.00	4.45 ^	output6/A (sky130_fd_sc_hd__clkbuf_4)
		0.08	0.13	4.59 ^	output6/X (sky130_fd_sc_hd__clkbuf_4)
1	0.03				pump_en (net)
		0.08	0.00	4.59 ^	pump_en (out)
				4.59	data arrival time
			20.00	20.00	clock clk (rise edge)
			0.00	20.00	clock network delay (propagated)
			-0.25	19.75	clock uncertainty
			0.00	19.75	clock reconvergence pessimism
			-4.00	15.75	output external delay
				15.75	data required time
				15.75	data required time
				-4.59	data arrival time
				11.16	slack (MET)

========== Slowest Corner ==========

```
Startpoint: rst_n (input port clocked by clk)
Endpoint: _20_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
```

```
openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/26-mca/rcx_min_sta.checks.rpt ? pa
================================================================================

Path Type: max
Corner: Slowest

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                  0.03    0.02    4.02 ^ rst_n (in)
     1    0.00                           rst_n (net)
                  0.03    0.00    4.02 ^ input4/A (sky130_fd_sc_hd__buf_1)
                  0.28    0.30    4.31 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.01                           net4 (net)
                  0.28    0.00    4.31 ^ _20_/SET_B (sky130_fd_sc_hd__dfstp_1)
                                  4.31   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock source latency
                  0.09    0.05   20.05 ^ clk (in)
     1    0.01                           clk (net)
                  0.09    0.00   20.05 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.07    0.22   20.27 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                  0.07    0.00   20.27 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.05    0.19   20.46 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                  0.05    0.00   20.46 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_1)
                         -0.25   20.21   clock uncertainty
                          0.00   20.21   clock reconvergence pessimism
                          0.24   20.45   library recovery time
                                 20.45   data required time
---------------------------------------------------------------------------
                                 20.45   data required time
                                 -4.31   data arrival time
---------------------------------------------------------------------------
                                 16.14   slack (MET)


Startpoint: level_full (input port clocked by clk)
Endpoint: pump_en (output port clocked by clk)
Path Group: clk
Path Type: max
Corner: Slowest
```

```
Fanout     Cap     Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/26-mca/rcx_min_sta.checks.rpt ? pa
==

```
                        4.00    4.00 v input external delay
                0.02    0.01    4.01 v level_full (in)
     1    0.00                        level_full (net)
                0.02    0.00    4.01 v input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
                0.18    0.37    4.38 v input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
     3    0.02                        net2 (net)
                0.18    0.00    4.38 v _09_/A2 (sky130_fd_sc_hd__a21oi_2)
                0.35    0.45    4.83 ^ _09_/Y (sky130_fd_sc_hd__a21oi_2)
     4    0.02                        _03_ (net)
                0.35    0.00    4.83 ^ _18_/B (sky130_fd_sc_hd__and2_1)
                0.08    0.33    5.17 ^ _18_/X (sky130_fd_sc_hd__and2_1)
     1    0.00                        _08_ (net)
                0.08    0.00    5.17 ^ _19_/A (sky130_fd_sc_hd__clkbuf_1)
                0.08    0.16    5.33 ^ _19_/X (sky130_fd_sc_hd__clkbuf_1)
     1    0.00                        net6 (net)
                0.08    0.00    5.33 ^ output6/A (sky130_fd_sc_hd__clkbuf_4)
                0.18    0.35    5.68 ^ output6/X (sky130_fd_sc_hd__clkbuf_4)
     1    0.03                        pump_en (net)
                0.18    0.00    5.68 ^ pump_en (out)
                        5.68    data arrival time

                       20.00   20.00   clock clk (rise edge)
                        0.00   20.00   clock network delay (propagated)
                       -0.25   19.75   clock uncertainty
                        0.00   19.75   clock reconvergence pessimism
                       -4.00   15.75   output external delay
                               15.75   data required time
-------------------------------------------------------------------------------
                               15.75   data required time
                               -5.68   data arrival time
-------------------------------------------------------------------------------
                               10.07   slack (MET)



====================== Typical Corner ===================================

Startpoint: rst_n (input port clocked by clk)
Endpoint: _20_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
Path Type: max
Corner: Typical

Fanout     Cap    Slew   Delay    Time   Description
-------------------------------------------------------------------------
```

```
0.00    0.00   clock clk (rise edge)
0.00    0.00   clock network delay (propagated)
4.00    4.00 ^ input external delay
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/26-mca/rcx_min_sta.checks.rpt ? pa
==========

```
                  0.02    0.01    4.01 ^ rst_n (in)
     1    0.00                           rst_n (net)
                  0.02    0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                  0.19    0.18    4.19 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.02                           net4 (net)
                  0.19    0.00    4.19 ^ _20_/SET_B (sky130_fd_sc_hd__dfstp_1)
                                  4.19   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock source latency
                  0.06    0.04   20.04 ^ clk (in)
     1    0.01                           clk (net)
                  0.06    0.00   20.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.04    0.12   20.16 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                  0.04    0.00   20.16 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.03    0.11   20.26 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                  0.03    0.00   20.27 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_1)
                         -0.25   20.02   clock uncertainty
                          0.00   20.02   clock reconvergence pessimism
                          0.12   20.13   library recovery time
                                 20.13   data required time
-------------------------------------------------------------------------------
                                 20.13   data required time
                                 -4.19   data arrival time
-------------------------------------------------------------------------------
                                 15.94   slack (MET)


Startpoint: level_full (input port clocked by clk)
Endpoint: pump_en (output port clocked by clk)
Path Group: clk
Path Type: max
Corner: Typical

Fanout     Cap    Slew   Delay    Time   Description
-------------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 v input external delay
                  0.01    0.00    4.00 v level_full (in)
     1    0.00                           level_full (net)
```

```
              0.01    0.00    4.00 v input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
              0.09    0.18    4.19 v input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
3    0.02                             net2 (net)
              0.09    0.00    4.19 v _09_/A2 (sky130_fd_sc_hd__a21oi_2)
              0.22    0.26    4.44 ^ _09_/Y (sky130_fd_sc_hd__a21oi_2)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/26-mca/rcx_min_sta.checks.rpt ? pa
==

```
     4    0.02                           _03_ (net)
                  0.22    0.00    4.44 ^ _18_/B (sky130_fd_sc_hd__and2_1)
                  0.05    0.16    4.61 ^ _18_/X (sky130_fd_sc_hd__and2_1)
     1    0.00                           _08_ (net)
                  0.05    0.00    4.61 ^ _19_/A (sky130_fd_sc_hd__clkbuf_1)
                  0.05    0.09    4.69 ^ _19_/X (sky130_fd_sc_hd__clkbuf_1)
     1    0.00                           net6 (net)
                  0.05    0.00    4.69 ^ output6/A (sky130_fd_sc_hd__clkbuf_4)
                  0.11    0.20    4.89 ^ output6/X (sky130_fd_sc_hd__clkbuf_4)
     1    0.03                           pump_en (net)
                  0.11    0.00    4.89 ^ pump_en (out)
                                  4.89   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock network delay (propagated)
                         -0.25   19.75   clock uncertainty
                          0.00   19.75   clock reconvergence pessimism
                         -4.00   15.75   output external delay
                                 15.75   data required time
-------------------------------------------------------------------------------
                                 15.75   data required time
                                 -4.89   data arrival time
-------------------------------------------------------------------------------
                                 10.86   slack (MET)
```

```
================================================================================
report_checks --slack_max -0.01
================================================================================
====================== Fastest Corner ==================================

No paths found.

====================== Slowest Corner ==================================

No paths found.

====================== Typical Corner ==================================

No paths found.


================================================================================
```

```
 report_check_types -max_slew -max_cap -max_fanout -violators
================================================================================
======================== Fastest Corner ========================================
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/26-mca/rcx_min_sta.checks.rpt ? pa
==

========================= Slowest Corner ===================================

========================= Typical Corner ===================================

===
report_parasitic_annotation -report_unannotated
===
Found 0 unannotated drivers.
Found 0 partially unannotated drivers.

===
max slew violation count 0
max fanout violation count 0
max cap violation count 0
===

===
check_setup -verbose -unconstrained_endpoints -multiple_clock -no_clock -no_input_delay
-loops -generated_clocks
===

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/26-mca/rcx_min_sta.max.rpt ? part
==

==
report_checks -path_delay max (Setup)
==
======================== Fastest Corner ===================================

Startpoint: level_full (input port clocked by clk)
Endpoint: pump_en (output port clocked by clk)
Path Group: clk
Path Type: max
Corner: Fastest

```
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 v input external delay
                 0.01     0.00    4.00 v level_full (in)
     1    0.00                           level_full (net)
                 0.01     0.00    4.00 v input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
                 0.07     0.12    4.12 v input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
     3    0.02                           net2 (net)
                 0.07     0.00    4.12 v _09_/A2 (sky130_fd_sc_hd__a21oi_2)
                 0.16     0.18    4.30 ^ _09_/Y (sky130_fd_sc_hd__a21oi_2)
     4    0.02                           _03_ (net)
                 0.16     0.00    4.30 ^ _18_/B (sky130_fd_sc_hd__and2_1)
                 0.03     0.09    4.40 ^ _18_/X (sky130_fd_sc_hd__and2_1)
     1    0.00                           _08_ (net)
                 0.03     0.00    4.40 ^ _19_/A (sky130_fd_sc_hd__clkbuf_1)
                 0.03     0.06    4.45 ^ _19_/X (sky130_fd_sc_hd__clkbuf_1)
     1    0.00                           net6 (net)
                 0.03     0.00    4.45 ^ output6/A (sky130_fd_sc_hd__clkbuf_4)
                 0.08     0.13    4.59 ^ output6/X (sky130_fd_sc_hd__clkbuf_4)
     1    0.03                           pump_en (net)
                 0.08     0.00    4.59 ^ pump_en (out)
                           4.59   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock network delay (propagated)
                         -0.25   19.75   clock uncertainty
                          0.00   19.75   clock reconvergence pessimism
                         -4.00   15.75   output external delay
                                 15.75   data required time
---------------------------------------------------------------------------
                                 15.75   data required time
                                 -4.59   data arrival time
```

11.16 slack (MET)

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/26-mca/rcx_min_sta.max.rpt ? part
==========

Startpoint: level_full (input port clocked by clk)
Endpoint: alarm (output port clocked by clk)
Path Group: clk
Path Type: max
Corner: Fastest

Fanout	Cap	Slew	Delay	Time		Description
			0.00	0.00		clock clk (rise edge)
			0.00	0.00		clock network delay (propagated)
			4.00	4.00	v	input external delay
		0.01	0.00	4.00	v	level_full (in)
1	0.00					level_full (net)
		0.01	0.00	4.00	v	input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
		0.07	0.12	4.12	v	input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
3	0.02					net2 (net)
		0.07	0.00	4.12	v	_09_/A2 (sky130_fd_sc_hd__a21oi_2)
		0.16	0.18	4.30	^	_09_/Y (sky130_fd_sc_hd__a21oi_2)
4	0.02					_03_ (net)
		0.16	0.00	4.30	^	_10_/A (sky130_fd_sc_hd__inv_2)
		0.04	0.02	4.33	v	_10_/Y (sky130_fd_sc_hd__inv_2)
2	0.01					net5 (net)
		0.04	0.00	4.33	v	output5/A (sky130_fd_sc_hd__buf_2)
		0.06	0.13	4.45	v	output5/X (sky130_fd_sc_hd__buf_2)
1	0.03					alarm (net)
		0.06	0.00	4.45	v	alarm (out)
				4.45		data arrival time
			20.00	20.00		clock clk (rise edge)
			0.00	20.00		clock network delay (propagated)
			-0.25	19.75		clock uncertainty
			0.00	19.75		clock reconvergence pessimism
			-4.00	15.75		output external delay
				15.75		data required time
				15.75		data required time
				-4.45		data arrival time
				11.30		slack (MET)

Startpoint: level_full (input port clocked by clk)
Endpoint: _21_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk

Path Type: max
Corner: Fastest

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/26-mca/rcx_min_sta.max.rpt ? part
==========

```
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 v input external delay
                   0.01   0.00    4.00 v level_full (in)
     1    0.00                           level_full (net)
                   0.01   0.00    4.00 v input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
                   0.07   0.12    4.12 v input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
     3    0.02                           net2 (net)
                   0.07   0.00    4.12 v _09_/A2 (sky130_fd_sc_hd__a21oi_2)
                   0.16   0.18    4.30 ^ _09_/Y (sky130_fd_sc_hd__a21oi_2)
     4    0.02                           _03_ (net)
                   0.16   0.00    4.30 ^ _17_/B1 (sky130_fd_sc_hd__o211a_1)
                   0.03   0.08    4.38 ^ _17_/X (sky130_fd_sc_hd__o211a_1)
     1    0.00                           _01_ (net)
                   0.03   0.00    4.38 ^ hold2/A (sky130_fd_sc_hd__dlygate4sd3_1)
                   0.04   0.39    4.77 ^ hold2/X (sky130_fd_sc_hd__dlygate4sd3_1)
     1    0.00                           net8 (net)
                   0.04   0.00    4.77 ^ _21_/D (sky130_fd_sc_hd__dfrtp_1)
                                  4.77   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock source latency
                   0.04   0.03   20.03 ^ clk (in)
     1    0.01                           clk (net)
                   0.04   0.00   20.03 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                   0.03   0.08   20.11 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                   0.03   0.00   20.11 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                   0.02   0.07   20.18 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                   0.02   0.00   20.18 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                         -0.25   19.93   clock uncertainty
                          0.00   19.93   clock reconvergence pessimism
                         -0.03   19.90   library setup time
                                 19.90   data required time
---------------------------------------------------------------------------
                                 19.90   data required time
                                 -4.77   data arrival time
---------------------------------------------------------------------------
                                 15.13   slack (MET)
```

```
Startpoint: level_full (input port clocked by clk)
Endpoint: _20_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/26-mca/rcx_min_sta.max.rpt ? part
==========

Corner: Fastest

```
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                   0.02   0.01    4.01 ^ level_full (in)
     1    0.00                           level_full (net)
                   0.02   0.00    4.01 ^ input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
                   0.14   0.14    4.15 ^ input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
     3    0.02                           net2 (net)
                   0.14   0.00    4.15 ^ _11_/A (sky130_fd_sc_hd__inv_2)
                   0.03   0.02    4.17 v _11_/Y (sky130_fd_sc_hd__inv_2)
     2    0.01                           _04_ (net)
                   0.03   0.00    4.17 v _15_/A1 (sky130_fd_sc_hd__a21oi_1)
                   0.07   0.08    4.25 ^ _15_/Y (sky130_fd_sc_hd__a21oi_1)
     1    0.00                           _07_ (net)
                   0.07   0.00    4.25 ^ hold4/A (sky130_fd_sc_hd__dlygate4sd3_1)
                   0.04   0.40    4.65 ^ hold4/X (sky130_fd_sc_hd__dlygate4sd3_1)
     1    0.00                           net10 (net)
                   0.04   0.00    4.65 ^ _16_/A2 (sky130_fd_sc_hd__o21ai_1)
                   0.04   0.04    4.69 v _16_/Y (sky130_fd_sc_hd__o21ai_1)
     1    0.00                           _00_ (net)
                   0.04   0.00    4.69 v _20_/D (sky130_fd_sc_hd__dfstp_1)
                                  4.69   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock source latency
                   0.04   0.03   20.03 ^ clk (in)
     1    0.01                           clk (net)
                   0.04   0.00   20.03 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                   0.03   0.08   20.11 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                   0.03   0.00   20.11 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                   0.02   0.07   20.18 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                   0.02   0.00   20.18 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_1)
                         -0.25   19.93   clock uncertainty
                          0.00   19.93   clock reconvergence pessimism
                         -0.05   19.88   library setup time
                                 19.88   data required time
---------------------------------------------------------------------------
```

```
                                19.88   data required time
                                -4.69   data arrival time
-----------------------------------------------------------------------------
                                15.19   slack (MET)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/26-mca/rcx_min_sta.max.rpt ? part
==

Startpoint: level_full (input port clocked by clk)
Endpoint: _22_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: Fastest

Fanout	Cap	Slew	Delay	Time	Description
			0.00	0.00	clock clk (rise edge)
			0.00	0.00	clock network delay (propagated)
			4.00	4.00 v	input external delay
		0.01	0.00	4.00 v	level_full (in)
1	0.00				level_full (net)
		0.01	0.00	4.00 v	input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
		0.07	0.12	4.12 v	input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
3	0.02				net2 (net)
		0.07	0.00	4.12 v	_09_/A2 (sky130_fd_sc_hd__a21oi_2)
		0.16	0.18	4.30 ^	_09_/Y (sky130_fd_sc_hd__a21oi_2)
4	0.02				_03_ (net)
		0.16	0.00	4.30 ^	_10_/A (sky130_fd_sc_hd__inv_2)
		0.04	0.02	4.33 v	_10_/Y (sky130_fd_sc_hd__inv_2)
2	0.01				net5 (net)
		0.04	0.00	4.33 v	_14_/A (sky130_fd_sc_hd__nor2_1)
		0.05	0.07	4.40 ^	_14_/Y (sky130_fd_sc_hd__nor2_1)
1	0.00				_02_ (net)
		0.05	0.00	4.40 ^	_22_/D (sky130_fd_sc_hd__dfrtp_1)
				4.40	data arrival time
			20.00	20.00	clock clk (rise edge)
			0.00	20.00	clock source latency
		0.04	0.03	20.03 ^	clk (in)
1	0.01				clk (net)
		0.04	0.00	20.03 ^	clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
		0.03	0.08	20.11 ^	clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
2	0.02				clknet_0_clk (net)
		0.03	0.00	20.11 ^	clkbuf_1_1__f_clk/A (sky130_fd_sc_hd__clkbuf_16)
		0.02	0.07	20.18 ^	clkbuf_1_1__f_clk/X (sky130_fd_sc_hd__clkbuf_16)
1	0.00				clknet_1_1__leaf_clk (net)
		0.02	0.00	20.18 ^	_22_/CLK (sky130_fd_sc_hd__dfrtp_1)
			-0.25	19.93	clock uncertainty
			0.00	19.93	clock reconvergence pessimism
			-0.04	19.89	library setup time

```
                                19.89   data required time
-----------------------------------------------------------------------------
                                19.89   data required time
                                -4.40   data arrival time
-----------------------------------------------------------------------------
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/26-mca/rcx_min_sta.max.rpt ? part
==

                                15.50   slack (MET)

Startpoint: rst_n (input port clocked by clk)
Endpoint: _20_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
Path Type: max
Corner: Fastest

Fanout   Cap   Slew   Delay   Time   Description

                          0.00   0.00   clock clk (rise edge)
                          0.00   0.00   clock network delay (propagated)
                          4.00   4.00 ^ input external delay
                 0.01   0.01   4.01 ^ rst_n (in)
     1   0.00   rst_n (net)
                 0.01   0.00   4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                 0.14   0.13   4.14 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3   0.02   net4 (net)
                 0.14   0.00   4.14 ^ _20_/SET_B (sky130_fd_sc_hd__dfstp_1)
                                   4.14   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock source latency
                 0.04   0.03   20.03 ^ clk (in)
     1   0.01   clk (net)
                 0.04   0.00   20.03 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                 0.03   0.08   20.11 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2   0.02   clknet_0_clk (net)
                 0.03   0.00   20.11 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                 0.02   0.07   20.18 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2   0.01   clknet_1_0__leaf_clk (net)
                 0.02   0.00   20.18 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_1)
                         -0.25   19.93   clock uncertainty
                          0.00   19.93   clock reconvergence pessimism
                          0.09   20.02   library recovery time
                                 20.02   data required time

                                 20.02   data required time
                                 -4.14   data arrival time

                                 15.88   slack (MET)

```
Startpoint: rst_n (input port clocked by clk)
Endpoint: _22_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
Path Type: max
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/26-mca/rcx_min_sta.max.rpt ? part
==========

Corner: Fastest

```
Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                 0.01     0.01    4.01 ^ rst_n (in)
     1    0.00                           rst_n (net)
                 0.01     0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                 0.14     0.13    4.14 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.02                           net4 (net)
                 0.14     0.00    4.14 ^ _22_/RESET_B (sky130_fd_sc_hd__dfrtp_1)
                                  4.14   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock source latency
                 0.04     0.03   20.03 ^ clk (in)
     1    0.01                           clk (net)
                 0.04     0.00   20.03 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                 0.03     0.08   20.11 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                 0.03     0.00   20.11 ^ clkbuf_1_1__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                 0.02     0.07   20.18 ^ clkbuf_1_1__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     1    0.00                           clknet_1_1__leaf_clk (net)
                 0.02     0.00   20.18 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_1)
                         -0.25   19.93   clock uncertainty
                          0.00   19.93   clock reconvergence pessimism
                          0.12   20.05   library recovery time
                                 20.05   data required time
-----------------------------------------------------------------------------
                                 20.05   data required time
                                 -4.14   data arrival time
-----------------------------------------------------------------------------
                                 15.92   slack (MET)


Startpoint: rst_n (input port clocked by clk)
Endpoint: _21_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
Path Type: max
Corner: Fastest
```

```
Fanout     Cap     Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/26-mca/rcx_min_sta.max.rpt ? part
==

```
                0.01    0.01    4.01 ^ rst_n (in)
     1    0.00                        rst_n (net)
                0.01    0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                0.14    0.13    4.14 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.02                        net4 (net)
                0.14    0.00    4.14 ^ _21_/RESET_B (sky130_fd_sc_hd__dfrtp_1)
                                4.14   data arrival time

                       20.00   20.00   clock clk (rise edge)
                        0.00   20.00   clock source latency
                0.04    0.03   20.03 ^ clk (in)
     1    0.01                        clk (net)
                0.04    0.00   20.03 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                0.03    0.08   20.11 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                        clknet_0_clk (net)
                0.03    0.00   20.11 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                0.02    0.07   20.18 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                        clknet_1_0__leaf_clk (net)
                0.02    0.00   20.18 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                       -0.25   19.93   clock uncertainty
                        0.00   19.93   clock reconvergence pessimism
                        0.12   20.05   library recovery time
                               20.05   data required time
------------------------------------------------------------------------------
                               20.05   data required time
                               -4.14   data arrival time
------------------------------------------------------------------------------
                               15.92   slack (MET)



====================== Slowest Corner ===================================

Startpoint: level_full (input port clocked by clk)
Endpoint: pump_en (output port clocked by clk)
Path Group: clk
Path Type: max
Corner: Slowest

Fanout     Cap    Slew   Delay    Time   Description
------------------------------------------------------------------------------
                        0.00    0.00   clock clk (rise edge)
                        0.00    0.00   clock network delay (propagated)
```

```
                    4.00    4.00 v input external delay
              0.02  0.01    4.01 v level_full (in)
1     0.00                       level_full (net)
              0.02  0.00    4.01 v input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
              0.18  0.37    4.38 v input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/26-mca/rcx_min_sta.max.rpt ? part
==========

```
     3    0.02                           net2 (net)
                  0.18    0.00    4.38 v _09_/A2 (sky130_fd_sc_hd__a21oi_2)
                  0.35    0.45    4.83 ^ _09_/Y (sky130_fd_sc_hd__a21oi_2)
     4    0.02                           _03_ (net)
                  0.35    0.00    4.83 ^ _18_/B (sky130_fd_sc_hd__and2_1)
                  0.08    0.33    5.17 ^ _18_/X (sky130_fd_sc_hd__and2_1)
     1    0.00                           _08_ (net)
                  0.08    0.00    5.17 ^ _19_/A (sky130_fd_sc_hd__clkbuf_1)
                  0.08    0.16    5.33 ^ _19_/X (sky130_fd_sc_hd__clkbuf_1)
     1    0.00                           net6 (net)
                  0.08    0.00    5.33 ^ output6/A (sky130_fd_sc_hd__clkbuf_4)
                  0.18    0.35    5.68 ^ output6/X (sky130_fd_sc_hd__clkbuf_4)
     1    0.03                           pump_en (net)
                  0.18    0.00    5.68 ^ pump_en (out)
                                  5.68   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock network delay (propagated)
                         -0.25   19.75   clock uncertainty
                          0.00   19.75   clock reconvergence pessimism
                         -4.00   15.75   output external delay
                                 15.75   data required time
-----------------------------------------------------------------------------
                                 15.75   data required time
                                 -5.68   data arrival time
-----------------------------------------------------------------------------
                                 10.07   slack (MET)


Startpoint: level_full (input port clocked by clk)
Endpoint: alarm (output port clocked by clk)
Path Group: clk
Path Type: max
Corner: Slowest

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 v input external delay
                  0.02    0.01    4.01 v level_full (in)
     1    0.00                           level_full (net)
                  0.02    0.00    4.01 v input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
                  0.18    0.37    4.38 v input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
     3    0.02                           net2 (net)
```

```
              0.18    0.00    4.38 v _09_/A2 (sky130_fd_sc_hd__a21oi_2)
              0.35    0.45    4.83 ^ _09_/Y (sky130_fd_sc_hd__a21oi_2)
4    0.02                           _03_ (net)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/26-mca/rcx_min_sta.max.rpt ? part
==

```
                  0.35    0.00    4.83 ^ _10_/A (sky130_fd_sc_hd__inv_2)
                  0.08    0.16    4.99 v _10_/Y (sky130_fd_sc_hd__inv_2)
     2    0.01                           net5 (net)
                  0.08    0.00    4.99 v output5/A (sky130_fd_sc_hd__buf_2)
                  0.17    0.37    5.36 v output5/X (sky130_fd_sc_hd__buf_2)
     1    0.03                           alarm (net)
                  0.17    0.00    5.36 v alarm (out)
                                  5.36   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock network delay (propagated)
                         -0.25   19.75   clock uncertainty
                          0.00   19.75   clock reconvergence pessimism
                         -4.00   15.75   output external delay
                                 15.75   data required time
-------------------------------------------------------------------------------
                                 15.75   data required time
                                 -5.36   data arrival time
-------------------------------------------------------------------------------
                                 10.39   slack (MET)


Startpoint: level_full (input port clocked by clk)
Endpoint: _21_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: Slowest

Fanout     Cap    Slew   Delay    Time   Description
-------------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 v input external delay
                  0.02    0.01    4.01 v level_full (in)
     1    0.00                           level_full (net)
                  0.02    0.00    4.01 v input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
                  0.18    0.37    4.38 v input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
     3    0.02                           net2 (net)
                  0.18    0.00    4.38 v _09_/A2 (sky130_fd_sc_hd__a21oi_2)
                  0.35    0.45    4.83 ^ _09_/Y (sky130_fd_sc_hd__a21oi_2)
     4    0.02                           _03_ (net)
                  0.35    0.00    4.83 ^ _17_/B1 (sky130_fd_sc_hd__o211a_1)
                  0.07    0.40    5.24 ^ _17_/X (sky130_fd_sc_hd__o211a_1)
     1    0.00                           _01_ (net)
                  0.07    0.00    5.24 ^ hold2/A (sky130_fd_sc_hd__dlygate4sd3_1)
```

```
             0.09    1.09    6.32 ^ hold2/X (sky130_fd_sc_hd__dlygate4sd3_1)
1    0.00                           net8 (net)
             0.09    0.00    6.32 ^ _21_/D (sky130_fd_sc_hd__dfrtp_1)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/26-mca/rcx_min_sta.max.rpt ? part
==========

```
                                6.32   data arrival time

                       20.00   20.00   clock clk (rise edge)
                        0.00   20.00   clock source latency
                0.09    0.05   20.05 ^ clk (in)
     1    0.01                        clk (net)
                0.09    0.00   20.05 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                0.07    0.22   20.27 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                        clknet_0_clk (net)
                0.07    0.00   20.27 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                0.05    0.19   20.46 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                        clknet_1_0__leaf_clk (net)
                0.05    0.00   20.46 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                       -0.25   20.21   clock uncertainty
                        0.00   20.21   clock reconvergence pessimism
                       -0.14   20.07   library setup time
                               20.07   data required time
-----------------------------------------------------------------------------
                               20.07   data required time
                               -6.32   data arrival time
-----------------------------------------------------------------------------
                               13.75   slack (MET)


Startpoint: level_full (input port clocked by clk)
Endpoint: _20_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: Slowest

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                        0.00    0.00   clock clk (rise edge)
                        0.00    0.00   clock network delay (propagated)
                        4.00    4.00 v input external delay
                0.02    0.01    4.01 v level_full (in)
     1    0.00                        level_full (net)
                0.02    0.00    4.01 v input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
                0.18    0.37    4.38 v input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
     3    0.02                        net2 (net)
                0.18    0.00    4.38 v _11_/A (sky130_fd_sc_hd__inv_2)
                0.08    0.14    4.53 ^ _11_/Y (sky130_fd_sc_hd__inv_2)
     2    0.01                        _04_ (net)
```

```
             0.08    0.00    4.53 ^ _15_/A1 (sky130_fd_sc_hd__a21oi_1)
             0.08    0.12    4.64 v _15_/Y (sky130_fd_sc_hd__a21oi_1)
1    0.00                           _07_ (net)
             0.08    0.00    4.64 v hold4/A (sky130_fd_sc_hd__dlygate4sd3_1)
             0.11    1.19    5.83 v hold4/X (sky130_fd_sc_hd__dlygate4sd3_1)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/26-mca/rcx_min_sta.max.rpt ? part
==

```
     1    0.00                           net10 (net)
                  0.11    0.00    5.83 v _16_/A2 (sky130_fd_sc_hd__o21ai_1)
                  0.18    0.23    6.06 ^ _16_/Y (sky130_fd_sc_hd__o21ai_1)
     1    0.00                           _00_ (net)
                  0.18    0.00    6.06 ^ _20_/D (sky130_fd_sc_hd__dfstp_1)
                                  6.06   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock source latency
                  0.09    0.05   20.05 ^ clk (in)
     1    0.01                           clk (net)
                  0.09    0.00   20.05 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.07    0.22   20.27 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                  0.07    0.00   20.27 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.05    0.19   20.46 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                  0.05    0.00   20.46 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_1)
                         -0.25   20.21   clock uncertainty
                          0.00   20.21   clock reconvergence pessimism
                         -0.15   20.06   library setup time
                                 20.06   data required time
--------------------------------------------------------------------------------
                                 20.06   data required time
                                 -6.06   data arrival time
--------------------------------------------------------------------------------
                                 14.00   slack (MET)


Startpoint: level_full (input port clocked by clk)
Endpoint: _22_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: Slowest

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 v input external delay
                  0.02    0.01    4.01 v level_full (in)
     1    0.00                           level_full (net)
                  0.02    0.00    4.01 v input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
```

```
              0.18    0.37    4.38 v input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
3    0.02                             net2 (net)
              0.18    0.00    4.38 v _09_/A2 (sky130_fd_sc_hd__a21oi_2)
              0.35    0.45    4.83 ^ _09_/Y (sky130_fd_sc_hd__a21oi_2)
4    0.02                             _03_ (net)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/26-mca/rcx_min_sta.max.rpt ? part
==

```
                 0.35    0.00    4.83 ^ _10_/A (sky130_fd_sc_hd__inv_2)
                 0.08    0.16    4.99 v _10_/Y (sky130_fd_sc_hd__inv_2)
     2    0.01                         net5 (net)
                 0.08    0.00    4.99 v _14_/A (sky130_fd_sc_hd__nor2_1)
                 0.13    0.18    5.17 ^ _14_/Y (sky130_fd_sc_hd__nor2_1)
     1    0.00                         _02_ (net)
                 0.13    0.00    5.17 ^ _22_/D (sky130_fd_sc_hd__dfrtp_1)
                                 5.17   data arrival time

                        20.00   20.00   clock clk (rise edge)
                         0.00   20.00   clock source latency
                 0.09    0.05   20.05 ^ clk (in)
     1    0.01                         clk (net)
                 0.09    0.00   20.05 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                 0.07    0.22   20.27 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                         clknet_0_clk (net)
                 0.07    0.00   20.27 ^ clkbuf_1_1__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                 0.04    0.18   20.46 ^ clkbuf_1_1__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     1    0.00                         clknet_1_1__leaf_clk (net)
                 0.04    0.00   20.46 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_1)
                        -0.25   20.21   clock uncertainty
                         0.00   20.21   clock reconvergence pessimism
                        -0.15   20.05   library setup time
                                20.05   data required time
---------------------------------------------------------------------------
                                20.05   data required time
                                -5.17   data arrival time
---------------------------------------------------------------------------
                                14.88   slack (MET)


Startpoint: rst_n (input port clocked by clk)
Endpoint: _20_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
Path Type: max
Corner: Slowest

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                         0.00    0.00   clock clk (rise edge)
                         0.00    0.00   clock network delay (propagated)
                         4.00    4.00 ^ input external delay
                 0.03    0.02    4.02 ^ rst_n (in)
```

```
1    0.00                           rst_n (net)
             0.03    0.00    4.02 ^ input4/A (sky130_fd_sc_hd__buf_1)
             0.28    0.30    4.31 ^ input4/X (sky130_fd_sc_hd__buf_1)
3    0.01                           net4 (net)
             0.28    0.00    4.31 ^ _20_/SET_B (sky130_fd_sc_hd__dfstp_1)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/26-mca/rcx_min_sta.max.rpt ? part
==========

```
                                4.31   data arrival time

                       20.00   20.00   clock clk (rise edge)
                        0.00   20.00   clock source latency
                0.09    0.05   20.05 ^ clk (in)
     1    0.01                        clk (net)
                0.09    0.00   20.05 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                0.07    0.22   20.27 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                        clknet_0_clk (net)
                0.07    0.00   20.27 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                0.05    0.19   20.46 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                        clknet_1_0__leaf_clk (net)
                0.05    0.00   20.46 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_1)
                       -0.25   20.21   clock uncertainty
                        0.00   20.21   clock reconvergence pessimism
                        0.24   20.45   library recovery time
                               20.45   data required time
-------------------------------------------------------------------------
                               20.45   data required time
                               -4.31   data arrival time
-------------------------------------------------------------------------
                               16.14   slack (MET)


Startpoint: rst_n (input port clocked by clk)
Endpoint: _22_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
Path Type: max
Corner: Slowest

Fanout     Cap    Slew   Delay    Time   Description
-------------------------------------------------------------------------
                        0.00    0.00   clock clk (rise edge)
                        0.00    0.00   clock network delay (propagated)
                        4.00    4.00 ^ input external delay
                0.03    0.02    4.02 ^ rst_n (in)
     1    0.00                        rst_n (net)
                0.03    0.00    4.02 ^ input4/A (sky130_fd_sc_hd__buf_1)
                0.28    0.30    4.31 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.01                        net4 (net)
                0.28    0.00    4.31 ^ _22_/RESET_B (sky130_fd_sc_hd__dfrtp_1)
                               4.31   data arrival time
```

```
                  20.00   20.00   clock clk (rise edge)
                   0.00   20.00   clock source latency
            0.09   0.05   20.05 ^ clk (in)
1   0.01                          clk (net)
            0.09   0.00   20.05 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
```

```
openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/26-mca/rcx_min_sta.max.rpt ? part
================================================================================

                  0.07    0.22   20.27 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                  0.07    0.00   20.27 ^ clkbuf_1_1__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.04    0.18   20.46 ^ clkbuf_1_1__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     1    0.00                           clknet_1_1__leaf_clk (net)
                  0.04    0.00   20.46 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_1)
                         -0.25   20.21   clock uncertainty
                          0.00   20.21   clock reconvergence pessimism
                          0.38   20.58   library recovery time
                                 20.58   data required time
-------------------------------------------------------------------------------
                                 20.58   data required time
                                 -4.31   data arrival time
-------------------------------------------------------------------------------
                                 16.27   slack (MET)


Startpoint: rst_n (input port clocked by clk)
Endpoint: _21_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
Path Type: max
Corner: Slowest

Fanout     Cap    Slew   Delay    Time   Description
-------------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                  0.03    0.02    4.02 ^ rst_n (in)
     1    0.00                           rst_n (net)
                  0.03    0.00    4.02 ^ input4/A (sky130_fd_sc_hd__buf_1)
                  0.28    0.30    4.31 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.01                           net4 (net)
                  0.28    0.00    4.31 ^ _21_/RESET_B (sky130_fd_sc_hd__dfrtp_1)
                                  4.31   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock source latency
                  0.09    0.05   20.05 ^ clk (in)
     1    0.01                           clk (net)
                  0.09    0.00   20.05 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.07    0.22   20.27 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
```

```
                  0.07    0.00   20.27 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.05    0.19   20.46 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                  0.05    0.00   20.46 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                         -0.25   20.21   clock uncertainty
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/26-mca/rcx_min_sta.max.rpt ? part
==

```
                          0.00   20.21   clock reconvergence pessimism
                          0.38   20.59   library recovery time
                                 20.59   data required time
-------------------------------------------------------------------------------
                                 20.59   data required time
                                 -4.31   data arrival time
-------------------------------------------------------------------------------
                                 16.28   slack (MET)



====================== Typical Corner ===================================

Startpoint: level_full (input port clocked by clk)
Endpoint: pump_en (output port clocked by clk)
Path Group: clk
Path Type: max
Corner: Typical

Fanout     Cap    Slew   Delay    Time   Description
-------------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 v input external delay
                   0.01   0.00    4.00 v level_full (in)
     1    0.00                           level_full (net)
                   0.01   0.00    4.00 v input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
                   0.09   0.18    4.19 v input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
     3    0.02                           net2 (net)
                   0.09   0.00    4.19 v _09_/A2 (sky130_fd_sc_hd__a21oi_2)
                   0.22   0.26    4.44 ^ _09_/Y (sky130_fd_sc_hd__a21oi_2)
     4    0.02                           _03_ (net)
                   0.22   0.00    4.44 ^ _18_/B (sky130_fd_sc_hd__and2_1)
                   0.05   0.16    4.61 ^ _18_/X (sky130_fd_sc_hd__and2_1)
     1    0.00                           _08_ (net)
                   0.05   0.00    4.61 ^ _19_/A (sky130_fd_sc_hd__clkbuf_1)
                   0.05   0.09    4.69 ^ _19_/X (sky130_fd_sc_hd__clkbuf_1)
     1    0.00                           net6 (net)
                   0.05   0.00    4.69 ^ output6/A (sky130_fd_sc_hd__clkbuf_4)
                   0.11   0.20    4.89 ^ output6/X (sky130_fd_sc_hd__clkbuf_4)
     1    0.03                           pump_en (net)
                   0.11   0.00    4.89 ^ pump_en (out)
                                  4.89   data arrival time

                         20.00   20.00   clock clk (rise edge)
```

```
 0.00   20.00   clock network delay (propagated)
-0.25   19.75   clock uncertainty
 0.00   19.75   clock reconvergence pessimism
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/26-mca/rcx_min_sta.max.rpt ? part
==

```
                          -4.00   15.75   output external delay
                                  15.75   data required time
-------------------------------------------------------------------------------
                                  15.75   data required time
                                  -4.89   data arrival time
-------------------------------------------------------------------------------
                                  10.86   slack (MET)


Startpoint: level_full (input port clocked by clk)
Endpoint: alarm (output port clocked by clk)
Path Group: clk
Path Type: max
Corner: Typical

Fanout     Cap    Slew   Delay    Time   Description
-------------------------------------------------------------------------------
                           0.00    0.00   clock clk (rise edge)
                           0.00    0.00   clock network delay (propagated)
                           4.00    4.00 v input external delay
                  0.01     0.00    4.00 v level_full (in)
     1    0.00                            level_full (net)
                  0.01     0.00    4.00 v input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
                  0.09     0.18    4.19 v input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
     3    0.02                            net2 (net)
                  0.09     0.00    4.19 v _09_/A2 (sky130_fd_sc_hd__a21oi_2)
                  0.22     0.26    4.44 ^ _09_/Y (sky130_fd_sc_hd__a21oi_2)
     4    0.02                            _03_ (net)
                  0.22     0.00    4.44 ^ _10_/A (sky130_fd_sc_hd__inv_2)
                  0.05     0.06    4.51 v _10_/Y (sky130_fd_sc_hd__inv_2)
     2    0.01                            net5 (net)
                  0.05     0.00    4.51 v output5/A (sky130_fd_sc_hd__buf_2)
                  0.09     0.19    4.70 v output5/X (sky130_fd_sc_hd__buf_2)
     1    0.03                            alarm (net)
                  0.09     0.00    4.70 v alarm (out)
                                   4.70   data arrival time

                          20.00   20.00   clock clk (rise edge)
                           0.00   20.00   clock network delay (propagated)
                          -0.25   19.75   clock uncertainty
                           0.00   19.75   clock reconvergence pessimism
                          -4.00   15.75   output external delay
                                  15.75   data required time
-------------------------------------------------------------------------------
                                  15.75   data required time
```

```
                                  -4.70   data arrival time
-----------------------------------------------------------------------------
                                  11.05   slack (MET)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/26-mca/rcx_min_sta.max.rpt ? part
==========

Startpoint: level_full (input port clocked by clk)
Endpoint: _21_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: Typical

Fanout	Cap	Slew	Delay	Time		Description
			0.00	0.00		clock clk (rise edge)
			0.00	0.00		clock network delay (propagated)
			4.00	4.00	v	input external delay
		0.01	0.00	4.00	v	level_full (in)
1	0.00					level_full (net)
		0.01	0.00	4.00	v	input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
		0.09	0.18	4.19	v	input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
3	0.02					net2 (net)
		0.09	0.00	4.19	v	_09_/A2 (sky130_fd_sc_hd__a21oi_2)
		0.22	0.26	4.44	^	_09_/Y (sky130_fd_sc_hd__a21oi_2)
4	0.02					_03_ (net)
		0.22	0.00	4.44	^	_17_/B1 (sky130_fd_sc_hd__o211a_1)
		0.04	0.18	4.63	^	_17_/X (sky130_fd_sc_hd__o211a_1)
1	0.00					_01_ (net)
		0.04	0.00	4.63	^	hold2/A (sky130_fd_sc_hd__dlygate4sd3_1)
		0.05	0.56	5.19	^	hold2/X (sky130_fd_sc_hd__dlygate4sd3_1)
1	0.00					net8 (net)
		0.05	0.00	5.19	^	_21_/D (sky130_fd_sc_hd__dfrtp_1)
				5.19		data arrival time
			20.00	20.00		clock clk (rise edge)
			0.00	20.00		clock source latency
		0.06	0.04	20.04	^	clk (in)
1	0.01					clk (net)
		0.06	0.00	20.04	^	clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
		0.04	0.12	20.16	^	clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
2	0.02					clknet_0_clk (net)
		0.04	0.00	20.16	^	clkbuf_1_0__f_clk/A (sky130_fd_sc_hd__clkbuf_16)
		0.03	0.11	20.26	^	clkbuf_1_0__f_clk/X (sky130_fd_sc_hd__clkbuf_16)
2	0.01					clknet_1_0__leaf_clk (net)
		0.03	0.00	20.26	^	_21_/CLK (sky130_fd_sc_hd__dfrtp_1)
			-0.25	20.01		clock uncertainty
			0.00	20.01		clock reconvergence pessimism

```
                          -0.06   19.95   library setup time
                                  19.95   data required time
-----------------------------------------------------------------------------
                                  19.95   data required time
                                  -5.19   data arrival time
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/26-mca/rcx_min_sta.max.rpt ? part
==

```
                                14.76   slack (MET)


Startpoint: level_full (input port clocked by clk)
Endpoint: _20_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: Typical

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                  0.02    0.01    4.01 ^ level_full (in)
     1    0.00                           level_full (net)
                  0.02    0.00    4.01 ^ input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
                  0.19    0.19    4.20 ^ input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
     3    0.02                           net2 (net)
                  0.19    0.00    4.20 ^ _11_/A (sky130_fd_sc_hd__inv_2)
                  0.05    0.06    4.26 v _11_/Y (sky130_fd_sc_hd__inv_2)
     2    0.01                           _04_ (net)
                  0.05    0.00    4.26 v _15_/A1 (sky130_fd_sc_hd__a21oi_1)
                  0.10    0.11    4.38 ^ _15_/Y (sky130_fd_sc_hd__a21oi_1)
     1    0.00                           _07_ (net)
                  0.10    0.00    4.38 ^ hold4/A (sky130_fd_sc_hd__dlygate4sd3_1)
                  0.06    0.58    4.96 ^ hold4/X (sky130_fd_sc_hd__dlygate4sd3_1)
     1    0.00                           net10 (net)
                  0.06    0.00    4.96 ^ _16_/A2 (sky130_fd_sc_hd__o21ai_1)
                  0.07    0.06    5.02 v _16_/Y (sky130_fd_sc_hd__o21ai_1)
     1    0.00                           _00_ (net)
                  0.07    0.00    5.02 v _20_/D (sky130_fd_sc_hd__dfstp_1)
                                  5.02   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock source latency
                  0.06    0.04   20.04 ^ clk (in)
     1    0.01                           clk (net)
                  0.06    0.00   20.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.04    0.12   20.16 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                  0.04    0.00   20.16 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.03    0.11   20.26 ^ clkbuf_1_0__f_clk/X
```

```
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                  0.03    0.00   20.27 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_1)
                         -0.25   20.02   clock uncertainty
                          0.00   20.02   clock reconvergence pessimism
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/26-mca/rcx_min_sta.max.rpt ? part
==========

```
                          -0.08   19.93   library setup time
                                  19.93   data required time
-----------------------------------------------------------------------------
                                  19.93   data required time
                                  -5.02   data arrival time
-----------------------------------------------------------------------------
                                  14.92   slack (MET)


Startpoint: level_full (input port clocked by clk)
Endpoint: _22_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: Typical

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                           0.00    0.00   clock clk (rise edge)
                           0.00    0.00   clock network delay (propagated)
                           4.00    4.00 v input external delay
                  0.01     0.00    4.00 v level_full (in)
     1    0.00                            level_full (net)
                  0.01     0.00    4.00 v input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
                  0.09     0.18    4.19 v input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
     3    0.02                            net2 (net)
                  0.09     0.00    4.19 v _09_/A2 (sky130_fd_sc_hd__a21oi_2)
                  0.22     0.26    4.44 ^ _09_/Y (sky130_fd_sc_hd__a21oi_2)
     4    0.02                            _03_ (net)
                  0.22     0.00    4.44 ^ _10_/A (sky130_fd_sc_hd__inv_2)
                  0.05     0.06    4.51 v _10_/Y (sky130_fd_sc_hd__inv_2)
     2    0.01                            net5 (net)
                  0.05     0.00    4.51 v _14_/A (sky130_fd_sc_hd__nor2_1)
                  0.08     0.10    4.61 ^ _14_/Y (sky130_fd_sc_hd__nor2_1)
     1    0.00                            _02_ (net)
                  0.08     0.00    4.61 ^ _22_/D (sky130_fd_sc_hd__dfrtp_1)
                           4.61    4.61   data arrival time

                          20.00   20.00   clock clk (rise edge)
                           0.00   20.00   clock source latency
                  0.06     0.04   20.04 ^ clk (in)
     1    0.01                            clk (net)
                  0.06     0.00   20.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.04     0.12   20.16 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                            clknet_0_clk (net)
                  0.04     0.00   20.16 ^ clkbuf_1_1__f_clk/A
```

```
(sky130_fd_sc_hd__clkbuf_16)
                  0.03    0.10   20.26 ^ clkbuf_1_1__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     1    0.00                           clknet_1_1__leaf_clk (net)
                  0.03    0.00   20.26 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_1)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/26-mca/rcx_min_sta.max.rpt ? part
==

```
                          -0.25   20.01   clock uncertainty
                           0.00   20.01   clock reconvergence pessimism
                          -0.07   19.94   library setup time
                                  19.94   data required time
-----------------------------------------------------------------------------
                                  19.94   data required time
                                  -4.61   data arrival time
-----------------------------------------------------------------------------
                                  15.33   slack (MET)


Startpoint: rst_n (input port clocked by clk)
Endpoint: _20_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
Path Type: max
Corner: Typical

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                           0.00    0.00   clock clk (rise edge)
                           0.00    0.00   clock network delay (propagated)
                           4.00    4.00 ^ input external delay
                  0.02     0.01    4.01 ^ rst_n (in)
     1    0.00                            rst_n (net)
                  0.02     0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                  0.19     0.18    4.19 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.02                            net4 (net)
                  0.19     0.00    4.19 ^ _20_/SET_B (sky130_fd_sc_hd__dfstp_1)
                                    4.19   data arrival time

                          20.00   20.00   clock clk (rise edge)
                           0.00   20.00   clock source latency
                  0.06     0.04   20.04 ^ clk (in)
     1    0.01                            clk (net)
                  0.06     0.00   20.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.04     0.12   20.16 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                            clknet_0_clk (net)
                  0.04     0.00   20.16 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.03     0.11   20.26 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                            clknet_1_0__leaf_clk (net)
                  0.03     0.00   20.27 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_1)
                          -0.25   20.02   clock uncertainty
                           0.00   20.02   clock reconvergence pessimism
```

```
                          0.12   20.13   library recovery time
                                 20.13   data required time
-----------------------------------------------------------------------------
                                 20.13   data required time
                                 -4.19   data arrival time
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/26-mca/rcx_min_sta.max.rpt ? part
==========

```
---------------------------------------------------------------------------
                                15.94   slack (MET)


Startpoint: rst_n (input port clocked by clk)
Endpoint: _22_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
Path Type: max
Corner: Typical

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                   0.02   0.01    4.01 ^ rst_n (in)
     1    0.00                           rst_n (net)
                   0.02   0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                   0.19   0.18    4.19 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.02                           net4 (net)
                   0.19   0.00    4.19 ^ _22_/RESET_B (sky130_fd_sc_hd__dfrtp_1)
                                  4.19   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock source latency
                   0.06   0.04   20.04 ^ clk (in)
     1    0.01                           clk (net)
                   0.06   0.00   20.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                   0.04   0.12   20.16 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                   0.04   0.00   20.16 ^ clkbuf_1_1__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                   0.03   0.10   20.26 ^ clkbuf_1_1__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     1    0.00                           clknet_1_1__leaf_clk (net)
                   0.03   0.00   20.26 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_1)
                         -0.25   20.01   clock uncertainty
                          0.00   20.01   clock reconvergence pessimism
                          0.19   20.20   library recovery time
                                 20.20   data required time
---------------------------------------------------------------------------
                                20.20   data required time
                                -4.19   data arrival time
---------------------------------------------------------------------------
                                16.01   slack (MET)
```

```
Startpoint: rst_n (input port clocked by clk)
Endpoint: _21_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/26-mca/rcx_min_sta.max.rpt ? part
==========

Path Type: max
Corner: Typical

```
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                  0.02    0.01    4.01 ^ rst_n (in)
     1    0.00                           rst_n (net)
                  0.02    0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                  0.19    0.18    4.19 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.02                           net4 (net)
                  0.19    0.00    4.19 ^ _21_/RESET_B (sky130_fd_sc_hd__dfrtp_1)
                                  4.19   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock source latency
                  0.06    0.04   20.04 ^ clk (in)
     1    0.01                           clk (net)
                  0.06    0.00   20.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.04    0.12   20.16 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                  0.04    0.00   20.16 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.03    0.11   20.26 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                  0.03    0.00   20.26 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                         -0.25   20.01   clock uncertainty
                          0.00   20.01   clock reconvergence pessimism
                          0.19   20.20   library recovery time
                                 20.20   data required time
---------------------------------------------------------------------------
                                 20.20   data required time
                                 -4.19   data arrival time
---------------------------------------------------------------------------
                                 16.01   slack (MET)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/26-mca/rcx_min_sta.min.rpt ? part
==

==
report_checks -path_delay min (Hold)
==
======================== Fastest Corner ===================================

Startpoint: _21_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _22_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
Corner: Fastest

```
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                  0.04    0.03    0.03 ^ clk (in)
     1    0.01                          clk (net)
                  0.04    0.00    0.03 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.03    0.08    0.11 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                          clknet_0_clk (net)
                  0.03    0.00    0.11 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.02    0.07    0.18 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                          clknet_1_0__leaf_clk (net)
                  0.02    0.00    0.18 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                  0.08    0.22    0.40 ^ _21_/Q (sky130_fd_sc_hd__dfrtp_1)
     3    0.01                          state[1] (net)
                  0.08    0.00    0.40 ^ _12_/A1 (sky130_fd_sc_hd__o21a_1)
                  0.03    0.08    0.48 ^ _12_/X (sky130_fd_sc_hd__o21a_1)
     1    0.00                          _05_ (net)
                  0.03    0.00    0.48 ^ _13_/B1 (sky130_fd_sc_hd__a21oi_1)
                  0.02    0.02    0.50 v _13_/Y (sky130_fd_sc_hd__a21oi_1)
     1    0.00                          _06_ (net)
                  0.02    0.00    0.50 v _14_/B (sky130_fd_sc_hd__nor2_1)
                  0.05    0.05    0.55 ^ _14_/Y (sky130_fd_sc_hd__nor2_1)
     1    0.00                          _02_ (net)
                  0.05    0.00    0.55 ^ _22_/D (sky130_fd_sc_hd__dfrtp_1)
                                  0.55   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                  0.04    0.03    0.03 ^ clk (in)
     1    0.01                          clk (net)
                  0.04    0.00    0.03 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
```

```
                0.03    0.09    0.12 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                       clknet_0_clk (net)
                0.03    0.00    0.12 ^ clkbuf_1_1__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                0.02    0.08    0.20 ^ clkbuf_1_1__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     1    0.00                       clknet_1_1__leaf_clk (net)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/26-mca/rcx_min_sta.min.rpt ? part
==

```
                  0.02    0.00    0.20 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_1)
                          0.25    0.45   clock uncertainty
                         -0.01    0.44   clock reconvergence pessimism
                         -0.02    0.41   library hold time
                                  0.41   data required time
------------------------------------------------------------------------------
                                  0.41   data required time
                                 -0.55   data arrival time
------------------------------------------------------------------------------
                                  0.13   slack (MET)


Startpoint: _21_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _21_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
Corner: Fastest

Fanout     Cap    Slew   Delay    Time   Description
------------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                  0.04    0.03    0.03 ^ clk (in)
     1    0.01                          clk (net)
                  0.04    0.00    0.03 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.03    0.08    0.11 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                          clknet_0_clk (net)
                  0.03    0.00    0.11 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.02    0.07    0.18 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                          clknet_1_0__leaf_clk (net)
                  0.02    0.00    0.18 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                  0.08    0.22    0.40 ^ _21_/Q (sky130_fd_sc_hd__dfrtp_1)
     3    0.01                          state[1] (net)
                  0.08    0.00    0.40 ^ _17_/A1 (sky130_fd_sc_hd__o211a_1)
                  0.03    0.09    0.49 ^ _17_/X (sky130_fd_sc_hd__o211a_1)
     1    0.00                          _01_ (net)
                  0.03    0.00    0.49 ^ hold2/A (sky130_fd_sc_hd__dlygate4sd3_1)
                  0.04    0.35    0.84 ^ hold2/X (sky130_fd_sc_hd__dlygate4sd3_1)
     1    0.00                          net8 (net)
                  0.04    0.00    0.84 ^ _21_/D (sky130_fd_sc_hd__dfrtp_1)
                                  0.84   data arrival time

                          0.00    0.00   clock clk (rise edge)
```

```
                   0.00    0.00   clock source latency
             0.04    0.03    0.03 ^ clk (in)
1    0.01                          clk (net)
             0.04    0.00    0.03 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
             0.03    0.09    0.12 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
```

```
openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/26-mca/rcx_min_sta.min.rpt ? part
================================================================================

     2    0.02                           clknet_0_clk (net)
                  0.03    0.00    0.12 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.02    0.08    0.20 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                  0.02    0.00    0.20 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                          0.25    0.45   clock uncertainty
                         -0.02    0.43   clock reconvergence pessimism
                         -0.02    0.41   library hold time
                                  0.41   data required time
-----------------------------------------------------------------------------
                                  0.41   data required time
                                 -0.84   data arrival time
-----------------------------------------------------------------------------
                                  0.43   slack (MET)


Startpoint: _21_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _20_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
Corner: Fastest

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                  0.04    0.03    0.03 ^ clk (in)
     1    0.01                           clk (net)
                  0.04    0.00    0.03 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.03    0.08    0.11 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                  0.03    0.00    0.11 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.02    0.07    0.18 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                  0.02    0.00    0.18 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                  0.08    0.22    0.40 ^ _21_/Q (sky130_fd_sc_hd__dfrtp_1)
     3    0.01                           state[1] (net)
                  0.08    0.00    0.40 ^ _15_/A2 (sky130_fd_sc_hd__a21oi_1)
                  0.02    0.04    0.44 v _15_/Y (sky130_fd_sc_hd__a21oi_1)
     1    0.00                           _07_ (net)
                  0.02    0.00    0.44 v hold4/A (sky130_fd_sc_hd__dlygate4sd3_1)
```

```
             0.04    0.34    0.78 v hold4/X (sky130_fd_sc_hd__dlygate4sd3_1)
1    0.00                            net10 (net)
             0.04    0.00    0.78 v _16_/A2 (sky130_fd_sc_hd__o21ai_1)
             0.04    0.08    0.86 ^ _16_/Y (sky130_fd_sc_hd__o21ai_1)
1    0.00                            _00_ (net)
             0.04    0.00    0.86 ^ _20_/D (sky130_fd_sc_hd__dfstp_1)
                             0.86   data arrival time
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/26-mca/rcx_min_sta.min.rpt ? part
==

```
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                 0.04     0.03    0.03 ^ clk (in)
     1    0.01                           clk (net)
                 0.04     0.00    0.03 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                 0.03     0.09    0.12 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                 0.03     0.00    0.12 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                 0.02     0.08    0.20 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                 0.02     0.00    0.20 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_1)
                          0.25    0.45   clock uncertainty
                         -0.02    0.43   clock reconvergence pessimism
                         -0.02    0.41   library hold time
                                  0.41   data required time
-----------------------------------------------------------------------------
                                  0.41   data required time
                                 -0.86   data arrival time
-----------------------------------------------------------------------------
                                  0.45   slack (MET)


Startpoint: rst_n (input port clocked by clk)
Endpoint: _21_ (removal check against rising-edge clock clk)
Path Group: asynchronous
Path Type: min
Corner: Fastest

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                 0.01     0.01    4.01 ^ rst_n (in)
     1    0.00                           rst_n (net)
                 0.01     0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                 0.14     0.12    4.12 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.02                           net4 (net)
                 0.14     0.00    4.12 ^ _21_/RESET_B (sky130_fd_sc_hd__dfrtp_1)
                                  4.12   data arrival time

                          0.00    0.00   clock clk (rise edge)
```

```
                   0.00    0.00   clock source latency
             0.04    0.03    0.03 ^ clk (in)
1    0.01                          clk (net)
             0.04    0.00    0.03 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
             0.03    0.09    0.12 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/26-mca/rcx_min_sta.min.rpt ? part
==

```
     2    0.02                           clknet_0_clk (net)
                  0.03    0.00    0.12 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.02    0.08    0.20 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                  0.02    0.00    0.20 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                          0.25    0.45   clock uncertainty
                          0.00    0.45   clock reconvergence pessimism
                          0.22    0.68   library removal time
                                  0.68   data required time
-----------------------------------------------------------------------------
                                  0.68   data required time
                                 -4.12   data arrival time
-----------------------------------------------------------------------------
                                  3.45   slack (MET)


Startpoint: rst_n (input port clocked by clk)
Endpoint: _22_ (removal check against rising-edge clock clk)
Path Group: asynchronous
Path Type: min
Corner: Fastest

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                  0.01    0.01    4.01 ^ rst_n (in)
     1    0.00                           rst_n (net)
                  0.01    0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                  0.14    0.12    4.12 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.02                           net4 (net)
                  0.14    0.00    4.12 ^ _22_/RESET_B (sky130_fd_sc_hd__dfrtp_1)
                                  4.12   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                  0.04    0.03    0.03 ^ clk (in)
     1    0.01                           clk (net)
                  0.04    0.00    0.03 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.03    0.09    0.12 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                  0.03    0.00    0.12 ^ clkbuf_1_1__f_clk/A
```

```
(sky130_fd_sc_hd__clkbuf_16)
                  0.02    0.08    0.20 ^ clkbuf_1_1__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     1    0.00                           clknet_1_1__leaf_clk (net)
                  0.02    0.00    0.20 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_1)
                          0.25    0.45   clock uncertainty
                          0.00    0.45   clock reconvergence pessimism
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/26-mca/rcx_min_sta.min.rpt ? part
==

```
                          0.22    0.67   library removal time
                                  0.67   data required time
-----------------------------------------------------------------------------
                                  0.67   data required time
                                 -4.12   data arrival time
-----------------------------------------------------------------------------
                                  3.45   slack (MET)


Startpoint: rst_n (input port clocked by clk)
Endpoint: _20_ (removal check against rising-edge clock clk)
Path Group: asynchronous
Path Type: min
Corner: Fastest

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                   0.01   0.01    4.01 ^ rst_n (in)
     1    0.00                           rst_n (net)
                   0.01   0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                   0.14   0.12    4.12 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.02                           net4 (net)
                   0.14   0.00    4.12 ^ _20_/SET_B (sky130_fd_sc_hd__dfstp_1)
                                  4.12   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                   0.04   0.03    0.03 ^ clk (in)
     1    0.01                           clk (net)
                   0.04   0.00    0.03 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                   0.03   0.09    0.12 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                   0.03   0.00    0.12 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                   0.02   0.08    0.20 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                   0.02   0.00    0.20 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_1)
                          0.25    0.45   clock uncertainty
                          0.00    0.45   clock reconvergence pessimism
                          0.10    0.55   library removal time
                                  0.55   data required time
```

```
------------------------------------------------------------------------------
                                0.55   data required time
                               -4.12   data arrival time
------------------------------------------------------------------------------
                                3.57   slack (MET)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/26-mca/rcx_min_sta.min.rpt ? part
==

Startpoint: _22_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: pump_en (output port clocked by clk)
Path Group: clk
Path Type: min
Corner: Fastest

Fanout	Cap	Slew	Delay	Time	Description
			0.00	0.00	clock clk (rise edge)
			0.00	0.00	clock source latency
		0.04	0.03	0.03 ^	clk (in)
1	0.01				clk (net)
		0.04	0.00	0.03 ^	clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
		0.03	0.08	0.11 ^	clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
2	0.02				clknet_0_clk (net)
		0.03	0.00	0.11 ^	clkbuf_1_1__f_clk/A (sky130_fd_sc_hd__clkbuf_16)
		0.02	0.07	0.18 ^	clkbuf_1_1__f_clk/X (sky130_fd_sc_hd__clkbuf_16)
1	0.00				clknet_1_1__leaf_clk (net)
		0.02	0.00	0.18 ^	_22_/CLK (sky130_fd_sc_hd__dfrtp_1)
		0.04	0.19	0.37 ^	_22_/Q (sky130_fd_sc_hd__dfrtp_1)
2	0.00				state[2] (net)
		0.04	0.00	0.37 ^	_18_/A (sky130_fd_sc_hd__and2_1)
		0.03	0.07	0.44 ^	_18_/X (sky130_fd_sc_hd__and2_1)
1	0.00				_08_ (net)
		0.03	0.00	0.44 ^	_19_/A (sky130_fd_sc_hd__clkbuf_1)
		0.03	0.05	0.49 ^	_19_/X (sky130_fd_sc_hd__clkbuf_1)
1	0.00				net6 (net)
		0.03	0.00	0.49 ^	output6/A (sky130_fd_sc_hd__clkbuf_4)
		0.08	0.12	0.61 ^	output6/X (sky130_fd_sc_hd__clkbuf_4)
1	0.03				pump_en (net)
		0.08	0.00	0.61 ^	pump_en (out)
				0.61	data arrival time
			0.00	0.00	clock clk (rise edge)
			0.00	0.00	clock network delay (propagated)
			0.25	0.25	clock uncertainty
			0.00	0.25	clock reconvergence pessimism
			-4.00	-3.75	output external delay
				-3.75	data required time
				-3.75	data required time

```
                                  -0.61   data arrival time
-----------------------------------------------------------------------------
                                   4.36   slack (MET)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/26-mca/rcx_min_sta.min.rpt ? part
==

Startpoint: dry_run_fault (input port clocked by clk)
Endpoint: alarm (output port clocked by clk)
Path Group: clk
Path Type: min
Corner: Fastest

Fanout	Cap	Slew	Delay	Time	Description
			0.00	0.00	clock clk (rise edge)
			0.00	0.00	clock network delay (propagated)
			4.00	4.00 ^	input external delay
		0.01	0.01	4.01 ^	dry_run_fault (in)
1	0.00				dry_run_fault (net)
		0.01	0.00	4.01 ^	input1/A (sky130_fd_sc_hd__clkbuf_1)
		0.05	0.06	4.07 ^	input1/X (sky130_fd_sc_hd__clkbuf_1)
1	0.01				net1 (net)
		0.05	0.00	4.07 ^	_09_/B1 (sky130_fd_sc_hd__a21oi_2)
		0.03	0.04	4.10 v	_09_/Y (sky130_fd_sc_hd__a21oi_2)
4	0.02				_03_ (net)
		0.03	0.00	4.10 v	_10_/A (sky130_fd_sc_hd__inv_2)
		0.03	0.04	4.14 ^	_10_/Y (sky130_fd_sc_hd__inv_2)
2	0.01				net5 (net)
		0.03	0.00	4.14 ^	output5/A (sky130_fd_sc_hd__buf_2)
		0.13	0.13	4.27 ^	output5/X (sky130_fd_sc_hd__buf_2)
1	0.03				alarm (net)
		0.13	0.00	4.27 ^	alarm (out)
				4.27	data arrival time
			0.00	0.00	clock clk (rise edge)
			0.00	0.00	clock network delay (propagated)
			0.25	0.25	clock uncertainty
			0.00	0.25	clock reconvergence pessimism
			-4.00	-3.75	output external delay
				-3.75	data required time
				-3.75	data required time
				-4.27	data arrival time
				8.02	slack (MET)

========== Slowest Corner ==========

Startpoint: _21_ (rising edge-triggered flip-flop clocked by clk)

Endpoint: _22_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/26-mca/rcx_min_sta.min.rpt ? part
==

Corner: Slowest

```
Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                  0.09    0.05    0.05 ^ clk (in)
     1    0.01                          clk (net)
                  0.09    0.00    0.05 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.07    0.22    0.27 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                          clknet_0_clk (net)
                  0.07    0.00    0.27 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.05    0.19    0.46 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                          clknet_1_0__leaf_clk (net)
                  0.05    0.00    0.46 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                  0.17    0.67    1.14 ^ _21_/Q (sky130_fd_sc_hd__dfrtp_1)
     3    0.01                          state[1] (net)
                  0.17    0.00    1.14 ^ _12_/A1 (sky130_fd_sc_hd__o21a_1)
                  0.07    0.26    1.40 ^ _12_/X (sky130_fd_sc_hd__o21a_1)
     1    0.00                          _05_ (net)
                  0.07    0.00    1.40 ^ _13_/B1 (sky130_fd_sc_hd__a21oi_1)
                  0.04    0.07    1.46 v _13_/Y (sky130_fd_sc_hd__a21oi_1)
     1    0.00                          _06_ (net)
                  0.04    0.00    1.46 v _14_/B (sky130_fd_sc_hd__nor2_1)
                  0.13    0.12    1.59 ^ _14_/Y (sky130_fd_sc_hd__nor2_1)
     1    0.00                          _02_ (net)
                  0.13    0.00    1.59 ^ _22_/D (sky130_fd_sc_hd__dfrtp_1)
                                  1.59   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                  0.09    0.06    0.06 ^ clk (in)
     1    0.01                          clk (net)
                  0.09    0.00    0.06 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.07    0.24    0.30 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                          clknet_0_clk (net)
                  0.07    0.00    0.30 ^ clkbuf_1_1__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.04    0.20    0.51 ^ clkbuf_1_1__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     1    0.00                          clknet_1_1__leaf_clk (net)
                  0.04    0.00    0.51 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_1)
                          0.25    0.76   clock uncertainty
```

```
                          -0.03    0.73   clock reconvergence pessimism
                          -0.09    0.63   library hold time
                                   0.63   data required time
-----------------------------------------------------------------------------
                                   0.63   data required time
                                  -1.59   data arrival time
-----------------------------------------------------------------------------
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/26-mca/rcx_min_sta.min.rpt ? part
==

                                0.95   slack (MET)

Startpoint: _21_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _21_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
Corner: Slowest

Fanout   Cap   Slew   Delay   Time   Description

                         0.00   0.00   clock clk (rise edge)
                         0.00   0.00   clock source latency
                  0.09   0.05   0.05 ^ clk (in)
     1   0.01   clk (net)
                  0.09   0.00   0.05 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.07   0.22   0.27 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2   0.02   clknet_0_clk (net)
                  0.07   0.00   0.27 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.05   0.19   0.46 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2   0.01   clknet_1_0__leaf_clk (net)
                  0.05   0.00   0.46 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                  0.17   0.67   1.14 ^ _21_/Q (sky130_fd_sc_hd__dfrtp_1)
     3   0.01   state[1] (net)
                  0.17   0.00   1.14 ^ _17_/A1 (sky130_fd_sc_hd__o211a_1)
                  0.07   0.33   1.46 ^ _17_/X (sky130_fd_sc_hd__o211a_1)
     1   0.00   _01_ (net)
                  0.07   0.00   1.46 ^ hold2/A (sky130_fd_sc_hd__dlygate4sd3_1)
                  0.09   0.98   2.44 ^ hold2/X (sky130_fd_sc_hd__dlygate4sd3_1)
     1   0.00   net8 (net)
                  0.09   0.00   2.44 ^ _21_/D (sky130_fd_sc_hd__dfrtp_1)
                                  2.44   data arrival time

                         0.00   0.00   clock clk (rise edge)
                         0.00   0.00   clock source latency
                  0.09   0.06   0.06 ^ clk (in)
     1   0.01   clk (net)
                  0.09   0.00   0.06 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.07   0.24   0.30 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2   0.02   clknet_0_clk (net)
                  0.07   0.00   0.30 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.05   0.21   0.51 ^ clkbuf_1_0__f_clk/X

```
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                  0.05    0.00    0.51 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                          0.25    0.76   clock uncertainty
                         -0.05    0.71   clock reconvergence pessimism
                         -0.09    0.63   library hold time
                                  0.63   data required time
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/26-mca/rcx_min_sta.min.rpt ? part
==

```
--------------------------------------------------------------------------
                                0.63   data required time
                               -2.44   data arrival time
--------------------------------------------------------------------------
                                1.82   slack (MET)


Startpoint: _21_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _20_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
Corner: Slowest

Fanout     Cap    Slew   Delay    Time   Description
--------------------------------------------------------------------------
                         0.00    0.00   clock clk (rise edge)
                         0.00    0.00   clock source latency
                 0.09    0.05    0.05 ^ clk (in)
     1    0.01                          clk (net)
                 0.09    0.00    0.05 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                 0.07    0.22    0.27 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                          clknet_0_clk (net)
                 0.07    0.00    0.27 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                 0.05    0.19    0.46 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                          clknet_1_0__leaf_clk (net)
                 0.05    0.00    0.46 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                 0.13    0.75    1.21 v _21_/Q (sky130_fd_sc_hd__dfrtp_1)
     3    0.01                          state[1] (net)
                 0.13    0.00    1.21 v _15_/A2 (sky130_fd_sc_hd__a21oi_1)
                 0.13    0.22    1.44 ^ _15_/Y (sky130_fd_sc_hd__a21oi_1)
     1    0.00                          _07_ (net)
                 0.13    0.00    1.44 ^ hold4/A (sky130_fd_sc_hd__dlygate4sd3_1)
                 0.10    1.02    2.46 ^ hold4/X (sky130_fd_sc_hd__dlygate4sd3_1)
     1    0.00                          net10 (net)
                 0.10    0.00    2.46 ^ _16_/A2 (sky130_fd_sc_hd__o21ai_1)
                 0.07    0.12    2.58 v _16_/Y (sky130_fd_sc_hd__o21ai_1)
     1    0.00                          _00_ (net)
                 0.07    0.00    2.58 v _20_/D (sky130_fd_sc_hd__dfstp_1)
                                 2.58   data arrival time

                         0.00    0.00   clock clk (rise edge)
                         0.00    0.00   clock source latency
                 0.09    0.06    0.06 ^ clk (in)
```

```
     1    0.01                           clk (net)
                  0.09    0.00    0.06 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.07    0.24    0.30 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                  0.07    0.00    0.30 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/26-mca/rcx_min_sta.min.rpt ? part
==========

```
                  0.05    0.21    0.51 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                  0.05    0.00    0.51 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_1)
                          0.25    0.76   clock uncertainty
                         -0.05    0.71   clock reconvergence pessimism
                         -0.01    0.70   library hold time
                                  0.70   data required time
-----------------------------------------------------------------------------
                                  0.70   data required time
                                 -2.58   data arrival time
-----------------------------------------------------------------------------
                                  1.88   slack (MET)


Startpoint: rst_n (input port clocked by clk)
Endpoint: _21_ (removal check against rising-edge clock clk)
Path Group: asynchronous
Path Type: min
Corner: Slowest

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                  0.03    0.01    4.01 ^ rst_n (in)
     1    0.00                           rst_n (net)
                  0.03    0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                  0.28    0.27    4.28 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.01                           net4 (net)
                  0.28    0.00    4.28 ^ _21_/RESET_B (sky130_fd_sc_hd__dfrtp_1)
                                  4.28   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                  0.09    0.06    0.06 ^ clk (in)
     1    0.01                           clk (net)
                  0.09    0.00    0.06 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.07    0.24    0.30 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                  0.07    0.00    0.30 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.05    0.21    0.51 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
```

```
2    0.01                           clknet_1_0__leaf_clk (net)
             0.05    0.00    0.51 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                     0.25    0.76   clock uncertainty
                     0.00    0.76   clock reconvergence pessimism
                     0.72    1.48   library removal time
                             1.48   data required time
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/26-mca/rcx_min_sta.min.rpt ? part
==

```
--------------------------------------------------------------------------
                                1.48   data required time
                               -4.28   data arrival time
--------------------------------------------------------------------------
                                2.81   slack (MET)


Startpoint: rst_n (input port clocked by clk)
Endpoint: _22_ (removal check against rising-edge clock clk)
Path Group: asynchronous
Path Type: min
Corner: Slowest

Fanout     Cap    Slew   Delay    Time   Description
--------------------------------------------------------------------------
                         0.00    0.00   clock clk (rise edge)
                         0.00    0.00   clock network delay (propagated)
                         4.00    4.00 ^ input external delay
                0.03     0.01    4.01 ^ rst_n (in)
     1    0.00                          rst_n (net)
                0.03     0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                0.28     0.27    4.28 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.01                          net4 (net)
                0.28     0.00    4.28 ^ _22_/RESET_B (sky130_fd_sc_hd__dfrtp_1)
                                 4.28   data arrival time

                         0.00    0.00   clock clk (rise edge)
                         0.00    0.00   clock source latency
                0.09     0.06    0.06 ^ clk (in)
     1    0.01                          clk (net)
                0.09     0.00    0.06 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                0.07     0.24    0.30 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                          clknet_0_clk (net)
                0.07     0.00    0.30 ^ clkbuf_1_1__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                0.04     0.20    0.51 ^ clkbuf_1_1__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     1    0.00                          clknet_1_1__leaf_clk (net)
                0.04     0.00    0.51 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_1)
                         0.25    0.76   clock uncertainty
                         0.00    0.76   clock reconvergence pessimism
                         0.71    1.47   library removal time
                                 1.47   data required time
--------------------------------------------------------------------------
                                1.47   data required time
```

```
                                  -4.28   data arrival time
-----------------------------------------------------------------------------
                                   2.81   slack (MET)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/26-mca/rcx_min_sta.min.rpt ? part
==

Startpoint: rst_n (input port clocked by clk)
Endpoint: _20_ (removal check against rising-edge clock clk)
Path Group: asynchronous
Path Type: min
Corner: Slowest

```
Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                   0.03   0.01    4.01 ^ rst_n (in)
     1    0.00                           rst_n (net)
                   0.03   0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                   0.28   0.27    4.28 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.01                           net4 (net)
                   0.28   0.00    4.28 ^ _20_/SET_B (sky130_fd_sc_hd__dfstp_1)
                                  4.28   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                   0.09   0.06    0.06 ^ clk (in)
     1    0.01                           clk (net)
                   0.09   0.00    0.06 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                   0.07   0.24    0.30 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                   0.07   0.00    0.30 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                   0.05   0.21    0.51 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                   0.05   0.00    0.51 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_1)
                          0.25    0.76   clock uncertainty
                          0.00    0.76   clock reconvergence pessimism
                          0.27    1.03   library removal time
                                  1.03   data required time
-----------------------------------------------------------------------------
                                  1.03   data required time
                                 -4.28   data arrival time
-----------------------------------------------------------------------------
                                  3.25   slack (MET)


Startpoint: _22_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: pump_en (output port clocked by clk)
```

Path Group: clk
Path Type: min
Corner: Slowest

Fanout	Cap	Slew	Delay	Time	Description

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/26-mca/rcx_min_sta.min.rpt ? part
==

```
--------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                 0.09     0.05    0.05 ^ clk (in)
     1    0.01                           clk (net)
                 0.09     0.00    0.05 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                 0.07     0.22    0.27 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                 0.07     0.00    0.27 ^ clkbuf_1_1__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                 0.04     0.18    0.46 ^ clkbuf_1_1__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     1    0.00                           clknet_1_1__leaf_clk (net)
                 0.04     0.00    0.46 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_1)
                 0.09     0.61    1.07 ^ _22_/Q (sky130_fd_sc_hd__dfrtp_1)
     2    0.00                           state[2] (net)
                 0.09     0.00    1.07 ^ _18_/A (sky130_fd_sc_hd__and2_1)
                 0.07     0.21    1.27 ^ _18_/X (sky130_fd_sc_hd__and2_1)
     1    0.00                           _08_ (net)
                 0.07     0.00    1.27 ^ _19_/A (sky130_fd_sc_hd__clkbuf_1)
                 0.08     0.15    1.42 ^ _19_/X (sky130_fd_sc_hd__clkbuf_1)
     1    0.00                           net6 (net)
                 0.08     0.00    1.42 ^ output6/A (sky130_fd_sc_hd__clkbuf_4)
                 0.18     0.32    1.74 ^ output6/X (sky130_fd_sc_hd__clkbuf_4)
     1    0.03                           pump_en (net)
                 0.18     0.00    1.74 ^ pump_en (out)
                                  1.74   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          0.25    0.25   clock uncertainty
                          0.00    0.25   clock reconvergence pessimism
                         -4.00   -3.75   output external delay
                                 -3.75   data required time
--------------------------------------------------------------------------
                                 -3.75   data required time
                                 -1.74   data arrival time
--------------------------------------------------------------------------
                                  5.49   slack (MET)


Startpoint: dry_run_fault (input port clocked by clk)
Endpoint: alarm (output port clocked by clk)
Path Group: clk
Path Type: min
```

Corner: Slowest

Fanout	Cap	Slew	Delay	Time	Description
			0.00	0.00	clock clk (rise edge)

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/26-mca/rcx_min_sta.min.rpt ? part
==

```
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                 0.03     0.01    4.01 ^ dry_run_fault (in)
     1    0.00                           dry_run_fault (net)
                 0.03     0.00    4.01 ^ input1/A (sky130_fd_sc_hd__clkbuf_1)
                 0.11     0.15    4.17 ^ input1/X (sky130_fd_sc_hd__clkbuf_1)
     1    0.00                           net1 (net)
                 0.11     0.00    4.17 ^ _09_/B1 (sky130_fd_sc_hd__a21oi_2)
                 0.08     0.11    4.28 v _09_/Y (sky130_fd_sc_hd__a21oi_2)
     4    0.02                           _03_ (net)
                 0.08     0.00    4.28 v _10_/A (sky130_fd_sc_hd__inv_2)
                 0.06     0.09    4.36 ^ _10_/Y (sky130_fd_sc_hd__inv_2)
     2    0.01                           net5 (net)
                 0.06     0.00    4.36 ^ output5/A (sky130_fd_sc_hd__buf_2)
                 0.27     0.32    4.68 ^ output5/X (sky130_fd_sc_hd__buf_2)
     1    0.03                           alarm (net)
                 0.27     0.00    4.68 ^ alarm (out)
                          4.68    4.68   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          0.25    0.25   clock uncertainty
                          0.00    0.25   clock reconvergence pessimism
                         -4.00   -3.75   output external delay
                                 -3.75   data required time
------------------------------------------------------------------------------
                                 -3.75   data required time
                                 -4.68   data arrival time
------------------------------------------------------------------------------
                                  8.43   slack (MET)



====================== Typical Corner ===================================

Startpoint: _21_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _22_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
Corner: Typical

Fanout     Cap    Slew   Delay    Time   Description
------------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
```

```
             0.06    0.04    0.04 ^ clk (in)
1    0.01                           clk (net)
             0.06    0.00    0.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
```

```
openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/26-mca/rcx_min_sta.min.rpt ? part
================================================================================

                  0.04    0.12    0.16 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                  0.04    0.00    0.16 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.03    0.11    0.26 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                  0.03    0.00    0.26 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                  0.10    0.34    0.61 ^ _21_/Q (sky130_fd_sc_hd__dfrtp_1)
     3    0.01                           state[1] (net)
                  0.10    0.00    0.61 ^ _12_/A1 (sky130_fd_sc_hd__o21a_1)
                  0.04    0.12    0.73 ^ _12_/X (sky130_fd_sc_hd__o21a_1)
     1    0.00                           _05_ (net)
                  0.04    0.00    0.73 ^ _13_/B1 (sky130_fd_sc_hd__a21oi_1)
                  0.03    0.03    0.76 v _13_/Y (sky130_fd_sc_hd__a21oi_1)
     1    0.00                           _06_ (net)
                  0.03    0.00    0.76 v _14_/B (sky130_fd_sc_hd__nor2_1)
                  0.08    0.07    0.84 ^ _14_/Y (sky130_fd_sc_hd__nor2_1)
     1    0.00                           _02_ (net)
                  0.08    0.00    0.84 ^ _22_/D (sky130_fd_sc_hd__dfrtp_1)
                                  0.84   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                  0.06    0.04    0.04 ^ clk (in)
     1    0.01                           clk (net)
                  0.06    0.00    0.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.04    0.14    0.18 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                  0.04    0.00    0.18 ^ clkbuf_1_1__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.03    0.11    0.29 ^ clkbuf_1_1__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     1    0.00                           clknet_1_1__leaf_clk (net)
                  0.03    0.00    0.29 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_1)
                          0.25    0.54   clock uncertainty
                         -0.02    0.52   clock reconvergence pessimism
                         -0.04    0.48   library hold time
                                  0.48   data required time
--------------------------------------------------------------------------------
                                  0.48   data required time
                                 -0.84   data arrival time
--------------------------------------------------------------------------------
                                  0.36   slack (MET)
```

Startpoint: _21_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _21_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
Corner: Typical

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/26-mca/rcx_min_sta.min.rpt ? part
==

```
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                  0.06    0.04    0.04 ^ clk (in)
     1    0.01                          clk (net)
                  0.06    0.00    0.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.04    0.12    0.16 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                          clknet_0_clk (net)
                  0.04    0.00    0.16 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.03    0.11    0.26 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                          clknet_1_0__leaf_clk (net)
                  0.03    0.00    0.26 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                  0.10    0.34    0.61 ^ _21_/Q (sky130_fd_sc_hd__dfrtp_1)
     3    0.01                          state[1] (net)
                  0.10    0.00    0.61 ^ _17_/A1 (sky130_fd_sc_hd__o211a_1)
                  0.04    0.15    0.75 ^ _17_/X (sky130_fd_sc_hd__o211a_1)
     1    0.00                          _01_ (net)
                  0.04    0.00    0.75 ^ hold2/A (sky130_fd_sc_hd__dlygate4sd3_1)
                  0.05    0.50    1.26 ^ hold2/X (sky130_fd_sc_hd__dlygate4sd3_1)
     1    0.00                          net8 (net)
                  0.05    0.00    1.26 ^ _21_/D (sky130_fd_sc_hd__dfrtp_1)
                                  1.26   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                  0.06    0.04    0.04 ^ clk (in)
     1    0.01                          clk (net)
                  0.06    0.00    0.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.04    0.14    0.18 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                          clknet_0_clk (net)
                  0.04    0.00    0.18 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.03    0.12    0.29 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                          clknet_1_0__leaf_clk (net)
                  0.03    0.00    0.29 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                          0.25    0.54   clock uncertainty
                         -0.03    0.51   clock reconvergence pessimism
                         -0.04    0.48   library hold time
                                  0.48   data required time
---------------------------------------------------------------------------
                                  0.48   data required time
```

```
                                  -1.26   data arrival time
-----------------------------------------------------------------------------
                                   0.78   slack (MET)


Startpoint: _21_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _20_ (rising edge-triggered flip-flop clocked by clk)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/26-mca/rcx_min_sta.min.rpt ? part
==

Path Group: clk
Path Type: min
Corner: Typical

```
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                  0.06    0.04    0.04 ^ clk (in)
     1    0.01                          clk (net)
                  0.06    0.00    0.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.04    0.12    0.16 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                          clknet_0_clk (net)
                  0.04    0.00    0.16 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.03    0.11    0.26 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                          clknet_1_0__leaf_clk (net)
                  0.03    0.00    0.26 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                  0.10    0.34    0.61 ^ _21_/Q (sky130_fd_sc_hd__dfrtp_1)
     3    0.01                          state[1] (net)
                  0.10    0.00    0.61 ^ _15_/A2 (sky130_fd_sc_hd__a21oi_1)
                  0.03    0.07    0.67 v _15_/Y (sky130_fd_sc_hd__a21oi_1)
     1    0.00                          _07_ (net)
                  0.03    0.00    0.67 v hold4/A (sky130_fd_sc_hd__dlygate4sd3_1)
                  0.05    0.52    1.19 v hold4/X (sky130_fd_sc_hd__dlygate4sd3_1)
     1    0.00                          net10 (net)
                  0.05    0.00    1.19 v _16_/A2 (sky130_fd_sc_hd__o21ai_1)
                  0.08    0.12    1.30 ^ _16_/Y (sky130_fd_sc_hd__o21ai_1)
     1    0.00                          _00_ (net)
                  0.08    0.00    1.30 ^ _20_/D (sky130_fd_sc_hd__dfstp_1)
                                  1.30   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                  0.06    0.04    0.04 ^ clk (in)
     1    0.01                          clk (net)
                  0.06    0.00    0.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.04    0.14    0.18 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                          clknet_0_clk (net)
                  0.04    0.00    0.18 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.03    0.12    0.29 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                          clknet_1_0__leaf_clk (net)
```

```
                  0.03    0.00    0.29 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_1)
                          0.25    0.54   clock uncertainty
                         -0.03    0.52   clock reconvergence pessimism
                         -0.04    0.47   library hold time
                                  0.47   data required time
-----------------------------------------------------------------------------
                                  0.47   data required time
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/26-mca/rcx_min_sta.min.rpt ? part
==

                                -1.30   data arrival time

                                 0.83   slack (MET)

Startpoint: rst_n (input port clocked by clk)
Endpoint: _21_ (removal check against rising-edge clock clk)
Path Group: asynchronous
Path Type: min
Corner: Typical

Fanout   Cap   Slew   Delay   Time   Description

                          0.00   0.00   clock clk (rise edge)
                          0.00   0.00   clock network delay (propagated)
                          4.00   4.00 ^ input external delay
                  0.02   0.01   4.01 ^ rst_n (in)
     1   0.00   rst_n (net)
                  0.02   0.00   4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                  0.19   0.16   4.17 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3   0.02   net4 (net)
                  0.19   0.00   4.17 ^ _21_/RESET_B (sky130_fd_sc_hd__dfrtp_1)
                                  4.17   data arrival time

                          0.00   0.00   clock clk (rise edge)
                          0.00   0.00   clock source latency
                  0.06   0.04   0.04 ^ clk (in)
     1   0.01   clk (net)
                  0.06   0.00   0.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.04   0.14   0.18 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2   0.02   clknet_0_clk (net)
                  0.04   0.00   0.18 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.03   0.12   0.29 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2   0.01   clknet_1_0__leaf_clk (net)
                  0.03   0.00   0.29 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                          0.25   0.54   clock uncertainty
                          0.00   0.54   clock reconvergence pessimism
                          0.35   0.90   library removal time
                                  0.90   data required time

                                  0.90   data required time
                                 -4.17   data arrival time

3.27   slack (MET)

Startpoint: rst_n (input port clocked by clk)
Endpoint: _22_ (removal check against rising-edge clock clk)

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/26-mca/rcx_min_sta.min.rpt ? part
==

Path Group: asynchronous
Path Type: min
Corner: Typical

```
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                   0.02   0.01    4.01 ^ rst_n (in)
     1    0.00                           rst_n (net)
                   0.02   0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                   0.19   0.16    4.17 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.02                           net4 (net)
                   0.19   0.00    4.17 ^ _22_/RESET_B (sky130_fd_sc_hd__dfrtp_1)
                                  4.17   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                   0.06   0.04    0.04 ^ clk (in)
     1    0.01                           clk (net)
                   0.06   0.00    0.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                   0.04   0.14    0.18 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                   0.04   0.00    0.18 ^ clkbuf_1_1__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                   0.03   0.11    0.29 ^ clkbuf_1_1__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     1    0.00                           clknet_1_1__leaf_clk (net)
                   0.03   0.00    0.29 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_1)
                          0.25    0.54   clock uncertainty
                          0.00    0.54   clock reconvergence pessimism
                          0.35    0.89   library removal time
                                  0.89   data required time
---------------------------------------------------------------------------
                                  0.89   data required time
                                 -4.17   data arrival time
---------------------------------------------------------------------------
                                  3.28   slack (MET)


Startpoint: rst_n (input port clocked by clk)
Endpoint: _20_ (removal check against rising-edge clock clk)
Path Group: asynchronous
Path Type: min
```

Corner: Typical

Fanout	Cap	Slew	Delay	Time	Description
			0.00	0.00	clock clk (rise edge)

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/26-mca/rcx_min_sta.min.rpt ? part
==

```
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                 0.02     0.01    4.01 ^ rst_n (in)
     1    0.00                           rst_n (net)
                 0.02     0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                 0.19     0.16    4.17 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.02                           net4 (net)
                 0.19     0.00    4.17 ^ _20_/SET_B (sky130_fd_sc_hd__dfstp_1)
                                  4.17   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                 0.06     0.04    0.04 ^ clk (in)
     1    0.01                           clk (net)
                 0.06     0.00    0.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                 0.04     0.14    0.18 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                 0.04     0.00    0.18 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                 0.03     0.12    0.29 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                 0.03     0.00    0.29 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_1)
                          0.25    0.54   clock uncertainty
                          0.00    0.54   clock reconvergence pessimism
                          0.14    0.68   library removal time
                                  0.68   data required time
------------------------------------------------------------------------------
                                  0.68   data required time
                                 -4.17   data arrival time
------------------------------------------------------------------------------
                                  3.49   slack (MET)


Startpoint: _22_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: pump_en (output port clocked by clk)
Path Group: clk
Path Type: min
Corner: Typical

Fanout     Cap    Slew   Delay    Time   Description
------------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                 0.06     0.04    0.04 ^ clk (in)
```

```
     1    0.01                           clk (net)
                  0.06    0.00    0.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.04    0.12    0.16 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                  0.04    0.00    0.16 ^ clkbuf_1_1__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/26-mca/rcx_min_sta.min.rpt ? part
==

```
                  0.03    0.10    0.26 ^ clkbuf_1_1__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     1    0.00                           clknet_1_1__leaf_clk (net)
                  0.03    0.00    0.26 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_1)
                  0.06    0.30    0.56 ^ _22_/Q (sky130_fd_sc_hd__dfrtp_1)
     2    0.00                           state[2] (net)
                  0.06    0.00    0.56 ^ _18_/A (sky130_fd_sc_hd__and2_1)
                  0.04    0.11    0.67 ^ _18_/X (sky130_fd_sc_hd__and2_1)
     1    0.00                           _08_ (net)
                  0.04    0.00    0.67 ^ _19_/A (sky130_fd_sc_hd__clkbuf_1)
                  0.05    0.08    0.75 ^ _19_/X (sky130_fd_sc_hd__clkbuf_1)
     1    0.00                           net6 (net)
                  0.05    0.00    0.75 ^ output6/A (sky130_fd_sc_hd__clkbuf_4)
                  0.11    0.18    0.92 ^ output6/X (sky130_fd_sc_hd__clkbuf_4)
     1    0.03                           pump_en (net)
                  0.11    0.00    0.92 ^ pump_en (out)
                                  0.92   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          0.25    0.25   clock uncertainty
                          0.00    0.25   clock reconvergence pessimism
                         -4.00   -3.75   output external delay
                                 -3.75   data required time
---------------------------------------------------------------------------
                                 -3.75   data required time
                                 -0.92   data arrival time
---------------------------------------------------------------------------
                                  4.67   slack (MET)


Startpoint: dry_run_fault (input port clocked by clk)
Endpoint: alarm (output port clocked by clk)
Path Group: clk
Path Type: min
Corner: Typical

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                  0.02    0.01    4.01 ^ dry_run_fault (in)
     1    0.00                           dry_run_fault (net)
                  0.02    0.00    4.01 ^ input1/A (sky130_fd_sc_hd__clkbuf_1)
```

```
             0.07    0.08    4.09 ^ input1/X (sky130_fd_sc_hd__clkbuf_1)
1    0.00                           net1 (net)
             0.07    0.00    4.09 ^ _09_/B1 (sky130_fd_sc_hd__a21oi_2)
             0.05    0.06    4.15 v _09_/Y (sky130_fd_sc_hd__a21oi_2)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/26-mca/rcx_min_sta.min.rpt ? part
==========

```
     4    0.02                           _03_ (net)
                  0.05    0.00    4.15 v _10_/A (sky130_fd_sc_hd__inv_2)
                  0.04    0.05    4.20 ^ _10_/Y (sky130_fd_sc_hd__inv_2)
     2    0.01                           net5 (net)
                  0.04    0.00    4.20 ^ output5/A (sky130_fd_sc_hd__buf_2)
                  0.17    0.19    4.39 ^ output5/X (sky130_fd_sc_hd__buf_2)
     1    0.03                           alarm (net)
                  0.17    0.00    4.39 ^ alarm (out)
                                  4.39   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          0.25    0.25   clock uncertainty
                          0.00    0.25   clock reconvergence pessimism
                         -4.00   -3.75   output external delay
                                 -3.75   data required time
-----------------------------------------------------------------------------
                                 -3.75   data required time
                                 -4.39   data arrival time
-----------------------------------------------------------------------------
                                  8.14   slack (MET)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/26-mca/rcx_min_sta.power.rpt
==

==
 report_power
==
======================== Fastest Corner ===================================

Group	Internal Power	Switching Power	Leakage Power	Total Power (Watts)	
Sequential	7.62e-06	2.62e-07	1.47e-08	7.90e-06	16.6%
Combinational	3.15e-05	8.17e-06	5.48e-10	3.96e-05	83.4%
Macro	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Pad	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Total	3.91e-05	8.43e-06	1.52e-08	4.75e-05	100.0%
	82.2%	17.7%	0.0%		

======================== Slowest Corner ===================================

Group	Internal Power	Switching Power	Leakage Power	Total Power (Watts)	
Sequential	5.16e-06	1.69e-07	5.86e-08	5.39e-06	16.3%
Combinational	2.22e-05	5.39e-06	1.61e-07	2.78e-05	83.7%
Macro	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Pad	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Total	2.74e-05	5.56e-06	2.20e-07	3.31e-05	100.0%
	82.6%	16.8%	0.7%		

======================== Typical Corner ===================================

Group	Internal Power	Switching Power	Leakage Power	Total Power (Watts)	
Sequential	6.60e-06	2.18e-07	1.47e-08	6.83e-06	16.6%
Combinational	2.74e-05	6.88e-06	2.85e-10	3.43e-05	83.4%
Macro	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Pad	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Total	3.40e-05	7.10e-06	1.50e-08	4.11e-05	100.0%
	82.7%	17.3%	0.0%		

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/26-mca/rcx_min_sta.skew.rpt
==

==
report_clock_skew
==
Clock clk
Latency   CRPR   Skew
_20_/CLK ^
   0.51
_22_/CLK ^
   0.46   -0.03   0.03

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/26-mca/rcx_min_sta.summary.rpt
==

==
report_tns
==
tns 0.00

==
report_wns
==
wns 0.00

==
report_worst_slack -max (Setup)
==
worst slack 10.07

==
report_worst_slack -min (Hold)
==
worst slack 0.13

```
openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/28-mca/rcx_max_sta.checks.rpt ? pa
========================================================================================

========================================================================
report_checks -unconstrained
========================================================================
======================= Fastest Corner ===================================

Startpoint: rst_n (input port clocked by clk)
Endpoint: _20_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
Path Type: max
Corner: Fastest

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                0.01      0.01    4.01 ^ rst_n (in)
     1    0.00                           rst_n (net)
                0.01      0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                0.15      0.13    4.14 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.02                           net4 (net)
                0.15      0.00    4.14 ^ _20_/SET_B (sky130_fd_sc_hd__dfstp_1)
                                  4.14   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock source latency
                0.04      0.03   20.03 ^ clk (in)
     1    0.01                           clk (net)
                0.04      0.00   20.03 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                0.03      0.08   20.11 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                0.03      0.00   20.11 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                0.02      0.07   20.19 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                0.02      0.00   20.19 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_1)
                         -0.25   19.94   clock uncertainty
                          0.00   19.94   clock reconvergence pessimism
                          0.09   20.02   library recovery time
                                 20.02   data required time
---------------------------------------------------------------------------
                                 20.02   data required time
                                 -4.14   data arrival time
---------------------------------------------------------------------------
```

15.88   slack (MET)

Startpoint: level_full (input port clocked by clk)
Endpoint: pump_en (output port clocked by clk)

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/28-mca/rcx_max_sta.checks.rpt ? pa
==

Path Group: clk
Path Type: max
Corner: Fastest

Fanout	Cap	Slew	Delay	Time	Description
			0.00	0.00	clock clk (rise edge)
			0.00	0.00	clock network delay (propagated)
			4.00	4.00 v	input external delay
		0.01	0.00	4.00 v	level_full (in)
1	0.00				level_full (net)
		0.01	0.00	4.00 v	input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
		0.07	0.12	4.12 v	input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
3	0.02				net2 (net)
		0.07	0.00	4.12 v	_09_/A2 (sky130_fd_sc_hd__a21oi_2)
		0.17	0.19	4.31 ^	_09_/Y (sky130_fd_sc_hd__a21oi_2)
4	0.02				_03_ (net)
		0.17	0.00	4.31 ^	_18_/B (sky130_fd_sc_hd__and2_1)
		0.03	0.10	4.41 ^	_18_/X (sky130_fd_sc_hd__and2_1)
1	0.00				_08_ (net)
		0.03	0.00	4.41 ^	_19_/A (sky130_fd_sc_hd__clkbuf_1)
		0.04	0.06	4.47 ^	_19_/X (sky130_fd_sc_hd__clkbuf_1)
1	0.00				net6 (net)
		0.04	0.00	4.47 ^	output6/A (sky130_fd_sc_hd__clkbuf_4)
		0.08	0.13	4.60 ^	output6/X (sky130_fd_sc_hd__clkbuf_4)
1	0.03				pump_en (net)
		0.08	0.00	4.60 ^	pump_en (out)
				4.60	data arrival time
			20.00	20.00	clock clk (rise edge)
			0.00	20.00	clock network delay (propagated)
			-0.25	19.75	clock uncertainty
			0.00	19.75	clock reconvergence pessimism
			-4.00	15.75	output external delay
				15.75	data required time
				15.75	data required time
				-4.60	data arrival time
				11.15	slack (MET)

========== Slowest Corner ==========

```
Startpoint: rst_n (input port clocked by clk)
Endpoint: _20_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
```

```
openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/28-mca/rcx_max_sta.checks.rpt ? pa
========================================================================================

Path Type: max
Corner: Slowest

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                   0.03    0.02    4.02 ^ rst_n (in)
     1    0.00                           rst_n (net)
                   0.03    0.00    4.02 ^ input4/A (sky130_fd_sc_hd__buf_1)
                   0.29    0.31    4.32 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.02                           net4 (net)
                   0.29    0.00    4.33 ^ _20_/SET_B (sky130_fd_sc_hd__dfstp_1)
                                   4.33   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock source latency
                   0.09    0.06   20.06 ^ clk (in)
     1    0.01                           clk (net)
                   0.09    0.00   20.06 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                   0.07    0.22   20.28 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                   0.07    0.00   20.28 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                   0.05    0.19   20.47 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                   0.05    0.00   20.47 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_1)
                         -0.25   20.22   clock uncertainty
                          0.00   20.22   clock reconvergence pessimism
                          0.23   20.46   library recovery time
                                 20.46   data required time
---------------------------------------------------------------------------
                                 20.46   data required time
                                 -4.33   data arrival time
---------------------------------------------------------------------------
                                 16.13   slack (MET)


Startpoint: level_full (input port clocked by clk)
Endpoint: pump_en (output port clocked by clk)
Path Group: clk
Path Type: max
Corner: Slowest
```

```
Fanout     Cap     Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/28-mca/rcx_max_sta.checks.rpt ? pa
==

```
                        4.00    4.00 v input external delay
                0.02    0.01    4.01 v level_full (in)
     1    0.00                        level_full (net)
                0.02    0.00    4.01 v input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
                0.19    0.38    4.39 v input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
     3    0.02                        net2 (net)
                0.19    0.00    4.39 v _09_/A2 (sky130_fd_sc_hd__a21oi_2)
                0.37    0.47    4.86 ^ _09_/Y (sky130_fd_sc_hd__a21oi_2)
     4    0.02                        _03_ (net)
                0.37    0.00    4.86 ^ _18_/B (sky130_fd_sc_hd__and2_1)
                0.08    0.34    5.20 ^ _18_/X (sky130_fd_sc_hd__and2_1)
     1    0.00                        _08_ (net)
                0.08    0.00    5.20 ^ _19_/A (sky130_fd_sc_hd__clkbuf_1)
                0.08    0.17    5.37 ^ _19_/X (sky130_fd_sc_hd__clkbuf_1)
     1    0.00                        net6 (net)
                0.08    0.00    5.37 ^ output6/A (sky130_fd_sc_hd__clkbuf_4)
                0.18    0.35    5.72 ^ output6/X (sky130_fd_sc_hd__clkbuf_4)
     1    0.03                        pump_en (net)
                0.18    0.00    5.72 ^ pump_en (out)
                                5.72   data arrival time

                       20.00   20.00   clock clk (rise edge)
                        0.00   20.00   clock network delay (propagated)
                       -0.25   19.75   clock uncertainty
                        0.00   19.75   clock reconvergence pessimism
                       -4.00   15.75   output external delay
                               15.75   data required time
-------------------------------------------------------------------------------
                               15.75   data required time
                               -5.72   data arrival time
-------------------------------------------------------------------------------
                               10.03   slack (MET)



====================== Typical Corner ===================================

Startpoint: rst_n (input port clocked by clk)
Endpoint: _20_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
Path Type: max
Corner: Typical

Fanout     Cap    Slew   Delay    Time   Description
-------------------------------------------------------------------------------
```

```
0.00    0.00   clock clk (rise edge)
0.00    0.00   clock network delay (propagated)
4.00    4.00 ^ input external delay
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/28-mca/rcx_max_sta.checks.rpt ? pa
==========

```
                  0.02    0.01    4.01 ^ rst_n (in)
     1    0.00                           rst_n (net)
                  0.02    0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                  0.20    0.19    4.20 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.02                           net4 (net)
                  0.20    0.00    4.20 ^ _20_/SET_B (sky130_fd_sc_hd__dfstp_1)
                                  4.20   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock source latency
                  0.06    0.04   20.04 ^ clk (in)
     1    0.01                           clk (net)
                  0.06    0.00   20.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.04    0.13   20.16 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                  0.04    0.00   20.16 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.03    0.11   20.27 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                  0.03    0.00   20.27 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_1)
                         -0.25   20.02   clock uncertainty
                          0.00   20.02   clock reconvergence pessimism
                          0.12   20.14   library recovery time
                                 20.14   data required time
-------------------------------------------------------------------------------
                                 20.14   data required time
                                 -4.20   data arrival time
-------------------------------------------------------------------------------
                                 15.94   slack (MET)


Startpoint: level_full (input port clocked by clk)
Endpoint: pump_en (output port clocked by clk)
Path Group: clk
Path Type: max
Corner: Typical

Fanout     Cap    Slew   Delay    Time   Description
-------------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 v input external delay
                  0.01    0.01    4.01 v level_full (in)
     1    0.00                           level_full (net)
```

```
              0.01    0.00    4.01 v input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
              0.10    0.18    4.19 v input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
3    0.02                             net2 (net)
              0.10    0.00    4.19 v _09_/A2 (sky130_fd_sc_hd__a21oi_2)
              0.23    0.27    4.46 ^ _09_/Y (sky130_fd_sc_hd__a21oi_2)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/28-mca/rcx_max_sta.checks.rpt ? pa
==

```
     4    0.02                           _03_ (net)
                  0.23    0.00    4.46 ^ _18_/B (sky130_fd_sc_hd__and2_1)
                  0.05    0.17    4.63 ^ _18_/X (sky130_fd_sc_hd__and2_1)
     1    0.00                           _08_ (net)
                  0.05    0.00    4.63 ^ _19_/A (sky130_fd_sc_hd__clkbuf_1)
                  0.05    0.09    4.71 ^ _19_/X (sky130_fd_sc_hd__clkbuf_1)
     1    0.00                           net6 (net)
                  0.05    0.00    4.71 ^ output6/A (sky130_fd_sc_hd__clkbuf_4)
                  0.11    0.20    4.91 ^ output6/X (sky130_fd_sc_hd__clkbuf_4)
     1    0.03                           pump_en (net)
                  0.11    0.00    4.91 ^ pump_en (out)
                                  4.91   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock network delay (propagated)
                         -0.25   19.75   clock uncertainty
                          0.00   19.75   clock reconvergence pessimism
                         -4.00   15.75   output external delay
                                 15.75   data required time
-------------------------------------------------------------------------------
                                 15.75   data required time
                                 -4.91   data arrival time
-------------------------------------------------------------------------------
                                 10.84   slack (MET)
```

```
================================================================================
report_checks --slack_max -0.01
================================================================================
====================== Fastest Corner ==================================

No paths found.

====================== Slowest Corner ==================================

No paths found.

====================== Typical Corner ==================================

No paths found.


================================================================================
```

```
 report_check_types -max_slew -max_cap -max_fanout -violators
================================================================================
======================== Fastest Corner ========================================
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/28-mca/rcx_max_sta.checks.rpt ? pa
==

====================== Slowest Corner ===================================

====================== Typical Corner ===================================

==
report_parasitic_annotation -report_unannotated
==
Found 0 unannotated drivers.
Found 0 partially unannotated drivers.

==
max slew violation count 0
max fanout violation count 0
max cap violation count 0
==

==
check_setup -verbose -unconstrained_endpoints -multiple_clock -no_clock -no_input_delay
-loops -generated_clocks
==

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/28-mca/rcx_max_sta.max.rpt ? part
==

==
report_checks -path_delay max (Setup)
==
======================= Fastest Corner ===================================

Startpoint: level_full (input port clocked by clk)
Endpoint: pump_en (output port clocked by clk)
Path Group: clk
Path Type: max
Corner: Fastest

Fanout	Cap	Slew	Delay	Time	Description
			0.00	0.00	clock clk (rise edge)
			0.00	0.00	clock network delay (propagated)
			4.00	4.00 v	input external delay
		0.01	0.00	4.00 v	level_full (in)
1	0.00				level_full (net)
		0.01	0.00	4.00 v	input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
		0.07	0.12	4.12 v	input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
3	0.02				net2 (net)
		0.07	0.00	4.12 v	_09_/A2 (sky130_fd_sc_hd__a21oi_2)
		0.17	0.19	4.31 ^	_09_/Y (sky130_fd_sc_hd__a21oi_2)
4	0.02				_03_ (net)
		0.17	0.00	4.31 ^	_18_/B (sky130_fd_sc_hd__and2_1)
		0.03	0.10	4.41 ^	_18_/X (sky130_fd_sc_hd__and2_1)
1	0.00				_08_ (net)
		0.03	0.00	4.41 ^	_19_/A (sky130_fd_sc_hd__clkbuf_1)
		0.04	0.06	4.47 ^	_19_/X (sky130_fd_sc_hd__clkbuf_1)
1	0.00				net6 (net)
		0.04	0.00	4.47 ^	output6/A (sky130_fd_sc_hd__clkbuf_4)
		0.08	0.13	4.60 ^	output6/X (sky130_fd_sc_hd__clkbuf_4)
1	0.03				pump_en (net)
		0.08	0.00	4.60 ^	pump_en (out)
				4.60	data arrival time
			20.00	20.00	clock clk (rise edge)
			0.00	20.00	clock network delay (propagated)
			-0.25	19.75	clock uncertainty
			0.00	19.75	clock reconvergence pessimism
			-4.00	15.75	output external delay
				15.75	data required time

15.75	data required time
-4.60	data arrival time

11.15 slack (MET)

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/28-mca/rcx_max_sta.max.rpt ? part
==

Startpoint: level_full (input port clocked by clk)
Endpoint: alarm (output port clocked by clk)
Path Group: clk
Path Type: max
Corner: Fastest

Fanout	Cap	Slew	Delay	Time		Description
			0.00	0.00		clock clk (rise edge)
			0.00	0.00		clock network delay (propagated)
			4.00	4.00	v	input external delay
		0.01	0.00	4.00	v	level_full (in)
1	0.00					level_full (net)
		0.01	0.00	4.00	v	input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
		0.07	0.12	4.12	v	input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
3	0.02					net2 (net)
		0.07	0.00	4.12	v	_09_/A2 (sky130_fd_sc_hd__a21oi_2)
		0.17	0.19	4.31	^	_09_/Y (sky130_fd_sc_hd__a21oi_2)
4	0.02					_03_ (net)
		0.17	0.00	4.31	^	_10_/A (sky130_fd_sc_hd__inv_2)
		0.04	0.02	4.34	v	_10_/Y (sky130_fd_sc_hd__inv_2)
2	0.01					net5 (net)
		0.04	0.00	4.34	v	output5/A (sky130_fd_sc_hd__buf_2)
		0.06	0.13	4.46	v	output5/X (sky130_fd_sc_hd__buf_2)
1	0.03					alarm (net)
		0.06	0.00	4.46	v	alarm (out)
				4.46		data arrival time
			20.00	20.00		clock clk (rise edge)
			0.00	20.00		clock network delay (propagated)
			-0.25	19.75		clock uncertainty
			0.00	19.75		clock reconvergence pessimism
			-4.00	15.75		output external delay
				15.75		data required time
				15.75		data required time
				-4.46		data arrival time
				11.29		slack (MET)

Startpoint: level_full (input port clocked by clk)
Endpoint: _21_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk

Path Type: max
Corner: Fastest

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/28-mca/rcx_max_sta.max.rpt ? part
==========

```
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 v input external delay
                   0.01    0.00    4.00 v level_full (in)
     1    0.00                           level_full (net)
                   0.01    0.00    4.00 v input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
                   0.07    0.12    4.12 v input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
     3    0.02                           net2 (net)
                   0.07    0.00    4.12 v _09_/A2 (sky130_fd_sc_hd__a21oi_2)
                   0.17    0.19    4.31 ^ _09_/Y (sky130_fd_sc_hd__a21oi_2)
     4    0.02                           _03_ (net)
                   0.17    0.00    4.31 ^ _17_/B1 (sky130_fd_sc_hd__o211a_1)
                   0.03    0.08    4.39 ^ _17_/X (sky130_fd_sc_hd__o211a_1)
     1    0.00                           _01_ (net)
                   0.03    0.00    4.39 ^ hold2/A (sky130_fd_sc_hd__dlygate4sd3_1)
                   0.04    0.39    4.78 ^ hold2/X (sky130_fd_sc_hd__dlygate4sd3_1)
     1    0.00                           net8 (net)
                   0.04    0.00    4.78 ^ _21_/D (sky130_fd_sc_hd__dfrtp_1)
                           4.78    4.78   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock source latency
                   0.04    0.03   20.03 ^ clk (in)
     1    0.01                           clk (net)
                   0.04    0.00   20.03 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                   0.03    0.08   20.11 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                   0.03    0.00   20.11 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                   0.02    0.07   20.19 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                   0.02    0.00   20.19 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                         -0.25   19.94   clock uncertainty
                          0.00   19.94   clock reconvergence pessimism
                         -0.03   19.90   library setup time
                                 19.90   data required time
---------------------------------------------------------------------------
                                 19.90   data required time
                                 -4.78   data arrival time
---------------------------------------------------------------------------
                                 15.12   slack (MET)
```

```
Startpoint: level_full (input port clocked by clk)
Endpoint: _20_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/28-mca/rcx_max_sta.max.rpt ? part
==

Corner: Fastest

```
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                   0.02   0.01    4.01 ^ level_full (in)
     1    0.00                           level_full (net)
                   0.02   0.00    4.01 ^ input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
                   0.15   0.14    4.15 ^ input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
     3    0.02                           net2 (net)
                   0.15   0.00    4.15 ^ _11_/A (sky130_fd_sc_hd__inv_2)
                   0.04   0.03    4.18 v _11_/Y (sky130_fd_sc_hd__inv_2)
     2    0.01                           _04_ (net)
                   0.04   0.00    4.18 v _15_/A1 (sky130_fd_sc_hd__a21oi_1)
                   0.07   0.08    4.26 ^ _15_/Y (sky130_fd_sc_hd__a21oi_1)
     1    0.00                           _07_ (net)
                   0.07   0.00    4.26 ^ hold4/A (sky130_fd_sc_hd__dlygate4sd3_1)
                   0.04   0.40    4.66 ^ hold4/X (sky130_fd_sc_hd__dlygate4sd3_1)
     1    0.00                           net10 (net)
                   0.04   0.00    4.66 ^ _16_/A2 (sky130_fd_sc_hd__o21ai_1)
                   0.05   0.04    4.70 v _16_/Y (sky130_fd_sc_hd__o21ai_1)
     1    0.00                           _00_ (net)
                   0.05   0.00    4.70 v _20_/D (sky130_fd_sc_hd__dfstp_1)
                                  4.70   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock source latency
                   0.04   0.03   20.03 ^ clk (in)
     1    0.01                           clk (net)
                   0.04   0.00   20.03 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                   0.03   0.08   20.11 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                   0.03   0.00   20.11 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                   0.02   0.07   20.19 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                   0.02   0.00   20.19 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_1)
                         -0.25   19.94   clock uncertainty
                          0.00   19.94   clock reconvergence pessimism
                         -0.05   19.89   library setup time
                                 19.89   data required time
---------------------------------------------------------------------------
```

```
                                19.89   data required time
                                -4.70   data arrival time
-------------------------------------------------------------------------------
                                15.19   slack (MET)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/28-mca/rcx_max_sta.max.rpt ? part
==========

Startpoint: level_full (input port clocked by clk)
Endpoint: _22_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: Fastest

Fanout	Cap	Slew	Delay	Time	Description
			0.00	0.00	clock clk (rise edge)
			0.00	0.00	clock network delay (propagated)
			4.00	4.00 v	input external delay
		0.01	0.00	4.00 v	level_full (in)
1	0.00				level_full (net)
		0.01	0.00	4.00 v	input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
		0.07	0.12	4.12 v	input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
3	0.02				net2 (net)
		0.07	0.00	4.12 v	_09_/A2 (sky130_fd_sc_hd__a21oi_2)
		0.17	0.19	4.31 ^	_09_/Y (sky130_fd_sc_hd__a21oi_2)
4	0.02				_03_ (net)
		0.17	0.00	4.31 ^	_10_/A (sky130_fd_sc_hd__inv_2)
		0.04	0.02	4.34 v	_10_/Y (sky130_fd_sc_hd__inv_2)
2	0.01				net5 (net)
		0.04	0.00	4.34 v	_14_/A (sky130_fd_sc_hd__nor2_1)
		0.06	0.07	4.41 ^	_14_/Y (sky130_fd_sc_hd__nor2_1)
1	0.00				_02_ (net)
		0.06	0.00	4.41 ^	_22_/D (sky130_fd_sc_hd__dfrtp_1)
				4.41	data arrival time
			20.00	20.00	clock clk (rise edge)
			0.00	20.00	clock source latency
		0.04	0.03	20.03 ^	clk (in)
1	0.01				clk (net)
		0.04	0.00	20.03 ^	clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
		0.03	0.08	20.11 ^	clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
2	0.02				clknet_0_clk (net)
		0.03	0.00	20.11 ^	clkbuf_1_1__f_clk/A (sky130_fd_sc_hd__clkbuf_16)
		0.02	0.07	20.18 ^	clkbuf_1_1__f_clk/X (sky130_fd_sc_hd__clkbuf_16)
1	0.00				clknet_1_1__leaf_clk (net)
		0.02	0.00	20.18 ^	_22_/CLK (sky130_fd_sc_hd__dfrtp_1)
			-0.25	19.93	clock uncertainty
			0.00	19.93	clock reconvergence pessimism
			-0.04	19.90	library setup time

```
                                19.90   data required time
-----------------------------------------------------------------------------
                                19.90   data required time
                                -4.41   data arrival time
-----------------------------------------------------------------------------
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/28-mca/rcx_max_sta.max.rpt ? part
==

                                15.49   slack (MET)

Startpoint: rst_n (input port clocked by clk)
Endpoint: _20_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
Path Type: max
Corner: Fastest

Fanout   Cap   Slew   Delay   Time   Description

                          0.00   0.00   clock clk (rise edge)
                          0.00   0.00   clock network delay (propagated)
                          4.00   4.00 ^ input external delay
                 0.01   0.01   4.01 ^ rst_n (in)
     1   0.00   rst_n (net)
                 0.01   0.00   4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                 0.15   0.13   4.14 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3   0.02   net4 (net)
                 0.15   0.00   4.14 ^ _20_/SET_B (sky130_fd_sc_hd__dfstp_1)
                                   4.14   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock source latency
                 0.04   0.03   20.03 ^ clk (in)
     1   0.01   clk (net)
                 0.04   0.00   20.03 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                 0.03   0.08   20.11 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2   0.02   clknet_0_clk (net)
                 0.03   0.00   20.11 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                 0.02   0.07   20.19 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2   0.01   clknet_1_0__leaf_clk (net)
                 0.02   0.00   20.19 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_1)
                         -0.25   19.94   clock uncertainty
                          0.00   19.94   clock reconvergence pessimism
                          0.09   20.02   library recovery time
                                 20.02   data required time

                                 20.02   data required time
                                 -4.14   data arrival time

                                 15.88   slack (MET)

```
Startpoint: rst_n (input port clocked by clk)
Endpoint: _22_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
Path Type: max
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/28-mca/rcx_max_sta.max.rpt ? part
==========

Corner: Fastest

```
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                 0.01     0.01    4.01 ^ rst_n (in)
     1    0.00                           rst_n (net)
                 0.01     0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                 0.15     0.13    4.14 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.02                           net4 (net)
                 0.15     0.00    4.14 ^ _22_/RESET_B (sky130_fd_sc_hd__dfrtp_1)
                                  4.14   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock source latency
                 0.04     0.03   20.03 ^ clk (in)
     1    0.01                           clk (net)
                 0.04     0.00   20.03 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                 0.03     0.08   20.11 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                 0.03     0.00   20.11 ^ clkbuf_1_1__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                 0.02     0.07   20.18 ^ clkbuf_1_1__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     1    0.00                           clknet_1_1__leaf_clk (net)
                 0.02     0.00   20.18 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_1)
                         -0.25   19.93   clock uncertainty
                          0.00   19.93   clock reconvergence pessimism
                          0.12   20.05   library recovery time
                                 20.05   data required time
---------------------------------------------------------------------------
                                 20.05   data required time
                                 -4.14   data arrival time
---------------------------------------------------------------------------
                                 15.91   slack (MET)


Startpoint: rst_n (input port clocked by clk)
Endpoint: _21_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
Path Type: max
Corner: Fastest
```

```
Fanout     Cap     Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/28-mca/rcx_max_sta.max.rpt ? part
==

```
                  0.01    0.01    4.01 ^ rst_n (in)
     1    0.00                           rst_n (net)
                  0.01    0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                  0.15    0.13    4.14 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.02                           net4 (net)
                  0.15    0.00    4.14 ^ _21_/RESET_B (sky130_fd_sc_hd__dfrtp_1)
                                  4.14   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock source latency
                  0.04    0.03   20.03 ^ clk (in)
     1    0.01                           clk (net)
                  0.04    0.00   20.03 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.03    0.08   20.11 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                  0.03    0.00   20.11 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.02    0.07   20.19 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                  0.02    0.00   20.19 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                         -0.25   19.94   clock uncertainty
                          0.00   19.94   clock reconvergence pessimism
                          0.12   20.06   library recovery time
                                 20.06   data required time
------------------------------------------------------------------------------
                                 20.06   data required time
                                 -4.14   data arrival time
------------------------------------------------------------------------------
                                 15.92   slack (MET)



====================== Slowest Corner ===================================

Startpoint: level_full (input port clocked by clk)
Endpoint: pump_en (output port clocked by clk)
Path Group: clk
Path Type: max
Corner: Slowest

Fanout     Cap    Slew   Delay    Time   Description
------------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
```

```
                    4.00    4.00 v input external delay
              0.02    0.01    4.01 v level_full (in)
1     0.00                         level_full (net)
              0.02    0.00    4.01 v input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
              0.19    0.38    4.39 v input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/28-mca/rcx_max_sta.max.rpt ? part
==

```
     3    0.02                           net2 (net)
                  0.19    0.00    4.39 v _09_/A2 (sky130_fd_sc_hd__a21oi_2)
                  0.37    0.47    4.86 ^ _09_/Y (sky130_fd_sc_hd__a21oi_2)
     4    0.02                           _03_ (net)
                  0.37    0.00    4.86 ^ _18_/B (sky130_fd_sc_hd__and2_1)
                  0.08    0.34    5.20 ^ _18_/X (sky130_fd_sc_hd__and2_1)
     1    0.00                           _08_ (net)
                  0.08    0.00    5.20 ^ _19_/A (sky130_fd_sc_hd__clkbuf_1)
                  0.08    0.17    5.37 ^ _19_/X (sky130_fd_sc_hd__clkbuf_1)
     1    0.00                           net6 (net)
                  0.08    0.00    5.37 ^ output6/A (sky130_fd_sc_hd__clkbuf_4)
                  0.18    0.35    5.72 ^ output6/X (sky130_fd_sc_hd__clkbuf_4)
     1    0.03                           pump_en (net)
                  0.18    0.00    5.72 ^ pump_en (out)
                                  5.72   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock network delay (propagated)
                         -0.25   19.75   clock uncertainty
                          0.00   19.75   clock reconvergence pessimism
                         -4.00   15.75   output external delay
                                 15.75   data required time
-----------------------------------------------------------------------------
                                 15.75   data required time
                                 -5.72   data arrival time
-----------------------------------------------------------------------------
                                 10.03   slack (MET)


Startpoint: level_full (input port clocked by clk)
Endpoint: alarm (output port clocked by clk)
Path Group: clk
Path Type: max
Corner: Slowest

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 v input external delay
                  0.02    0.01    4.01 v level_full (in)
     1    0.00                           level_full (net)
                  0.02    0.00    4.01 v input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
                  0.19    0.38    4.39 v input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
     3    0.02                           net2 (net)
```

```
              0.19    0.00    4.39 v _09_/A2 (sky130_fd_sc_hd__a21oi_2)
              0.37    0.47    4.86 ^ _09_/Y (sky130_fd_sc_hd__a21oi_2)
4    0.02                           _03_ (net)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/28-mca/rcx_max_sta.max.rpt ? part
==

```
                  0.37    0.00    4.86 ^ _10_/A (sky130_fd_sc_hd__inv_2)
                  0.09    0.17    5.03 v _10_/Y (sky130_fd_sc_hd__inv_2)
     2    0.01                           net5 (net)
                  0.09    0.00    5.03 v output5/A (sky130_fd_sc_hd__buf_2)
                  0.17    0.37    5.39 v output5/X (sky130_fd_sc_hd__buf_2)
     1    0.03                           alarm (net)
                  0.17    0.00    5.40 v alarm (out)
                                  5.40   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock network delay (propagated)
                         -0.25   19.75   clock uncertainty
                          0.00   19.75   clock reconvergence pessimism
                         -4.00   15.75   output external delay
                                 15.75   data required time
---------------------------------------------------------------------------
                                 15.75   data required time
                                 -5.40   data arrival time
---------------------------------------------------------------------------
                                 10.35   slack (MET)


Startpoint: level_full (input port clocked by clk)
Endpoint: _21_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: Slowest

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 v input external delay
                  0.02    0.01    4.01 v level_full (in)
     1    0.00                           level_full (net)
                  0.02    0.00    4.01 v input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
                  0.19    0.38    4.39 v input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
     3    0.02                           net2 (net)
                  0.19    0.00    4.39 v _09_/A2 (sky130_fd_sc_hd__a21oi_2)
                  0.37    0.47    4.86 ^ _09_/Y (sky130_fd_sc_hd__a21oi_2)
     4    0.02                           _03_ (net)
                  0.37    0.00    4.86 ^ _17_/B1 (sky130_fd_sc_hd__o211a_1)
                  0.07    0.41    5.27 ^ _17_/X (sky130_fd_sc_hd__o211a_1)
     1    0.00                           _01_ (net)
                  0.07    0.00    5.27 ^ hold2/A (sky130_fd_sc_hd__dlygate4sd3_1)
```

```
             0.09    1.09    6.36 ^ hold2/X (sky130_fd_sc_hd__dlygate4sd3_1)
1    0.00                           net8 (net)
             0.09    0.00    6.36 ^ _21_/D (sky130_fd_sc_hd__dfrtp_1)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/28-mca/rcx_max_sta.max.rpt ? part
==

```
                                6.36   data arrival time

                       20.00   20.00   clock clk (rise edge)
                        0.00   20.00   clock source latency
                0.09    0.06   20.06 ^ clk (in)
     1    0.01                        clk (net)
                0.09    0.00   20.06 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                0.07    0.22   20.28 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                        clknet_0_clk (net)
                0.07    0.00   20.28 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                0.05    0.19   20.47 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                        clknet_1_0__leaf_clk (net)
                0.05    0.00   20.47 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                       -0.25   20.22   clock uncertainty
                        0.00   20.22   clock reconvergence pessimism
                       -0.14   20.08   library setup time
                               20.08   data required time
---------------------------------------------------------------------------
                               20.08   data required time
                               -6.36   data arrival time
---------------------------------------------------------------------------
                               13.72   slack (MET)


Startpoint: level_full (input port clocked by clk)
Endpoint: _20_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: Slowest

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                        0.00    0.00   clock clk (rise edge)
                        0.00    0.00   clock network delay (propagated)
                        4.00    4.00 v input external delay
                0.02    0.01    4.01 v level_full (in)
     1    0.00                        level_full (net)
                0.02    0.00    4.01 v input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
                0.19    0.38    4.39 v input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
     3    0.02                        net2 (net)
                0.19    0.00    4.39 v _11_/A (sky130_fd_sc_hd__inv_2)
                0.08    0.15    4.54 ^ _11_/Y (sky130_fd_sc_hd__inv_2)
     2    0.01                        _04_ (net)
```

```
             0.08    0.00    4.54 ^ _15_/A1 (sky130_fd_sc_hd__a21oi_1)
             0.08    0.12    4.66 v _15_/Y (sky130_fd_sc_hd__a21oi_1)
1    0.00                           _07_ (net)
             0.08    0.00    4.66 v hold4/A (sky130_fd_sc_hd__dlygate4sd3_1)
             0.11    1.19    5.85 v hold4/X (sky130_fd_sc_hd__dlygate4sd3_1)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/28-mca/rcx_max_sta.max.rpt ? part
==

```
     1    0.00                           net10 (net)
                  0.11    0.00    5.85 v _16_/A2 (sky130_fd_sc_hd__o21ai_1)
                  0.19    0.24    6.09 ^ _16_/Y (sky130_fd_sc_hd__o21ai_1)
     1    0.00                           _00_ (net)
                  0.19    0.00    6.09 ^ _20_/D (sky130_fd_sc_hd__dfstp_1)
                                  6.09   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock source latency
                  0.09    0.06   20.06 ^ clk (in)
     1    0.01                           clk (net)
                  0.09    0.00   20.06 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.07    0.22   20.28 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                  0.07    0.00   20.28 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.05    0.19   20.47 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                  0.05    0.00   20.47 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_1)
                         -0.25   20.22   clock uncertainty
                          0.00   20.22   clock reconvergence pessimism
                         -0.15   20.07   library setup time
                                 20.07   data required time
------------------------------------------------------------------------------
                                 20.07   data required time
                                 -6.09   data arrival time
------------------------------------------------------------------------------
                                 13.99   slack (MET)


Startpoint: level_full (input port clocked by clk)
Endpoint: _22_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: Slowest

Fanout     Cap    Slew   Delay    Time   Description
------------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 v input external delay
                  0.02    0.01    4.01 v level_full (in)
     1    0.00                           level_full (net)
                  0.02    0.00    4.01 v input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
```

```
              0.19    0.38    4.39 v input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
3    0.02                             net2 (net)
              0.19    0.00    4.39 v _09_/A2 (sky130_fd_sc_hd__a21oi_2)
              0.37    0.47    4.86 ^ _09_/Y (sky130_fd_sc_hd__a21oi_2)
4    0.02                             _03_ (net)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/28-mca/rcx_max_sta.max.rpt ? part
==

```
                  0.37    0.00    4.86 ^ _10_/A (sky130_fd_sc_hd__inv_2)
                  0.09    0.17    5.03 v _10_/Y (sky130_fd_sc_hd__inv_2)
     2    0.01                           net5 (net)
                  0.09    0.00    5.03 v _14_/A (sky130_fd_sc_hd__nor2_1)
                  0.13    0.18    5.21 ^ _14_/Y (sky130_fd_sc_hd__nor2_1)
     1    0.00                           _02_ (net)
                  0.13    0.00    5.21 ^ _22_/D (sky130_fd_sc_hd__dfrtp_1)
                                  5.21   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock source latency
                  0.09    0.06   20.06 ^ clk (in)
     1    0.01                           clk (net)
                  0.09    0.00   20.06 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.07    0.22   20.28 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                  0.07    0.00   20.28 ^ clkbuf_1_1__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.04    0.18   20.47 ^ clkbuf_1_1__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     1    0.00                           clknet_1_1__leaf_clk (net)
                  0.04    0.00   20.47 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_1)
                         -0.25   20.22   clock uncertainty
                          0.00   20.22   clock reconvergence pessimism
                         -0.15   20.06   library setup time
                                 20.06   data required time
------------------------------------------------------------------------------
                                 20.06   data required time
                                 -5.21   data arrival time
------------------------------------------------------------------------------
                                 14.85   slack (MET)


Startpoint: rst_n (input port clocked by clk)
Endpoint: _20_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
Path Type: max
Corner: Slowest

Fanout     Cap    Slew   Delay    Time   Description
------------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                  0.03    0.02    4.02 ^ rst_n (in)
```

```
1    0.00                           rst_n (net)
             0.03    0.00    4.02 ^ input4/A (sky130_fd_sc_hd__buf_1)
             0.29    0.31    4.32 ^ input4/X (sky130_fd_sc_hd__buf_1)
3    0.02                           net4 (net)
             0.29    0.00    4.33 ^ _20_/SET_B (sky130_fd_sc_hd__dfstp_1)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/28-mca/rcx_max_sta.max.rpt ? part
==

```
                                4.33   data arrival time

                       20.00   20.00   clock clk (rise edge)
                        0.00   20.00   clock source latency
                0.09    0.06   20.06 ^ clk (in)
     1    0.01                        clk (net)
                0.09    0.00   20.06 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                0.07    0.22   20.28 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                        clknet_0_clk (net)
                0.07    0.00   20.28 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                0.05    0.19   20.47 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                        clknet_1_0__leaf_clk (net)
                0.05    0.00   20.47 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_1)
                       -0.25   20.22   clock uncertainty
                        0.00   20.22   clock reconvergence pessimism
                        0.23   20.46   library recovery time
                               20.46   data required time
---------------------------------------------------------------------------
                               20.46   data required time
                               -4.33   data arrival time
---------------------------------------------------------------------------
                               16.13   slack (MET)


Startpoint: rst_n (input port clocked by clk)
Endpoint: _22_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
Path Type: max
Corner: Slowest

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                        0.00    0.00   clock clk (rise edge)
                        0.00    0.00   clock network delay (propagated)
                        4.00    4.00 ^ input external delay
                0.03    0.02    4.02 ^ rst_n (in)
     1    0.00                        rst_n (net)
                0.03    0.00    4.02 ^ input4/A (sky130_fd_sc_hd__buf_1)
                0.29    0.31    4.32 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.02                        net4 (net)
                0.29    0.00    4.33 ^ _22_/RESET_B (sky130_fd_sc_hd__dfrtp_1)
                               4.33   data arrival time
```

```
                  20.00   20.00   clock clk (rise edge)
                   0.00   20.00   clock source latency
            0.09   0.06   20.06 ^ clk (in)
1   0.01                          clk (net)
            0.09   0.00   20.06 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/28-mca/rcx_max_sta.max.rpt ? part
==

```
                  0.07    0.22   20.28 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                  0.07    0.00   20.28 ^ clkbuf_1_1__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.04    0.18   20.47 ^ clkbuf_1_1__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     1    0.00                           clknet_1_1__leaf_clk (net)
                  0.04    0.00   20.47 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_1)
                         -0.25   20.22   clock uncertainty
                          0.00   20.22   clock reconvergence pessimism
                          0.37   20.59   library recovery time
                                 20.59   data required time
-----------------------------------------------------------------------------
                                 20.59   data required time
                                 -4.33   data arrival time
-----------------------------------------------------------------------------
                                 16.26   slack (MET)


Startpoint: rst_n (input port clocked by clk)
Endpoint: _21_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
Path Type: max
Corner: Slowest

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                  0.03    0.02    4.02 ^ rst_n (in)
     1    0.00                           rst_n (net)
                  0.03    0.00    4.02 ^ input4/A (sky130_fd_sc_hd__buf_1)
                  0.29    0.31    4.32 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.02                           net4 (net)
                  0.29    0.00    4.33 ^ _21_/RESET_B (sky130_fd_sc_hd__dfrtp_1)
                                  4.33   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock source latency
                  0.09    0.06   20.06 ^ clk (in)
     1    0.01                           clk (net)
                  0.09    0.00   20.06 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.07    0.22   20.28 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
```

```
                  0.07    0.00   20.28 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.05    0.19   20.47 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                  0.05    0.00   20.47 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                         -0.25   20.22   clock uncertainty
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/28-mca/rcx_max_sta.max.rpt ? part
==

```
                          0.00   20.22   clock reconvergence pessimism
                          0.37   20.60   library recovery time
                                 20.60   data required time
-------------------------------------------------------------------------------
                                 20.60   data required time
                                 -4.33   data arrival time
-------------------------------------------------------------------------------
                                 16.27   slack (MET)



====================== Typical Corner ===================================

Startpoint: level_full (input port clocked by clk)
Endpoint: pump_en (output port clocked by clk)
Path Group: clk
Path Type: max
Corner: Typical

Fanout     Cap    Slew   Delay    Time   Description
-------------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 v input external delay
                   0.01   0.01    4.01 v level_full (in)
     1    0.00                           level_full (net)
                   0.01   0.00    4.01 v input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
                   0.10   0.18    4.19 v input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
     3    0.02                           net2 (net)
                   0.10   0.00    4.19 v _09_/A2 (sky130_fd_sc_hd__a21oi_2)
                   0.23   0.27    4.46 ^ _09_/Y (sky130_fd_sc_hd__a21oi_2)
     4    0.02                           _03_ (net)
                   0.23   0.00    4.46 ^ _18_/B (sky130_fd_sc_hd__and2_1)
                   0.05   0.17    4.63 ^ _18_/X (sky130_fd_sc_hd__and2_1)
     1    0.00                           _08_ (net)
                   0.05   0.00    4.63 ^ _19_/A (sky130_fd_sc_hd__clkbuf_1)
                   0.05   0.09    4.71 ^ _19_/X (sky130_fd_sc_hd__clkbuf_1)
     1    0.00                           net6 (net)
                   0.05   0.00    4.71 ^ output6/A (sky130_fd_sc_hd__clkbuf_4)
                   0.11   0.20    4.91 ^ output6/X (sky130_fd_sc_hd__clkbuf_4)
     1    0.03                           pump_en (net)
                   0.11   0.00    4.91 ^ pump_en (out)
                                  4.91   data arrival time

                         20.00   20.00   clock clk (rise edge)
```

```
 0.00   20.00   clock network delay (propagated)
-0.25   19.75   clock uncertainty
 0.00   19.75   clock reconvergence pessimism
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/28-mca/rcx_max_sta.max.rpt ? part
==

```
                          -4.00   15.75   output external delay
                                  15.75   data required time
-------------------------------------------------------------------------------
                                  15.75   data required time
                                  -4.91   data arrival time
-------------------------------------------------------------------------------
                                  10.84   slack (MET)


Startpoint: level_full (input port clocked by clk)
Endpoint: alarm (output port clocked by clk)
Path Group: clk
Path Type: max
Corner: Typical

Fanout     Cap    Slew   Delay    Time   Description
-------------------------------------------------------------------------------
                           0.00    0.00   clock clk (rise edge)
                           0.00    0.00   clock network delay (propagated)
                           4.00    4.00 v input external delay
                  0.01     0.01    4.01 v level_full (in)
     1    0.00                            level_full (net)
                  0.01     0.00    4.01 v input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
                  0.10     0.18    4.19 v input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
     3    0.02                            net2 (net)
                  0.10     0.00    4.19 v _09_/A2 (sky130_fd_sc_hd__a21oi_2)
                  0.23     0.27    4.46 ^ _09_/Y (sky130_fd_sc_hd__a21oi_2)
     4    0.02                            _03_ (net)
                  0.23     0.00    4.46 ^ _10_/A (sky130_fd_sc_hd__inv_2)
                  0.05     0.07    4.53 v _10_/Y (sky130_fd_sc_hd__inv_2)
     2    0.01                            net5 (net)
                  0.05     0.00    4.53 v output5/A (sky130_fd_sc_hd__buf_2)
                  0.09     0.19    4.72 v output5/X (sky130_fd_sc_hd__buf_2)
     1    0.03                            alarm (net)
                  0.09     0.00    4.72 v alarm (out)
                                   4.72   data arrival time

                          20.00   20.00   clock clk (rise edge)
                           0.00   20.00   clock network delay (propagated)
                          -0.25   19.75   clock uncertainty
                           0.00   19.75   clock reconvergence pessimism
                          -4.00   15.75   output external delay
                                  15.75   data required time
-------------------------------------------------------------------------------
                                  15.75   data required time
```

```
                                  -4.72   data arrival time
-----------------------------------------------------------------------------
                                  11.03   slack (MET)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/28-mca/rcx_max_sta.max.rpt ? part
==========

Startpoint: level_full (input port clocked by clk)
Endpoint: _21_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: Typical

Fanout	Cap	Slew	Delay	Time		Description
			0.00	0.00		clock clk (rise edge)
			0.00	0.00		clock network delay (propagated)
			4.00	4.00	v	input external delay
		0.01	0.01	4.01	v	level_full (in)
1	0.00					level_full (net)
		0.01	0.00	4.01	v	input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
		0.10	0.18	4.19	v	input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
3	0.02					net2 (net)
		0.10	0.00	4.19	v	_09_/A2 (sky130_fd_sc_hd__a21oi_2)
		0.23	0.27	4.46	^	_09_/Y (sky130_fd_sc_hd__a21oi_2)
4	0.02					_03_ (net)
		0.23	0.00	4.46	^	_17_/B1 (sky130_fd_sc_hd__o211a_1)
		0.04	0.19	4.65	^	_17_/X (sky130_fd_sc_hd__o211a_1)
1	0.00					_01_ (net)
		0.04	0.00	4.65	^	hold2/A (sky130_fd_sc_hd__dlygate4sd3_1)
		0.05	0.56	5.21	^	hold2/X (sky130_fd_sc_hd__dlygate4sd3_1)
1	0.00					net8 (net)
		0.05	0.00	5.21	^	_21_/D (sky130_fd_sc_hd__dfrtp_1)
				5.21		data arrival time
			20.00	20.00		clock clk (rise edge)
			0.00	20.00		clock source latency
		0.06	0.04	20.04	^	clk (in)
1	0.01					clk (net)
		0.06	0.00	20.04	^	clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
		0.04	0.13	20.16	^	clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
2	0.02					clknet_0_clk (net)
		0.04	0.00	20.16	^	clkbuf_1_0__f_clk/A (sky130_fd_sc_hd__clkbuf_16)
		0.03	0.11	20.27	^	clkbuf_1_0__f_clk/X (sky130_fd_sc_hd__clkbuf_16)
2	0.01					clknet_1_0__leaf_clk (net)
		0.03	0.00	20.27	^	_21_/CLK (sky130_fd_sc_hd__dfrtp_1)
			-0.25	20.02		clock uncertainty
			0.00	20.02		clock reconvergence pessimism

```
                          -0.06   19.96   library setup time
                                  19.96   data required time
-----------------------------------------------------------------------------
                                  19.96   data required time
                                  -5.21   data arrival time
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/28-mca/rcx_max_sta.max.rpt ? part
==

14.75 slack (MET)

Startpoint: level_full (input port clocked by clk)
Endpoint: _20_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: Typical

Fanout	Cap	Slew	Delay	Time	Description
			0.00	0.00	clock clk (rise edge)
			0.00	0.00	clock network delay (propagated)
			4.00	4.00 ^	input external delay
		0.02	0.01	4.01 ^	level_full (in)
1	0.00				level_full (net)
		0.02	0.00	4.01 ^	input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
		0.20	0.20	4.21 ^	input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
3	0.02				net2 (net)
		0.20	0.00	4.21 ^	_11_/A (sky130_fd_sc_hd__inv_2)
		0.05	0.06	4.27 v	_11_/Y (sky130_fd_sc_hd__inv_2)
2	0.01				_04_ (net)
		0.05	0.00	4.27 v	_15_/A1 (sky130_fd_sc_hd__a21oi_1)
		0.10	0.12	4.39 ^	_15_/Y (sky130_fd_sc_hd__a21oi_1)
1	0.00				_07_ (net)
		0.10	0.00	4.39 ^	hold4/A (sky130_fd_sc_hd__dlygate4sd3_1)
		0.06	0.58	4.97 ^	hold4/X (sky130_fd_sc_hd__dlygate4sd3_1)
1	0.00				net10 (net)
		0.06	0.00	4.97 ^	_16_/A2 (sky130_fd_sc_hd__o21ai_1)
		0.08	0.06	5.03 v	_16_/Y (sky130_fd_sc_hd__o21ai_1)
1	0.00				_00_ (net)
		0.08	0.00	5.03 v	_20_/D (sky130_fd_sc_hd__dfstp_1)
				5.03	data arrival time
			20.00	20.00	clock clk (rise edge)
			0.00	20.00	clock source latency
		0.06	0.04	20.04 ^	clk (in)
1	0.01				clk (net)
		0.06	0.00	20.04 ^	clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
		0.04	0.13	20.16 ^	clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
2	0.02				clknet_0_clk (net)
		0.04	0.00	20.16 ^	clkbuf_1_0__f_clk/A (sky130_fd_sc_hd__clkbuf_16)
		0.03	0.11	20.27 ^	clkbuf_1_0__f_clk/X

```
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                  0.03    0.00   20.27 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_1)
                         -0.25   20.02   clock uncertainty
                          0.00   20.02   clock reconvergence pessimism
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/28-mca/rcx_max_sta.max.rpt ? part
==

```
                          -0.08   19.94   library setup time
                                  19.94   data required time
-----------------------------------------------------------------------------
                                  19.94   data required time
                                  -5.03   data arrival time
-----------------------------------------------------------------------------
                                  14.90   slack (MET)


Startpoint: level_full (input port clocked by clk)
Endpoint: _22_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: Typical

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                           0.00    0.00   clock clk (rise edge)
                           0.00    0.00   clock network delay (propagated)
                           4.00    4.00 v input external delay
                  0.01     0.01    4.01 v level_full (in)
     1    0.00                            level_full (net)
                  0.01     0.00    4.01 v input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
                  0.10     0.18    4.19 v input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
     3    0.02                            net2 (net)
                  0.10     0.00    4.19 v _09_/A2 (sky130_fd_sc_hd__a21oi_2)
                  0.23     0.27    4.46 ^ _09_/Y (sky130_fd_sc_hd__a21oi_2)
     4    0.02                            _03_ (net)
                  0.23     0.00    4.46 ^ _10_/A (sky130_fd_sc_hd__inv_2)
                  0.05     0.07    4.53 v _10_/Y (sky130_fd_sc_hd__inv_2)
     2    0.01                            net5 (net)
                  0.05     0.00    4.53 v _14_/A (sky130_fd_sc_hd__nor2_1)
                  0.08     0.11    4.63 ^ _14_/Y (sky130_fd_sc_hd__nor2_1)
     1    0.00                            _02_ (net)
                  0.08     0.00    4.63 ^ _22_/D (sky130_fd_sc_hd__dfrtp_1)
                           4.63    4.63   data arrival time

                          20.00   20.00   clock clk (rise edge)
                           0.00   20.00   clock source latency
                  0.06     0.04   20.04 ^ clk (in)
     1    0.01                            clk (net)
                  0.06     0.00   20.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.04     0.13   20.16 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                            clknet_0_clk (net)
                  0.04     0.00   20.16 ^ clkbuf_1_1__f_clk/A
```

```
(sky130_fd_sc_hd__clkbuf_16)
                  0.03    0.10   20.27 ^ clkbuf_1_1__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     1    0.00                           clknet_1_1__leaf_clk (net)
                  0.03    0.00   20.27 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_1)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/28-mca/rcx_max_sta.max.rpt ? part
==

```
                          -0.25   20.02   clock uncertainty
                           0.00   20.02   clock reconvergence pessimism
                          -0.07   19.95   library setup time
                                  19.95   data required time
-----------------------------------------------------------------------------
                                  19.95   data required time
                                  -4.63   data arrival time
-----------------------------------------------------------------------------
                                  15.31   slack (MET)


Startpoint: rst_n (input port clocked by clk)
Endpoint: _20_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
Path Type: max
Corner: Typical

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                           0.00    0.00   clock clk (rise edge)
                           0.00    0.00   clock network delay (propagated)
                           4.00    4.00 ^ input external delay
                  0.02     0.01    4.01 ^ rst_n (in)
     1    0.00                            rst_n (net)
                  0.02     0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                  0.20     0.19    4.20 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.02                            net4 (net)
                  0.20     0.00    4.20 ^ _20_/SET_B (sky130_fd_sc_hd__dfstp_1)
                                    4.20   data arrival time

                          20.00   20.00   clock clk (rise edge)
                           0.00   20.00   clock source latency
                  0.06     0.04   20.04 ^ clk (in)
     1    0.01                            clk (net)
                  0.06     0.00   20.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.04     0.13   20.16 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                            clknet_0_clk (net)
                  0.04     0.00   20.16 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.03     0.11   20.27 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                            clknet_1_0__leaf_clk (net)
                  0.03     0.00   20.27 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_1)
                          -0.25   20.02   clock uncertainty
                           0.00   20.02   clock reconvergence pessimism
```

```
                          0.12   20.14   library recovery time
                                 20.14   data required time
-----------------------------------------------------------------------------
                                 20.14   data required time
                                 -4.20   data arrival time
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/28-mca/rcx_max_sta.max.rpt ? part
==========

```
                                15.94   slack (MET)


Startpoint: rst_n (input port clocked by clk)
Endpoint: _22_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
Path Type: max
Corner: Typical

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                 0.02     0.01    4.01 ^ rst_n (in)
     1    0.00                           rst_n (net)
                 0.02     0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                 0.20     0.19    4.20 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.02                           net4 (net)
                 0.20     0.00    4.20 ^ _22_/RESET_B (sky130_fd_sc_hd__dfrtp_1)
                                  4.20   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock source latency
                 0.06     0.04   20.04 ^ clk (in)
     1    0.01                           clk (net)
                 0.06     0.00   20.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                 0.04     0.13   20.16 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                 0.04     0.00   20.16 ^ clkbuf_1_1__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                 0.03     0.10   20.27 ^ clkbuf_1_1__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     1    0.00                           clknet_1_1__leaf_clk (net)
                 0.03     0.00   20.27 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_1)
                         -0.25   20.02   clock uncertainty
                          0.00   20.02   clock reconvergence pessimism
                          0.19   20.20   library recovery time
                                 20.20   data required time
---------------------------------------------------------------------------
                                 20.20   data required time
                                 -4.20   data arrival time
---------------------------------------------------------------------------
                                 16.00   slack (MET)
```

```
Startpoint: rst_n (input port clocked by clk)
Endpoint: _21_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/28-mca/rcx_max_sta.max.rpt ? part
==

Path Type: max
Corner: Typical

```
Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                  0.02    0.01    4.01 ^ rst_n (in)
     1    0.00                           rst_n (net)
                  0.02    0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                  0.20    0.19    4.20 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.02                           net4 (net)
                  0.20    0.00    4.20 ^ _21_/RESET_B (sky130_fd_sc_hd__dfrtp_1)
                                  4.20   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock source latency
                  0.06    0.04   20.04 ^ clk (in)
     1    0.01                           clk (net)
                  0.06    0.00   20.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.04    0.13   20.16 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                  0.04    0.00   20.16 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.03    0.11   20.27 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                  0.03    0.00   20.27 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                         -0.25   20.02   clock uncertainty
                          0.00   20.02   clock reconvergence pessimism
                          0.19   20.21   library recovery time
                                 20.21   data required time
-----------------------------------------------------------------------------
                                 20.21   data required time
                                 -4.20   data arrival time
-----------------------------------------------------------------------------
                                 16.01   slack (MET)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/28-mca/rcx_max_sta.min.rpt ? part
==

==
report_checks -path_delay min (Hold)
==
======================= Fastest Corner ===================================

Startpoint: _21_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _22_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
Corner: Fastest

Fanout	Cap	Slew	Delay	Time	Description
			0.00	0.00	clock clk (rise edge)
			0.00	0.00	clock source latency
		0.04	0.03	0.03 ^	clk (in)
1	0.01				clk (net)
		0.04	0.00	0.03 ^	clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
		0.03	0.08	0.11 ^	clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
2	0.02				clknet_0_clk (net)
		0.03	0.00	0.11 ^	clkbuf_1_0__f_clk/A (sky130_fd_sc_hd__clkbuf_16)
		0.02	0.07	0.19 ^	clkbuf_1_0__f_clk/X (sky130_fd_sc_hd__clkbuf_16)
2	0.01				clknet_1_0__leaf_clk (net)
		0.02	0.00	0.19 ^	_21_/CLK (sky130_fd_sc_hd__dfrtp_1)
		0.08	0.22	0.41 ^	_21_/Q (sky130_fd_sc_hd__dfrtp_1)
3	0.01				state[1] (net)
		0.08	0.00	0.41 ^	_12_/A1 (sky130_fd_sc_hd__o21a_1)
		0.03	0.08	0.49 ^	_12_/X (sky130_fd_sc_hd__o21a_1)
1	0.00				_05_ (net)
		0.03	0.00	0.49 ^	_13_/B1 (sky130_fd_sc_hd__a21oi_1)
		0.02	0.02	0.51 v	_13_/Y (sky130_fd_sc_hd__a21oi_1)
1	0.00				_06_ (net)
		0.02	0.00	0.51 v	_14_/B (sky130_fd_sc_hd__nor2_1)
		0.06	0.05	0.56 ^	_14_/Y (sky130_fd_sc_hd__nor2_1)
1	0.00				_02_ (net)
		0.06	0.00	0.56 ^	_22_/D (sky130_fd_sc_hd__dfrtp_1)
				0.56	data arrival time
			0.00	0.00	clock clk (rise edge)
			0.00	0.00	clock source latency
		0.04	0.03	0.03 ^	clk (in)
1	0.01				clk (net)
		0.04	0.00	0.03 ^	clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)

```
                0.03    0.09    0.12 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                       clknet_0_clk (net)
                0.03    0.00    0.13 ^ clkbuf_1_1__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                0.02    0.08    0.20 ^ clkbuf_1_1__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     1    0.00                       clknet_1_1__leaf_clk (net)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/28-mca/rcx_max_sta.min.rpt ? part
==========

```
                  0.02    0.00    0.20 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_1)
                          0.25    0.45   clock uncertainty
                         -0.01    0.44   clock reconvergence pessimism
                         -0.02    0.42   library hold time
                                  0.42   data required time
-----------------------------------------------------------------------------
                                  0.42   data required time
                                 -0.56   data arrival time
-----------------------------------------------------------------------------
                                  0.14   slack (MET)


Startpoint: _21_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _21_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
Corner: Fastest

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                  0.04    0.03    0.03 ^ clk (in)
     1    0.01                          clk (net)
                  0.04    0.00    0.03 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.03    0.08    0.11 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                          clknet_0_clk (net)
                  0.03    0.00    0.11 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.02    0.07    0.19 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                          clknet_1_0__leaf_clk (net)
                  0.02    0.00    0.19 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                  0.08    0.22    0.41 ^ _21_/Q (sky130_fd_sc_hd__dfrtp_1)
     3    0.01                          state[1] (net)
                  0.08    0.00    0.41 ^ _17_/A1 (sky130_fd_sc_hd__o211a_1)
                  0.03    0.09    0.50 ^ _17_/X (sky130_fd_sc_hd__o211a_1)
     1    0.00                          _01_ (net)
                  0.03    0.00    0.50 ^ hold2/A (sky130_fd_sc_hd__dlygate4sd3_1)
                  0.04    0.35    0.85 ^ hold2/X (sky130_fd_sc_hd__dlygate4sd3_1)
     1    0.00                          net8 (net)
                  0.04    0.00    0.85 ^ _21_/D (sky130_fd_sc_hd__dfrtp_1)
                                  0.85   data arrival time

                          0.00    0.00   clock clk (rise edge)
```

```
                   0.00    0.00   clock source latency
             0.04    0.03    0.03 ^ clk (in)
1    0.01                          clk (net)
             0.04    0.00    0.03 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
             0.03    0.09    0.12 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/28-mca/rcx_max_sta.min.rpt ? part
==

```
     2    0.02                           clknet_0_clk (net)
                  0.03    0.00    0.13 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.02    0.08    0.21 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                  0.02    0.00    0.21 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                          0.25    0.46   clock uncertainty
                         -0.02    0.44   clock reconvergence pessimism
                         -0.02    0.42   library hold time
                                  0.42   data required time
-----------------------------------------------------------------------------
                                  0.42   data required time
                                 -0.85   data arrival time
-----------------------------------------------------------------------------
                                  0.43   slack (MET)


Startpoint: _21_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _20_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
Corner: Fastest

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                  0.04    0.03    0.03 ^ clk (in)
     1    0.01                           clk (net)
                  0.04    0.00    0.03 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.03    0.08    0.11 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                  0.03    0.00    0.11 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.02    0.07    0.19 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                  0.02    0.00    0.19 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                  0.08    0.22    0.41 ^ _21_/Q (sky130_fd_sc_hd__dfrtp_1)
     3    0.01                           state[1] (net)
                  0.08    0.00    0.41 ^ _15_/A2 (sky130_fd_sc_hd__a21oi_1)
                  0.02    0.04    0.45 v _15_/Y (sky130_fd_sc_hd__a21oi_1)
     1    0.00                           _07_ (net)
                  0.02    0.00    0.45 v hold4/A (sky130_fd_sc_hd__dlygate4sd3_1)
```

```
             0.04    0.34    0.79 v hold4/X (sky130_fd_sc_hd__dlygate4sd3_1)
1    0.00                            net10 (net)
             0.04    0.00    0.79 v _16_/A2 (sky130_fd_sc_hd__o21ai_1)
             0.05    0.08    0.87 ^ _16_/Y (sky130_fd_sc_hd__o21ai_1)
1    0.00                            _00_ (net)
             0.05    0.00    0.87 ^ _20_/D (sky130_fd_sc_hd__dfstp_1)
                             0.87   data arrival time
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/28-mca/rcx_max_sta.min.rpt ? part
==

```
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                 0.04     0.03    0.03 ^ clk (in)
     1    0.01                           clk (net)
                 0.04     0.00    0.03 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                 0.03     0.09    0.12 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                 0.03     0.00    0.13 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                 0.02     0.08    0.21 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                 0.02     0.00    0.21 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_1)
                          0.25    0.46   clock uncertainty
                         -0.02    0.44   clock reconvergence pessimism
                         -0.02    0.42   library hold time
                                  0.42   data required time
-----------------------------------------------------------------------------
                                  0.42   data required time
                                 -0.87   data arrival time
-----------------------------------------------------------------------------
                                  0.45   slack (MET)


Startpoint: rst_n (input port clocked by clk)
Endpoint: _21_ (removal check against rising-edge clock clk)
Path Group: asynchronous
Path Type: min
Corner: Fastest

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                 0.01     0.01    4.01 ^ rst_n (in)
     1    0.00                           rst_n (net)
                 0.01     0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                 0.15     0.12    4.13 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.02                           net4 (net)
                 0.15     0.00    4.13 ^ _21_/RESET_B (sky130_fd_sc_hd__dfrtp_1)
                                  4.13   data arrival time

                          0.00    0.00   clock clk (rise edge)
```

```
                   0.00    0.00   clock source latency
             0.04    0.03    0.03 ^ clk (in)
1    0.01                          clk (net)
             0.04    0.00    0.03 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
             0.03    0.09    0.12 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/28-mca/rcx_max_sta.min.rpt ? part
==

```
     2    0.02                           clknet_0_clk (net)
                  0.03    0.00    0.13 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.02    0.08    0.21 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                  0.02    0.00    0.21 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                          0.25    0.46   clock uncertainty
                          0.00    0.46   clock reconvergence pessimism
                          0.23    0.68   library removal time
                                  0.68   data required time
-----------------------------------------------------------------------------
                                  0.68   data required time
                                 -4.13   data arrival time
-----------------------------------------------------------------------------
                                  3.45   slack (MET)


Startpoint: rst_n (input port clocked by clk)
Endpoint: _22_ (removal check against rising-edge clock clk)
Path Group: asynchronous
Path Type: min
Corner: Fastest

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                  0.01    0.01    4.01 ^ rst_n (in)
     1    0.00                           rst_n (net)
                  0.01    0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                  0.15    0.12    4.13 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.02                           net4 (net)
                  0.15    0.00    4.13 ^ _22_/RESET_B (sky130_fd_sc_hd__dfrtp_1)
                                  4.13   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                  0.04    0.03    0.03 ^ clk (in)
     1    0.01                           clk (net)
                  0.04    0.00    0.03 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.03    0.09    0.12 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                  0.03    0.00    0.13 ^ clkbuf_1_1__f_clk/A
```

```
(sky130_fd_sc_hd__clkbuf_16)
                  0.02    0.08    0.20 ^ clkbuf_1_1__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     1    0.00                           clknet_1_1__leaf_clk (net)
                  0.02    0.00    0.20 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_1)
                          0.25    0.45   clock uncertainty
                          0.00    0.45   clock reconvergence pessimism
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/28-mca/rcx_max_sta.min.rpt ? part
==

```
                          0.23    0.68   library removal time
                                  0.68   data required time
-----------------------------------------------------------------------------
                                  0.68   data required time
                                 -4.13   data arrival time
-----------------------------------------------------------------------------
                                  3.45   slack (MET)


Startpoint: rst_n (input port clocked by clk)
Endpoint: _20_ (removal check against rising-edge clock clk)
Path Group: asynchronous
Path Type: min
Corner: Fastest

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                  0.01    0.01    4.01 ^ rst_n (in)
     1    0.00                           rst_n (net)
                  0.01    0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                  0.15    0.12    4.13 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.02                           net4 (net)
                  0.15    0.00    4.13 ^ _20_/SET_B (sky130_fd_sc_hd__dfstp_1)
                                  4.13   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                  0.04    0.03    0.03 ^ clk (in)
     1    0.01                           clk (net)
                  0.04    0.00    0.03 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.03    0.09    0.12 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                  0.03    0.00    0.13 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.02    0.08    0.21 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                  0.02    0.00    0.21 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_1)
                          0.25    0.46   clock uncertainty
                          0.00    0.46   clock reconvergence pessimism
                          0.10    0.56   library removal time
                                  0.56   data required time
```

```
--------------------------------------------------------------------------------
                                0.56   data required time
                               -4.13   data arrival time
--------------------------------------------------------------------------------
                                3.57   slack (MET)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/28-mca/rcx_max_sta.min.rpt ? part
==

Startpoint: _22_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: pump_en (output port clocked by clk)
Path Group: clk
Path Type: min
Corner: Fastest

Fanout	Cap	Slew	Delay	Time	Description
			0.00	0.00	clock clk (rise edge)
			0.00	0.00	clock source latency
		0.04	0.03	0.03 ^	clk (in)
1	0.01				clk (net)
		0.04	0.00	0.03 ^	clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
		0.03	0.08	0.11 ^	clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
2	0.02				clknet_0_clk (net)
		0.03	0.00	0.11 ^	clkbuf_1_1__f_clk/A (sky130_fd_sc_hd__clkbuf_16)
		0.02	0.07	0.18 ^	clkbuf_1_1__f_clk/X (sky130_fd_sc_hd__clkbuf_16)
1	0.00				clknet_1_1__leaf_clk (net)
		0.02	0.00	0.18 ^	_22_/CLK (sky130_fd_sc_hd__dfrtp_1)
		0.04	0.19	0.38 ^	_22_/Q (sky130_fd_sc_hd__dfrtp_1)
2	0.00				state[2] (net)
		0.04	0.00	0.38 ^	_18_/A (sky130_fd_sc_hd__and2_1)
		0.03	0.07	0.44 ^	_18_/X (sky130_fd_sc_hd__and2_1)
1	0.00				_08_ (net)
		0.03	0.00	0.44 ^	_19_/A (sky130_fd_sc_hd__clkbuf_1)
		0.04	0.05	0.49 ^	_19_/X (sky130_fd_sc_hd__clkbuf_1)
1	0.00				net6 (net)
		0.04	0.00	0.49 ^	output6/A (sky130_fd_sc_hd__clkbuf_4)
		0.08	0.12	0.61 ^	output6/X (sky130_fd_sc_hd__clkbuf_4)
1	0.03				pump_en (net)
		0.08	0.00	0.62 ^	pump_en (out)
				0.62	data arrival time
			0.00	0.00	clock clk (rise edge)
			0.00	0.00	clock network delay (propagated)
			0.25	0.25	clock uncertainty
			0.00	0.25	clock reconvergence pessimism
			-4.00	-3.75	output external delay
				-3.75	data required time
				-3.75	data required time

```
                                  -0.62   data arrival time
-----------------------------------------------------------------------------
                                   4.37   slack (MET)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/28-mca/rcx_max_sta.min.rpt ? part
==

Startpoint: dry_run_fault (input port clocked by clk)
Endpoint: alarm (output port clocked by clk)
Path Group: clk
Path Type: min
Corner: Fastest

Fanout	Cap	Slew	Delay	Time	Description
			0.00	0.00	clock clk (rise edge)
			0.00	0.00	clock network delay (propagated)
			4.00	4.00 ^	input external delay
		0.01	0.01	4.01 ^	dry_run_fault (in)
1	0.00				dry_run_fault (net)
		0.01	0.00	4.01 ^	input1/A (sky130_fd_sc_hd__clkbuf_1)
		0.05	0.06	4.07 ^	input1/X (sky130_fd_sc_hd__clkbuf_1)
1	0.01				net1 (net)
		0.05	0.00	4.07 ^	_09_/B1 (sky130_fd_sc_hd__a21oi_2)
		0.03	0.04	4.10 v	_09_/Y (sky130_fd_sc_hd__a21oi_2)
4	0.02				_03_ (net)
		0.03	0.00	4.10 v	_10_/A (sky130_fd_sc_hd__inv_2)
		0.03	0.04	4.14 ^	_10_/Y (sky130_fd_sc_hd__inv_2)
2	0.01				net5 (net)
		0.03	0.00	4.14 ^	output5/A (sky130_fd_sc_hd__buf_2)
		0.13	0.13	4.27 ^	output5/X (sky130_fd_sc_hd__buf_2)
1	0.03				alarm (net)
		0.13	0.00	4.27 ^	alarm (out)
				4.27	data arrival time
			0.00	0.00	clock clk (rise edge)
			0.00	0.00	clock network delay (propagated)
			0.25	0.25	clock uncertainty
			0.00	0.25	clock reconvergence pessimism
			-4.00	-3.75	output external delay
				-3.75	data required time
				-3.75	data required time
				-4.27	data arrival time
				8.02	slack (MET)

========== Slowest Corner ==========

Startpoint: _21_ (rising edge-triggered flip-flop clocked by clk)

Endpoint: _22_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/28-mca/rcx_max_sta.min.rpt ? part
==

Corner: Slowest

```
Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                   0.09    0.06    0.06 ^ clk (in)
     1    0.01                           clk (net)
                   0.09    0.00    0.06 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                   0.07    0.22    0.28 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                   0.07    0.00    0.28 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                   0.05    0.19    0.47 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                   0.05    0.00    0.47 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                   0.17    0.68    1.15 ^ _21_/Q (sky130_fd_sc_hd__dfrtp_1)
     3    0.01                           state[1] (net)
                   0.17    0.00    1.15 ^ _12_/A1 (sky130_fd_sc_hd__o21a_1)
                   0.07    0.26    1.41 ^ _12_/X (sky130_fd_sc_hd__o21a_1)
     1    0.00                           _05_ (net)
                   0.07    0.00    1.42 ^ _13_/B1 (sky130_fd_sc_hd__a21oi_1)
                   0.05    0.07    1.48 v _13_/Y (sky130_fd_sc_hd__a21oi_1)
     1    0.00                           _06_ (net)
                   0.05    0.00    1.48 v _14_/B (sky130_fd_sc_hd__nor2_1)
                   0.13    0.13    1.61 ^ _14_/Y (sky130_fd_sc_hd__nor2_1)
     1    0.00                           _02_ (net)
                   0.13    0.00    1.61 ^ _22_/D (sky130_fd_sc_hd__dfrtp_1)
                                   1.61   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                   0.09    0.06    0.06 ^ clk (in)
     1    0.01                           clk (net)
                   0.09    0.00    0.06 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                   0.07    0.25    0.31 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                   0.07    0.00    0.31 ^ clkbuf_1_1__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                   0.04    0.20    0.51 ^ clkbuf_1_1__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     1    0.00                           clknet_1_1__leaf_clk (net)
                   0.04    0.00    0.51 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_1)
                          0.25    0.76   clock uncertainty
```

```
                          -0.03    0.74   clock reconvergence pessimism
                          -0.10    0.64   library hold time
                                   0.64   data required time
-----------------------------------------------------------------------------
                                   0.64   data required time
                                  -1.61   data arrival time
-----------------------------------------------------------------------------
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/28-mca/rcx_max_sta.min.rpt ? part
==

                                  0.97   slack (MET)

Startpoint: _21_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _21_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
Corner: Slowest

Fanout   Cap   Slew   Delay   Time   Description

                          0.00   0.00   clock clk (rise edge)
                          0.00   0.00   clock source latency
                  0.09   0.06   0.06 ^ clk (in)
     1   0.01   clk (net)
                  0.09   0.00   0.06 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.07   0.22   0.28 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2   0.02   clknet_0_clk (net)
                  0.07   0.00   0.28 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.05   0.19   0.47 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2   0.01   clknet_1_0__leaf_clk (net)
                  0.05   0.00   0.47 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                  0.17   0.68   1.15 ^ _21_/Q (sky130_fd_sc_hd__dfrtp_1)
     3   0.01   state[1] (net)
                  0.17   0.00   1.15 ^ _17_/A1 (sky130_fd_sc_hd__o211a_1)
                  0.07   0.33   1.48 ^ _17_/X (sky130_fd_sc_hd__o211a_1)
     1   0.00   _01_ (net)
                  0.07   0.00   1.48 ^ hold2/A (sky130_fd_sc_hd__dlygate4sd3_1)
                  0.09   0.98   2.47 ^ hold2/X (sky130_fd_sc_hd__dlygate4sd3_1)
     1   0.00   net8 (net)
                  0.09   0.00   2.47 ^ _21_/D (sky130_fd_sc_hd__dfrtp_1)
                                  2.47   data arrival time

                          0.00   0.00   clock clk (rise edge)
                          0.00   0.00   clock source latency
                  0.09   0.06   0.06 ^ clk (in)
     1   0.01   clk (net)
                  0.09   0.00   0.06 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.07   0.25   0.31 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2   0.02   clknet_0_clk (net)
                  0.07   0.00   0.31 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.05   0.21   0.52 ^ clkbuf_1_0__f_clk/X

```
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                  0.05    0.00    0.52 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                          0.25    0.77   clock uncertainty
                         -0.05    0.72   clock reconvergence pessimism
                         -0.08    0.64   library hold time
                                  0.64   data required time
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/28-mca/rcx_max_sta.min.rpt ? part
==

```
                                  0.64   data required time
                                 -2.47   data arrival time
---------------------------------------------------------------------------
                                  1.83   slack (MET)


Startpoint: _21_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _20_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
Corner: Slowest

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                 0.09     0.06    0.06 ^ clk (in)
     1    0.01                           clk (net)
                 0.09     0.00    0.06 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                 0.07     0.22    0.28 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                 0.07     0.00    0.28 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                 0.05     0.19    0.47 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                 0.05     0.00    0.47 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                 0.14     0.76    1.23 v _21_/Q (sky130_fd_sc_hd__dfrtp_1)
     3    0.01                           state[1] (net)
                 0.14     0.00    1.23 v _15_/A2 (sky130_fd_sc_hd__a21oi_1)
                 0.13     0.23    1.46 ^ _15_/Y (sky130_fd_sc_hd__a21oi_1)
     1    0.00                           _07_ (net)
                 0.13     0.00    1.46 ^ hold4/A (sky130_fd_sc_hd__dlygate4sd3_1)
                 0.10     1.02    2.48 ^ hold4/X (sky130_fd_sc_hd__dlygate4sd3_1)
     1    0.00                           net10 (net)
                 0.10     0.00    2.48 ^ _16_/A2 (sky130_fd_sc_hd__o21ai_1)
                 0.08     0.13    2.60 v _16_/Y (sky130_fd_sc_hd__o21ai_1)
     1    0.00                           _00_ (net)
                 0.08     0.00    2.60 v _20_/D (sky130_fd_sc_hd__dfstp_1)
                                  2.60   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                 0.09     0.06    0.06 ^ clk (in)
```

```
     1    0.01                           clk (net)
                  0.09    0.00    0.06 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.07    0.25    0.31 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                  0.07    0.00    0.31 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/28-mca/rcx_max_sta.min.rpt ? part
==

```
                  0.05    0.21    0.52 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                  0.05    0.00    0.52 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_1)
                          0.25    0.77   clock uncertainty
                         -0.05    0.72   clock reconvergence pessimism
                         -0.01    0.71   library hold time
                                  0.71   data required time
-----------------------------------------------------------------------------
                                  0.71   data required time
                                 -2.60   data arrival time
-----------------------------------------------------------------------------
                                  1.89   slack (MET)


Startpoint: rst_n (input port clocked by clk)
Endpoint: _21_ (removal check against rising-edge clock clk)
Path Group: asynchronous
Path Type: min
Corner: Slowest

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                  0.03    0.02    4.02 ^ rst_n (in)
     1    0.00                           rst_n (net)
                  0.03    0.00    4.02 ^ input4/A (sky130_fd_sc_hd__buf_1)
                  0.29    0.28    4.29 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.02                           net4 (net)
                  0.29    0.00    4.29 ^ _21_/RESET_B (sky130_fd_sc_hd__dfrtp_1)
                                  4.29   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                  0.09    0.06    0.06 ^ clk (in)
     1    0.01                           clk (net)
                  0.09    0.00    0.06 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.07    0.25    0.31 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                  0.07    0.00    0.31 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.05    0.21    0.52 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
```

```
2    0.01                           clknet_1_0__leaf_clk (net)
             0.05    0.00    0.52 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                     0.25    0.77   clock uncertainty
                     0.00    0.77   clock reconvergence pessimism
                     0.72    1.49   library removal time
                             1.49   data required time
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/28-mca/rcx_max_sta.min.rpt ? part
==

```
---------------------------------------------------------------------------
                                1.49   data required time
                               -4.29   data arrival time
---------------------------------------------------------------------------
                                2.80   slack (MET)


Startpoint: rst_n (input port clocked by clk)
Endpoint: _22_ (removal check against rising-edge clock clk)
Path Group: asynchronous
Path Type: min
Corner: Slowest

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                  0.03    0.02    4.02 ^ rst_n (in)
     1    0.00                           rst_n (net)
                  0.03    0.00    4.02 ^ input4/A (sky130_fd_sc_hd__buf_1)
                  0.29    0.28    4.29 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.02                           net4 (net)
                  0.29    0.00    4.29 ^ _22_/RESET_B (sky130_fd_sc_hd__dfrtp_1)
                                  4.29   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                  0.09    0.06    0.06 ^ clk (in)
     1    0.01                           clk (net)
                  0.09    0.00    0.06 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.07    0.25    0.31 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                  0.07    0.00    0.31 ^ clkbuf_1_1__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.04    0.20    0.51 ^ clkbuf_1_1__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     1    0.00                           clknet_1_1__leaf_clk (net)
                  0.04    0.00    0.51 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_1)
                          0.25    0.76   clock uncertainty
                          0.00    0.76   clock reconvergence pessimism
                          0.72    1.48   library removal time
                                  1.48   data required time
---------------------------------------------------------------------------
                                1.48   data required time
```

```
                                  -4.29   data arrival time
-----------------------------------------------------------------------------
                                   2.81   slack (MET)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/28-mca/rcx_max_sta.min.rpt ? part
==

Startpoint: rst_n (input port clocked by clk)
Endpoint: _20_ (removal check against rising-edge clock clk)
Path Group: asynchronous
Path Type: min
Corner: Slowest

```
Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                  0.03    0.02    4.02 ^ rst_n (in)
     1    0.00                           rst_n (net)
                  0.03    0.00    4.02 ^ input4/A (sky130_fd_sc_hd__buf_1)
                  0.29    0.28    4.29 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.02                           net4 (net)
                  0.29    0.00    4.29 ^ _20_/SET_B (sky130_fd_sc_hd__dfstp_1)
                                  4.29   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                  0.09    0.06    0.06 ^ clk (in)
     1    0.01                           clk (net)
                  0.09    0.00    0.06 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.07    0.25    0.31 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                  0.07    0.00    0.31 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.05    0.21    0.52 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                  0.05    0.00    0.52 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_1)
                          0.25    0.77   clock uncertainty
                          0.00    0.77   clock reconvergence pessimism
                          0.27    1.04   library removal time
                                  1.04   data required time
-----------------------------------------------------------------------------
                                  1.04   data required time
                                 -4.29   data arrival time
-----------------------------------------------------------------------------
                                  3.25   slack (MET)


Startpoint: _22_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: pump_en (output port clocked by clk)
```

Path Group: clk
Path Type: min
Corner: Slowest

Fanout	Cap	Slew	Delay	Time	Description

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/28-mca/rcx_max_sta.min.rpt ? part
==

```
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                 0.09     0.06    0.06 ^ clk (in)
     1    0.01                           clk (net)
                 0.09     0.00    0.06 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                 0.07     0.22    0.28 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                 0.07     0.00    0.28 ^ clkbuf_1_1__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                 0.04     0.18    0.47 ^ clkbuf_1_1__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     1    0.00                           clknet_1_1__leaf_clk (net)
                 0.04     0.00    0.47 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_1)
                 0.10     0.61    1.08 ^ _22_/Q (sky130_fd_sc_hd__dfrtp_1)
     2    0.00                           state[2] (net)
                 0.10     0.00    1.08 ^ _18_/A (sky130_fd_sc_hd__and2_1)
                 0.08     0.21    1.29 ^ _18_/X (sky130_fd_sc_hd__and2_1)
     1    0.00                           _08_ (net)
                 0.08     0.00    1.29 ^ _19_/A (sky130_fd_sc_hd__clkbuf_1)
                 0.08     0.15    1.44 ^ _19_/X (sky130_fd_sc_hd__clkbuf_1)
     1    0.00                           net6 (net)
                 0.08     0.00    1.44 ^ output6/A (sky130_fd_sc_hd__clkbuf_4)
                 0.18     0.32    1.75 ^ output6/X (sky130_fd_sc_hd__clkbuf_4)
     1    0.03                           pump_en (net)
                 0.18     0.00    1.75 ^ pump_en (out)
                                  1.75   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          0.25    0.25   clock uncertainty
                          0.00    0.25   clock reconvergence pessimism
                         -4.00   -3.75   output external delay
                                 -3.75   data required time
---------------------------------------------------------------------------
                                 -3.75   data required time
                                 -1.75   data arrival time
---------------------------------------------------------------------------
                                  5.50   slack (MET)


Startpoint: dry_run_fault (input port clocked by clk)
Endpoint: alarm (output port clocked by clk)
Path Group: clk
Path Type: min
```

Corner: Slowest

Fanout	Cap	Slew	Delay	Time	Description
			0.00	0.00	clock clk (rise edge)

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/28-mca/rcx_max_sta.min.rpt ? part
==

```
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                 0.03     0.02    4.02 ^ dry_run_fault (in)
     1    0.00                           dry_run_fault (net)
                 0.03     0.00    4.02 ^ input1/A (sky130_fd_sc_hd__clkbuf_1)
                 0.11     0.15    4.17 ^ input1/X (sky130_fd_sc_hd__clkbuf_1)
     1    0.01                           net1 (net)
                 0.11     0.00    4.17 ^ _09_/B1 (sky130_fd_sc_hd__a21oi_2)
                 0.09     0.11    4.28 v _09_/Y (sky130_fd_sc_hd__a21oi_2)
     4    0.02                           _03_ (net)
                 0.09     0.00    4.28 v _10_/A (sky130_fd_sc_hd__inv_2)
                 0.06     0.09    4.37 ^ _10_/Y (sky130_fd_sc_hd__inv_2)
     2    0.01                           net5 (net)
                 0.06     0.00    4.37 ^ output5/A (sky130_fd_sc_hd__buf_2)
                 0.27     0.32    4.69 ^ output5/X (sky130_fd_sc_hd__buf_2)
     1    0.03                           alarm (net)
                 0.27     0.00    4.70 ^ alarm (out)
                          4.70    4.70   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          0.25    0.25   clock uncertainty
                          0.00    0.25   clock reconvergence pessimism
                         -4.00   -3.75   output external delay
                                 -3.75   data required time
------------------------------------------------------------------------------
                                 -3.75   data required time
                                 -4.70   data arrival time
------------------------------------------------------------------------------
                                  8.45   slack (MET)



====================== Typical Corner ===================================

Startpoint: _21_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _22_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
Corner: Typical

Fanout     Cap    Slew   Delay    Time   Description
------------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
```

```
             0.06    0.04    0.04 ^ clk (in)
1    0.01                           clk (net)
             0.06    0.00    0.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
```

```
openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/28-mca/rcx_max_sta.min.rpt ? part
===========================================================================================

                  0.04    0.13    0.16 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                  0.04    0.00    0.16 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.03    0.11    0.27 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                  0.03    0.00    0.27 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                  0.11    0.34    0.61 ^ _21_/Q (sky130_fd_sc_hd__dfrtp_1)
     3    0.01                           state[1] (net)
                  0.11    0.00    0.62 ^ _12_/A1 (sky130_fd_sc_hd__o21a_1)
                  0.04    0.13    0.74 ^ _12_/X (sky130_fd_sc_hd__o21a_1)
     1    0.00                           _05_ (net)
                  0.04    0.00    0.74 ^ _13_/B1 (sky130_fd_sc_hd__a21oi_1)
                  0.03    0.03    0.78 v _13_/Y (sky130_fd_sc_hd__a21oi_1)
     1    0.00                           _06_ (net)
                  0.03    0.00    0.78 v _14_/B (sky130_fd_sc_hd__nor2_1)
                  0.08    0.07    0.85 ^ _14_/Y (sky130_fd_sc_hd__nor2_1)
     1    0.00                           _02_ (net)
                  0.08    0.00    0.85 ^ _22_/D (sky130_fd_sc_hd__dfrtp_1)
                                  0.85   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                  0.06    0.04    0.04 ^ clk (in)
     1    0.01                           clk (net)
                  0.06    0.00    0.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.04    0.14    0.18 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                  0.04    0.00    0.18 ^ clkbuf_1_1__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.03    0.11    0.29 ^ clkbuf_1_1__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     1    0.00                           clknet_1_1__leaf_clk (net)
                  0.03    0.00    0.29 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_1)
                          0.25    0.54   clock uncertainty
                         -0.02    0.53   clock reconvergence pessimism
                         -0.04    0.48   library hold time
                                  0.48   data required time
---------------------------------------------------------------------------
                                  0.48   data required time
                                 -0.85   data arrival time
---------------------------------------------------------------------------
                                  0.37   slack (MET)
```

Startpoint: _21_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _21_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
Corner: Typical

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/28-mca/rcx_max_sta.min.rpt ? part
==

```
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                  0.06    0.04    0.04 ^ clk (in)
     1    0.01                          clk (net)
                  0.06    0.00    0.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.04    0.13    0.16 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                          clknet_0_clk (net)
                  0.04    0.00    0.16 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.03    0.11    0.27 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                          clknet_1_0__leaf_clk (net)
                  0.03    0.00    0.27 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                  0.11    0.34    0.61 ^ _21_/Q (sky130_fd_sc_hd__dfrtp_1)
     3    0.01                          state[1] (net)
                  0.11    0.00    0.62 ^ _17_/A1 (sky130_fd_sc_hd__o211a_1)
                  0.04    0.15    0.76 ^ _17_/X (sky130_fd_sc_hd__o211a_1)
     1    0.00                          _01_ (net)
                  0.04    0.00    0.76 ^ hold2/A (sky130_fd_sc_hd__dlygate4sd3_1)
                  0.05    0.51    1.27 ^ hold2/X (sky130_fd_sc_hd__dlygate4sd3_1)
     1    0.00                          net8 (net)
                  0.05    0.00    1.27 ^ _21_/D (sky130_fd_sc_hd__dfrtp_1)
                                  1.27   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                  0.06    0.04    0.04 ^ clk (in)
     1    0.01                          clk (net)
                  0.06    0.00    0.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.04    0.14    0.18 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                          clknet_0_clk (net)
                  0.04    0.00    0.18 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.03    0.12    0.30 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                          clknet_1_0__leaf_clk (net)
                  0.03    0.00    0.30 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                          0.25    0.55   clock uncertainty
                         -0.03    0.52   clock reconvergence pessimism
                         -0.04    0.48   library hold time
                                  0.48   data required time
---------------------------------------------------------------------------
                                  0.48   data required time
```

```
                                  -1.27   data arrival time
-----------------------------------------------------------------------------
                                   0.79   slack (MET)


Startpoint: _21_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _20_ (rising edge-triggered flip-flop clocked by clk)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/28-mca/rcx_max_sta.min.rpt ? part
==

Path Group: clk
Path Type: min
Corner: Typical

```
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                  0.06    0.04    0.04 ^ clk (in)
     1    0.01                          clk (net)
                  0.06    0.00    0.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.04    0.13    0.16 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                          clknet_0_clk (net)
                  0.04    0.00    0.16 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.03    0.11    0.27 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                          clknet_1_0__leaf_clk (net)
                  0.03    0.00    0.27 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                  0.07    0.37    0.64 v _21_/Q (sky130_fd_sc_hd__dfrtp_1)
     3    0.01                          state[1] (net)
                  0.07    0.00    0.64 v _15_/A2 (sky130_fd_sc_hd__a21oi_1)
                  0.08    0.13    0.77 ^ _15_/Y (sky130_fd_sc_hd__a21oi_1)
     1    0.00                          _07_ (net)
                  0.08    0.00    0.77 ^ hold4/A (sky130_fd_sc_hd__dlygate4sd3_1)
                  0.06    0.52    1.29 ^ hold4/X (sky130_fd_sc_hd__dlygate4sd3_1)
     1    0.00                          net10 (net)
                  0.06    0.00    1.29 ^ _16_/A2 (sky130_fd_sc_hd__o21ai_1)
                  0.03    0.06    1.35 v _16_/Y (sky130_fd_sc_hd__o21ai_1)
     1    0.00                          _00_ (net)
                  0.03    0.00    1.35 v _20_/D (sky130_fd_sc_hd__dfstp_1)
                                  1.35   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                  0.06    0.04    0.04 ^ clk (in)
     1    0.01                          clk (net)
                  0.06    0.00    0.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.04    0.14    0.18 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                          clknet_0_clk (net)
                  0.04    0.00    0.18 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.03    0.12    0.30 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                          clknet_1_0__leaf_clk (net)
```

```
                  0.03    0.00    0.30 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_1)
                          0.25    0.55   clock uncertainty
                         -0.03    0.52   clock reconvergence pessimism
                         -0.01    0.51   library hold time
                                  0.51   data required time
-----------------------------------------------------------------------------
                                  0.51   data required time
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/28-mca/rcx_max_sta.min.rpt ? part
==

                                -1.35   data arrival time

                                 0.84   slack (MET)

Startpoint: rst_n (input port clocked by clk)
Endpoint: _21_ (removal check against rising-edge clock clk)
Path Group: asynchronous
Path Type: min
Corner: Typical

Fanout   Cap   Slew   Delay   Time   Description

                          0.00   0.00   clock clk (rise edge)
                          0.00   0.00   clock network delay (propagated)
                          4.00   4.00 ^ input external delay
                 0.02   0.01   4.01 ^ rst_n (in)
     1   0.00   rst_n (net)
                 0.02   0.00   4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                 0.20   0.17   4.18 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3   0.02   net4 (net)
                 0.20   0.00   4.18 ^ _21_/RESET_B (sky130_fd_sc_hd__dfrtp_1)
                                  4.18   data arrival time

                          0.00   0.00   clock clk (rise edge)
                          0.00   0.00   clock source latency
                 0.06   0.04   0.04 ^ clk (in)
     1   0.01   clk (net)
                 0.06   0.00   0.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                 0.04   0.14   0.18 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2   0.02   clknet_0_clk (net)
                 0.04   0.00   0.18 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                 0.03   0.12   0.30 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2   0.01   clknet_1_0__leaf_clk (net)
                 0.03   0.00   0.30 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                          0.25   0.55   clock uncertainty
                          0.00   0.55   clock reconvergence pessimism
                          0.36   0.91   library removal time
                                  0.91   data required time

                                  0.91   data required time
                                 -4.18   data arrival time

3.27   slack (MET)

Startpoint: rst_n (input port clocked by clk)
Endpoint: _22_ (removal check against rising-edge clock clk)

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/28-mca/rcx_max_sta.min.rpt ? part
==========

Path Group: asynchronous
Path Type: min
Corner: Typical

```
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                   0.02   0.01    4.01 ^ rst_n (in)
     1    0.00                           rst_n (net)
                   0.02   0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                   0.20   0.17    4.18 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.02                           net4 (net)
                   0.20   0.00    4.18 ^ _22_/RESET_B (sky130_fd_sc_hd__dfrtp_1)
                                  4.18   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                   0.06   0.04    0.04 ^ clk (in)
     1    0.01                           clk (net)
                   0.06   0.00    0.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                   0.04   0.14    0.18 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                   0.04   0.00    0.18 ^ clkbuf_1_1__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                   0.03   0.11    0.29 ^ clkbuf_1_1__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     1    0.00                           clknet_1_1__leaf_clk (net)
                   0.03   0.00    0.29 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_1)
                          0.25    0.54   clock uncertainty
                          0.00    0.54   clock reconvergence pessimism
                          0.36    0.90   library removal time
                                  0.90   data required time
---------------------------------------------------------------------------
                                  0.90   data required time
                                 -4.18   data arrival time
---------------------------------------------------------------------------
                                  3.28   slack (MET)


Startpoint: rst_n (input port clocked by clk)
Endpoint: _20_ (removal check against rising-edge clock clk)
Path Group: asynchronous
Path Type: min
```

Corner: Typical

Fanout	Cap	Slew	Delay	Time	Description
			0.00	0.00	clock clk (rise edge)

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/28-mca/rcx_max_sta.min.rpt ? part
==

```
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                 0.02     0.01    4.01 ^ rst_n (in)
     1    0.00                           rst_n (net)
                 0.02     0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                 0.20     0.17    4.18 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.02                           net4 (net)
                 0.20     0.00    4.18 ^ _20_/SET_B (sky130_fd_sc_hd__dfstp_1)
                                  4.18   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                 0.06     0.04    0.04 ^ clk (in)
     1    0.01                           clk (net)
                 0.06     0.00    0.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                 0.04     0.14    0.18 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                 0.04     0.00    0.18 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                 0.03     0.12    0.30 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                 0.03     0.00    0.30 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_1)
                          0.25    0.55   clock uncertainty
                          0.00    0.55   clock reconvergence pessimism
                          0.14    0.69   library removal time
                                  0.69   data required time
------------------------------------------------------------------------------
                                  0.69   data required time
                                 -4.18   data arrival time
------------------------------------------------------------------------------
                                  3.49   slack (MET)


Startpoint: _22_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: pump_en (output port clocked by clk)
Path Group: clk
Path Type: min
Corner: Typical

Fanout     Cap    Slew   Delay    Time   Description
------------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                 0.06     0.04    0.04 ^ clk (in)
```

```
     1    0.01                           clk (net)
                  0.06    0.00    0.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.04    0.13    0.16 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                  0.04    0.00    0.16 ^ clkbuf_1_1__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/28-mca/rcx_max_sta.min.rpt ? part
==

```
                  0.03    0.10    0.27 ^ clkbuf_1_1__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     1    0.00                           clknet_1_1__leaf_clk (net)
                  0.03    0.00    0.27 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_1)
                  0.06    0.30    0.57 ^ _22_/Q (sky130_fd_sc_hd__dfrtp_1)
     2    0.00                           state[2] (net)
                  0.06    0.00    0.57 ^ _18_/A (sky130_fd_sc_hd__and2_1)
                  0.04    0.11    0.68 ^ _18_/X (sky130_fd_sc_hd__and2_1)
     1    0.00                           _08_ (net)
                  0.04    0.00    0.68 ^ _19_/A (sky130_fd_sc_hd__clkbuf_1)
                  0.05    0.08    0.76 ^ _19_/X (sky130_fd_sc_hd__clkbuf_1)
     1    0.00                           net6 (net)
                  0.05    0.00    0.76 ^ output6/A (sky130_fd_sc_hd__clkbuf_4)
                  0.11    0.18    0.93 ^ output6/X (sky130_fd_sc_hd__clkbuf_4)
     1    0.03                           pump_en (net)
                  0.11    0.00    0.93 ^ pump_en (out)
                                  0.93   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          0.25    0.25   clock uncertainty
                          0.00    0.25   clock reconvergence pessimism
                         -4.00   -3.75   output external delay
                                 -3.75   data required time
------------------------------------------------------------------------------
                                 -3.75   data required time
                                 -0.93   data arrival time
------------------------------------------------------------------------------
                                  4.68   slack (MET)


Startpoint: dry_run_fault (input port clocked by clk)
Endpoint: alarm (output port clocked by clk)
Path Group: clk
Path Type: min
Corner: Typical

Fanout     Cap    Slew   Delay    Time   Description
------------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                  0.02    0.01    4.01 ^ dry_run_fault (in)
     1    0.00                           dry_run_fault (net)
                  0.02    0.00    4.01 ^ input1/A (sky130_fd_sc_hd__clkbuf_1)
```

```
             0.07    0.09    4.10 ^ input1/X (sky130_fd_sc_hd__clkbuf_1)
1    0.01                           net1 (net)
             0.07    0.00    4.10 ^ _09_/B1 (sky130_fd_sc_hd__a21oi_2)
             0.05    0.06    4.15 v _09_/Y (sky130_fd_sc_hd__a21oi_2)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/28-mca/rcx_max_sta.min.rpt ? part
==========

```
     4    0.02                           _03_ (net)
                  0.05    0.00    4.15 v _10_/A (sky130_fd_sc_hd__inv_2)
                  0.04    0.05    4.21 ^ _10_/Y (sky130_fd_sc_hd__inv_2)
     2    0.01                           net5 (net)
                  0.04    0.00    4.21 ^ output5/A (sky130_fd_sc_hd__buf_2)
                  0.17    0.19    4.39 ^ output5/X (sky130_fd_sc_hd__buf_2)
     1    0.03                           alarm (net)
                  0.17    0.00    4.39 ^ alarm (out)
                                  4.39   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          0.25    0.25   clock uncertainty
                          0.00    0.25   clock reconvergence pessimism
                         -4.00   -3.75   output external delay
                                 -3.75   data required time
-----------------------------------------------------------------------------
                                 -3.75   data required time
                                 -4.39   data arrival time
-----------------------------------------------------------------------------
                                  8.14   slack (MET)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/28-mca/rcx_max_sta.power.rpt
==

==
 report_power
==
======================= Fastest Corner ===================================

Group	Internal Power	Switching Power	Leakage Power	Total Power (Watts)	
Sequential	7.62e-06	2.78e-07	1.47e-08	7.92e-06	16.5%
Combinational	3.15e-05	8.65e-06	5.48e-10	4.01e-05	83.5%
Macro	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Pad	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Total	3.91e-05	8.92e-06	1.52e-08	4.81e-05	100.0%
	81.4%	18.6%	0.0%		

======================= Slowest Corner ===================================

Group	Internal Power	Switching Power	Leakage Power	Total Power (Watts)	
Sequential	5.16e-06	1.80e-07	5.86e-08	5.40e-06	16.1%
Combinational	2.22e-05	5.71e-06	1.61e-07	2.81e-05	83.9%
Macro	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Pad	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Total	2.74e-05	5.89e-06	2.20e-07	3.35e-05	100.0%
	81.8%	17.6%	0.7%		

======================= Typical Corner ===================================

Group	Internal Power	Switching Power	Leakage Power	Total Power (Watts)	
Sequential	6.60e-06	2.32e-07	1.47e-08	6.85e-06	16.5%
Combinational	2.74e-05	7.29e-06	2.85e-10	3.47e-05	83.5%
Macro	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Pad	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Total	3.40e-05	7.52e-06	1.50e-08	4.16e-05	100.0%
	81.9%	18.1%	0.0%		

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/28-mca/rcx_max_sta.skew.rpt
==

==
report_clock_skew
==
Clock clk
Latency   CRPR   Skew
_20_/CLK ^
   0.52
_22_/CLK ^
   0.47   -0.03   0.03

```
openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/28-mca/rcx_max_sta.summary.rpt
==========================================================================================

===========================================================================
report_tns
===========================================================================
tns 0.00

===========================================================================
report_wns
===========================================================================
wns 0.00

===========================================================================
report_worst_slack -max (Setup)
===========================================================================
worst slack 10.03

===========================================================================
report_worst_slack -min (Hold)
===========================================================================
worst slack 0.14
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/30-mca/rcx_nom_sta.checks.rpt ? pa
==

==
report_checks -unconstrained
==
======================== Fastest Corner ==================================

Startpoint: rst_n (input port clocked by clk)
Endpoint: _20_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
Path Type: max
Corner: Fastest

```
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                 0.01     0.01    4.01 ^ rst_n (in)
     1    0.00                           rst_n (net)
                 0.01     0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                 0.14     0.13    4.14 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.02                           net4 (net)
                 0.14     0.00    4.14 ^ _20_/SET_B (sky130_fd_sc_hd__dfstp_1)
                                   4.14   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock source latency
                 0.04     0.03   20.03 ^ clk (in)
     1    0.01                           clk (net)
                 0.04     0.00   20.03 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                 0.03     0.08   20.11 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                 0.03     0.00   20.11 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                 0.02     0.07   20.18 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                 0.02     0.00   20.18 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_1)
                         -0.25   19.93   clock uncertainty
                          0.00   19.93   clock reconvergence pessimism
                          0.09   20.02   library recovery time
                                 20.02   data required time
---------------------------------------------------------------------------
                                 20.02   data required time
                                 -4.14   data arrival time
---------------------------------------------------------------------------
```

15.88   slack (MET)

Startpoint: level_full (input port clocked by clk)
Endpoint: pump_en (output port clocked by clk)

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/30-mca/rcx_nom_sta.checks.rpt ? pa
==

Path Group: clk
Path Type: max
Corner: Fastest

Fanout	Cap	Slew	Delay	Time	Description
			0.00	0.00	clock clk (rise edge)
			0.00	0.00	clock network delay (propagated)
			4.00	4.00 v	input external delay
		0.01	0.00	4.00 v	level_full (in)
1	0.00				level_full (net)
		0.01	0.00	4.00 v	input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
		0.07	0.12	4.12 v	input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
3	0.02				net2 (net)
		0.07	0.00	4.12 v	_09_/A2 (sky130_fd_sc_hd__a21oi_2)
		0.17	0.18	4.31 ^	_09_/Y (sky130_fd_sc_hd__a21oi_2)
4	0.02				_03_ (net)
		0.17	0.00	4.31 ^	_18_/B (sky130_fd_sc_hd__and2_1)
		0.03	0.09	4.40 ^	_18_/X (sky130_fd_sc_hd__and2_1)
1	0.00				_08_ (net)
		0.03	0.00	4.40 ^	_19_/A (sky130_fd_sc_hd__clkbuf_1)
		0.04	0.06	4.46 ^	_19_/X (sky130_fd_sc_hd__clkbuf_1)
1	0.00				net6 (net)
		0.04	0.00	4.46 ^	output6/A (sky130_fd_sc_hd__clkbuf_4)
		0.08	0.13	4.59 ^	output6/X (sky130_fd_sc_hd__clkbuf_4)
1	0.03				pump_en (net)
		0.09	0.00	4.59 ^	pump_en (out)
				4.59	data arrival time
			20.00	20.00	clock clk (rise edge)
			0.00	20.00	clock network delay (propagated)
			-0.25	19.75	clock uncertainty
			0.00	19.75	clock reconvergence pessimism
			-4.00	15.75	output external delay
				15.75	data required time
				15.75	data required time
				-4.59	data arrival time
				11.16	slack (MET)

========== Slowest Corner ==========

```
Startpoint: rst_n (input port clocked by clk)
Endpoint: _20_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
```

```
openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/30-mca/rcx_nom_sta.checks.rpt ? pa
========================================================================================

Path Type: max
Corner: Slowest

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                   0.03    0.02    4.02 ^ rst_n (in)
     1    0.00                           rst_n (net)
                   0.03    0.00    4.02 ^ input4/A (sky130_fd_sc_hd__buf_1)
                   0.29    0.30    4.32 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.01                           net4 (net)
                   0.29    0.00    4.32 ^ _20_/SET_B (sky130_fd_sc_hd__dfstp_1)
                                   4.32   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock source latency
                   0.09    0.06   20.06 ^ clk (in)
     1    0.01                           clk (net)
                   0.09    0.00   20.06 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                   0.07    0.22   20.28 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                   0.07    0.00   20.28 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                   0.05    0.19   20.47 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                   0.05    0.00   20.47 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_1)
                         -0.25   20.22   clock uncertainty
                          0.00   20.22   clock reconvergence pessimism
                          0.23   20.45   library recovery time
                                 20.45   data required time
---------------------------------------------------------------------------
                                 20.45   data required time
                                 -4.32   data arrival time
---------------------------------------------------------------------------
                                 16.13   slack (MET)


Startpoint: level_full (input port clocked by clk)
Endpoint: pump_en (output port clocked by clk)
Path Group: clk
Path Type: max
Corner: Slowest
```

```
Fanout     Cap     Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/30-mca/rcx_nom_sta.checks.rpt ? pa
==

```
                          4.00    4.00 v input external delay
                 0.02     0.01    4.01 v level_full (in)
     1    0.00                           level_full (net)
                 0.02     0.00    4.01 v input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
                 0.18     0.38    4.39 v input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
     3    0.02                           net2 (net)
                 0.18     0.00    4.39 v _09_/A2 (sky130_fd_sc_hd__a21oi_2)
                 0.36     0.46    4.85 ^ _09_/Y (sky130_fd_sc_hd__a21oi_2)
     4    0.02                           _03_ (net)
                 0.36     0.00    4.85 ^ _18_/B (sky130_fd_sc_hd__and2_1)
                 0.08     0.34    5.19 ^ _18_/X (sky130_fd_sc_hd__and2_1)
     1    0.00                           _08_ (net)
                 0.08     0.00    5.19 ^ _19_/A (sky130_fd_sc_hd__clkbuf_1)
                 0.08     0.17    5.35 ^ _19_/X (sky130_fd_sc_hd__clkbuf_1)
     1    0.00                           net6 (net)
                 0.08     0.00    5.35 ^ output6/A (sky130_fd_sc_hd__clkbuf_4)
                 0.18     0.35    5.70 ^ output6/X (sky130_fd_sc_hd__clkbuf_4)
     1    0.03                           pump_en (net)
                 0.18     0.00    5.70 ^ pump_en (out)
                           5.70    5.70   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock network delay (propagated)
                         -0.25   19.75   clock uncertainty
                          0.00   19.75   clock reconvergence pessimism
                         -4.00   15.75   output external delay
                                 15.75   data required time
------------------------------------------------------------------------------
                                 15.75   data required time
                                 -5.70   data arrival time
------------------------------------------------------------------------------
                                 10.05   slack (MET)



======================= Typical Corner ===================================

Startpoint: rst_n (input port clocked by clk)
Endpoint: _20_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
Path Type: max
Corner: Typical

Fanout     Cap    Slew   Delay    Time   Description
------------------------------------------------------------------------------
```

```
0.00    0.00   clock clk (rise edge)
0.00    0.00   clock network delay (propagated)
4.00    4.00 ^ input external delay
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/30-mca/rcx_nom_sta.checks.rpt ? pa
==

```
                0.02    0.01    4.01 ^ rst_n (in)
     1    0.00                        rst_n (net)
                0.02    0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                0.19    0.18    4.19 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.02                        net4 (net)
                0.19    0.00    4.19 ^ _20_/SET_B (sky130_fd_sc_hd__dfstp_1)
                                4.19   data arrival time

                       20.00   20.00   clock clk (rise edge)
                        0.00   20.00   clock source latency
                0.06    0.04   20.04 ^ clk (in)
     1    0.01                        clk (net)
                0.06    0.00   20.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                0.04    0.12   20.16 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                        clknet_0_clk (net)
                0.04    0.00   20.16 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                0.03    0.11   20.27 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                        clknet_1_0__leaf_clk (net)
                0.03    0.00   20.27 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_1)
                       -0.25   20.02   clock uncertainty
                        0.00   20.02   clock reconvergence pessimism
                        0.12   20.14   library recovery time
                               20.14   data required time
------------------------------------------------------------------------------
                               20.14   data required time
                               -4.19   data arrival time
------------------------------------------------------------------------------
                               15.94   slack (MET)


Startpoint: level_full (input port clocked by clk)
Endpoint: pump_en (output port clocked by clk)
Path Group: clk
Path Type: max
Corner: Typical

Fanout     Cap    Slew   Delay    Time   Description
------------------------------------------------------------------------------
                        0.00    0.00   clock clk (rise edge)
                        0.00    0.00   clock network delay (propagated)
                        4.00    4.00 v input external delay
                0.01    0.01    4.01 v level_full (in)
     1    0.00                        level_full (net)
```

```
              0.01    0.00    4.01 v input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
              0.09    0.18    4.19 v input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
3    0.02                             net2 (net)
              0.09    0.00    4.19 v _09_/A2 (sky130_fd_sc_hd__a21oi_2)
              0.23    0.26    4.45 ^ _09_/Y (sky130_fd_sc_hd__a21oi_2)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/30-mca/rcx_nom_sta.checks.rpt ? pa
==

```
     4    0.02                           _03_ (net)
                  0.23    0.00    4.45 ^ _18_/B (sky130_fd_sc_hd__and2_1)
                  0.05    0.16    4.62 ^ _18_/X (sky130_fd_sc_hd__and2_1)
     1    0.00                           _08_ (net)
                  0.05    0.00    4.62 ^ _19_/A (sky130_fd_sc_hd__clkbuf_1)
                  0.05    0.09    4.71 ^ _19_/X (sky130_fd_sc_hd__clkbuf_1)
     1    0.00                           net6 (net)
                  0.05    0.00    4.71 ^ output6/A (sky130_fd_sc_hd__clkbuf_4)
                  0.11    0.20    4.90 ^ output6/X (sky130_fd_sc_hd__clkbuf_4)
     1    0.03                           pump_en (net)
                  0.11    0.00    4.90 ^ pump_en (out)
                                  4.90   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock network delay (propagated)
                         -0.25   19.75   clock uncertainty
                          0.00   19.75   clock reconvergence pessimism
                         -4.00   15.75   output external delay
                                 15.75   data required time
-------------------------------------------------------------------------------
                                 15.75   data required time
                                 -4.90   data arrival time
-------------------------------------------------------------------------------
                                 10.85   slack (MET)
```

```
===========================================================================
report_checks --slack_max -0.01
===========================================================================
======================== Fastest Corner ===================================

No paths found.

======================== Slowest Corner ===================================

No paths found.

======================== Typical Corner ===================================

No paths found.


===========================================================================
```

```
 report_check_types -max_slew -max_cap -max_fanout -violators
================================================================================
======================== Fastest Corner ========================================
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/30-mca/rcx_nom_sta.checks.rpt ? pa
==

====================== Slowest Corner ===================================

====================== Typical Corner ===================================

==
report_parasitic_annotation -report_unannotated
==
Found 0 unannotated drivers.
Found 0 partially unannotated drivers.

==
max slew violation count 0
max fanout violation count 0
max cap violation count 0
==

==
check_setup -verbose -unconstrained_endpoints -multiple_clock -no_clock -no_input_delay
-loops -generated_clocks
==

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/30-mca/rcx_nom_sta.max.rpt ? part
==========

```
===========================================================================
report_checks -path_delay max (Setup)
===========================================================================
======================== Fastest Corner ===================================

Startpoint: level_full (input port clocked by clk)
Endpoint: pump_en (output port clocked by clk)
Path Group: clk
Path Type: max
Corner: Fastest

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 v input external delay
                  0.01    0.00    4.00 v level_full (in)
     1    0.00                           level_full (net)
                  0.01    0.00    4.00 v input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
                  0.07    0.12    4.12 v input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
     3    0.02                           net2 (net)
                  0.07    0.00    4.12 v _09_/A2 (sky130_fd_sc_hd__a21oi_2)
                  0.17    0.18    4.31 ^ _09_/Y (sky130_fd_sc_hd__a21oi_2)
     4    0.02                           _03_ (net)
                  0.17    0.00    4.31 ^ _18_/B (sky130_fd_sc_hd__and2_1)
                  0.03    0.09    4.40 ^ _18_/X (sky130_fd_sc_hd__and2_1)
     1    0.00                           _08_ (net)
                  0.03    0.00    4.40 ^ _19_/A (sky130_fd_sc_hd__clkbuf_1)
                  0.04    0.06    4.46 ^ _19_/X (sky130_fd_sc_hd__clkbuf_1)
     1    0.00                           net6 (net)
                  0.04    0.00    4.46 ^ output6/A (sky130_fd_sc_hd__clkbuf_4)
                  0.08    0.13    4.59 ^ output6/X (sky130_fd_sc_hd__clkbuf_4)
     1    0.03                           pump_en (net)
                  0.09    0.00    4.59 ^ pump_en (out)
                                  4.59   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock network delay (propagated)
                         -0.25   19.75   clock uncertainty
                          0.00   19.75   clock reconvergence pessimism
                         -4.00   15.75   output external delay
                                 15.75   data required time
---------------------------------------------------------------------------
                                 15.75   data required time
                                 -4.59   data arrival time
```

11.16 slack (MET)

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/30-mca/rcx_nom_sta.max.rpt ? part
==========

Startpoint: level_full (input port clocked by clk)
Endpoint: alarm (output port clocked by clk)
Path Group: clk
Path Type: max
Corner: Fastest

Fanout	Cap	Slew	Delay	Time		Description
			0.00	0.00		clock clk (rise edge)
			0.00	0.00		clock network delay (propagated)
			4.00	4.00	v	input external delay
		0.01	0.00	4.00	v	level_full (in)
1	0.00					level_full (net)
		0.01	0.00	4.00	v	input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
		0.07	0.12	4.12	v	input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
3	0.02					net2 (net)
		0.07	0.00	4.12	v	_09_/A2 (sky130_fd_sc_hd__a21oi_2)
		0.17	0.18	4.31	^	_09_/Y (sky130_fd_sc_hd__a21oi_2)
4	0.02					_03_ (net)
		0.17	0.00	4.31	^	_10_/A (sky130_fd_sc_hd__inv_2)
		0.04	0.02	4.33	v	_10_/Y (sky130_fd_sc_hd__inv_2)
2	0.01					net5 (net)
		0.04	0.00	4.33	v	output5/A (sky130_fd_sc_hd__buf_2)
		0.06	0.13	4.46	v	output5/X (sky130_fd_sc_hd__buf_2)
1	0.03					alarm (net)
		0.06	0.00	4.46	v	alarm (out)
				4.46		data arrival time
			20.00	20.00		clock clk (rise edge)
			0.00	20.00		clock network delay (propagated)
			-0.25	19.75		clock uncertainty
			0.00	19.75		clock reconvergence pessimism
			-4.00	15.75		output external delay
				15.75		data required time
				15.75		data required time
				-4.46		data arrival time
				11.29		slack (MET)

Startpoint: level_full (input port clocked by clk)
Endpoint: _21_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk

Path Type: max
Corner: Fastest

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/30-mca/rcx_nom_sta.max.rpt ? part
==========

```
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 v input external delay
                   0.01    0.00    4.00 v level_full (in)
     1    0.00                           level_full (net)
                   0.01    0.00    4.00 v input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
                   0.07    0.12    4.12 v input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
     3    0.02                           net2 (net)
                   0.07    0.00    4.12 v _09_/A2 (sky130_fd_sc_hd__a21oi_2)
                   0.17    0.18    4.31 ^ _09_/Y (sky130_fd_sc_hd__a21oi_2)
     4    0.02                           _03_ (net)
                   0.17    0.00    4.31 ^ _17_/B1 (sky130_fd_sc_hd__o211a_1)
                   0.03    0.08    4.39 ^ _17_/X (sky130_fd_sc_hd__o211a_1)
     1    0.00                           _01_ (net)
                   0.03    0.00    4.39 ^ hold2/A (sky130_fd_sc_hd__dlygate4sd3_1)
                   0.04    0.39    4.78 ^ hold2/X (sky130_fd_sc_hd__dlygate4sd3_1)
     1    0.00                           net8 (net)
                   0.04    0.00    4.78 ^ _21_/D (sky130_fd_sc_hd__dfrtp_1)
                           4.78    4.78   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock source latency
                   0.04    0.03   20.03 ^ clk (in)
     1    0.01                           clk (net)
                   0.04    0.00   20.03 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                   0.03    0.08   20.11 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                   0.03    0.00   20.11 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                   0.02    0.07   20.18 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                   0.02    0.00   20.18 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                         -0.25   19.93   clock uncertainty
                          0.00   19.93   clock reconvergence pessimism
                         -0.03   19.90   library setup time
                                 19.90   data required time
---------------------------------------------------------------------------
                                 19.90   data required time
                                 -4.78   data arrival time
---------------------------------------------------------------------------
                                 15.12   slack (MET)
```

```
Startpoint: level_full (input port clocked by clk)
Endpoint: _20_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/30-mca/rcx_nom_sta.max.rpt ? part
==========

Corner: Fastest

```
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                   0.02   0.01    4.01 ^ level_full (in)
     1    0.00                           level_full (net)
                   0.02   0.00    4.01 ^ input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
                   0.15   0.14    4.15 ^ input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
     3    0.02                           net2 (net)
                   0.15   0.00    4.15 ^ _11_/A (sky130_fd_sc_hd__inv_2)
                   0.04   0.03    4.18 v _11_/Y (sky130_fd_sc_hd__inv_2)
     2    0.01                           _04_ (net)
                   0.04   0.00    4.18 v _15_/A1 (sky130_fd_sc_hd__a21oi_1)
                   0.07   0.08    4.26 ^ _15_/Y (sky130_fd_sc_hd__a21oi_1)
     1    0.00                           _07_ (net)
                   0.07   0.00    4.26 ^ hold4/A (sky130_fd_sc_hd__dlygate4sd3_1)
                   0.04   0.40    4.66 ^ hold4/X (sky130_fd_sc_hd__dlygate4sd3_1)
     1    0.00                           net10 (net)
                   0.04   0.00    4.66 ^ _16_/A2 (sky130_fd_sc_hd__o21ai_1)
                   0.04   0.04    4.69 v _16_/Y (sky130_fd_sc_hd__o21ai_1)
     1    0.00                           _00_ (net)
                   0.04   0.00    4.69 v _20_/D (sky130_fd_sc_hd__dfstp_1)
                                  4.69   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock source latency
                   0.04   0.03   20.03 ^ clk (in)
     1    0.01                           clk (net)
                   0.04   0.00   20.03 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                   0.03   0.08   20.11 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                   0.03   0.00   20.11 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                   0.02   0.07   20.18 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                   0.02   0.00   20.18 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_1)
                         -0.25   19.93   clock uncertainty
                          0.00   19.93   clock reconvergence pessimism
                         -0.05   19.88   library setup time
                                 19.88   data required time
---------------------------------------------------------------------------
```

```
                                19.88   data required time
                                -4.69   data arrival time
-----------------------------------------------------------------------------
                                15.19   slack (MET)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/30-mca/rcx_nom_sta.max.rpt ? part
==

Startpoint: level_full (input port clocked by clk)
Endpoint: _22_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: Fastest

Fanout	Cap	Slew	Delay	Time	Description
			0.00	0.00	clock clk (rise edge)
			0.00	0.00	clock network delay (propagated)
			4.00	4.00 v	input external delay
		0.01	0.00	4.00 v	level_full (in)
1	0.00				level_full (net)
		0.01	0.00	4.00 v	input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
		0.07	0.12	4.12 v	input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
3	0.02				net2 (net)
		0.07	0.00	4.12 v	_09_/A2 (sky130_fd_sc_hd__a21oi_2)
		0.17	0.18	4.31 ^	_09_/Y (sky130_fd_sc_hd__a21oi_2)
4	0.02				_03_ (net)
		0.17	0.00	4.31 ^	_10_/A (sky130_fd_sc_hd__inv_2)
		0.04	0.02	4.33 v	_10_/Y (sky130_fd_sc_hd__inv_2)
2	0.01				net5 (net)
		0.04	0.00	4.33 v	_14_/A (sky130_fd_sc_hd__nor2_1)
		0.06	0.07	4.40 ^	_14_/Y (sky130_fd_sc_hd__nor2_1)
1	0.00				_02_ (net)
		0.06	0.00	4.40 ^	_22_/D (sky130_fd_sc_hd__dfrtp_1)
				4.40	data arrival time
			20.00	20.00	clock clk (rise edge)
			0.00	20.00	clock source latency
		0.04	0.03	20.03 ^	clk (in)
1	0.01				clk (net)
		0.04	0.00	20.03 ^	clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
		0.03	0.08	20.11 ^	clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
2	0.02				clknet_0_clk (net)
		0.03	0.00	20.11 ^	clkbuf_1_1__f_clk/A (sky130_fd_sc_hd__clkbuf_16)
		0.02	0.07	20.18 ^	clkbuf_1_1__f_clk/X (sky130_fd_sc_hd__clkbuf_16)
1	0.00				clknet_1_1__leaf_clk (net)
		0.02	0.00	20.18 ^	_22_/CLK (sky130_fd_sc_hd__dfrtp_1)
			-0.25	19.93	clock uncertainty
			0.00	19.93	clock reconvergence pessimism
			-0.04	19.89	library setup time

```
                                19.89   data required time
-----------------------------------------------------------------------------
                                19.89   data required time
                                -4.40   data arrival time
-----------------------------------------------------------------------------
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/30-mca/rcx_nom_sta.max.rpt ? part
==

                                15.49   slack (MET)

Startpoint: rst_n (input port clocked by clk)
Endpoint: _20_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
Path Type: max
Corner: Fastest

Fanout   Cap   Slew   Delay   Time   Description

                          0.00   0.00   clock clk (rise edge)
                          0.00   0.00   clock network delay (propagated)
                          4.00   4.00 ^ input external delay
                 0.01   0.01   4.01 ^ rst_n (in)
     1   0.00   rst_n (net)
                 0.01   0.00   4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                 0.14   0.13   4.14 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3   0.02   net4 (net)
                 0.14   0.00   4.14 ^ _20_/SET_B (sky130_fd_sc_hd__dfstp_1)
                                   4.14   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock source latency
                 0.04   0.03   20.03 ^ clk (in)
     1   0.01   clk (net)
                 0.04   0.00   20.03 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                 0.03   0.08   20.11 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2   0.02   clknet_0_clk (net)
                 0.03   0.00   20.11 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                 0.02   0.07   20.18 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2   0.01   clknet_1_0__leaf_clk (net)
                 0.02   0.00   20.18 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_1)
                         -0.25   19.93   clock uncertainty
                          0.00   19.93   clock reconvergence pessimism
                          0.09   20.02   library recovery time
                                 20.02   data required time

                                 20.02   data required time
                                 -4.14   data arrival time

                                 15.88   slack (MET)

```
Startpoint: rst_n (input port clocked by clk)
Endpoint: _22_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
Path Type: max
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/30-mca/rcx_nom_sta.max.rpt ? part
==========

Corner: Fastest

```
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                   0.01   0.01    4.01 ^ rst_n (in)
     1    0.00                           rst_n (net)
                   0.01   0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                   0.14   0.13    4.14 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.02                           net4 (net)
                   0.14   0.00    4.14 ^ _22_/RESET_B (sky130_fd_sc_hd__dfrtp_1)
                                  4.14   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock source latency
                   0.04   0.03   20.03 ^ clk (in)
     1    0.01                           clk (net)
                   0.04   0.00   20.03 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                   0.03   0.08   20.11 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                   0.03   0.00   20.11 ^ clkbuf_1_1__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                   0.02   0.07   20.18 ^ clkbuf_1_1__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     1    0.00                           clknet_1_1__leaf_clk (net)
                   0.02   0.00   20.18 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_1)
                         -0.25   19.93   clock uncertainty
                          0.00   19.93   clock reconvergence pessimism
                          0.12   20.05   library recovery time
                                 20.05   data required time
---------------------------------------------------------------------------
                                 20.05   data required time
                                 -4.14   data arrival time
---------------------------------------------------------------------------
                                 15.91   slack (MET)


Startpoint: rst_n (input port clocked by clk)
Endpoint: _21_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
Path Type: max
Corner: Fastest
```

```
Fanout     Cap     Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/30-mca/rcx_nom_sta.max.rpt ? part
==

```
                0.01    0.01    4.01 ^ rst_n (in)
     1    0.00                        rst_n (net)
                0.01    0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                0.14    0.13    4.14 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.02                        net4 (net)
                0.14    0.00    4.14 ^ _21_/RESET_B (sky130_fd_sc_hd__dfrtp_1)
                                4.14   data arrival time

                       20.00   20.00   clock clk (rise edge)
                        0.00   20.00   clock source latency
                0.04    0.03   20.03 ^ clk (in)
     1    0.01                        clk (net)
                0.04    0.00   20.03 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                0.03    0.08   20.11 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                        clknet_0_clk (net)
                0.03    0.00   20.11 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                0.02    0.07   20.18 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                        clknet_1_0__leaf_clk (net)
                0.02    0.00   20.18 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                       -0.25   19.93   clock uncertainty
                        0.00   19.93   clock reconvergence pessimism
                        0.12   20.06   library recovery time
                               20.06   data required time
------------------------------------------------------------------------------
                               20.06   data required time
                               -4.14   data arrival time
------------------------------------------------------------------------------
                               15.92   slack (MET)



====================== Slowest Corner ===================================

Startpoint: level_full (input port clocked by clk)
Endpoint: pump_en (output port clocked by clk)
Path Group: clk
Path Type: max
Corner: Slowest

Fanout     Cap    Slew   Delay    Time   Description
------------------------------------------------------------------------------
                        0.00    0.00   clock clk (rise edge)
                        0.00    0.00   clock network delay (propagated)
```

```
                    4.00    4.00 v input external delay
              0.02    0.01    4.01 v level_full (in)
1     0.00                         level_full (net)
              0.02    0.00    4.01 v input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
              0.18    0.38    4.39 v input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/30-mca/rcx_nom_sta.max.rpt ? part
==

```
     3    0.02                           net2 (net)
                  0.18    0.00    4.39 v _09_/A2 (sky130_fd_sc_hd__a21oi_2)
                  0.36    0.46    4.85 ^ _09_/Y (sky130_fd_sc_hd__a21oi_2)
     4    0.02                           _03_ (net)
                  0.36    0.00    4.85 ^ _18_/B (sky130_fd_sc_hd__and2_1)
                  0.08    0.34    5.19 ^ _18_/X (sky130_fd_sc_hd__and2_1)
     1    0.00                           _08_ (net)
                  0.08    0.00    5.19 ^ _19_/A (sky130_fd_sc_hd__clkbuf_1)
                  0.08    0.17    5.35 ^ _19_/X (sky130_fd_sc_hd__clkbuf_1)
     1    0.00                           net6 (net)
                  0.08    0.00    5.35 ^ output6/A (sky130_fd_sc_hd__clkbuf_4)
                  0.18    0.35    5.70 ^ output6/X (sky130_fd_sc_hd__clkbuf_4)
     1    0.03                           pump_en (net)
                  0.18    0.00    5.70 ^ pump_en (out)
                                  5.70   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock network delay (propagated)
                         -0.25   19.75   clock uncertainty
                          0.00   19.75   clock reconvergence pessimism
                         -4.00   15.75   output external delay
                                 15.75   data required time
-----------------------------------------------------------------------------
                                 15.75   data required time
                                 -5.70   data arrival time
-----------------------------------------------------------------------------
                                 10.05   slack (MET)


Startpoint: level_full (input port clocked by clk)
Endpoint: alarm (output port clocked by clk)
Path Group: clk
Path Type: max
Corner: Slowest

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 v input external delay
                  0.02    0.01    4.01 v level_full (in)
     1    0.00                           level_full (net)
                  0.02    0.00    4.01 v input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
                  0.18    0.38    4.39 v input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
     3    0.02                           net2 (net)
```

```
              0.18    0.00    4.39 v _09_/A2 (sky130_fd_sc_hd__a21oi_2)
              0.36    0.46    4.85 ^ _09_/Y (sky130_fd_sc_hd__a21oi_2)
4    0.02                           _03_ (net)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/30-mca/rcx_nom_sta.max.rpt ? part
==========

```
                  0.36    0.00    4.85 ^ _10_/A (sky130_fd_sc_hd__inv_2)
                  0.08    0.16    5.01 v _10_/Y (sky130_fd_sc_hd__inv_2)
     2    0.01                           net5 (net)
                  0.08    0.00    5.01 v output5/A (sky130_fd_sc_hd__buf_2)
                  0.17    0.37    5.38 v output5/X (sky130_fd_sc_hd__buf_2)
     1    0.03                           alarm (net)
                  0.17    0.00    5.38 v alarm (out)
                                  5.38   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock network delay (propagated)
                         -0.25   19.75   clock uncertainty
                          0.00   19.75   clock reconvergence pessimism
                         -4.00   15.75   output external delay
                                 15.75   data required time
-------------------------------------------------------------------------------
                                 15.75   data required time
                                 -5.38   data arrival time
-------------------------------------------------------------------------------
                                 10.37   slack (MET)


Startpoint: level_full (input port clocked by clk)
Endpoint: _21_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: Slowest

Fanout     Cap    Slew   Delay    Time   Description
-------------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 v input external delay
                  0.02    0.01    4.01 v level_full (in)
     1    0.00                           level_full (net)
                  0.02    0.00    4.01 v input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
                  0.18    0.38    4.39 v input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
     3    0.02                           net2 (net)
                  0.18    0.00    4.39 v _09_/A2 (sky130_fd_sc_hd__a21oi_2)
                  0.36    0.46    4.85 ^ _09_/Y (sky130_fd_sc_hd__a21oi_2)
     4    0.02                           _03_ (net)
                  0.36    0.00    4.85 ^ _17_/B1 (sky130_fd_sc_hd__o211a_1)
                  0.07    0.41    5.25 ^ _17_/X (sky130_fd_sc_hd__o211a_1)
     1    0.00                           _01_ (net)
                  0.07    0.00    5.25 ^ hold2/A (sky130_fd_sc_hd__dlygate4sd3_1)
```

```
             0.09    1.09    6.34 ^ hold2/X (sky130_fd_sc_hd__dlygate4sd3_1)
1    0.00                           net8 (net)
             0.09    0.00    6.34 ^ _21_/D (sky130_fd_sc_hd__dfrtp_1)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/30-mca/rcx_nom_sta.max.rpt ? part
==

```
                                 6.34   data arrival time

                          20.00   20.00   clock clk (rise edge)
                           0.00   20.00   clock source latency
                  0.09    0.06   20.06 ^ clk (in)
     1    0.01                           clk (net)
                  0.09    0.00   20.06 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.07    0.22   20.28 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                  0.07    0.00   20.28 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.05    0.19   20.47 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                  0.05    0.00   20.47 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                         -0.25   20.22   clock uncertainty
                          0.00   20.22   clock reconvergence pessimism
                         -0.14   20.08   library setup time
                                 20.08   data required time
---------------------------------------------------------------------------
                                 20.08   data required time
                                 -6.34   data arrival time
---------------------------------------------------------------------------
                                 13.73   slack (MET)


Startpoint: level_full (input port clocked by clk)
Endpoint: _20_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: Slowest

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 v input external delay
                  0.02    0.01    4.01 v level_full (in)
     1    0.00                           level_full (net)
                  0.02    0.00    4.01 v input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
                  0.18    0.38    4.39 v input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
     3    0.02                           net2 (net)
                  0.18    0.00    4.39 v _11_/A (sky130_fd_sc_hd__inv_2)
                  0.08    0.14    4.53 ^ _11_/Y (sky130_fd_sc_hd__inv_2)
     2    0.01                           _04_ (net)
```

```
             0.08    0.00    4.53 ^ _15_/A1 (sky130_fd_sc_hd__a21oi_1)
             0.08    0.12    4.65 v _15_/Y (sky130_fd_sc_hd__a21oi_1)
1    0.00                           _07_ (net)
             0.08    0.00    4.65 v hold4/A (sky130_fd_sc_hd__dlygate4sd3_1)
             0.11    1.19    5.84 v hold4/X (sky130_fd_sc_hd__dlygate4sd3_1)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/30-mca/rcx_nom_sta.max.rpt ? part
==========

```
     1    0.00                           net10 (net)
                  0.11    0.00    5.84 v _16_/A2 (sky130_fd_sc_hd__o21ai_1)
                  0.18    0.23    6.07 ^ _16_/Y (sky130_fd_sc_hd__o21ai_1)
     1    0.00                           _00_ (net)
                  0.18    0.00    6.07 ^ _20_/D (sky130_fd_sc_hd__dfstp_1)
                                  6.07   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock source latency
                  0.09    0.06   20.06 ^ clk (in)
     1    0.01                           clk (net)
                  0.09    0.00   20.06 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.07    0.22   20.28 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                  0.07    0.00   20.28 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.05    0.19   20.47 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                  0.05    0.00   20.47 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_1)
                         -0.25   20.22   clock uncertainty
                          0.00   20.22   clock reconvergence pessimism
                         -0.15   20.07   library setup time
                                 20.07   data required time
-----------------------------------------------------------------------------
                                 20.07   data required time
                                 -6.07   data arrival time
-----------------------------------------------------------------------------
                                 14.00   slack (MET)


Startpoint: level_full (input port clocked by clk)
Endpoint: _22_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: Slowest

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 v input external delay
                  0.02    0.01    4.01 v level_full (in)
     1    0.00                           level_full (net)
                  0.02    0.00    4.01 v input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
```

```
              0.18    0.38    4.39 v input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
3    0.02                             net2 (net)
              0.18    0.00    4.39 v _09_/A2 (sky130_fd_sc_hd__a21oi_2)
              0.36    0.46    4.85 ^ _09_/Y (sky130_fd_sc_hd__a21oi_2)
4    0.02                             _03_ (net)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/30-mca/rcx_nom_sta.max.rpt ? part
==

```
                 0.36    0.00    4.85 ^ _10_/A (sky130_fd_sc_hd__inv_2)
                 0.08    0.16    5.01 v _10_/Y (sky130_fd_sc_hd__inv_2)
     2    0.01                         net5 (net)
                 0.08    0.00    5.01 v _14_/A (sky130_fd_sc_hd__nor2_1)
                 0.13    0.18    5.19 ^ _14_/Y (sky130_fd_sc_hd__nor2_1)
     1    0.00                         _02_ (net)
                 0.13    0.00    5.19 ^ _22_/D (sky130_fd_sc_hd__dfrtp_1)
                                 5.19   data arrival time

                        20.00   20.00   clock clk (rise edge)
                         0.00   20.00   clock source latency
                 0.09    0.06   20.06 ^ clk (in)
     1    0.01                         clk (net)
                 0.09    0.00   20.06 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                 0.07    0.22   20.28 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                         clknet_0_clk (net)
                 0.07    0.00   20.28 ^ clkbuf_1_1__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                 0.04    0.18   20.46 ^ clkbuf_1_1__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     1    0.00                         clknet_1_1__leaf_clk (net)
                 0.04    0.00   20.46 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_1)
                        -0.25   20.21   clock uncertainty
                         0.00   20.21   clock reconvergence pessimism
                        -0.15   20.06   library setup time
                                20.06   data required time
-------------------------------------------------------------------------------
                                20.06   data required time
                                -5.19   data arrival time
-------------------------------------------------------------------------------
                                14.87   slack (MET)


Startpoint: rst_n (input port clocked by clk)
Endpoint: _20_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
Path Type: max
Corner: Slowest

Fanout     Cap    Slew   Delay    Time   Description
-------------------------------------------------------------------------------
                         0.00    0.00   clock clk (rise edge)
                         0.00    0.00   clock network delay (propagated)
                         4.00    4.00 ^ input external delay
                 0.03    0.02    4.02 ^ rst_n (in)
```

```
1    0.00                           rst_n (net)
             0.03    0.00    4.02 ^ input4/A (sky130_fd_sc_hd__buf_1)
             0.29    0.30    4.32 ^ input4/X (sky130_fd_sc_hd__buf_1)
3    0.01                           net4 (net)
             0.29    0.00    4.32 ^ _20_/SET_B (sky130_fd_sc_hd__dfstp_1)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/30-mca/rcx_nom_sta.max.rpt ? part
==

```
                                 4.32   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock source latency
                 0.09    0.06   20.06 ^ clk (in)
     1    0.01                          clk (net)
                 0.09    0.00   20.06 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                 0.07    0.22   20.28 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                          clknet_0_clk (net)
                 0.07    0.00   20.28 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                 0.05    0.19   20.47 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                          clknet_1_0__leaf_clk (net)
                 0.05    0.00   20.47 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_1)
                        -0.25   20.22   clock uncertainty
                         0.00   20.22   clock reconvergence pessimism
                         0.23   20.45   library recovery time
                                20.45   data required time
---------------------------------------------------------------------------
                                20.45   data required time
                                -4.32   data arrival time
---------------------------------------------------------------------------
                                16.13   slack (MET)


Startpoint: rst_n (input port clocked by clk)
Endpoint: _22_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
Path Type: max
Corner: Slowest

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                 0.03    0.02    4.02 ^ rst_n (in)
     1    0.00                          rst_n (net)
                 0.03    0.00    4.02 ^ input4/A (sky130_fd_sc_hd__buf_1)
                 0.29    0.30    4.32 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.01                          net4 (net)
                 0.29    0.00    4.32 ^ _22_/RESET_B (sky130_fd_sc_hd__dfrtp_1)
                                 4.32   data arrival time
```

```
                  20.00   20.00   clock clk (rise edge)
                   0.00   20.00   clock source latency
            0.09   0.06   20.06 ^ clk (in)
1   0.01                          clk (net)
            0.09   0.00   20.06 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/30-mca/rcx_nom_sta.max.rpt ? part
==

```
                  0.07    0.22   20.28 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                  0.07    0.00   20.28 ^ clkbuf_1_1__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.04    0.18   20.46 ^ clkbuf_1_1__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     1    0.00                           clknet_1_1__leaf_clk (net)
                  0.04    0.00   20.46 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_1)
                         -0.25   20.21   clock uncertainty
                          0.00   20.21   clock reconvergence pessimism
                          0.37   20.58   library recovery time
                                 20.58   data required time
-----------------------------------------------------------------------------
                                 20.58   data required time
                                 -4.32   data arrival time
-----------------------------------------------------------------------------
                                 16.27   slack (MET)


Startpoint: rst_n (input port clocked by clk)
Endpoint: _21_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
Path Type: max
Corner: Slowest

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                  0.03    0.02    4.02 ^ rst_n (in)
     1    0.00                           rst_n (net)
                  0.03    0.00    4.02 ^ input4/A (sky130_fd_sc_hd__buf_1)
                  0.29    0.30    4.32 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.01                           net4 (net)
                  0.29    0.00    4.32 ^ _21_/RESET_B (sky130_fd_sc_hd__dfrtp_1)
                                  4.32   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock source latency
                  0.09    0.06   20.06 ^ clk (in)
     1    0.01                           clk (net)
                  0.09    0.00   20.06 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.07    0.22   20.28 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
```

```
                  0.07    0.00   20.28 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.05    0.19   20.47 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                  0.05    0.00   20.47 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                         -0.25   20.22   clock uncertainty
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/30-mca/rcx_nom_sta.max.rpt ? part
==

```
                          0.00   20.22   clock reconvergence pessimism
                          0.38   20.59   library recovery time
                                 20.59   data required time
-------------------------------------------------------------------------------
                                 20.59   data required time
                                 -4.32   data arrival time
-------------------------------------------------------------------------------
                                 16.27   slack (MET)



====================== Typical Corner ===================================

Startpoint: level_full (input port clocked by clk)
Endpoint: pump_en (output port clocked by clk)
Path Group: clk
Path Type: max
Corner: Typical

Fanout     Cap    Slew   Delay    Time   Description
-------------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 v input external delay
                   0.01   0.01    4.01 v level_full (in)
     1    0.00                           level_full (net)
                   0.01   0.00    4.01 v input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
                   0.09   0.18    4.19 v input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
     3    0.02                           net2 (net)
                   0.09   0.00    4.19 v _09_/A2 (sky130_fd_sc_hd__a21oi_2)
                   0.23   0.26    4.45 ^ _09_/Y (sky130_fd_sc_hd__a21oi_2)
     4    0.02                           _03_ (net)
                   0.23   0.00    4.45 ^ _18_/B (sky130_fd_sc_hd__and2_1)
                   0.05   0.16    4.62 ^ _18_/X (sky130_fd_sc_hd__and2_1)
     1    0.00                           _08_ (net)
                   0.05   0.00    4.62 ^ _19_/A (sky130_fd_sc_hd__clkbuf_1)
                   0.05   0.09    4.71 ^ _19_/X (sky130_fd_sc_hd__clkbuf_1)
     1    0.00                           net6 (net)
                   0.05   0.00    4.71 ^ output6/A (sky130_fd_sc_hd__clkbuf_4)
                   0.11   0.20    4.90 ^ output6/X (sky130_fd_sc_hd__clkbuf_4)
     1    0.03                           pump_en (net)
                   0.11   0.00    4.90 ^ pump_en (out)
                                  4.90   data arrival time

                         20.00   20.00   clock clk (rise edge)
```

```
 0.00   20.00   clock network delay (propagated)
-0.25   19.75   clock uncertainty
 0.00   19.75   clock reconvergence pessimism
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/30-mca/rcx_nom_sta.max.rpt ? part
==

```
                          -4.00   15.75   output external delay
                                  15.75   data required time
-------------------------------------------------------------------------------
                                  15.75   data required time
                                  -4.90   data arrival time
-------------------------------------------------------------------------------
                                  10.85   slack (MET)


Startpoint: level_full (input port clocked by clk)
Endpoint: alarm (output port clocked by clk)
Path Group: clk
Path Type: max
Corner: Typical

Fanout     Cap    Slew   Delay    Time   Description
-------------------------------------------------------------------------------
                           0.00    0.00   clock clk (rise edge)
                           0.00    0.00   clock network delay (propagated)
                           4.00    4.00 v input external delay
                  0.01     0.01    4.01 v level_full (in)
     1    0.00                            level_full (net)
                  0.01     0.00    4.01 v input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
                  0.09     0.18    4.19 v input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
     3    0.02                            net2 (net)
                  0.09     0.00    4.19 v _09_/A2 (sky130_fd_sc_hd__a21oi_2)
                  0.23     0.26    4.45 ^ _09_/Y (sky130_fd_sc_hd__a21oi_2)
     4    0.02                            _03_ (net)
                  0.23     0.00    4.45 ^ _10_/A (sky130_fd_sc_hd__inv_2)
                  0.05     0.06    4.52 v _10_/Y (sky130_fd_sc_hd__inv_2)
     2    0.01                            net5 (net)
                  0.05     0.00    4.52 v output5/A (sky130_fd_sc_hd__buf_2)
                  0.09     0.19    4.71 v output5/X (sky130_fd_sc_hd__buf_2)
     1    0.03                            alarm (net)
                  0.09     0.00    4.71 v alarm (out)
                                   4.71   data arrival time

                          20.00   20.00   clock clk (rise edge)
                           0.00   20.00   clock network delay (propagated)
                          -0.25   19.75   clock uncertainty
                           0.00   19.75   clock reconvergence pessimism
                          -4.00   15.75   output external delay
                                  15.75   data required time
-------------------------------------------------------------------------------
                                  15.75   data required time
```

```
                                  -4.71   data arrival time
-----------------------------------------------------------------------------
                                  11.04   slack (MET)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/30-mca/rcx_nom_sta.max.rpt ? part
==========

Startpoint: level_full (input port clocked by clk)
Endpoint: _21_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: Typical

Fanout	Cap	Slew	Delay	Time		Description
			0.00	0.00		clock clk (rise edge)
			0.00	0.00		clock network delay (propagated)
			4.00	4.00	v	input external delay
		0.01	0.01	4.01	v	level_full (in)
1	0.00					level_full (net)
		0.01	0.00	4.01	v	input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
		0.09	0.18	4.19	v	input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
3	0.02					net2 (net)
		0.09	0.00	4.19	v	_09_/A2 (sky130_fd_sc_hd__a21oi_2)
		0.23	0.26	4.45	^	_09_/Y (sky130_fd_sc_hd__a21oi_2)
4	0.02					_03_ (net)
		0.23	0.00	4.45	^	_17_/B1 (sky130_fd_sc_hd__o211a_1)
		0.04	0.19	4.64	^	_17_/X (sky130_fd_sc_hd__o211a_1)
1	0.00					_01_ (net)
		0.04	0.00	4.64	^	hold2/A (sky130_fd_sc_hd__dlygate4sd3_1)
		0.05	0.56	5.20	^	hold2/X (sky130_fd_sc_hd__dlygate4sd3_1)
1	0.00					net8 (net)
		0.05	0.00	5.20	^	_21_/D (sky130_fd_sc_hd__dfrtp_1)
				5.20		data arrival time
			20.00	20.00		clock clk (rise edge)
			0.00	20.00		clock source latency
		0.06	0.04	20.04	^	clk (in)
1	0.01					clk (net)
		0.06	0.00	20.04	^	clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
		0.04	0.12	20.16	^	clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
2	0.02					clknet_0_clk (net)
		0.04	0.00	20.16	^	clkbuf_1_0__f_clk/A (sky130_fd_sc_hd__clkbuf_16)
		0.03	0.11	20.27	^	clkbuf_1_0__f_clk/X (sky130_fd_sc_hd__clkbuf_16)
2	0.01					clknet_1_0__leaf_clk (net)
		0.03	0.00	20.27	^	_21_/CLK (sky130_fd_sc_hd__dfrtp_1)
			-0.25	20.02		clock uncertainty
			0.00	20.02		clock reconvergence pessimism

```
                          -0.06   19.95   library setup time
                                  19.95   data required time
-----------------------------------------------------------------------------
                                  19.95   data required time
                                  -5.20   data arrival time
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/30-mca/rcx_nom_sta.max.rpt ? part
==========

```
                                14.76   slack (MET)


Startpoint: level_full (input port clocked by clk)
Endpoint: _20_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: Typical

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                  0.02    0.01    4.01 ^ level_full (in)
     1    0.00                           level_full (net)
                  0.02    0.00    4.01 ^ input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
                  0.19    0.19    4.21 ^ input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
     3    0.02                           net2 (net)
                  0.19    0.00    4.21 ^ _11_/A (sky130_fd_sc_hd__inv_2)
                  0.05    0.06    4.27 v _11_/Y (sky130_fd_sc_hd__inv_2)
     2    0.01                           _04_ (net)
                  0.05    0.00    4.27 v _15_/A1 (sky130_fd_sc_hd__a21oi_1)
                  0.09    0.12    4.38 ^ _15_/Y (sky130_fd_sc_hd__a21oi_1)
     1    0.00                           _07_ (net)
                  0.09    0.00    4.38 ^ hold4/A (sky130_fd_sc_hd__dlygate4sd3_1)
                  0.06    0.58    4.96 ^ hold4/X (sky130_fd_sc_hd__dlygate4sd3_1)
     1    0.00                           net10 (net)
                  0.06    0.00    4.96 ^ _16_/A2 (sky130_fd_sc_hd__o21ai_1)
                  0.07    0.06    5.03 v _16_/Y (sky130_fd_sc_hd__o21ai_1)
     1    0.00                           _00_ (net)
                  0.07    0.00    5.03 v _20_/D (sky130_fd_sc_hd__dfstp_1)
                                  5.03   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock source latency
                  0.06    0.04   20.04 ^ clk (in)
     1    0.01                           clk (net)
                  0.06    0.00   20.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.04    0.12   20.16 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                  0.04    0.00   20.16 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.03    0.11   20.27 ^ clkbuf_1_0__f_clk/X
```

```
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                  0.03    0.00   20.27 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_1)
                         -0.25   20.02   clock uncertainty
                          0.00   20.02   clock reconvergence pessimism
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/30-mca/rcx_nom_sta.max.rpt ? part
==

```
                          -0.08   19.93   library setup time
                                  19.93   data required time
------------------------------------------------------------------------------
                                  19.93   data required time
                                  -5.03   data arrival time
------------------------------------------------------------------------------
                                  14.91   slack (MET)


Startpoint: level_full (input port clocked by clk)
Endpoint: _22_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: Typical

Fanout     Cap    Slew   Delay    Time   Description
------------------------------------------------------------------------------
                           0.00    0.00   clock clk (rise edge)
                           0.00    0.00   clock network delay (propagated)
                           4.00    4.00 v input external delay
                  0.01     0.01    4.01 v level_full (in)
     1    0.00                            level_full (net)
                  0.01     0.00    4.01 v input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
                  0.09     0.18    4.19 v input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
     3    0.02                            net2 (net)
                  0.09     0.00    4.19 v _09_/A2 (sky130_fd_sc_hd__a21oi_2)
                  0.23     0.26    4.45 ^ _09_/Y (sky130_fd_sc_hd__a21oi_2)
     4    0.02                            _03_ (net)
                  0.23     0.00    4.45 ^ _10_/A (sky130_fd_sc_hd__inv_2)
                  0.05     0.06    4.52 v _10_/Y (sky130_fd_sc_hd__inv_2)
     2    0.01                            net5 (net)
                  0.05     0.00    4.52 v _14_/A (sky130_fd_sc_hd__nor2_1)
                  0.08     0.11    4.62 ^ _14_/Y (sky130_fd_sc_hd__nor2_1)
     1    0.00                            _02_ (net)
                  0.08     0.00    4.62 ^ _22_/D (sky130_fd_sc_hd__dfrtp_1)
                           4.62    4.62   data arrival time

                          20.00   20.00   clock clk (rise edge)
                           0.00   20.00   clock source latency
                  0.06     0.04   20.04 ^ clk (in)
     1    0.01                            clk (net)
                  0.06     0.00   20.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.04     0.12   20.16 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                            clknet_0_clk (net)
                  0.04     0.00   20.16 ^ clkbuf_1_1__f_clk/A
```

```
(sky130_fd_sc_hd__clkbuf_16)
                  0.03    0.10   20.26 ^ clkbuf_1_1__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     1    0.00                           clknet_1_1__leaf_clk (net)
                  0.03    0.00   20.26 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_1)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/30-mca/rcx_nom_sta.max.rpt ? part
==

```
                          -0.25   20.01   clock uncertainty
                           0.00   20.01   clock reconvergence pessimism
                          -0.07   19.94   library setup time
                                  19.94   data required time
-----------------------------------------------------------------------------
                                  19.94   data required time
                                  -4.62   data arrival time
-----------------------------------------------------------------------------
                                  15.32   slack (MET)


Startpoint: rst_n (input port clocked by clk)
Endpoint: _20_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
Path Type: max
Corner: Typical

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                           0.00    0.00   clock clk (rise edge)
                           0.00    0.00   clock network delay (propagated)
                           4.00    4.00 ^ input external delay
                  0.02     0.01    4.01 ^ rst_n (in)
     1    0.00                            rst_n (net)
                  0.02     0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                  0.19     0.18    4.19 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.02                            net4 (net)
                  0.19     0.00    4.19 ^ _20_/SET_B (sky130_fd_sc_hd__dfstp_1)
                                    4.19   data arrival time

                          20.00   20.00   clock clk (rise edge)
                           0.00   20.00   clock source latency
                  0.06     0.04   20.04 ^ clk (in)
     1    0.01                            clk (net)
                  0.06     0.00   20.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.04     0.12   20.16 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                            clknet_0_clk (net)
                  0.04     0.00   20.16 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.03     0.11   20.27 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                            clknet_1_0__leaf_clk (net)
                  0.03     0.00   20.27 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_1)
                          -0.25   20.02   clock uncertainty
                           0.00   20.02   clock reconvergence pessimism
```

```
                           0.12   20.14   library recovery time
                                  20.14   data required time
-----------------------------------------------------------------------------
                                  20.14   data required time
                                  -4.19   data arrival time
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/30-mca/rcx_nom_sta.max.rpt ? part
==========

```
---------------------------------------------------------------------------
                                15.94   slack (MET)


Startpoint: rst_n (input port clocked by clk)
Endpoint: _22_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
Path Type: max
Corner: Typical

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                 0.02     0.01    4.01 ^ rst_n (in)
     1    0.00                           rst_n (net)
                 0.02     0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                 0.19     0.18    4.19 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.02                           net4 (net)
                 0.19     0.00    4.19 ^ _22_/RESET_B (sky130_fd_sc_hd__dfrtp_1)
                                  4.19   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock source latency
                 0.06     0.04   20.04 ^ clk (in)
     1    0.01                           clk (net)
                 0.06     0.00   20.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                 0.04     0.12   20.16 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                 0.04     0.00   20.16 ^ clkbuf_1_1__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                 0.03     0.10   20.26 ^ clkbuf_1_1__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     1    0.00                           clknet_1_1__leaf_clk (net)
                 0.03     0.00   20.26 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_1)
                         -0.25   20.01   clock uncertainty
                          0.00   20.01   clock reconvergence pessimism
                          0.19   20.20   library recovery time
                                 20.20   data required time
---------------------------------------------------------------------------
                                 20.20   data required time
                                 -4.19   data arrival time
---------------------------------------------------------------------------
                                 16.01   slack (MET)
```

```
Startpoint: rst_n (input port clocked by clk)
Endpoint: _21_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/30-mca/rcx_nom_sta.max.rpt ? part
==========

Path Type: max
Corner: Typical

```
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                  0.02    0.01    4.01 ^ rst_n (in)
     1    0.00                           rst_n (net)
                  0.02    0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                  0.19    0.18    4.19 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.02                           net4 (net)
                  0.19    0.00    4.19 ^ _21_/RESET_B (sky130_fd_sc_hd__dfrtp_1)
                                  4.19   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock source latency
                  0.06    0.04   20.04 ^ clk (in)
     1    0.01                           clk (net)
                  0.06    0.00   20.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.04    0.12   20.16 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                  0.04    0.00   20.16 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.03    0.11   20.27 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                  0.03    0.00   20.27 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                         -0.25   20.02   clock uncertainty
                          0.00   20.02   clock reconvergence pessimism
                          0.19   20.21   library recovery time
                                 20.21   data required time
---------------------------------------------------------------------------
                                 20.21   data required time
                                 -4.19   data arrival time
---------------------------------------------------------------------------
                                 16.01   slack (MET)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/30-mca/rcx_nom_sta.min.rpt ? part
==

==
report_checks -path_delay min (Hold)
==
======================== Fastest Corner ===================================

Startpoint: _21_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _22_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
Corner: Fastest

```
Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                  0.04    0.03    0.03 ^ clk (in)
     1    0.01                          clk (net)
                  0.04    0.00    0.03 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.03    0.08    0.11 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                          clknet_0_clk (net)
                  0.03    0.00    0.11 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.02    0.07    0.18 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                          clknet_1_0__leaf_clk (net)
                  0.02    0.00    0.18 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                  0.08    0.22    0.40 ^ _21_/Q (sky130_fd_sc_hd__dfrtp_1)
     3    0.01                          state[1] (net)
                  0.08    0.00    0.40 ^ _12_/A1 (sky130_fd_sc_hd__o21a_1)
                  0.03    0.08    0.48 ^ _12_/X (sky130_fd_sc_hd__o21a_1)
     1    0.00                          _05_ (net)
                  0.03    0.00    0.48 ^ _13_/B1 (sky130_fd_sc_hd__a21oi_1)
                  0.02    0.02    0.50 v _13_/Y (sky130_fd_sc_hd__a21oi_1)
     1    0.00                          _06_ (net)
                  0.02    0.00    0.50 v _14_/B (sky130_fd_sc_hd__nor2_1)
                  0.06    0.05    0.55 ^ _14_/Y (sky130_fd_sc_hd__nor2_1)
     1    0.00                          _02_ (net)
                  0.06    0.00    0.55 ^ _22_/D (sky130_fd_sc_hd__dfrtp_1)
                                  0.55   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                  0.04    0.03    0.03 ^ clk (in)
     1    0.01                          clk (net)
                  0.04    0.00    0.03 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
```

```
                0.03    0.09    0.12 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                       clknet_0_clk (net)
                0.03    0.00    0.12 ^ clkbuf_1_1__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                0.02    0.08    0.20 ^ clkbuf_1_1__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     1    0.00                       clknet_1_1__leaf_clk (net)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/30-mca/rcx_nom_sta.min.rpt ? part
==========

```
                  0.02    0.00    0.20 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_1)
                          0.25    0.45   clock uncertainty
                         -0.01    0.44   clock reconvergence pessimism
                         -0.02    0.42   library hold time
                                  0.42   data required time
-----------------------------------------------------------------------------
                                  0.42   data required time
                                 -0.55   data arrival time
-----------------------------------------------------------------------------
                                  0.14   slack (MET)


Startpoint: _21_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _21_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
Corner: Fastest

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                  0.04    0.03    0.03 ^ clk (in)
     1    0.01                          clk (net)
                  0.04    0.00    0.03 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.03    0.08    0.11 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                          clknet_0_clk (net)
                  0.03    0.00    0.11 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.02    0.07    0.18 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                          clknet_1_0__leaf_clk (net)
                  0.02    0.00    0.18 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                  0.08    0.22    0.40 ^ _21_/Q (sky130_fd_sc_hd__dfrtp_1)
     3    0.01                          state[1] (net)
                  0.08    0.00    0.40 ^ _17_/A1 (sky130_fd_sc_hd__o211a_1)
                  0.03    0.09    0.49 ^ _17_/X (sky130_fd_sc_hd__o211a_1)
     1    0.00                          _01_ (net)
                  0.03    0.00    0.49 ^ hold2/A (sky130_fd_sc_hd__dlygate4sd3_1)
                  0.04    0.35    0.84 ^ hold2/X (sky130_fd_sc_hd__dlygate4sd3_1)
     1    0.00                          net8 (net)
                  0.04    0.00    0.84 ^ _21_/D (sky130_fd_sc_hd__dfrtp_1)
                                  0.84   data arrival time

                          0.00    0.00   clock clk (rise edge)
```

```
                   0.00    0.00   clock source latency
             0.04    0.03    0.03 ^ clk (in)
1    0.01                          clk (net)
             0.04    0.00    0.03 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
             0.03    0.09    0.12 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
```

```
openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/30-mca/rcx_nom_sta.min.rpt ? part
================================================================================

     2    0.02                           clknet_0_clk (net)
                  0.03    0.00    0.12 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.02    0.08    0.20 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                  0.02    0.00    0.20 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                          0.25    0.45   clock uncertainty
                         -0.02    0.43   clock reconvergence pessimism
                         -0.02    0.41   library hold time
                                  0.41   data required time
-----------------------------------------------------------------------------
                                  0.41   data required time
                                 -0.84   data arrival time
-----------------------------------------------------------------------------
                                  0.43   slack (MET)


Startpoint: _21_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _20_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
Corner: Fastest

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                  0.04    0.03    0.03 ^ clk (in)
     1    0.01                           clk (net)
                  0.04    0.00    0.03 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.03    0.08    0.11 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                  0.03    0.00    0.11 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.02    0.07    0.18 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                  0.02    0.00    0.18 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                  0.08    0.22    0.40 ^ _21_/Q (sky130_fd_sc_hd__dfrtp_1)
     3    0.01                           state[1] (net)
                  0.08    0.00    0.40 ^ _15_/A2 (sky130_fd_sc_hd__a21oi_1)
                  0.02    0.04    0.44 v _15_/Y (sky130_fd_sc_hd__a21oi_1)
     1    0.00                           _07_ (net)
                  0.02    0.00    0.44 v hold4/A (sky130_fd_sc_hd__dlygate4sd3_1)
```

```
             0.04    0.34    0.78 v hold4/X (sky130_fd_sc_hd__dlygate4sd3_1)
1    0.00                           net10 (net)
             0.04    0.00    0.78 v _16_/A2 (sky130_fd_sc_hd__o21ai_1)
             0.05    0.08    0.86 ^ _16_/Y (sky130_fd_sc_hd__o21ai_1)
1    0.00                           _00_ (net)
             0.05    0.00    0.86 ^ _20_/D (sky130_fd_sc_hd__dfstp_1)
                             0.86   data arrival time
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/30-mca/rcx_nom_sta.min.rpt ? part
==

```
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                 0.04     0.03    0.03 ^ clk (in)
     1    0.01                           clk (net)
                 0.04     0.00    0.03 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                 0.03     0.09    0.12 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                 0.03     0.00    0.12 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                 0.02     0.08    0.20 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                 0.02     0.00    0.20 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_1)
                          0.25    0.45   clock uncertainty
                         -0.02    0.43   clock reconvergence pessimism
                         -0.02    0.41   library hold time
                                  0.41   data required time
-----------------------------------------------------------------------------
                                  0.41   data required time
                                 -0.86   data arrival time
-----------------------------------------------------------------------------
                                  0.45   slack (MET)


Startpoint: rst_n (input port clocked by clk)
Endpoint: _21_ (removal check against rising-edge clock clk)
Path Group: asynchronous
Path Type: min
Corner: Fastest

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                 0.01     0.01    4.01 ^ rst_n (in)
     1    0.00                           rst_n (net)
                 0.01     0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                 0.14     0.12    4.12 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.02                           net4 (net)
                 0.14     0.00    4.13 ^ _21_/RESET_B (sky130_fd_sc_hd__dfrtp_1)
                                  4.13   data arrival time

                          0.00    0.00   clock clk (rise edge)
```

```
                   0.00    0.00   clock source latency
             0.04    0.03    0.03 ^ clk (in)
1    0.01                          clk (net)
             0.04    0.00    0.03 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
             0.03    0.09    0.12 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/30-mca/rcx_nom_sta.min.rpt ? part
==

```
     2    0.02                           clknet_0_clk (net)
                  0.03    0.00    0.12 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.02    0.08    0.20 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                  0.02    0.00    0.20 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                          0.25    0.45   clock uncertainty
                          0.00    0.45   clock reconvergence pessimism
                          0.22    0.68   library removal time
                                  0.68   data required time
-----------------------------------------------------------------------------
                                  0.68   data required time
                                 -4.13   data arrival time
-----------------------------------------------------------------------------
                                  3.45   slack (MET)


Startpoint: rst_n (input port clocked by clk)
Endpoint: _22_ (removal check against rising-edge clock clk)
Path Group: asynchronous
Path Type: min
Corner: Fastest

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                  0.01    0.01    4.01 ^ rst_n (in)
     1    0.00                           rst_n (net)
                  0.01    0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                  0.14    0.12    4.12 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.02                           net4 (net)
                  0.14    0.00    4.13 ^ _22_/RESET_B (sky130_fd_sc_hd__dfrtp_1)
                                  4.13   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                  0.04    0.03    0.03 ^ clk (in)
     1    0.01                           clk (net)
                  0.04    0.00    0.03 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.03    0.09    0.12 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                  0.03    0.00    0.12 ^ clkbuf_1_1__f_clk/A
```

```
(sky130_fd_sc_hd__clkbuf_16)
                  0.02    0.08    0.20 ^ clkbuf_1_1__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     1    0.00                           clknet_1_1__leaf_clk (net)
                  0.02    0.00    0.20 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_1)
                          0.25    0.45   clock uncertainty
                          0.00    0.45   clock reconvergence pessimism
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/30-mca/rcx_nom_sta.min.rpt ? part
==

```
                          0.22    0.67   library removal time
                                  0.67   data required time
-----------------------------------------------------------------------------
                                  0.67   data required time
                                 -4.13   data arrival time
-----------------------------------------------------------------------------
                                  3.45   slack (MET)


Startpoint: rst_n (input port clocked by clk)
Endpoint: _20_ (removal check against rising-edge clock clk)
Path Group: asynchronous
Path Type: min
Corner: Fastest

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                   0.01   0.01    4.01 ^ rst_n (in)
     1    0.00                           rst_n (net)
                   0.01   0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                   0.14   0.12    4.12 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.02                           net4 (net)
                   0.14   0.00    4.13 ^ _20_/SET_B (sky130_fd_sc_hd__dfstp_1)
                                  4.13   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                   0.04   0.03    0.03 ^ clk (in)
     1    0.01                           clk (net)
                   0.04   0.00    0.03 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                   0.03   0.09    0.12 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                   0.03   0.00    0.12 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                   0.02   0.08    0.20 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                   0.02   0.00    0.20 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_1)
                          0.25    0.45   clock uncertainty
                          0.00    0.45   clock reconvergence pessimism
                          0.10    0.55   library removal time
                                  0.55   data required time
```

```
--------------------------------------------------------------------------------
                                0.55   data required time
                               -4.13   data arrival time
--------------------------------------------------------------------------------
                                3.57   slack (MET)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/30-mca/rcx_nom_sta.min.rpt ? part
==

Startpoint: _22_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: pump_en (output port clocked by clk)
Path Group: clk
Path Type: min
Corner: Fastest

Fanout	Cap	Slew	Delay	Time	Description
			0.00	0.00	clock clk (rise edge)
			0.00	0.00	clock source latency
		0.04	0.03	0.03 ^	clk (in)
1	0.01				clk (net)
		0.04	0.00	0.03 ^	clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
		0.03	0.08	0.11 ^	clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
2	0.02				clknet_0_clk (net)
		0.03	0.00	0.11 ^	clkbuf_1_1__f_clk/A (sky130_fd_sc_hd__clkbuf_16)
		0.02	0.07	0.18 ^	clkbuf_1_1__f_clk/X (sky130_fd_sc_hd__clkbuf_16)
1	0.00				clknet_1_1__leaf_clk (net)
		0.02	0.00	0.18 ^	_22_/CLK (sky130_fd_sc_hd__dfrtp_1)
		0.04	0.19	0.37 ^	_22_/Q (sky130_fd_sc_hd__dfrtp_1)
2	0.00				state[2] (net)
		0.04	0.00	0.37 ^	_18_/A (sky130_fd_sc_hd__and2_1)
		0.03	0.07	0.44 ^	_18_/X (sky130_fd_sc_hd__and2_1)
1	0.00				_08_ (net)
		0.03	0.00	0.44 ^	_19_/A (sky130_fd_sc_hd__clkbuf_1)
		0.04	0.05	0.49 ^	_19_/X (sky130_fd_sc_hd__clkbuf_1)
1	0.00				net6 (net)
		0.04	0.00	0.49 ^	output6/A (sky130_fd_sc_hd__clkbuf_4)
		0.08	0.12	0.61 ^	output6/X (sky130_fd_sc_hd__clkbuf_4)
1	0.03				pump_en (net)
		0.09	0.00	0.61 ^	pump_en (out)
				0.61	data arrival time
			0.00	0.00	clock clk (rise edge)
			0.00	0.00	clock network delay (propagated)
			0.25	0.25	clock uncertainty
			0.00	0.25	clock reconvergence pessimism
			-4.00	-3.75	output external delay
				-3.75	data required time
				-3.75	data required time

```
                                  -0.61   data arrival time
-----------------------------------------------------------------------------
                                   4.36   slack (MET)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/30-mca/rcx_nom_sta.min.rpt ? part
==

Startpoint: dry_run_fault (input port clocked by clk)
Endpoint: alarm (output port clocked by clk)
Path Group: clk
Path Type: min
Corner: Fastest

Fanout	Cap	Slew	Delay	Time	Description
			0.00	0.00	clock clk (rise edge)
			0.00	0.00	clock network delay (propagated)
			4.00	4.00 ^	input external delay
		0.01	0.01	4.01 ^	dry_run_fault (in)
1	0.00				dry_run_fault (net)
		0.01	0.00	4.01 ^	input1/A (sky130_fd_sc_hd__clkbuf_1)
		0.05	0.06	4.07 ^	input1/X (sky130_fd_sc_hd__clkbuf_1)
1	0.01				net1 (net)
		0.05	0.00	4.07 ^	_09_/B1 (sky130_fd_sc_hd__a21oi_2)
		0.03	0.04	4.10 v	_09_/Y (sky130_fd_sc_hd__a21oi_2)
4	0.02				_03_ (net)
		0.03	0.00	4.10 v	_10_/A (sky130_fd_sc_hd__inv_2)
		0.03	0.04	4.14 ^	_10_/Y (sky130_fd_sc_hd__inv_2)
2	0.01				net5 (net)
		0.03	0.00	4.14 ^	output5/A (sky130_fd_sc_hd__buf_2)
		0.13	0.13	4.27 ^	output5/X (sky130_fd_sc_hd__buf_2)
1	0.03				alarm (net)
		0.13	0.00	4.27 ^	alarm (out)
				4.27	data arrival time
			0.00	0.00	clock clk (rise edge)
			0.00	0.00	clock network delay (propagated)
			0.25	0.25	clock uncertainty
			0.00	0.25	clock reconvergence pessimism
			-4.00	-3.75	output external delay
				-3.75	data required time
				-3.75	data required time
				-4.27	data arrival time
				8.02	slack (MET)

========== Slowest Corner ==========

Startpoint: _21_ (rising edge-triggered flip-flop clocked by clk)

Endpoint: _22_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/30-mca/rcx_nom_sta.min.rpt ? part
==

Corner: Slowest

```
Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                   0.09    0.06    0.06 ^ clk (in)
     1    0.01                           clk (net)
                   0.09    0.00    0.06 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                   0.07    0.22    0.28 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                   0.07    0.00    0.28 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                   0.05    0.19    0.47 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                   0.05    0.00    0.47 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                   0.17    0.68    1.14 ^ _21_/Q (sky130_fd_sc_hd__dfrtp_1)
     3    0.01                           state[1] (net)
                   0.17    0.00    1.14 ^ _12_/A1 (sky130_fd_sc_hd__o21a_1)
                   0.07    0.26    1.41 ^ _12_/X (sky130_fd_sc_hd__o21a_1)
     1    0.00                           _05_ (net)
                   0.07    0.00    1.41 ^ _13_/B1 (sky130_fd_sc_hd__a21oi_1)
                   0.05    0.07    1.47 v _13_/Y (sky130_fd_sc_hd__a21oi_1)
     1    0.00                           _06_ (net)
                   0.05    0.00    1.47 v _14_/B (sky130_fd_sc_hd__nor2_1)
                   0.13    0.13    1.60 ^ _14_/Y (sky130_fd_sc_hd__nor2_1)
     1    0.00                           _02_ (net)
                   0.13    0.00    1.60 ^ _22_/D (sky130_fd_sc_hd__dfrtp_1)
                                   1.60   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                   0.09    0.06    0.06 ^ clk (in)
     1    0.01                           clk (net)
                   0.09    0.00    0.06 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                   0.07    0.24    0.31 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                   0.07    0.00    0.31 ^ clkbuf_1_1__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                   0.04    0.20    0.51 ^ clkbuf_1_1__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     1    0.00                           clknet_1_1__leaf_clk (net)
                   0.04    0.00    0.51 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_1)
                          0.25    0.76   clock uncertainty
```

```
                          -0.03    0.73   clock reconvergence pessimism
                          -0.09    0.64   library hold time
                                   0.64   data required time
-----------------------------------------------------------------------------
                                   0.64   data required time
                                  -1.60   data arrival time
-----------------------------------------------------------------------------
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/30-mca/rcx_nom_sta.min.rpt ? part
==

                                0.96   slack (MET)

Startpoint: _21_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _21_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
Corner: Slowest

Fanout   Cap   Slew   Delay   Time   Description

                         0.00   0.00   clock clk (rise edge)
                         0.00   0.00   clock source latency
                  0.09   0.06   0.06 ^ clk (in)
     1   0.01   clk (net)
                  0.09   0.00   0.06 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.07   0.22   0.28 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2   0.02   clknet_0_clk (net)
                  0.07   0.00   0.28 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.05   0.19   0.47 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2   0.01   clknet_1_0__leaf_clk (net)
                  0.05   0.00   0.47 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                  0.17   0.68   1.14 ^ _21_/Q (sky130_fd_sc_hd__dfrtp_1)
     3   0.01   state[1] (net)
                  0.17   0.00   1.14 ^ _17_/A1 (sky130_fd_sc_hd__o211a_1)
                  0.07   0.33   1.47 ^ _17_/X (sky130_fd_sc_hd__o211a_1)
     1   0.00   _01_ (net)
                  0.07   0.00   1.47 ^ hold2/A (sky130_fd_sc_hd__dlygate4sd3_1)
                  0.09   0.98   2.45 ^ hold2/X (sky130_fd_sc_hd__dlygate4sd3_1)
     1   0.00   net8 (net)
                  0.09   0.00   2.45 ^ _21_/D (sky130_fd_sc_hd__dfrtp_1)
                                  2.45   data arrival time

                         0.00   0.00   clock clk (rise edge)
                         0.00   0.00   clock source latency
                  0.09   0.06   0.06 ^ clk (in)
     1   0.01   clk (net)
                  0.09   0.00   0.06 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.07   0.24   0.31 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2   0.02   clknet_0_clk (net)
                  0.07   0.00   0.31 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.05   0.21   0.52 ^ clkbuf_1_0__f_clk/X

```
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                  0.05    0.00    0.52 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                          0.25    0.77   clock uncertainty
                         -0.05    0.72   clock reconvergence pessimism
                         -0.08    0.63   library hold time
                                  0.63   data required time
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/30-mca/rcx_nom_sta.min.rpt ? part
==

```
--------------------------------------------------------------------------
                                0.63   data required time
                               -2.45   data arrival time
--------------------------------------------------------------------------
                                1.82   slack (MET)


Startpoint: _21_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _20_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
Corner: Slowest

Fanout     Cap    Slew   Delay    Time   Description
--------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                  0.09    0.06    0.06 ^ clk (in)
     1    0.01                          clk (net)
                  0.09    0.00    0.06 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.07    0.22    0.28 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                          clknet_0_clk (net)
                  0.07    0.00    0.28 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.05    0.19    0.47 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                          clknet_1_0__leaf_clk (net)
                  0.05    0.00    0.47 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                  0.14    0.75    1.22 v _21_/Q (sky130_fd_sc_hd__dfrtp_1)
     3    0.01                          state[1] (net)
                  0.14    0.00    1.22 v _15_/A2 (sky130_fd_sc_hd__a21oi_1)
                  0.13    0.23    1.45 ^ _15_/Y (sky130_fd_sc_hd__a21oi_1)
     1    0.00                          _07_ (net)
                  0.13    0.00    1.45 ^ hold4/A (sky130_fd_sc_hd__dlygate4sd3_1)
                  0.10    1.02    2.47 ^ hold4/X (sky130_fd_sc_hd__dlygate4sd3_1)
     1    0.00                          net10 (net)
                  0.10    0.00    2.47 ^ _16_/A2 (sky130_fd_sc_hd__o21ai_1)
                  0.08    0.12    2.59 v _16_/Y (sky130_fd_sc_hd__o21ai_1)
     1    0.00                          _00_ (net)
                  0.08    0.00    2.59 v _20_/D (sky130_fd_sc_hd__dfstp_1)
                                  2.59   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                  0.09    0.06    0.06 ^ clk (in)
```

```
     1    0.01                           clk (net)
                  0.09    0.00    0.06 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.07    0.24    0.31 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                  0.07    0.00    0.31 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/30-mca/rcx_nom_sta.min.rpt ? part
==

```
                  0.05    0.21    0.52 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                  0.05    0.00    0.52 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_1)
                          0.25    0.77   clock uncertainty
                         -0.05    0.72   clock reconvergence pessimism
                         -0.01    0.70   library hold time
                                  0.70   data required time
-----------------------------------------------------------------------------
                                  0.70   data required time
                                 -2.59   data arrival time
-----------------------------------------------------------------------------
                                  1.88   slack (MET)


Startpoint: rst_n (input port clocked by clk)
Endpoint: _21_ (removal check against rising-edge clock clk)
Path Group: asynchronous
Path Type: min
Corner: Slowest

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                  0.03    0.01    4.01 ^ rst_n (in)
     1    0.00                           rst_n (net)
                  0.03    0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                  0.29    0.27    4.29 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.01                           net4 (net)
                  0.29    0.00    4.29 ^ _21_/RESET_B (sky130_fd_sc_hd__dfrtp_1)
                                  4.29   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                  0.09    0.06    0.06 ^ clk (in)
     1    0.01                           clk (net)
                  0.09    0.00    0.06 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.07    0.24    0.31 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                  0.07    0.00    0.31 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.05    0.21    0.52 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
```

```
2    0.01                           clknet_1_0__leaf_clk (net)
             0.05    0.00    0.52 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                     0.25    0.77   clock uncertainty
                     0.00    0.77   clock reconvergence pessimism
                     0.72    1.48   library removal time
                             1.48   data required time
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/30-mca/rcx_nom_sta.min.rpt ? part
==

```
---------------------------------------------------------------------------
                                1.48   data required time
                               -4.29   data arrival time
---------------------------------------------------------------------------
                                2.80   slack (MET)


Startpoint: rst_n (input port clocked by clk)
Endpoint: _22_ (removal check against rising-edge clock clk)
Path Group: asynchronous
Path Type: min
Corner: Slowest

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                  0.03    0.01    4.01 ^ rst_n (in)
     1    0.00                           rst_n (net)
                  0.03    0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                  0.29    0.27    4.29 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.01                           net4 (net)
                  0.29    0.00    4.29 ^ _22_/RESET_B (sky130_fd_sc_hd__dfrtp_1)
                                  4.29   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                  0.09    0.06    0.06 ^ clk (in)
     1    0.01                           clk (net)
                  0.09    0.00    0.06 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.07    0.24    0.31 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                  0.07    0.00    0.31 ^ clkbuf_1_1__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.04    0.20    0.51 ^ clkbuf_1_1__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     1    0.00                           clknet_1_1__leaf_clk (net)
                  0.04    0.00    0.51 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_1)
                          0.25    0.76   clock uncertainty
                          0.00    0.76   clock reconvergence pessimism
                          0.72    1.48   library removal time
                                  1.48   data required time
---------------------------------------------------------------------------
                                1.48   data required time
```

```
                                  -4.29   data arrival time
-----------------------------------------------------------------------------
                                   2.81   slack (MET)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/30-mca/rcx_nom_sta.min.rpt ? part
==

Startpoint: rst_n (input port clocked by clk)
Endpoint: _20_ (removal check against rising-edge clock clk)
Path Group: asynchronous
Path Type: min
Corner: Slowest

```
Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                  0.03    0.01    4.01 ^ rst_n (in)
     1    0.00                           rst_n (net)
                  0.03    0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                  0.29    0.27    4.29 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.01                           net4 (net)
                  0.29    0.00    4.29 ^ _20_/SET_B (sky130_fd_sc_hd__dfstp_1)
                                  4.29   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                  0.09    0.06    0.06 ^ clk (in)
     1    0.01                           clk (net)
                  0.09    0.00    0.06 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.07    0.24    0.31 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                  0.07    0.00    0.31 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.05    0.21    0.52 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                  0.05    0.00    0.52 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_1)
                          0.25    0.77   clock uncertainty
                          0.00    0.77   clock reconvergence pessimism
                          0.27    1.04   library removal time
                                  1.04   data required time
-----------------------------------------------------------------------------
                                  1.04   data required time
                                 -4.29   data arrival time
-----------------------------------------------------------------------------
                                  3.25   slack (MET)


Startpoint: _22_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: pump_en (output port clocked by clk)
```

Path Group: clk
Path Type: min
Corner: Slowest

Fanout	Cap	Slew	Delay	Time	Description

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/30-mca/rcx_nom_sta.min.rpt ? part
==

```
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                 0.09     0.06    0.06 ^ clk (in)
     1    0.01                           clk (net)
                 0.09     0.00    0.06 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                 0.07     0.22    0.28 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                 0.07     0.00    0.28 ^ clkbuf_1_1__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                 0.04     0.18    0.46 ^ clkbuf_1_1__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     1    0.00                           clknet_1_1__leaf_clk (net)
                 0.04     0.00    0.46 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_1)
                 0.09     0.61    1.07 ^ _22_/Q (sky130_fd_sc_hd__dfrtp_1)
     2    0.00                           state[2] (net)
                 0.09     0.00    1.07 ^ _18_/A (sky130_fd_sc_hd__and2_1)
                 0.07     0.21    1.28 ^ _18_/X (sky130_fd_sc_hd__and2_1)
     1    0.00                           _08_ (net)
                 0.07     0.00    1.28 ^ _19_/A (sky130_fd_sc_hd__clkbuf_1)
                 0.08     0.15    1.43 ^ _19_/X (sky130_fd_sc_hd__clkbuf_1)
     1    0.00                           net6 (net)
                 0.08     0.00    1.43 ^ output6/A (sky130_fd_sc_hd__clkbuf_4)
                 0.18     0.32    1.75 ^ output6/X (sky130_fd_sc_hd__clkbuf_4)
     1    0.03                           pump_en (net)
                 0.18     0.00    1.75 ^ pump_en (out)
                                  1.75   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          0.25    0.25   clock uncertainty
                          0.00    0.25   clock reconvergence pessimism
                         -4.00   -3.75   output external delay
                                 -3.75   data required time
---------------------------------------------------------------------------
                                 -3.75   data required time
                                 -1.75   data arrival time
---------------------------------------------------------------------------
                                  5.50   slack (MET)


Startpoint: dry_run_fault (input port clocked by clk)
Endpoint: alarm (output port clocked by clk)
Path Group: clk
Path Type: min
```

Corner: Slowest

Fanout	Cap	Slew	Delay	Time	Description
			0.00	0.00	clock clk (rise edge)

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/30-mca/rcx_nom_sta.min.rpt ? part
==

```
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                 0.03     0.02    4.02 ^ dry_run_fault (in)
     1    0.00                           dry_run_fault (net)
                 0.03     0.00    4.02 ^ input1/A (sky130_fd_sc_hd__clkbuf_1)
                 0.11     0.15    4.17 ^ input1/X (sky130_fd_sc_hd__clkbuf_1)
     1    0.00                           net1 (net)
                 0.11     0.00    4.17 ^ _09_/B1 (sky130_fd_sc_hd__a21oi_2)
                 0.08     0.11    4.28 v _09_/Y (sky130_fd_sc_hd__a21oi_2)
     4    0.02                           _03_ (net)
                 0.08     0.00    4.28 v _10_/A (sky130_fd_sc_hd__inv_2)
                 0.06     0.09    4.37 ^ _10_/Y (sky130_fd_sc_hd__inv_2)
     2    0.01                           net5 (net)
                 0.06     0.00    4.37 ^ output5/A (sky130_fd_sc_hd__buf_2)
                 0.27     0.32    4.69 ^ output5/X (sky130_fd_sc_hd__buf_2)
     1    0.03                           alarm (net)
                 0.27     0.00    4.69 ^ alarm (out)
                          4.69    4.69   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          0.25    0.25   clock uncertainty
                          0.00    0.25   clock reconvergence pessimism
                         -4.00   -3.75   output external delay
                                 -3.75   data required time
------------------------------------------------------------------------------
                                 -3.75   data required time
                                 -4.69   data arrival time
------------------------------------------------------------------------------
                                  8.44   slack (MET)



====================== Typical Corner ===================================

Startpoint: _21_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _22_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
Corner: Typical

Fanout     Cap    Slew   Delay    Time   Description
------------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
```

```
             0.06    0.04    0.04 ^ clk (in)
1    0.01                           clk (net)
             0.06    0.00    0.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
```

```
openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/30-mca/rcx_nom_sta.min.rpt ? part
================================================================================

                  0.04    0.12    0.16 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                  0.04    0.00    0.16 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.03    0.11    0.27 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                  0.03    0.00    0.27 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                  0.10    0.34    0.61 ^ _21_/Q (sky130_fd_sc_hd__dfrtp_1)
     3    0.01                           state[1] (net)
                  0.10    0.00    0.61 ^ _12_/A1 (sky130_fd_sc_hd__o21a_1)
                  0.04    0.13    0.74 ^ _12_/X (sky130_fd_sc_hd__o21a_1)
     1    0.00                           _05_ (net)
                  0.04    0.00    0.74 ^ _13_/B1 (sky130_fd_sc_hd__a21oi_1)
                  0.03    0.03    0.77 v _13_/Y (sky130_fd_sc_hd__a21oi_1)
     1    0.00                           _06_ (net)
                  0.03    0.00    0.77 v _14_/B (sky130_fd_sc_hd__nor2_1)
                  0.08    0.07    0.84 ^ _14_/Y (sky130_fd_sc_hd__nor2_1)
     1    0.00                           _02_ (net)
                  0.08    0.00    0.84 ^ _22_/D (sky130_fd_sc_hd__dfrtp_1)
                                  0.84   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                  0.06    0.04    0.04 ^ clk (in)
     1    0.01                           clk (net)
                  0.06    0.00    0.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.04    0.14    0.18 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                  0.04    0.00    0.18 ^ clkbuf_1_1__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.03    0.11    0.29 ^ clkbuf_1_1__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     1    0.00                           clknet_1_1__leaf_clk (net)
                  0.03    0.00    0.29 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_1)
                          0.25    0.54   clock uncertainty
                         -0.02    0.52   clock reconvergence pessimism
                         -0.04    0.48   library hold time
                                  0.48   data required time
--------------------------------------------------------------------------------
                                  0.48   data required time
                                 -0.84   data arrival time
--------------------------------------------------------------------------------
                                  0.36   slack (MET)
```

Startpoint: _21_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _21_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
Corner: Typical

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/30-mca/rcx_nom_sta.min.rpt ? part
==

```
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                  0.06    0.04    0.04 ^ clk (in)
     1    0.01                          clk (net)
                  0.06    0.00    0.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.04    0.12    0.16 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                          clknet_0_clk (net)
                  0.04    0.00    0.16 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.03    0.11    0.27 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                          clknet_1_0__leaf_clk (net)
                  0.03    0.00    0.27 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                  0.10    0.34    0.61 ^ _21_/Q (sky130_fd_sc_hd__dfrtp_1)
     3    0.01                          state[1] (net)
                  0.10    0.00    0.61 ^ _17_/A1 (sky130_fd_sc_hd__o211a_1)
                  0.04    0.15    0.76 ^ _17_/X (sky130_fd_sc_hd__o211a_1)
     1    0.00                          _01_ (net)
                  0.04    0.00    0.76 ^ hold2/A (sky130_fd_sc_hd__dlygate4sd3_1)
                  0.05    0.50    1.26 ^ hold2/X (sky130_fd_sc_hd__dlygate4sd3_1)
     1    0.00                          net8 (net)
                  0.05    0.00    1.26 ^ _21_/D (sky130_fd_sc_hd__dfrtp_1)
                                  1.26   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                  0.06    0.04    0.04 ^ clk (in)
     1    0.01                          clk (net)
                  0.06    0.00    0.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.04    0.14    0.18 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                          clknet_0_clk (net)
                  0.04    0.00    0.18 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.03    0.12    0.30 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                          clknet_1_0__leaf_clk (net)
                  0.03    0.00    0.30 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                          0.25    0.55   clock uncertainty
                         -0.03    0.52   clock reconvergence pessimism
                         -0.04    0.48   library hold time
                                  0.48   data required time
---------------------------------------------------------------------------
                                  0.48   data required time
```

```
                                  -1.26   data arrival time
-----------------------------------------------------------------------------
                                   0.78   slack (MET)


Startpoint: _21_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _20_ (rising edge-triggered flip-flop clocked by clk)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/30-mca/rcx_nom_sta.min.rpt ? part
==

Path Group: clk
Path Type: min
Corner: Typical

Fanout	Cap	Slew	Delay	Time	Description
			0.00	0.00	clock clk (rise edge)
			0.00	0.00	clock source latency
		0.06	0.04	0.04 ^	clk (in)
1	0.01				clk (net)
		0.06	0.00	0.04 ^	clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
		0.04	0.12	0.16 ^	clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
2	0.02				clknet_0_clk (net)
		0.04	0.00	0.16 ^	clkbuf_1_0__f_clk/A (sky130_fd_sc_hd__clkbuf_16)
		0.03	0.11	0.27 ^	clkbuf_1_0__f_clk/X (sky130_fd_sc_hd__clkbuf_16)
2	0.01				clknet_1_0__leaf_clk (net)
		0.03	0.00	0.27 ^	_21_/CLK (sky130_fd_sc_hd__dfrtp_1)
		0.07	0.37	0.63 v	_21_/Q (sky130_fd_sc_hd__dfrtp_1)
3	0.01				state[1] (net)
		0.07	0.00	0.63 v	_15_/A2 (sky130_fd_sc_hd__a21oi_1)
		0.08	0.13	0.76 ^	_15_/Y (sky130_fd_sc_hd__a21oi_1)
1	0.00				_07_ (net)
		0.08	0.00	0.76 ^	hold4/A (sky130_fd_sc_hd__dlygate4sd3_1)
		0.06	0.52	1.29 ^	hold4/X (sky130_fd_sc_hd__dlygate4sd3_1)
1	0.00				net10 (net)
		0.06	0.00	1.29 ^	_16_/A2 (sky130_fd_sc_hd__o21ai_1)
		0.03	0.06	1.34 v	_16_/Y (sky130_fd_sc_hd__o21ai_1)
1	0.00				_00_ (net)
		0.03	0.00	1.34 v	_20_/D (sky130_fd_sc_hd__dfstp_1)
				1.34	data arrival time
			0.00	0.00	clock clk (rise edge)
			0.00	0.00	clock source latency
		0.06	0.04	0.04 ^	clk (in)
1	0.01				clk (net)
		0.06	0.00	0.04 ^	clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
		0.04	0.14	0.18 ^	clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
2	0.02				clknet_0_clk (net)
		0.04	0.00	0.18 ^	clkbuf_1_0__f_clk/A (sky130_fd_sc_hd__clkbuf_16)
		0.03	0.12	0.30 ^	clkbuf_1_0__f_clk/X (sky130_fd_sc_hd__clkbuf_16)
2	0.01				clknet_1_0__leaf_clk (net)

```
                  0.03    0.00    0.30 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_1)
                          0.25    0.55   clock uncertainty
                         -0.03    0.52   clock reconvergence pessimism
                         -0.01    0.50   library hold time
                                  0.50   data required time
-----------------------------------------------------------------------------
                                  0.50   data required time
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/30-mca/rcx_nom_sta.min.rpt ? part
==

                                -1.34   data arrival time

                                 0.84   slack (MET)

Startpoint: rst_n (input port clocked by clk)
Endpoint: _21_ (removal check against rising-edge clock clk)
Path Group: asynchronous
Path Type: min
Corner: Typical

Fanout   Cap   Slew   Delay   Time   Description

                          0.00   0.00   clock clk (rise edge)
                          0.00   0.00   clock network delay (propagated)
                          4.00   4.00 ^ input external delay
                 0.02   0.01   4.01 ^ rst_n (in)
     1   0.00   rst_n (net)
                 0.02   0.00   4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                 0.19   0.17   4.18 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3   0.02   net4 (net)
                 0.19   0.00   4.18 ^ _21_/RESET_B (sky130_fd_sc_hd__dfrtp_1)
                                  4.18   data arrival time

                          0.00   0.00   clock clk (rise edge)
                          0.00   0.00   clock source latency
                 0.06   0.04   0.04 ^ clk (in)
     1   0.01   clk (net)
                 0.06   0.00   0.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                 0.04   0.14   0.18 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2   0.02   clknet_0_clk (net)
                 0.04   0.00   0.18 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                 0.03   0.12   0.30 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2   0.01   clknet_1_0__leaf_clk (net)
                 0.03   0.00   0.30 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                          0.25   0.55   clock uncertainty
                          0.00   0.55   clock reconvergence pessimism
                          0.36   0.90   library removal time
                                  0.90   data required time

                                  0.90   data required time
                                 -4.18   data arrival time

3.27   slack (MET)

Startpoint: rst_n (input port clocked by clk)
Endpoint: _22_ (removal check against rising-edge clock clk)

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/30-mca/rcx_nom_sta.min.rpt ? part
==

Path Group: asynchronous
Path Type: min
Corner: Typical

```
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                   0.02   0.01    4.01 ^ rst_n (in)
     1    0.00                           rst_n (net)
                   0.02   0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                   0.19   0.17    4.18 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.02                           net4 (net)
                   0.19   0.00    4.18 ^ _22_/RESET_B (sky130_fd_sc_hd__dfrtp_1)
                                  4.18   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                   0.06   0.04    0.04 ^ clk (in)
     1    0.01                           clk (net)
                   0.06   0.00    0.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                   0.04   0.14    0.18 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                   0.04   0.00    0.18 ^ clkbuf_1_1__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                   0.03   0.11    0.29 ^ clkbuf_1_1__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     1    0.00                           clknet_1_1__leaf_clk (net)
                   0.03   0.00    0.29 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_1)
                          0.25    0.54   clock uncertainty
                          0.00    0.54   clock reconvergence pessimism
                          0.36    0.90   library removal time
                                  0.90   data required time
---------------------------------------------------------------------------
                                  0.90   data required time
                                 -4.18   data arrival time
---------------------------------------------------------------------------
                                  3.28   slack (MET)


Startpoint: rst_n (input port clocked by clk)
Endpoint: _20_ (removal check against rising-edge clock clk)
Path Group: asynchronous
Path Type: min
```

Corner: Typical

Fanout	Cap	Slew	Delay	Time	Description
			0.00	0.00	clock clk (rise edge)

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/30-mca/rcx_nom_sta.min.rpt ? part
==

```
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                 0.02     0.01    4.01 ^ rst_n (in)
     1    0.00                           rst_n (net)
                 0.02     0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                 0.19     0.17    4.18 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.02                           net4 (net)
                 0.19     0.00    4.18 ^ _20_/SET_B (sky130_fd_sc_hd__dfstp_1)
                                  4.18   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                 0.06     0.04    0.04 ^ clk (in)
     1    0.01                           clk (net)
                 0.06     0.00    0.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                 0.04     0.14    0.18 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                 0.04     0.00    0.18 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                 0.03     0.12    0.30 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                 0.03     0.00    0.30 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_1)
                          0.25    0.55   clock uncertainty
                          0.00    0.55   clock reconvergence pessimism
                          0.14    0.68   library removal time
                                  0.68   data required time
--------------------------------------------------------------------------------
                                  0.68   data required time
                                 -4.18   data arrival time
--------------------------------------------------------------------------------
                                  3.49   slack (MET)


Startpoint: _22_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: pump_en (output port clocked by clk)
Path Group: clk
Path Type: min
Corner: Typical

Fanout     Cap    Slew   Delay    Time   Description
--------------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                 0.06     0.04    0.04 ^ clk (in)
```

```
     1    0.01                           clk (net)
                  0.06    0.00    0.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.04    0.12    0.16 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                  0.04    0.00    0.16 ^ clkbuf_1_1__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/30-mca/rcx_nom_sta.min.rpt ? part
==

```
                  0.03    0.10    0.26 ^ clkbuf_1_1__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     1    0.00                           clknet_1_1__leaf_clk (net)
                  0.03    0.00    0.26 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_1)
                  0.06    0.30    0.57 ^ _22_/Q (sky130_fd_sc_hd__dfrtp_1)
     2    0.00                           state[2] (net)
                  0.06    0.00    0.57 ^ _18_/A (sky130_fd_sc_hd__and2_1)
                  0.04    0.11    0.67 ^ _18_/X (sky130_fd_sc_hd__and2_1)
     1    0.00                           _08_ (net)
                  0.04    0.00    0.67 ^ _19_/A (sky130_fd_sc_hd__clkbuf_1)
                  0.05    0.08    0.75 ^ _19_/X (sky130_fd_sc_hd__clkbuf_1)
     1    0.00                           net6 (net)
                  0.05    0.00    0.75 ^ output6/A (sky130_fd_sc_hd__clkbuf_4)
                  0.11    0.18    0.93 ^ output6/X (sky130_fd_sc_hd__clkbuf_4)
     1    0.03                           pump_en (net)
                  0.11    0.00    0.93 ^ pump_en (out)
                                  0.93   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          0.25    0.25   clock uncertainty
                          0.00    0.25   clock reconvergence pessimism
                         -4.00   -3.75   output external delay
                                 -3.75   data required time
-----------------------------------------------------------------------------
                                 -3.75   data required time
                                 -0.93   data arrival time
-----------------------------------------------------------------------------
                                  4.68   slack (MET)


Startpoint: dry_run_fault (input port clocked by clk)
Endpoint: alarm (output port clocked by clk)
Path Group: clk
Path Type: min
Corner: Typical

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                  0.02    0.01    4.01 ^ dry_run_fault (in)
     1    0.00                           dry_run_fault (net)
                  0.02    0.00    4.01 ^ input1/A (sky130_fd_sc_hd__clkbuf_1)
```

```
             0.07    0.09    4.10 ^ input1/X (sky130_fd_sc_hd__clkbuf_1)
1    0.01                           net1 (net)
             0.07    0.00    4.10 ^ _09_/B1 (sky130_fd_sc_hd__a21oi_2)
             0.05    0.06    4.15 v _09_/Y (sky130_fd_sc_hd__a21oi_2)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/30-mca/rcx_nom_sta.min.rpt ? part
==

```
     4    0.02                           _03_ (net)
                  0.05    0.00    4.15 v _10_/A (sky130_fd_sc_hd__inv_2)
                  0.04    0.05    4.20 ^ _10_/Y (sky130_fd_sc_hd__inv_2)
     2    0.01                           net5 (net)
                  0.04    0.00    4.20 ^ output5/A (sky130_fd_sc_hd__buf_2)
                  0.17    0.19    4.39 ^ output5/X (sky130_fd_sc_hd__buf_2)
     1    0.03                           alarm (net)
                  0.17    0.00    4.39 ^ alarm (out)
                                  4.39   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          0.25    0.25   clock uncertainty
                          0.00    0.25   clock reconvergence pessimism
                         -4.00   -3.75   output external delay
                                 -3.75   data required time
---------------------------------------------------------------------------
                                 -3.75   data required time
                                 -4.39   data arrival time
---------------------------------------------------------------------------
                                  8.14   slack (MET)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/30-mca/rcx_nom_sta.power.rpt
==========

==========
 report_power
==========
======================= Fastest Corner ===================================

Group	Internal Power	Switching Power	Leakage Power	Total Power (Watts)	
Sequential	7.62e-06	2.71e-07	1.47e-08	7.91e-06	16.5%
Combinational	3.15e-05	8.40e-06	5.48e-10	3.99e-05	83.5%
Macro	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Pad	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Total	3.91e-05	8.67e-06	1.52e-08	4.78e-05	100.0%
	81.8%	18.1%	0.0%		

======================= Slowest Corner ===================================

Group	Internal Power	Switching Power	Leakage Power	Total Power (Watts)	
Sequential	5.16e-06	1.75e-07	5.86e-08	5.39e-06	16.2%
Combinational	2.22e-05	5.55e-06	1.61e-07	2.79e-05	83.8%
Macro	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Pad	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Total	2.74e-05	5.72e-06	2.20e-07	3.33e-05	100.0%
	82.2%	17.2%	0.7%		

======================= Typical Corner ===================================

Group	Internal Power	Switching Power	Leakage Power	Total Power (Watts)	
Sequential	6.60e-06	2.26e-07	1.47e-08	6.84e-06	16.6%
Combinational	2.74e-05	7.08e-06	2.85e-10	3.45e-05	83.4%
Macro	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Pad	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Total	3.40e-05	7.31e-06	1.50e-08	4.13e-05	100.0%
	82.3%	17.7%	0.0%		

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/30-mca/rcx_nom_sta.skew.rpt
==

==
report_clock_skew
==
Clock clk
Latency   CRPR   Skew
_20_/CLK ^
   0.52
_22_/CLK ^
   0.46   -0.03   0.03

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/30-mca/rcx_nom_sta.summary.rpt
==

==
report_tns
==
tns 0.00

==
report_wns
==
wns 0.00

==
report_worst_slack -max (Setup)
==
worst slack 10.05

==
report_worst_slack -min (Hold)
==
worst slack 0.14

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/31-rcx_sta.checks.rpt ? part 1
==

```
===========================================================================
report_checks -unconstrained
===========================================================================
======================== Typical Corner ===================================

Startpoint: rst_n (input port clocked by clk)
Endpoint: _20_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
Path Type: max

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                0.02      0.01    4.01 ^ rst_n (in)
     1    0.00                           rst_n (net)
                0.02      0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                0.19      0.18    4.19 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.02                           net4 (net)
                0.19      0.00    4.19 ^ _20_/SET_B (sky130_fd_sc_hd__dfstp_1)
                                  4.19   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock source latency
                0.06      0.04   20.04 ^ clk (in)
     1    0.01                           clk (net)
                0.06      0.00   20.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                0.04      0.12   20.16 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                0.04      0.00   20.16 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                0.03      0.11   20.27 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                0.03      0.00   20.27 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_1)
                         -0.25   20.02   clock uncertainty
                          0.00   20.02   clock reconvergence pessimism
                          0.12   20.14   library recovery time
                                 20.14   data required time
---------------------------------------------------------------------------
                                 20.14   data required time
                                 -4.19   data arrival time
---------------------------------------------------------------------------
                                 15.94   slack (MET)
```

```
Startpoint: level_full (input port clocked by clk)
Endpoint: pump_en (output port clocked by clk)
Path Group: clk
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/31-rcx_sta.checks.rpt ? part 2
==========

Path Type: max

```
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 v input external delay
                   0.01   0.01    4.01 v level_full (in)
     1    0.00                           level_full (net)
                   0.01   0.00    4.01 v input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
                   0.09   0.18    4.19 v input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
     3    0.02                           net2 (net)
                   0.09   0.00    4.19 v _09_/A2 (sky130_fd_sc_hd__a21oi_2)
                   0.23   0.26    4.45 ^ _09_/Y (sky130_fd_sc_hd__a21oi_2)
     4    0.02                           _03_ (net)
                   0.23   0.00    4.45 ^ _18_/B (sky130_fd_sc_hd__and2_1)
                   0.05   0.16    4.62 ^ _18_/X (sky130_fd_sc_hd__and2_1)
     1    0.00                           _08_ (net)
                   0.05   0.00    4.62 ^ _19_/A (sky130_fd_sc_hd__clkbuf_1)
                   0.05   0.09    4.71 ^ _19_/X (sky130_fd_sc_hd__clkbuf_1)
     1    0.00                           net6 (net)
                   0.05   0.00    4.71 ^ output6/A (sky130_fd_sc_hd__clkbuf_4)
                   0.11   0.20    4.90 ^ output6/X (sky130_fd_sc_hd__clkbuf_4)
     1    0.03                           pump_en (net)
                   0.11   0.00    4.90 ^ pump_en (out)
                                  4.90   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock network delay (propagated)
                         -0.25   19.75   clock uncertainty
                          0.00   19.75   clock reconvergence pessimism
                         -4.00   15.75   output external delay
                                 15.75   data required time
---------------------------------------------------------------------------
                                 15.75   data required time
                                 -4.90   data arrival time
---------------------------------------------------------------------------
                                 10.85   slack (MET)
```

==========
report_checks --slack_max -0.01
==========

========== Typical Corner ==========

No paths found.

```
================================================================================


================================================================================
 report_check_types -max_slew -max_cap -max_fanout -violators
================================================================================
======================== Typical Corner ====================================



================================================================================
report_parasitic_annotation -report_unannotated
================================================================================
Found 0 unannotated drivers.
Found 0 partially unannotated drivers.

================================================================================
max slew violation count 0
max fanout violation count 0
max cap violation count 0
================================================================================

================================================================================
check_setup -verbose -unconstrained_endpoints -multiple_clock -no_clock -no_input_delay
-loops -generated_clocks
================================================================================
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/31-rcx_sta.max.rpt ? part 1
==========

```
============================================================================
report_checks -path_delay max (Setup)
============================================================================
======================== Typical Corner ===================================

Startpoint: level_full (input port clocked by clk)
Endpoint: pump_en (output port clocked by clk)
Path Group: clk
Path Type: max

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 v input external delay
                   0.01   0.01    4.01 v level_full (in)
     1    0.00                           level_full (net)
                   0.01   0.00    4.01 v input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
                   0.09   0.18    4.19 v input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
     3    0.02                           net2 (net)
                   0.09   0.00    4.19 v _09_/A2 (sky130_fd_sc_hd__a21oi_2)
                   0.23   0.26    4.45 ^ _09_/Y (sky130_fd_sc_hd__a21oi_2)
     4    0.02                           _03_ (net)
                   0.23   0.00    4.45 ^ _18_/B (sky130_fd_sc_hd__and2_1)
                   0.05   0.16    4.62 ^ _18_/X (sky130_fd_sc_hd__and2_1)
     1    0.00                           _08_ (net)
                   0.05   0.00    4.62 ^ _19_/A (sky130_fd_sc_hd__clkbuf_1)
                   0.05   0.09    4.71 ^ _19_/X (sky130_fd_sc_hd__clkbuf_1)
     1    0.00                           net6 (net)
                   0.05   0.00    4.71 ^ output6/A (sky130_fd_sc_hd__clkbuf_4)
                   0.11   0.20    4.90 ^ output6/X (sky130_fd_sc_hd__clkbuf_4)
     1    0.03                           pump_en (net)
                   0.11   0.00    4.90 ^ pump_en (out)
                          4.90   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock network delay (propagated)
                         -0.25   19.75   clock uncertainty
                          0.00   19.75   clock reconvergence pessimism
                         -4.00   15.75   output external delay
                                 15.75   data required time
---------------------------------------------------------------------------
                                 15.75   data required time
                                 -4.90   data arrival time
---------------------------------------------------------------------------
```

10.85 slack (MET)

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/31-rcx_sta.max.rpt ? part 2
==========

Startpoint: level_full (input port clocked by clk)
Endpoint: alarm (output port clocked by clk)
Path Group: clk
Path Type: max

Fanout	Cap	Slew	Delay	Time		Description
			0.00	0.00		clock clk (rise edge)
			0.00	0.00		clock network delay (propagated)
			4.00	4.00	v	input external delay
		0.01	0.01	4.01	v	level_full (in)
1	0.00					level_full (net)
		0.01	0.00	4.01	v	input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
		0.09	0.18	4.19	v	input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
3	0.02					net2 (net)
		0.09	0.00	4.19	v	_09_/A2 (sky130_fd_sc_hd__a21oi_2)
		0.23	0.26	4.45	^	_09_/Y (sky130_fd_sc_hd__a21oi_2)
4	0.02					_03_ (net)
		0.23	0.00	4.45	^	_10_/A (sky130_fd_sc_hd__inv_2)
		0.05	0.06	4.52	v	_10_/Y (sky130_fd_sc_hd__inv_2)
2	0.01					net5 (net)
		0.05	0.00	4.52	v	output5/A (sky130_fd_sc_hd__buf_2)
		0.09	0.19	4.71	v	output5/X (sky130_fd_sc_hd__buf_2)
1	0.03					alarm (net)
		0.09	0.00	4.71	v	alarm (out)
				4.71		data arrival time
			20.00	20.00		clock clk (rise edge)
			0.00	20.00		clock network delay (propagated)
			-0.25	19.75		clock uncertainty
			0.00	19.75		clock reconvergence pessimism
			-4.00	15.75		output external delay
				15.75		data required time
				15.75		data required time
				-4.71		data arrival time
				11.04		slack (MET)

Startpoint: level_full (input port clocked by clk)
Endpoint: _21_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

Fanout	Cap	Slew	Delay	Time	Description
			0.00	0.00	clock clk (rise edge)

```
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 v input external delay
                 0.01     0.01    4.01 v level_full (in)
     1    0.00                           level_full (net)
                 0.01     0.00    4.01 v input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
                 0.09     0.18    4.19 v input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
     3    0.02                           net2 (net)
                 0.09     0.00    4.19 v _09_/A2 (sky130_fd_sc_hd__a21oi_2)
                 0.23     0.26    4.45 ^ _09_/Y (sky130_fd_sc_hd__a21oi_2)
     4    0.02                           _03_ (net)
                 0.23     0.00    4.45 ^ _17_/B1 (sky130_fd_sc_hd__o211a_1)
                 0.04     0.19    4.64 ^ _17_/X (sky130_fd_sc_hd__o211a_1)
     1    0.00                           _01_ (net)
                 0.04     0.00    4.64 ^ hold2/A (sky130_fd_sc_hd__dlygate4sd3_1)
                 0.05     0.56    5.20 ^ hold2/X (sky130_fd_sc_hd__dlygate4sd3_1)
     1    0.00                           net8 (net)
                 0.05     0.00    5.20 ^ _21_/D (sky130_fd_sc_hd__dfrtp_1)
                           0.00    5.20   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock source latency
                 0.06     0.04   20.04 ^ clk (in)
     1    0.01                           clk (net)
                 0.06     0.00   20.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                 0.04     0.12   20.16 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                 0.04     0.00   20.16 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                 0.03     0.11   20.27 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                 0.03     0.00   20.27 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                         -0.25   20.02   clock uncertainty
                          0.00   20.02   clock reconvergence pessimism
                         -0.06   19.95   library setup time
                                 19.95   data required time
-----------------------------------------------------------------------------
                                 19.95   data required time
                                 -5.20   data arrival time
-----------------------------------------------------------------------------
                                 14.76   slack (MET)


Startpoint: level_full (input port clocked by clk)
Endpoint: _20_ (rising edge-triggered flip-flop clocked by clk)
```

Path Group: clk
Path Type: max

Fanout	Cap	Slew	Delay	Time	Description

```
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                 0.02     0.01    4.01 ^ level_full (in)
     1    0.00                           level_full (net)
                 0.02     0.00    4.01 ^ input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
                 0.19     0.19    4.21 ^ input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
     3    0.02                           net2 (net)
                 0.19     0.00    4.21 ^ _11_/A (sky130_fd_sc_hd__inv_2)
                 0.05     0.06    4.27 v _11_/Y (sky130_fd_sc_hd__inv_2)
     2    0.01                           _04_ (net)
                 0.05     0.00    4.27 v _15_/A1 (sky130_fd_sc_hd__a21oi_1)
                 0.09     0.12    4.38 ^ _15_/Y (sky130_fd_sc_hd__a21oi_1)
     1    0.00                           _07_ (net)
                 0.09     0.00    4.38 ^ hold4/A (sky130_fd_sc_hd__dlygate4sd3_1)
                 0.06     0.58    4.96 ^ hold4/X (sky130_fd_sc_hd__dlygate4sd3_1)
     1    0.00                           net10 (net)
                 0.06     0.00    4.96 ^ _16_/A2 (sky130_fd_sc_hd__o21ai_1)
                 0.07     0.06    5.03 v _16_/Y (sky130_fd_sc_hd__o21ai_1)
     1    0.00                           _00_ (net)
                 0.07     0.00    5.03 v _20_/D (sky130_fd_sc_hd__dfstp_1)
                           5.03    5.03   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock source latency
                 0.06     0.04   20.04 ^ clk (in)
     1    0.01                           clk (net)
                 0.06     0.00   20.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                 0.04     0.12   20.16 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                 0.04     0.00   20.16 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                 0.03     0.11   20.27 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                 0.03     0.00   20.27 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_1)
                         -0.25   20.02   clock uncertainty
                          0.00   20.02   clock reconvergence pessimism
                         -0.08   19.93   library setup time
                                 19.93   data required time
-----------------------------------------------------------------------------
                                 19.93   data required time
                                 -5.03   data arrival time
-----------------------------------------------------------------------------
                                 14.91   slack (MET)
```

Startpoint: level_full (input port clocked by clk)
Endpoint: _22_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk

```
================================================================================

Path Type: max

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 v input external delay
                   0.01   0.01    4.01 v level_full (in)
     1    0.00                           level_full (net)
                   0.01   0.00    4.01 v input2/A (sky130_fd_sc_hd__dlymetal6s2s_1)
                   0.09   0.18    4.19 v input2/X (sky130_fd_sc_hd__dlymetal6s2s_1)
     3    0.02                           net2 (net)
                   0.09   0.00    4.19 v _09_/A2 (sky130_fd_sc_hd__a21oi_2)
                   0.23   0.26    4.45 ^ _09_/Y (sky130_fd_sc_hd__a21oi_2)
     4    0.02                           _03_ (net)
                   0.23   0.00    4.45 ^ _10_/A (sky130_fd_sc_hd__inv_2)
                   0.05   0.06    4.52 v _10_/Y (sky130_fd_sc_hd__inv_2)
     2    0.01                           net5 (net)
                   0.05   0.00    4.52 v _14_/A (sky130_fd_sc_hd__nor2_1)
                   0.08   0.11    4.62 ^ _14_/Y (sky130_fd_sc_hd__nor2_1)
     1    0.00                           _02_ (net)
                   0.08   0.00    4.62 ^ _22_/D (sky130_fd_sc_hd__dfrtp_1)
                                  4.62   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock source latency
                   0.06   0.04   20.04 ^ clk (in)
     1    0.01                           clk (net)
                   0.06   0.00   20.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                   0.04   0.12   20.16 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                   0.04   0.00   20.16 ^ clkbuf_1_1__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                   0.03   0.10   20.26 ^ clkbuf_1_1__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     1    0.00                           clknet_1_1__leaf_clk (net)
                   0.03   0.00   20.26 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_1)
                         -0.25   20.01   clock uncertainty
                          0.00   20.01   clock reconvergence pessimism
                         -0.07   19.94   library setup time
                                 19.94   data required time
---------------------------------------------------------------------------
                                 19.94   data required time
                                 -4.62   data arrival time
---------------------------------------------------------------------------
```

15.32   slack (MET)

Startpoint: rst_n (input port clocked by clk)
Endpoint: _20_ (recovery check against rising-edge clock clk)

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/31-rcx_sta.max.rpt ? part 6
==========

Path Group: asynchronous
Path Type: max

```
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                 0.02     0.01    4.01 ^ rst_n (in)
     1    0.00                           rst_n (net)
                 0.02     0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                 0.19     0.18    4.19 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.02                           net4 (net)
                 0.19     0.00    4.19 ^ _20_/SET_B (sky130_fd_sc_hd__dfstp_1)
                                  4.19   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock source latency
                 0.06     0.04   20.04 ^ clk (in)
     1    0.01                           clk (net)
                 0.06     0.00   20.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                 0.04     0.12   20.16 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                 0.04     0.00   20.16 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                 0.03     0.11   20.27 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                 0.03     0.00   20.27 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_1)
                         -0.25   20.02   clock uncertainty
                          0.00   20.02   clock reconvergence pessimism
                          0.12   20.14   library recovery time
                                 20.14   data required time
---------------------------------------------------------------------------
                                 20.14   data required time
                                 -4.19   data arrival time
---------------------------------------------------------------------------
                                 15.94   slack (MET)


Startpoint: rst_n (input port clocked by clk)
Endpoint: _22_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
Path Type: max
```

```
Fanout     Cap     Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
```

==========

```
                  0.02    0.01    4.01 ^ rst_n (in)
     1    0.00                           rst_n (net)
                  0.02    0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                  0.19    0.18    4.19 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.02                           net4 (net)
                  0.19    0.00    4.19 ^ _22_/RESET_B (sky130_fd_sc_hd__dfrtp_1)
                                  4.19   data arrival time

                         20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock source latency
                  0.06    0.04   20.04 ^ clk (in)
     1    0.01                           clk (net)
                  0.06    0.00   20.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.04    0.12   20.16 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                  0.04    0.00   20.16 ^ clkbuf_1_1__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.03    0.10   20.26 ^ clkbuf_1_1__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     1    0.00                           clknet_1_1__leaf_clk (net)
                  0.03    0.00   20.26 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_1)
                         -0.25   20.01   clock uncertainty
                          0.00   20.01   clock reconvergence pessimism
                          0.19   20.20   library recovery time
                                 20.20   data required time
-------------------------------------------------------------------------------
                                 20.20   data required time
                                 -4.19   data arrival time
-------------------------------------------------------------------------------
                                 16.01   slack (MET)


Startpoint: rst_n (input port clocked by clk)
Endpoint: _21_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
Path Type: max

Fanout     Cap    Slew   Delay    Time   Description
-------------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                  0.02    0.01    4.01 ^ rst_n (in)
     1    0.00                           rst_n (net)
                  0.02    0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
```

```
              0.19    0.18    4.19 ^ input4/X (sky130_fd_sc_hd__buf_1)
3    0.02                             net4 (net)
              0.19    0.00    4.19 ^ _21_/RESET_B (sky130_fd_sc_hd__dfrtp_1)
                              4.19   data arrival time
```

```
                          20.00   20.00   clock clk (rise edge)
                           0.00   20.00   clock source latency
                  0.06     0.04   20.04 ^ clk (in)
     1    0.01                            clk (net)
                  0.06     0.00   20.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.04     0.12   20.16 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                            clknet_0_clk (net)
                  0.04     0.00   20.16 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.03     0.11   20.27 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                            clknet_1_0__leaf_clk (net)
                  0.03     0.00   20.27 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                          -0.25   20.02   clock uncertainty
                           0.00   20.02   clock reconvergence pessimism
                           0.19   20.21   library recovery time
                                  20.21   data required time
-------------------------------------------------------------------------------
                                  20.21   data required time
                                  -4.19   data arrival time
-------------------------------------------------------------------------------
                                  16.01   slack (MET)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/31-rcx_sta.min.rpt ? part 1
==========

```
============================================================================
report_checks -path_delay min (Hold)
============================================================================
======================== Typical Corner ===================================

Startpoint: _21_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _22_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                   0.06    0.04    0.04 ^ clk (in)
     1    0.01                           clk (net)
                   0.06    0.00    0.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                   0.04    0.12    0.16 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                   0.04    0.00    0.16 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                   0.03    0.11    0.27 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                   0.03    0.00    0.27 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                   0.10    0.34    0.61 ^ _21_/Q (sky130_fd_sc_hd__dfrtp_1)
     3    0.01                           state[1] (net)
                   0.10    0.00    0.61 ^ _12_/A1 (sky130_fd_sc_hd__o21a_1)
                   0.04    0.13    0.74 ^ _12_/X (sky130_fd_sc_hd__o21a_1)
     1    0.00                           _05_ (net)
                   0.04    0.00    0.74 ^ _13_/B1 (sky130_fd_sc_hd__a21oi_1)
                   0.03    0.03    0.77 v _13_/Y (sky130_fd_sc_hd__a21oi_1)
     1    0.00                           _06_ (net)
                   0.03    0.00    0.77 v _14_/B (sky130_fd_sc_hd__nor2_1)
                   0.08    0.07    0.84 ^ _14_/Y (sky130_fd_sc_hd__nor2_1)
     1    0.00                           _02_ (net)
                   0.08    0.00    0.84 ^ _22_/D (sky130_fd_sc_hd__dfrtp_1)
                                   0.84   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                   0.06    0.04    0.04 ^ clk (in)
     1    0.01                           clk (net)
                   0.06    0.00    0.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                   0.04    0.14    0.18 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
```

```
     2    0.02                           clknet_0_clk (net)
                  0.04    0.00    0.18 ^ clkbuf_1_1__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.03    0.11    0.29 ^ clkbuf_1_1__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     1    0.00                           clknet_1_1__leaf_clk (net)
                  0.03    0.00    0.29 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_1)
```

```
                          0.25    0.54   clock uncertainty
                         -0.02    0.52   clock reconvergence pessimism
                         -0.04    0.48   library hold time
                                  0.48   data required time
-----------------------------------------------------------------------------
                                  0.48   data required time
                                 -0.84   data arrival time
-----------------------------------------------------------------------------
                                  0.36   slack (MET)


Startpoint: _21_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _21_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                 0.06     0.04    0.04 ^ clk (in)
     1    0.01                           clk (net)
                 0.06     0.00    0.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                 0.04     0.12    0.16 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                 0.04     0.00    0.16 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                 0.03     0.11    0.27 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                 0.03     0.00    0.27 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                 0.10     0.34    0.61 ^ _21_/Q (sky130_fd_sc_hd__dfrtp_1)
     3    0.01                           state[1] (net)
                 0.10     0.00    0.61 ^ _17_/A1 (sky130_fd_sc_hd__o211a_1)
                 0.04     0.15    0.76 ^ _17_/X (sky130_fd_sc_hd__o211a_1)
     1    0.00                           _01_ (net)
                 0.04     0.00    0.76 ^ hold2/A (sky130_fd_sc_hd__dlygate4sd3_1)
                 0.05     0.50    1.26 ^ hold2/X (sky130_fd_sc_hd__dlygate4sd3_1)
     1    0.00                           net8 (net)
                 0.05     0.00    1.26 ^ _21_/D (sky130_fd_sc_hd__dfrtp_1)
                                  1.26   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                 0.06     0.04    0.04 ^ clk (in)
```

```
     1    0.01                           clk (net)
                  0.06    0.00    0.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.04    0.14    0.18 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                  0.04    0.00    0.18 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
```

==========

```
                  0.03    0.12    0.30 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                  0.03    0.00    0.30 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                          0.25    0.55   clock uncertainty
                         -0.03    0.52   clock reconvergence pessimism
                         -0.04    0.48   library hold time
                                  0.48   data required time
-----------------------------------------------------------------------------
                                  0.48   data required time
                                 -1.26   data arrival time
-----------------------------------------------------------------------------
                                  0.78   slack (MET)


Startpoint: _21_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _20_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                  0.06    0.04    0.04 ^ clk (in)
     1    0.01                           clk (net)
                  0.06    0.00    0.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.04    0.12    0.16 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                  0.04    0.00    0.16 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.03    0.11    0.27 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                  0.03    0.00    0.27 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                  0.07    0.37    0.63 v _21_/Q (sky130_fd_sc_hd__dfrtp_1)
     3    0.01                           state[1] (net)
                  0.07    0.00    0.63 v _15_/A2 (sky130_fd_sc_hd__a21oi_1)
                  0.08    0.13    0.76 ^ _15_/Y (sky130_fd_sc_hd__a21oi_1)
     1    0.00                           _07_ (net)
                  0.08    0.00    0.76 ^ hold4/A (sky130_fd_sc_hd__dlygate4sd3_1)
                  0.06    0.52    1.29 ^ hold4/X (sky130_fd_sc_hd__dlygate4sd3_1)
     1    0.00                           net10 (net)
                  0.06    0.00    1.29 ^ _16_/A2 (sky130_fd_sc_hd__o21ai_1)
                  0.03    0.06    1.34 v _16_/Y (sky130_fd_sc_hd__o21ai_1)
```

```
1    0.00                           _00_ (net)
             0.03    0.00    1.34 v _20_/D (sky130_fd_sc_hd__dfstp_1)
                             1.34   data arrival time

                     0.00    0.00   clock clk (rise edge)
                     0.00    0.00   clock source latency
```

==========

```
                  0.06    0.04    0.04 ^ clk (in)
     1    0.01                           clk (net)
                  0.06    0.00    0.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.04    0.14    0.18 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                  0.04    0.00    0.18 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.03    0.12    0.30 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                  0.03    0.00    0.30 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_1)
                          0.25    0.55   clock uncertainty
                         -0.03    0.52   clock reconvergence pessimism
                         -0.01    0.50   library hold time
                                  0.50   data required time
-----------------------------------------------------------------------------
                                  0.50   data required time
                                 -1.34   data arrival time
-----------------------------------------------------------------------------
                                  0.84   slack (MET)


Startpoint: rst_n (input port clocked by clk)
Endpoint: _21_ (removal check against rising-edge clock clk)
Path Group: asynchronous
Path Type: min

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                  0.02    0.01    4.01 ^ rst_n (in)
     1    0.00                           rst_n (net)
                  0.02    0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                  0.19    0.17    4.18 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.02                           net4 (net)
                  0.19    0.00    4.18 ^ _21_/RESET_B (sky130_fd_sc_hd__dfrtp_1)
                                  4.18   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                  0.06    0.04    0.04 ^ clk (in)
     1    0.01                           clk (net)
                  0.06    0.00    0.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
```

```
                0.04    0.14    0.18 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                       clknet_0_clk (net)
                0.04    0.00    0.18 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                0.03    0.12    0.30 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                       clknet_1_0__leaf_clk (net)
```

==========

```
                  0.03    0.00    0.30 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_1)
                          0.25    0.55   clock uncertainty
                          0.00    0.55   clock reconvergence pessimism
                          0.36    0.90   library removal time
                                  0.90   data required time
-----------------------------------------------------------------------------
                                  0.90   data required time
                                 -4.18   data arrival time
-----------------------------------------------------------------------------
                                  3.27   slack (MET)


Startpoint: rst_n (input port clocked by clk)
Endpoint: _22_ (removal check against rising-edge clock clk)
Path Group: asynchronous
Path Type: min

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                  0.02    0.01    4.01 ^ rst_n (in)
     1    0.00                           rst_n (net)
                  0.02    0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                  0.19    0.17    4.18 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.02                           net4 (net)
                  0.19    0.00    4.18 ^ _22_/RESET_B (sky130_fd_sc_hd__dfrtp_1)
                                  4.18   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                  0.06    0.04    0.04 ^ clk (in)
     1    0.01                           clk (net)
                  0.06    0.00    0.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.04    0.14    0.18 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                  0.04    0.00    0.18 ^ clkbuf_1_1__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.03    0.11    0.29 ^ clkbuf_1_1__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     1    0.00                           clknet_1_1__leaf_clk (net)
                  0.03    0.00    0.29 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_1)
                          0.25    0.54   clock uncertainty
                          0.00    0.54   clock reconvergence pessimism
```

```
                          0.36    0.90   library removal time
                                  0.90   data required time
-----------------------------------------------------------------------------
                                  0.90   data required time
                                 -4.18   data arrival time
```

==========

```
--------------------------------------------------------------------------
                                3.28   slack (MET)


Startpoint: rst_n (input port clocked by clk)
Endpoint: _20_ (removal check against rising-edge clock clk)
Path Group: asynchronous
Path Type: min

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                  0.02    0.01    4.01 ^ rst_n (in)
     1    0.00                           rst_n (net)
                  0.02    0.00    4.01 ^ input4/A (sky130_fd_sc_hd__buf_1)
                  0.19    0.17    4.18 ^ input4/X (sky130_fd_sc_hd__buf_1)
     3    0.02                           net4 (net)
                  0.19    0.00    4.18 ^ _20_/SET_B (sky130_fd_sc_hd__dfstp_1)
                                  4.18   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                  0.06    0.04    0.04 ^ clk (in)
     1    0.01                           clk (net)
                  0.06    0.00    0.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.04    0.14    0.18 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                           clknet_0_clk (net)
                  0.04    0.00    0.18 ^ clkbuf_1_0__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.03    0.12    0.30 ^ clkbuf_1_0__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     2    0.01                           clknet_1_0__leaf_clk (net)
                  0.03    0.00    0.30 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_1)
                          0.25    0.55   clock uncertainty
                          0.00    0.55   clock reconvergence pessimism
                          0.14    0.68   library removal time
                                  0.68   data required time
---------------------------------------------------------------------------
                                  0.68   data required time
                                 -4.18   data arrival time
---------------------------------------------------------------------------
                                  3.49   slack (MET)
```

```
Startpoint: _22_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: pump_en (output port clocked by clk)
Path Group: clk
Path Type: min
```

==========

```
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock source latency
                  0.06    0.04    0.04 ^ clk (in)
     1    0.01                          clk (net)
                  0.06    0.00    0.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
                  0.04    0.12    0.16 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
     2    0.02                          clknet_0_clk (net)
                  0.04    0.00    0.16 ^ clkbuf_1_1__f_clk/A
(sky130_fd_sc_hd__clkbuf_16)
                  0.03    0.10    0.26 ^ clkbuf_1_1__f_clk/X
(sky130_fd_sc_hd__clkbuf_16)
     1    0.00                          clknet_1_1__leaf_clk (net)
                  0.03    0.00    0.26 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_1)
                  0.06    0.30    0.57 ^ _22_/Q (sky130_fd_sc_hd__dfrtp_1)
     2    0.00                          state[2] (net)
                  0.06    0.00    0.57 ^ _18_/A (sky130_fd_sc_hd__and2_1)
                  0.04    0.11    0.67 ^ _18_/X (sky130_fd_sc_hd__and2_1)
     1    0.00                          _08_ (net)
                  0.04    0.00    0.67 ^ _19_/A (sky130_fd_sc_hd__clkbuf_1)
                  0.05    0.08    0.75 ^ _19_/X (sky130_fd_sc_hd__clkbuf_1)
     1    0.00                          net6 (net)
                  0.05    0.00    0.75 ^ output6/A (sky130_fd_sc_hd__clkbuf_4)
                  0.11    0.18    0.93 ^ output6/X (sky130_fd_sc_hd__clkbuf_4)
     1    0.03                          pump_en (net)
                  0.11    0.00    0.93 ^ pump_en (out)
                                  0.93   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          0.25    0.25   clock uncertainty
                          0.00    0.25   clock reconvergence pessimism
                         -4.00   -3.75   output external delay
                                 -3.75   data required time
---------------------------------------------------------------------------
                                 -3.75   data required time
                                 -0.93   data arrival time
---------------------------------------------------------------------------
                                  4.68   slack (MET)


Startpoint: dry_run_fault (input port clocked by clk)
Endpoint: alarm (output port clocked by clk)
```

Path Group: clk
Path Type: min

Fanout	Cap	Slew	Delay	Time	Description

```
                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          4.00    4.00 ^ input external delay
                 0.02     0.01    4.01 ^ dry_run_fault (in)
     1    0.00                           dry_run_fault (net)
                 0.02     0.00    4.01 ^ input1/A (sky130_fd_sc_hd__clkbuf_1)
                 0.07     0.09    4.10 ^ input1/X (sky130_fd_sc_hd__clkbuf_1)
     1    0.01                           net1 (net)
                 0.07     0.00    4.10 ^ _09_/B1 (sky130_fd_sc_hd__a21oi_2)
                 0.05     0.06    4.15 v _09_/Y (sky130_fd_sc_hd__a21oi_2)
     4    0.02                           _03_ (net)
                 0.05     0.00    4.15 v _10_/A (sky130_fd_sc_hd__inv_2)
                 0.04     0.05    4.20 ^ _10_/Y (sky130_fd_sc_hd__inv_2)
     2    0.01                           net5 (net)
                 0.04     0.00    4.20 ^ output5/A (sky130_fd_sc_hd__buf_2)
                 0.17     0.19    4.39 ^ output5/X (sky130_fd_sc_hd__buf_2)
     1    0.03                           alarm (net)
                 0.17     0.00    4.39 ^ alarm (out)
                                  4.39   data arrival time

                          0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (propagated)
                          0.25    0.25   clock uncertainty
                          0.00    0.25   clock reconvergence pessimism
                         -4.00   -3.75   output external delay
                                 -3.75   data required time
-------------------------------------------------------------------------------
                                 -3.75   data required time
                                 -4.39   data arrival time
-------------------------------------------------------------------------------
                                  8.14   slack (MET)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/31-rcx_sta.power.rpt
==

```
===========================================================================
 report_power
===========================================================================
======================== Typical Corner ===================================

Group                  Internal  Switching    Leakage      Total
                          Power      Power      Power      Power (Watts)
----------------------------------------------------------------
Sequential             6.60e-06   2.26e-07   3.33e-11   6.83e-06  16.5%
Combinational          2.74e-05   7.08e-06   2.90e-10   3.45e-05  83.5%
Macro                  0.00e+00   0.00e+00   0.00e+00   0.00e+00   0.0%
Pad                    0.00e+00   0.00e+00   0.00e+00   0.00e+00   0.0%
----------------------------------------------------------------
Total                  3.40e-05   7.31e-06   3.24e-10   4.13e-05 100.0%
                          82.3%      17.7%       0.0%
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/31-rcx_sta.skew.rpt
==

==
report_clock_skew
==
Clock clk
Latency   CRPR   Skew
_20_/CLK ^
   0.30
_22_/CLK ^
   0.26   -0.02   0.02

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/31-rcx_sta.summary.rpt
==

==
report_tns
==
tns 0.00

==
report_wns
==
wns 0.00

==
report_worst_slack -max (Setup)
==
worst slack 10.85

==
report_worst_slack -min (Hold)
==
worst slack 0.36

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/39-water_tank_controller.lvs.rpt
==

LVS reports no net, device, pin, or property mismatches.

Total errors = 0

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/signoff/drc.rpt
==

water_tank_controller

[INFO]: COUNT: 0
[INFO]: Should be divided by 3 or 4

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/synthesis/1-synthesis.AREA_0.stat.rpt
==

61. Printing statistics.

=== water_tank_controller ===

   Number of wires: 19
   Number of wire bits: 19
   Number of public wires: 10
   Number of public wire bits: 10
   Number of memories: 0
   Number of memory bits: 0
   Number of processes: 0
   Number of cells: 14
     sky130_fd_sc_hd__a21oi_2   3
     sky130_fd_sc_hd__and2_2   1
     sky130_fd_sc_hd__buf_1   1
     sky130_fd_sc_hd__dfrtp_2   2
     sky130_fd_sc_hd__dfstp_2   1
     sky130_fd_sc_hd__inv_2   2
     sky130_fd_sc_hd__nor2_2   1
     sky130_fd_sc_hd__o211a_2   1
     sky130_fd_sc_hd__o21a_2   1
     sky130_fd_sc_hd__o21ai_2   1

   Chip area for module ’\water_tank_controller’: 157.651200

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/synthesis/2-syn_sta.checks.rpt ? part 1
==

===
report_checks -unconstrained
===
======================== Typical Corner ===================================

Startpoint: rst_n (input port clocked by clk)
Endpoint: _20_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
Path Type: max

```
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          4.00    4.00 ^ input external delay
                  0.06    0.04    4.04 ^ rst_n (in)
     3    0.01                          rst_n (net)
                  0.06    0.00    4.04 ^ _20_/SET_B (sky130_fd_sc_hd__dfstp_2)
                                  4.04   data arrival time

                  0.15   20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock network delay (ideal)
                         -0.25   19.75   clock uncertainty
                          0.00   19.75   clock reconvergence pessimism
                                 19.75 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_2)
                          0.16   19.91   library recovery time
                                 19.91   data required time
---------------------------------------------------------------------------
                                 19.91   data required time
                                 -4.04   data arrival time
---------------------------------------------------------------------------
                                 15.88   slack (MET)


Startpoint: level_full (input port clocked by clk)
Endpoint: pump_en (output port clocked by clk)
Path Group: clk
Path Type: max

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          4.00    4.00 v input external delay
                  0.03    0.02    4.02 v level_full (in)
```

```
3    0.01                           level_full (net)
             0.03    0.00    4.02 v _09_/A2 (sky130_fd_sc_hd__a21oi_2)
             0.19    0.20    4.22 ^ _09_/Y (sky130_fd_sc_hd__a21oi_2)
```

```
================================================================================

     4    0.01                           _03_ (net)
                  0.19    0.00    4.22 ^ _18_/B (sky130_fd_sc_hd__and2_2)
                  0.04    0.17    4.39 ^ _18_/X (sky130_fd_sc_hd__and2_2)
     1    0.00                           _08_ (net)
                  0.04    0.00    4.39 ^ _19_/A (sky130_fd_sc_hd__buf_1)
                  0.40    0.34    4.73 ^ _19_/X (sky130_fd_sc_hd__buf_1)
     1    0.03                           pump_en (net)
                  0.40    0.00    4.73 ^ pump_en (out)
                                  4.73   data arrival time

                  0.15   20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock network delay (ideal)
                         -0.25   19.75   clock uncertainty
                          0.00   19.75   clock reconvergence pessimism
                         -4.00   15.75   output external delay
                                 15.75   data required time
---------------------------------------------------------------------------
                                 15.75   data required time
                                 -4.73   data arrival time
---------------------------------------------------------------------------
                                 11.02   slack (MET)




===========================================================================
report_checks --slack_max -0.01
===========================================================================
====================== Typical Corner ===================================

No paths found.


===========================================================================
 report_check_types -max_slew -max_cap -max_fanout -violators
===========================================================================
====================== Typical Corner ===================================



===========================================================================
report_parasitic_annotation -report_unannotated
===========================================================================
Found 19 unannotated drivers.
 clk
```

```
dry_run_fault
level_full
level_low
```

```
================================================================================

 rst_n
 _09_/Y
 _10_/Y
 _11_/Y
 _12_/X
 _13_/Y
 _14_/Y
 _15_/Y
 _16_/Y
 _17_/X
 _18_/X
 _19_/X
 _20_/Q
 _21_/Q
 _22_/Q
Found 0 partially unannotated drivers.

================================================================================
max slew violation count 0
max fanout violation count 0
max cap violation count 0
================================================================================

================================================================================
check_setup -verbose -unconstrained_endpoints -multiple_clock -no_clock -no_input_delay
-loops -generated_clocks
================================================================================
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/synthesis/2-syn_sta.max.rpt ? part 1
==

==
report_checks -path_delay max (Setup)
==
======================== Typical Corner ===================================

Startpoint: level_full (input port clocked by clk)
Endpoint: pump_en (output port clocked by clk)
Path Group: clk
Path Type: max

Fanout	Cap	Slew	Delay	Time	Description
		0.15	0.00	0.00	clock clk (rise edge)
			0.00	0.00	clock network delay (ideal)
			4.00	4.00 v	input external delay
		0.03	0.02	4.02 v	level_full (in)
3	0.01				level_full (net)
		0.03	0.00	4.02 v	_09_/A2 (sky130_fd_sc_hd__a21oi_2)
		0.19	0.20	4.22 ^	_09_/Y (sky130_fd_sc_hd__a21oi_2)
4	0.01				_03_ (net)
		0.19	0.00	4.22 ^	_18_/B (sky130_fd_sc_hd__and2_2)
		0.04	0.17	4.39 ^	_18_/X (sky130_fd_sc_hd__and2_2)
1	0.00				_08_ (net)
		0.04	0.00	4.39 ^	_19_/A (sky130_fd_sc_hd__buf_1)
		0.40	0.34	4.73 ^	_19_/X (sky130_fd_sc_hd__buf_1)
1	0.03				pump_en (net)
		0.40	0.00	4.73 ^	pump_en (out)
				4.73	data arrival time
		0.15	20.00	20.00	clock clk (rise edge)
			0.00	20.00	clock network delay (ideal)
			-0.25	19.75	clock uncertainty
			0.00	19.75	clock reconvergence pessimism
			-4.00	15.75	output external delay
				15.75	data required time
				15.75	data required time
				-4.73	data arrival time
				11.02	slack (MET)

Startpoint: level_full (input port clocked by clk)
Endpoint: alarm (output port clocked by clk)
Path Group: clk

Path Type: max

Fanout	Cap	Slew	Delay	Time	Description

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/synthesis/2-syn_sta.max.rpt ? part 2
==========

```
---------------------------------------------------------------------------
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          4.00    4.00 v input external delay
                  0.03    0.02    4.02 v level_full (in)
     3    0.01                           level_full (net)
                  0.03    0.00    4.02 v _09_/A2 (sky130_fd_sc_hd__a21oi_2)
                  0.19    0.20    4.22 ^ _09_/Y (sky130_fd_sc_hd__a21oi_2)
     4    0.01                           _03_ (net)
                  0.19    0.00    4.22 ^ _10_/A (sky130_fd_sc_hd__inv_2)
                  0.10    0.14    4.37 v _10_/Y (sky130_fd_sc_hd__inv_2)
     2    0.04                           alarm (net)
                  0.10    0.00    4.37 v alarm (out)
                                  4.37   data arrival time

                  0.15   20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock network delay (ideal)
                         -0.25   19.75   clock uncertainty
                          0.00   19.75   clock reconvergence pessimism
                         -4.00   15.75   output external delay
                                 15.75   data required time
---------------------------------------------------------------------------
                                 15.75   data required time
                                 -4.37   data arrival time
---------------------------------------------------------------------------
                                 11.38   slack (MET)


Startpoint: level_full (input port clocked by clk)
Endpoint: _22_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          4.00    4.00 v input external delay
                  0.03    0.02    4.02 v level_full (in)
     3    0.01                           level_full (net)
                  0.03    0.00    4.02 v _09_/A2 (sky130_fd_sc_hd__a21oi_2)
                  0.19    0.20    4.22 ^ _09_/Y (sky130_fd_sc_hd__a21oi_2)
     4    0.01                           _03_ (net)
                  0.19    0.00    4.22 ^ _10_/A (sky130_fd_sc_hd__inv_2)
                  0.10    0.14    4.37 v _10_/Y (sky130_fd_sc_hd__inv_2)
```

```
2    0.04                           alarm (net)
             0.10    0.00    4.37 v _14_/A (sky130_fd_sc_hd__nor2_2)
             0.06    0.12    4.48 ^ _14_/Y (sky130_fd_sc_hd__nor2_2)
```

==========

```
     1    0.00                           _02_ (net)
                  0.06    0.00    4.48 ^ _22_/D (sky130_fd_sc_hd__dfrtp_2)
                                  4.48   data arrival time

                  0.15   20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock network delay (ideal)
                         -0.25   19.75   clock uncertainty
                          0.00   19.75   clock reconvergence pessimism
                                 19.75 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_2)
                         -0.05   19.70   library setup time
                                 19.70   data required time
-----------------------------------------------------------------------------
                                 19.70   data required time
                                 -4.48   data arrival time
-----------------------------------------------------------------------------
                                 15.22   slack (MET)


Startpoint: level_full (input port clocked by clk)
Endpoint: _21_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          4.00    4.00 v input external delay
                  0.03    0.02    4.02 v level_full (in)
     3    0.01                           level_full (net)
                  0.03    0.00    4.02 v _09_/A2 (sky130_fd_sc_hd__a21oi_2)
                  0.19    0.20    4.22 ^ _09_/Y (sky130_fd_sc_hd__a21oi_2)
     4    0.01                           _03_ (net)
                  0.19    0.00    4.22 ^ _17_/B1 (sky130_fd_sc_hd__o211a_2)
                  0.04    0.21    4.43 ^ _17_/X (sky130_fd_sc_hd__o211a_2)
     1    0.00                           _01_ (net)
                  0.04    0.00    4.43 ^ _21_/D (sky130_fd_sc_hd__dfrtp_2)
                                  4.43   data arrival time

                  0.15   20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock network delay (ideal)
                         -0.25   19.75   clock uncertainty
                          0.00   19.75   clock reconvergence pessimism
                                 19.75 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_2)
                         -0.05   19.70   library setup time
```

```
                                  19.70   data required time
-----------------------------------------------------------------------------
                                  19.70   data required time
```

```
                                -4.43   data arrival time
-----------------------------------------------------------------------------
                                15.27   slack (MET)


Startpoint: level_full (input port clocked by clk)
Endpoint: _20_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          4.00    4.00 v input external delay
                  0.03    0.02    4.02 v level_full (in)
     3    0.01                           level_full (net)
                  0.03    0.00    4.02 v _09_/A2 (sky130_fd_sc_hd__a21oi_2)
                  0.19    0.20    4.22 ^ _09_/Y (sky130_fd_sc_hd__a21oi_2)
     4    0.01                           _03_ (net)
                  0.19    0.00    4.22 ^ _16_/B1 (sky130_fd_sc_hd__o21ai_2)
                  0.05    0.07    4.30 v _16_/Y (sky130_fd_sc_hd__o21ai_2)
     1    0.00                           _00_ (net)
                  0.05    0.00    4.30 v _20_/D (sky130_fd_sc_hd__dfstp_2)
                                  4.30   data arrival time

                  0.15   20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock network delay (ideal)
                         -0.25   19.75   clock uncertainty
                          0.00   19.75   clock reconvergence pessimism
                                 19.75 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_2)
                         -0.05   19.70   library setup time
                                 19.70   data required time
-----------------------------------------------------------------------------
                                19.70   data required time
                                -4.30   data arrival time
-----------------------------------------------------------------------------
                                15.40   slack (MET)


Startpoint: rst_n (input port clocked by clk)
Endpoint: _20_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
Path Type: max
```

Fanout	Cap	Slew	Delay	Time	Description
		0.15	0.00	0.00	clock clk (rise edge)

==========

```
                          0.00    0.00   clock network delay (ideal)
                          4.00    4.00 ^ input external delay
                 0.06     0.04    4.04 ^ rst_n (in)
     3    0.01                           rst_n (net)
                 0.06     0.00    4.04 ^ _20_/SET_B (sky130_fd_sc_hd__dfstp_2)
                                  4.04   data arrival time

                 0.15    20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock network delay (ideal)
                         -0.25   19.75   clock uncertainty
                          0.00   19.75   clock reconvergence pessimism
                                 19.75 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_2)
                          0.16   19.91   library recovery time
                                 19.91   data required time
-----------------------------------------------------------------------------
                                 19.91   data required time
                                 -4.04   data arrival time
-----------------------------------------------------------------------------
                                 15.88   slack (MET)


Startpoint: rst_n (input port clocked by clk)
Endpoint: _21_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
Path Type: max

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                 0.15     0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          4.00    4.00 ^ input external delay
                 0.06     0.04    4.04 ^ rst_n (in)
     3    0.01                           rst_n (net)
                 0.06     0.00    4.04 ^ _21_/RESET_B (sky130_fd_sc_hd__dfrtp_2)
                                  4.04   data arrival time

                 0.15    20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock network delay (ideal)
                         -0.25   19.75   clock uncertainty
                          0.00   19.75   clock reconvergence pessimism
                                 19.75 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_2)
                          0.25   20.00   library recovery time
                                 20.00   data required time
-----------------------------------------------------------------------------
                                 20.00   data required time
```

```
                                  -4.04   data arrival time
-----------------------------------------------------------------------------
                                  15.96   slack (MET)
```

```
Startpoint: rst_n (input port clocked by clk)
Endpoint: _22_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
Path Type: max

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          4.00    4.00 ^ input external delay
                  0.06    0.04    4.04 ^ rst_n (in)
     3    0.01                           rst_n (net)
                  0.06    0.00    4.04 ^ _22_/RESET_B (sky130_fd_sc_hd__dfrtp_2)
                                  4.04   data arrival time

                  0.15   20.00   20.00   clock clk (rise edge)
                          0.00   20.00   clock network delay (ideal)
                         -0.25   19.75   clock uncertainty
                          0.00   19.75   clock reconvergence pessimism
                                 19.75 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_2)
                          0.25   20.00   library recovery time
                                 20.00   data required time
---------------------------------------------------------------------------
                                 20.00   data required time
                                 -4.04   data arrival time
---------------------------------------------------------------------------
                                 15.96   slack (MET)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/synthesis/2-syn_sta.min.rpt ? part 1
==

==
report_checks -path_delay min (Hold)
==
======================== Typical Corner ===================================

Startpoint: _20_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _20_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

```
Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.15    0.00    0.00 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_2)
                  0.03    0.38    0.38 v _20_/Q (sky130_fd_sc_hd__dfstp_2)
     2    0.01                           state[0] (net)
                  0.03    0.00    0.38 v _15_/B1 (sky130_fd_sc_hd__a21oi_2)
                  0.09    0.10    0.48 ^ _15_/Y (sky130_fd_sc_hd__a21oi_2)
     1    0.00                           _07_ (net)
                  0.09    0.00    0.48 ^ _16_/A2 (sky130_fd_sc_hd__o21ai_2)
                  0.03    0.05    0.52 v _16_/Y (sky130_fd_sc_hd__o21ai_2)
     1    0.00                           _00_ (net)
                  0.03    0.00    0.52 v _20_/D (sky130_fd_sc_hd__dfstp_2)
                                  0.52   data arrival time

                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          0.25    0.25   clock uncertainty
                          0.00    0.25   clock reconvergence pessimism
                                  0.25 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_2)
                          0.02    0.27   library hold time
                                  0.27   data required time
-----------------------------------------------------------------------------
                                  0.27   data required time
                                 -0.52   data arrival time
-----------------------------------------------------------------------------
                                  0.25   slack (MET)


Startpoint: _22_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _22_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
```

Fanout	Cap	Slew	Delay	Time	Description
		0.15	0.00	0.00	clock clk (rise edge)

```
                          0.00    0.00   clock network delay (ideal)
                 0.15     0.00    0.00 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_2)
                 0.07     0.38    0.38 ^ _22_/Q (sky130_fd_sc_hd__dfrtp_2)
     3    0.01                           state[2] (net)
                 0.07     0.00    0.38 ^ _13_/A2 (sky130_fd_sc_hd__a21oi_2)
                 0.02     0.05    0.43 v _13_/Y (sky130_fd_sc_hd__a21oi_2)
     1    0.00                           _06_ (net)
                 0.02     0.00    0.43 v _14_/B (sky130_fd_sc_hd__nor2_2)
                 0.05     0.05    0.48 ^ _14_/Y (sky130_fd_sc_hd__nor2_2)
     1    0.00                           _02_ (net)
                 0.05     0.00    0.48 ^ _22_/D (sky130_fd_sc_hd__dfrtp_2)
                                  0.48   data arrival time

                 0.15     0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          0.25    0.25   clock uncertainty
                          0.00    0.25   clock reconvergence pessimism
                                  0.25 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_2)
                         -0.02    0.23   library hold time
                                  0.23   data required time
-----------------------------------------------------------------------------
                                  0.23   data required time
                                 -0.48   data arrival time
-----------------------------------------------------------------------------
                                  0.25   slack (MET)


Startpoint: _22_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _21_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                 0.15     0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                 0.15     0.00    0.00 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_2)
                 0.07     0.38    0.38 ^ _22_/Q (sky130_fd_sc_hd__dfrtp_2)
     3    0.01                           state[2] (net)
                 0.07     0.00    0.38 ^ _17_/A2 (sky130_fd_sc_hd__o211a_2)
                 0.04     0.15    0.52 ^ _17_/X (sky130_fd_sc_hd__o211a_2)
     1    0.00                           _01_ (net)
                 0.04     0.00    0.52 ^ _21_/D (sky130_fd_sc_hd__dfrtp_2)
                                  0.52   data arrival time
```

```
0.15    0.00    0.00   clock clk (rise edge)
        0.00    0.00   clock network delay (ideal)
        0.25    0.25   clock uncertainty
```

```
                          0.00    0.25   clock reconvergence pessimism
                                  0.25 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_2)
                         -0.02    0.23   library hold time
                                  0.23   data required time
-----------------------------------------------------------------------------
                                  0.23   data required time
                                 -0.52   data arrival time
-----------------------------------------------------------------------------
                                  0.29   slack (MET)


Startpoint: rst_n (input port clocked by clk)
Endpoint: _21_ (removal check against rising-edge clock clk)
Path Group: asynchronous
Path Type: min

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          4.00    4.00 ^ input external delay
                  0.06    0.03    4.03 ^ rst_n (in)
     3    0.01                           rst_n (net)
                  0.06    0.00    4.03 ^ _21_/RESET_B (sky130_fd_sc_hd__dfrtp_2)
                                  4.03   data arrival time

                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          0.25    0.25   clock uncertainty
                          0.00    0.25   clock reconvergence pessimism
                                  0.25 ^ _21_/CLK (sky130_fd_sc_hd__dfrtp_2)
                          0.35    0.60   library removal time
                                  0.60   data required time
-----------------------------------------------------------------------------
                                  0.60   data required time
                                 -4.03   data arrival time
-----------------------------------------------------------------------------
                                  3.43   slack (MET)


Startpoint: rst_n (input port clocked by clk)
Endpoint: _22_ (removal check against rising-edge clock clk)
Path Group: asynchronous
Path Type: min
```

Fanout	Cap	Slew	Delay	Time	Description
		0.15	0.00	0.00	clock clk (rise edge)

```
==========================================================================================

                          0.00    0.00   clock network delay (ideal)
                          4.00    4.00 ^ input external delay
                  0.06    0.03    4.03 ^ rst_n (in)
     3    0.01                          rst_n (net)
                  0.06    0.00    4.03 ^ _22_/RESET_B (sky130_fd_sc_hd__dfrtp_2)
                                  4.03   data arrival time

                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          0.25    0.25   clock uncertainty
                          0.00    0.25   clock reconvergence pessimism
                                  0.25 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_2)
                          0.35    0.60   library removal time
                                  0.60   data required time
------------------------------------------------------------------------------
                                  0.60   data required time
                                 -4.03   data arrival time
------------------------------------------------------------------------------
                                  3.43   slack (MET)


Startpoint: rst_n (input port clocked by clk)
Endpoint: _20_ (removal check against rising-edge clock clk)
Path Group: asynchronous
Path Type: min

Fanout     Cap    Slew   Delay    Time   Description
------------------------------------------------------------------------------
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          4.00    4.00 ^ input external delay
                  0.06    0.03    4.03 ^ rst_n (in)
     3    0.01                          rst_n (net)
                  0.06    0.00    4.03 ^ _20_/SET_B (sky130_fd_sc_hd__dfstp_2)
                                  4.03   data arrival time

                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          0.25    0.25   clock uncertainty
                          0.00    0.25   clock reconvergence pessimism
                                  0.25 ^ _20_/CLK (sky130_fd_sc_hd__dfstp_2)
                          0.18    0.43   library removal time
                                  0.43   data required time
------------------------------------------------------------------------------
                                  0.43   data required time
```

```
                                  -4.03   data arrival time
-----------------------------------------------------------------------------
                                   3.60   slack (MET)
```

```
Startpoint: _22_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: pump_en (output port clocked by clk)
Path Group: clk
Path Type: min

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.15    0.00    0.00 ^ _22_/CLK (sky130_fd_sc_hd__dfrtp_2)
                  0.05    0.42    0.42 v _22_/Q (sky130_fd_sc_hd__dfrtp_2)
     3    0.01                           state[2] (net)
                  0.05    0.00    0.42 v _18_/A (sky130_fd_sc_hd__and2_2)
                  0.03    0.15    0.57 v _18_/X (sky130_fd_sc_hd__and2_2)
     1    0.00                           _08_ (net)
                  0.03    0.00    0.57 v _19_/A (sky130_fd_sc_hd__buf_1)
                  0.19    0.20    0.77 v _19_/X (sky130_fd_sc_hd__buf_1)
     1    0.03                           pump_en (net)
                  0.19    0.00    0.77 v pump_en (out)
                                  0.77   data arrival time

                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          0.25    0.25   clock uncertainty
                          0.00    0.25   clock reconvergence pessimism
                         -4.00   -3.75   output external delay
                                 -3.75   data required time
---------------------------------------------------------------------------
                                 -3.75   data required time
                                 -0.77   data arrival time
---------------------------------------------------------------------------
                                  4.52   slack (MET)


Startpoint: dry_run_fault (input port clocked by clk)
Endpoint: alarm (output port clocked by clk)
Path Group: clk
Path Type: min

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
```

```
                    4.00    4.00 ^ input external delay
              0.03    0.02    4.02 ^ dry_run_fault (in)
1     0.00                         dry_run_fault (net)
```

```
                0.03    0.00    4.02 ^ _09_/B1 (sky130_fd_sc_hd__a21oi_2)
                0.04    0.04    4.05 v _09_/Y (sky130_fd_sc_hd__a21oi_2)
     4    0.01                        _03_ (net)
                0.04    0.00    4.05 v _10_/A (sky130_fd_sc_hd__inv_2)
                0.18    0.14    4.19 ^ _10_/Y (sky130_fd_sc_hd__inv_2)
     2    0.04                        alarm (net)
                0.18    0.00    4.19 ^ alarm (out)
                                4.19   data arrival time

                0.15    0.00    0.00   clock clk (rise edge)
                        0.00    0.00   clock network delay (ideal)
                        0.25    0.25   clock uncertainty
                        0.00    0.25   clock reconvergence pessimism
                       -4.00   -3.75   output external delay
                               -3.75   data required time
-----------------------------------------------------------------------------
                               -3.75   data required time
                               -4.19   data arrival time
-----------------------------------------------------------------------------
                                7.94   slack (MET)
```

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/synthesis/2-syn_sta.power.rpt
==

==
 report_power
==
======================== Typical Corner ===================================

Group	Internal Power	Switching Power	Leakage Power	Total Power (Watts)	
Sequential	6.64e-06	2.91e-07	3.18e-11	6.94e-06	75.5%
Combinational	9.22e-07	1.33e-06	2.86e-11	2.25e-06	24.5%
Macro	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Pad	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Total	7.57e-06	1.62e-06	6.04e-11	9.19e-06	100.0%
	82.4%	17.6%	0.0%		

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/synthesis/2-syn_sta.skew.rpt
==

==
report_clock_skew
==
Clock clk
Latency   CRPR   Skew
_20_/CLK ^
   0.02
_20_/CLK ^
   0.02   0.00   0.00

openlane/water_tank_controller/runs/RUN_2026.10.10_02.11.36/reports/synthesis/2-syn_sta.summary.rpt
==

==
report_tns
==
tns 0.00

==
report_wns
==
wns 0.00

==
report_worst_slack -max (Setup)
==
worst slack 11.02

==
report_worst_slack -min (Hold)
==
worst slack 0.25

Corner interpretation

MAX and MIN timing analyses are analysis modes, not by themselves proof of SS and FF process corners. The installed timing libraries and timing configuration must confirm process corners before assigning labels.

Setup and hold interpretation

Setup and hold slack must be interpreted using the exact constraints and timing corner for each report. A nonnegative slack indicates no violation in that reported analysis; it does not prove all corners pass.

Physical design artifacts

The final GDS and the available DRC/LVS reports are separate physical-design artifacts.
This document does not replace the detailed signoff reports.

Submission checklist

Include this PDF, original OpenLane timing reports, DRC/LVS reports, the final GDSII, and a KLayout screenshot. Verify that each claimed timing corner has a matching source report.